



IEEE Standard for Ethernet

Amendment 4: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Electrical Interfaces Based on 100 Gb/s Signaling

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3ck™-2022
(Amendment to IEEE Std 802.3™-2022
as amended by IEEE Std 802.3dd™-2022,
IEEE Std 802.3cs™-2022,
and IEEE Std 802.3db™-2022)

IEEE Standard for Ethernet

Amendment 4: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Electrical Interfaces Based on 100 Gb/s Signaling

Developed by the
LAN/MAN Standards Committee
of the
IEEE Computer Society

Approved 21 September 2022
IEEE SA Standards Board

Abstract: This amendment to IEEE Std 802.3-2022 adds Clause 161 through Clause 163, Annex 120F, Annex 120G, and Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

Keywords: 100GAUI-1, 100GBASE-CR1, 100GBASE-KR1, 100 Gigabit Ethernet, 200GAUI-2, 200GBASE-CR2, 200GBASE-KR2, 200 Gigabit Ethernet, 400GAUI-4, 400GBASE-CR4, 400GBASE-KR4, 400 Gigabit Ethernet, amendment, Ethernet, FEC, forward error correction, IEEE 802.3™, IEEE 802.3ck™

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

Copyright © 2022 by The Institute of Electrical and Electronics Engineers, Inc.
All rights reserved. Published 28 December 2022. Printed in the United States of America.

IEEE and 802 are registered trademarks in the U.S. Patent & Trademark Office, owned by The Institute of Electrical and Electronics Engineers, Incorporated.

PDF: ISBN 978-1-5044-9294-2 STD25851
Print: ISBN 978-1-5044-9295-9 STDPD25851

IEEE prohibits discrimination, harassment and bullying.

For more information, visit <http://www.ieee.org/web/aboutus/whatis/policies/p9-26.html>.

No part of this publication may be reproduced in any form, in an electronic retrieval system or otherwise, without the prior written permission of the publisher.

Important Notices and Disclaimers Concerning IEEE Standards Documents

IEEE Standards documents are made available for use subject to important notices and legal disclaimers. These notices and disclaimers, or a reference to this page (<https://standards.ieee.org/ipr/disclaimers.html>), appear in all standards and may be found under the heading “Important Notices and Disclaimers Concerning IEEE Standards Documents.”

Notice and Disclaimer of Liability Concerning the Use of IEEE Standards Documents

IEEE Standards documents are developed within IEEE Societies and subcommittees of IEEE Standards Association (IEEE SA) Board of Governors. IEEE develops its standards through an accredited consensus development process, which brings together volunteers representing varied viewpoints and interests to achieve the final product. IEEE Standards are documents developed by volunteers with scientific, academic, and industry-based expertise in technical working groups. Volunteers are not necessarily members of IEEE or IEEE SA and participate without compensation from IEEE. While IEEE administers the process and establishes rules to promote fairness in the consensus development process, IEEE does not independently evaluate, test, or verify the accuracy of any of the information or the soundness of any judgments contained in its standards.

IEEE makes no warranties or representations concerning its standards, and expressly disclaims all warranties, express or implied, concerning this standard, including but not limited to the warranties of merchantability, fitness for a particular purpose and non-infringement. In addition, IEEE does not warrant or represent that the use of the material contained in its standards is free from patent infringement. IEEE standards documents are supplied “AS IS” and “WITH ALL FAULTS.”

Use of an IEEE standard is wholly voluntary. The existence of an IEEE Standard does not imply that there are no other ways to produce, test, measure, purchase, market, or provide other goods and services related to the scope of the IEEE standard. Furthermore, the viewpoint expressed at the time a standard is approved and issued is subject to change brought about through developments in the state of the art and comments received from users of the standard.

In publishing and making its standards available, IEEE is not suggesting or rendering professional or other services for, or on behalf of, any person or entity, nor is IEEE undertaking to perform any duty owed by any other person or entity to another. Any person utilizing any IEEE Standards document, should rely upon his or her own independent judgment in the exercise of reasonable care in any given circumstances or, as appropriate, seek the advice of a competent professional in determining the appropriateness of a given IEEE standard.

IN NO EVENT SHALL IEEE BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO: THE NEED TO PROCURE SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE PUBLICATION, USE OF, OR RELIANCE UPON ANY STANDARD, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE AND REGARDLESS OF WHETHER SUCH DAMAGE WAS FORESEEABLE.

Translations

The IEEE consensus development process involves the review of documents in English only. In the event that an IEEE standard is translated, only the English version published by IEEE is the approved IEEE standard.

Official statements

A statement, written or oral, that is not processed in accordance with the IEEE SA Standards Board Operations Manual shall not be considered or inferred to be the official position of IEEE or any of its committees and shall not be considered to be, nor be relied upon as, a formal position of IEEE. At lectures, symposia, seminars, or educational courses, an individual presenting information on IEEE standards shall make it clear that the presenter's views should be considered the personal views of that individual rather than the formal position of IEEE, IEEE SA, the Standards Committee, or the Working Group. Statements made by volunteers may not represent the formal position of their employer(s) or affiliation(s).

Comments on standards

Comments for revision of IEEE Standards documents are welcome from any interested party, regardless of membership affiliation with IEEE or IEEE SA. However, **IEEE does not provide interpretations, consulting information, or advice pertaining to IEEE Standards documents.**

Suggestions for changes in documents should be in the form of a proposed change of text, together with appropriate supporting comments. Since IEEE standards represent a consensus of concerned interests, it is important that any responses to comments and questions also receive the concurrence of a balance of interests. For this reason, IEEE and the members of its Societies and subcommittees of the IEEE SA Board of Governors are not able to provide an instant response to comments, or questions except in those cases where the matter has previously been addressed. For the same reason, IEEE does not respond to interpretation requests. Any person who would like to participate in evaluating comments or in revisions to an IEEE standard is welcome to join the relevant IEEE working group. You can indicate interest in a working group using the Interests tab in the Manage Profile & Interests area of the [IEEE SA myProject system](#).¹ An IEEE Account is needed to access the application.

Comments on standards should be submitted using the [Contact Us](#) form.²

Laws and regulations

Users of IEEE Standards documents should consult all applicable laws and regulations. Compliance with the provisions of any IEEE Standards document does not constitute compliance to any applicable regulatory requirements. Implementers of the standard are responsible for observing or referring to the applicable regulatory requirements. IEEE does not, by the publication of its standards, intend to urge action that is not in compliance with applicable laws, and these documents may not be construed as doing so.

Data privacy

Users of IEEE Standards documents should evaluate the standards for considerations of data privacy and data ownership in the context of assessing and using the standards in compliance with applicable laws and regulations.

¹ Available at: <https://development.standards.ieee.org/myproject-web/public/view.html#landing>.

² Available at: <https://standards.ieee.org/content/ieee-standards/en/about/contact/index.html>.

Copyrights

IEEE draft and approved standards are copyrighted by IEEE under US and international copyright laws. They are made available by IEEE and are adopted for a wide variety of both public and private uses. These include both use, by reference, in laws and regulations, and use in private self-regulation, standardization, and the promotion of engineering practices and methods. By making these documents available for use and adoption by public authorities and private users, neither IEEE nor its licensors waive any rights in copyright to the documents.

Photocopies

Subject to payment of the appropriate licensing fees, IEEE will grant users a limited, non-exclusive license to photocopy portions of any individual standard for company or organizational internal use or individual, non-commercial use only. To arrange for payment of licensing fees, please contact Copyright Clearance Center, Customer Service, 222 Rosewood Drive, Danvers, MA 01923 USA; +1 978 750 8400; <https://www.copyright.com/>. Permission to photocopy portions of any individual standard for educational classroom use can also be obtained through the Copyright Clearance Center.

Updating of IEEE Standards documents

Users of IEEE Standards documents should be aware that these documents may be superseded at any time by the issuance of new editions or may be amended from time to time through the issuance of amendments, corrigenda, or errata. An official IEEE document at any point in time consists of the current edition of the document together with any amendments, corrigenda, or errata then in effect.

Every IEEE standard is subjected to review at least every 10 years. When a document is more than 10 years old and has not undergone a revision process, it is reasonable to conclude that its contents, although still of some value, do not wholly reflect the present state of the art. Users are cautioned to check to determine that they have the latest edition of any IEEE standard.

In order to determine whether a given document is the current edition and whether it has been amended through the issuance of amendments, corrigenda, or errata, visit [IEEE Xplore](#) or [contact IEEE](#).³ For more information about the IEEE SA or IEEE's standards development process, visit the IEEE SA Website.

Errata

Errata, if any, for all IEEE standards can be accessed on the [IEEE SA Website](#).⁴ Search for standard number and year of approval to access the web page of the published standard. Errata links are located under the Additional Resources Details section. Errata are also available in [IEEE Xplore](#). Users are encouraged to periodically check for errata.

Patents

IEEE Standards are developed in compliance with the [IEEE SA Patent Policy](#).⁵

Attention is called to the possibility that implementation of this standard may require use of subject matter covered by patent rights. By publication of this standard, no position is taken by the IEEE with respect to the existence or validity of any patent rights in connection therewith. If a patent holder or patent applicant has

³ Available at: <https://ieeexplore.ieee.org/browse/standards/collection/ieee>.

⁴ Available at: <https://standards.ieee.org/standard/index.html>.

⁵ Available at: <https://standards.ieee.org/about/sasb/patcom/materials.html>.

filed a statement of assurance via an Accepted Letter of Assurance, then the statement is listed on the IEEE SA Website at <https://standards.ieee.org/about/sasb/patcom/patents.html>. Letters of Assurance may indicate whether the Submitter is willing or unwilling to grant licenses under patent rights without compensation or under reasonable rates, with reasonable terms and conditions that are demonstrably free of any unfair discrimination to applicants desiring to obtain such licenses.

Essential Patent Claims may exist for which a Letter of Assurance has not been received. The IEEE is not responsible for identifying Essential Patent Claims for which a license may be required, for conducting inquiries into the legal validity or scope of Patents Claims, or determining whether any licensing terms or conditions provided in connection with submission of a Letter of Assurance, if any, or in any licensing agreements are reasonable or non-discriminatory. Users of this standard are expressly advised that determination of the validity of any patent rights, and the risk of infringement of such rights, is entirely their own responsibility. Further information may be obtained from the IEEE Standards Association.

IMPORTANT NOTICE

IEEE Standards do not guarantee or ensure safety, security, health, or environmental protection, or ensure against interference with or from other devices or networks. IEEE Standards development activities consider research and information presented to the standards development group in developing any safety recommendations. Other information about safety practices, changes in technology or technology implementation, or impact by peripheral systems also may be pertinent to safety considerations during implementation of the standard. Implementers and users of IEEE Standards documents are responsible for determining and complying with all appropriate safety, security, environmental, health, and interference protection practices and all applicable laws and regulations.

Participants

The following individuals were officers and members of the IEEE 802.3 Working Group at the beginning of the IEEE P802.3ck Working Group ballot.

David J. Law, *IEEE 802.3 Working Group Chair*

Adam Healey, *IEEE 802.3 Working Group Vice-Chair*

Jon Lewis, *IEEE 802.3 Working Group Secretary*

Steven B. Carlson, *IEEE 802.3 Working Group Executive Secretary*

Valerie Maguire, *IEEE 802.3 Working Group Treasurer*

Beth Kochuparambil, *IEEE P802.3ck Task Force Chair*

Kent Lusted, *IEEE P802.3ck Task Force Vice Chair*

Matt Brown, *IEEE P802.3ck Task Force Editor-in-Chief*

Chris Diminico, *IEEE P802.3ck Task Force Clause Editor*

Mark Gustlin, *IEEE P802.3ck Task Force Clause Editor*

Howard Heck, *IEEE P802.3ck Task Force Clause Editor*

Arthur Marris, *IEEE P802.3ck Task Force Clause Editor*

Phil Sun, *IEEE P802.3ck Task Force Clause Editor*

John Abbott
Rob Aekins
Stefan Andrae
Pete Anslow
Michikazu Aono
Nobuyasu Araki
Joseph Aronson
Tim Baggett
Thananya Baldwin
Denis Beaudoin
Liav Ben-Artzi
Vipul Bhatt
Mark Bordogna
Rich Boyer
Paul Brooks
Leon Bruckman
Michal Brychta
Jairo Bustos Heredia
Clark Carty
Bruce Champion
Frank Chang
Jacky Chang
Xin Chang
Chan Chen
Golam Choudhury
Keng Hua Chuang
Emilio Cuesta
John D'Ambrosia
Piers Dawe
Fred Dawson
John Deandrea
Mark Dearing
Claudio DeSanti
Eric DiBiaso
Kathryn Dube
Mike Dudek

Mike Dudek
David Estes
John Ewen
Massad Eyal
Dawei Fan
Vincent Ferretti
German Feyh
Matthias Fritsche
Takashi Fukuoka
Lemon Geng
Ali Ghiasi
Martin Glanzner
Joel Goergen
Steven Gorshe
Hideki Goto
Steffen Graber
Robert Grow
Martin Gubow
Marek Hajduczenia
Stephan Hartmann
Alexandra Haser
Xiang He
Rajmohan Hegde
David Hess
Yasuo Hidaka
Hidenari Hirase
Rita Horner
Thomas Huber
Gergely Huszak
Yasuhiro Hyakutake
Jonathan Ingham
Kazuhiko Ishibe
Hideki Isono
Tom Issenhuth
Andrew Jimenez
Chad Jones

Peter Jones
Ragnar Jonsson
Lokesh Kabra
Haysam Kadry
Manabu Kagami
Tomohiro Kikuta
Inho Kim
Yong Kim
Mark Kimber
Michael Klempa
Sam Kocsis
Wojciech Koczwara
Erwin Koependoerfer
Taiji Kondo
Kishore Kota
Taketo Kumada
Hans Lackner
Mark Laubach
Sylvanus Lee
David Lewis
Mike-Peng Li
Jane Lim
Robert Lingle
Hai-Feng Liu
Yuanqiu Luo
Zahy Madgar
Jeffery Maki
David Malicoat
Eric Maniloff
Flavio Marques
Chris Mash
Takeo Masuda
Kirsten Matheus
Mick McCarthy
Brett McClellan
Larry McMillan

Greg McSorley
Richard Mellitz
Guangcan Mi
Sanaz Mortazavi
Harald Mueller
Thomas Mueller
Shimon Muller
Sean Murphy
Ramana Murty
Hideki Nakagawa
Edward Nakamoto
Makoto Nariya
Raymond Nering
Christian Neulinger
Anthony New
Gary Nicholl
Shawn Nicholl
Mark Nowell
David Ofelt
Ichiro Ogura
Kumi Omori
Tom Palkert
Sujan Pandey
Carlos Pardo
Earl Parsons
Gerald Pepper
Ruben Perez De Aranda Alonso
David Piehler
Rick Pimpinella
Fabio Pittala

Richard Pitwon
Christopher Pohl
Jason Potterf
William Powell
Roland Preis
Rick Rabinovich
Adee Ran
Alon Regev
Hao Ren
Victor Renteria
Toshiaki Sakai
Sam Sambasivan
Olindo Savi
Hiroshi Sawano
Matthew Schmitt
Hossein Sedarat
Steve Sekel
Qingya She
Masayuki Shigematsu
Masato Shiino
Priyank Shukla
Scott Sommers
Massimo Sorbara
Tom Souvignier
Edward Sprague
Peter Stassar
Heath Stewart
Steve Swanson
Tomoo Takahara
Satoshi Takahashi
Tadashi Takahashi

Kazuya Takayama
James Theodoras
Geoffrey Thompson
Pirooz Tooyserkani
Luis Torres
Nathan Tracy
Viet Tran
David Tremblay
Stephen Trowbridge
Mike Tu
Paul Vanderlaan
Robert Voss
Roy Wang
Ruoxu Wang
Tongtong Wang
Xinyuan Wang
Yuji Watanabe
James Weaver
Matthias Wendt
Natalie Wienckowski
James Withey
Mau-Lin Wu
Peter Wu
Dayin Xu
Yu Xu
Yumeng Yang
Kenji Yonezawa
James Young
Bo Zhang
Yan Zhuang
George Zimmerman

The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

Enis Akbaba	Tetsushi Ikegami	Carlos Pardo
Thomas Alexander	Tom Issenhuth	Earl Parsons
Butch Anton	Raj Jain	Bansi Patel
Philippe Astier	SangKwon Jeong	Arumugam Paventhan
Liav Ben-Artsi	Peter Jones	David Piehler
Rich Boyer	Lokesh Kabra	Rick Pimpinella
Theodore Brillhart	Piotr Karocki	Fabio Pittala
Matt Brown	John Kay	Richard Pitwon
William Byrd	Stuart Kerry	Adee Ran
John Calvin	Evgeny Khorov	R. K. Rannow
Steven B. Carlson	Kihong/Joshua Kim	Maximilian Riegel
Juan Carreon	Yong Kim	Alexander Rysin
Clark Carty	Sam Kocsis	Toshiaki Sakai
Pin Chang	Taiji Kondo	Qingya She
Chan Chen	Gavin Lai	Priyank Shukla
Piers Dawe	Mark Laubach	William Simms
John Deandrea	David J. Law	Jeff Slavick
Stephen Didde	Pi-Cheng Law	Scott Sommers
Mike Dudek	Hyeong Ho Lee	Walter Struppler
Avraham Freedman	David Lewis	Mitsutoshi Sugawara
Matthias Fritsche	Jon Lewis	James Theodoras
Devon Gayle	Ru Lin	Michael Thompson
Ali Ghiasi	Kent Lusted	Nathan Tracy
James Gilb	Valerie Maguire	David Tremblay
Brandon Gore	Jeffery Maki	Lisa Ward
James Graba	Arthur Marris	James Weaver
Robert Grow	Michael Maytum	Stephen Webb
Marek Hajduczenia	Brett McClellan	Matthias Wendt
Xiang He	Jonathon McLendon	Andreas Wolf
Adam Healey	Richard Mellitz	Mau-Lin Wu
Howard Heck	Ramana Murty	Xin Wu
Marco Hernandez	N. Kishor Narang	James Young
David Hess	Raymond Nering	Yu Yuan
Yasuo Hidaka	Shawn Nicholl	Oren Yuen
Werner Hoelzl	Satoshi Obara	Janusz Zalewski
Yasuhiro Hyakutake	Eugene Opsasnick	Pavel Zivny

When the IEEE SA Standards Board approved this standard on 21 September 2022, it had the following membership:

David J. Law, *Chair*
Ted Burse, *Vice Chair*
Gary Hoffman, *Past Chair*
Konstantinos Karachalios, *Secretary*

Edward A. Addy	Johnny Daozhuang Lin	Mark Siira
Ramy Ahmed Fathy	Kevin Lu	Dorothy V. Stanley
J. Travis Griffith	Daleep C. Mohla	Lei Wang
Guido R. Hiertz	Andrew Myles	F. Keith Waters
Yousef Kimiagar	Damir Novosel	Karl Weber
Joseph L. Koepfinger*	Annette D. Reilly	Sha Wei
Thomas Koshy	Robby Robson	Philip B. Winston
John D. Kulick	Jon Walter Rosdahl	Daidi Zhong

*Member Emeritus

Introduction

This introduction is not part of IEEE Std 802.3ck-2022, IEEE Standard for Ethernet—Amendment 4: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Electrical Interfaces Based on 100 Gb/s Signaling.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3ck-2022 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Media Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

IEEE Std 802.3ck™-2022

Amendment 4—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 161 through Clause 163, Annex 120F, Annex 120G, Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

Contents

1. Introduction.....	29
1.1 Overview.....	29
1.1.3 Architectural perspectives.....	29
1.1.3.2 Compatibility interfaces.....	29
1.3 Normative references.....	30
1.4 Definitions.....	30
1.5 Abbreviations.....	31
30. Management.....	32
30.5 Layer management for medium attachment units (MAUs).....	32
30.5.1 MAU managed object class.....	32
30.5.1.1 MAU attributes.....	32
30.5.1.1.2 aMAUType.....	32
30.5.1.1.16 aFECmode.....	32
30.5.1.1.17 aFECCorrectedBlocks.....	33
30.5.1.1.18 aFECUncorrectedBlocks.....	33
30.6 Management for link Auto-Negotiation.....	34
30.6.1 Auto-Negotiation managed object class.....	34
30.6.1.1 Auto-Negotiation attributes.....	34
30.6.1.1.5 aAutoNegLocalTechnologyAbility.....	34
45. Management Data Input/Output (MDIO) Interface.....	35
45.2 MDIO Interface Registers.....	35
45.2.1 PMA/PMD registers.....	35
45.2.1.6 PMA/PMD control 2 register (Register 1.7).....	35
45.2.1.7 PMA/PMD status 2 register (Register 1.8).....	36
45.2.1.7.4 Transmit fault (1.8.11).....	36
45.2.1.7.5 Receive fault (1.8.10).....	36
45.2.1.8 PMD transmit disable register (Register 1.9).....	37
45.2.1.21 200G PMA/PMD extended ability (Register 1.23).....	37
45.2.1.21.1c 200GBASE-CR2 ability (1.23.8).....	37
45.2.1.21.1d 200GBASE-KR2 ability (1.23.7).....	38
45.2.1.22 400G PMA/PMD extended ability register (Register 1.24).....	38
45.2.1.22.11 400GBASE-CR4 ability (1.24.1).....	38
45.2.1.22.12 400GBASE-KR4 ability (1.24.0).....	38
45.2.1.24 40G/100G PMA/PMD extended ability 2 (Register 1.26).....	39
45.2.1.24.9 100GBASE-CR1 ability (1.26.1).....	39
45.2.1.24.10 100GBASE-KR1 ability (1.26.0).....	39
45.2.1.95 BASE-R PMD control register (Register 1.150).....	39
45.2.1.96 BASE-R PMD status register (Register 1.151).....	39
45.2.1.116 RS-FEC control register (Register 1.200).....	40
45.2.1.116.a 100G RS-FEC enable (1.200.6).....	40
45.2.1.116.b 100G RS-FEC-Int enable (1.200.5).....	40
45.2.1.116.1 FEC degraded SER enable (1.200.4).....	40
45.2.1.116.4 FEC bypass indication enable (1.200.1).....	40
45.2.1.117 RS-FEC status register (Register 1.201).....	41
45.2.1.117.1 PCS align status (1.201.15).....	41
45.2.1.117.2 RS-FEC align status (1.201.14).....	41

45.2.1.117.3	FEC AM lock 3 (1.201.11)	41
45.2.1.117.4	FEC AM lock 2 (1.201.10)	41
45.2.1.117.5	FEC AM lock 1 (1.201.9)	41
45.2.1.117.6	FEC AM lock 0 (1.201.8)	41
45.2.1.117.8	FEC degraded SER (1.201.4).....	42
45.2.1.117.9	FEC degraded SER ability (1.201.3)	42
45.2.1.117.10	RS-FEC high SER (1.201.2).....	42
45.2.1.117.11	FEC bypass indication ability (1.201.1).....	42
45.2.1.118	RS-FEC corrected codewords counter (Register 1.202, 1.203)	42
45.2.1.119	RS-FEC uncorrected codewords counter (Register 1.204, 1.205)	42
45.2.1.120a	RS-FEC codeword counter register (Register 1.207, 1.208, 1.209).....	42
45.2.1.121	RS-FEC symbol error counter lane 0 (Register 1.210, 1.211)	43
45.2.1.123	RS-FEC BIP error counter lane 0 (Register 1.230)	43
45.2.1.127	RS-FEC PCS alignment status 1 register (Register 1.280)	43
45.2.1.127.1	Block lock 7 (1.280.7).....	43
45.2.1.127.2	Block lock 6 (1.280.6).....	43
45.2.1.127.3	Block lock 5 (1.280.5).....	43
45.2.1.127.4	Block lock 4 (1.280.4).....	43
45.2.1.127.5	Block lock 3 (1.280.3).....	44
45.2.1.127.6	Block lock 2 (1.280.2).....	44
45.2.1.127.7	Block lock 1 (1.280.1).....	44
45.2.1.127.8	Block lock 0 (1.280.0).....	44
45.2.1.128	RS-FEC PCS alignment status 2 register (Register 1.281)	44
45.2.1.128.1	Block lock 19 (1.281.11).....	44
45.2.1.128.2	Block lock 18 (1.281.10).....	44
45.2.1.128.3	Block lock 17 (1.281.9).....	44
45.2.1.128.4	Block lock 16 (1.281.8).....	44
45.2.1.128.5	Block lock 15 (1.281.7).....	45
45.2.1.128.6	Block lock 14 (1.281.6).....	45
45.2.1.128.7	Block lock 13 (1.281.5).....	45
45.2.1.128.8	Block lock 12 (1.281.4).....	45
45.2.1.128.9	Block lock 11 (1.281.3).....	45
45.2.1.128.10	Block lock 10 (1.281.2).....	45
45.2.1.128.11	Block lock 9 (1.281.1).....	45
45.2.1.128.12	Block lock 8 (1.281.0).....	45
45.2.1.129	RS-FEC PCS alignment status 3 register (Register 1.282)	46
45.2.1.129.1	Lane 7 aligned (1.282.7)	46
45.2.1.129.2	Lane 6 aligned (1.282.6)	46
45.2.1.129.3	Lane 5 aligned (1.282.5)	46
45.2.1.129.4	Lane 4 aligned (1.282.4)	46
45.2.1.129.5	Lane 3 aligned (1.282.3)	46
45.2.1.129.6	Lane 2 aligned (1.282.2)	46
45.2.1.129.7	Lane 1 aligned (1.282.1)	46
45.2.1.129.8	Lane 0 aligned (1.282.0)	46
45.2.1.130	RS-FEC PCS alignment status 4 register (Register 1.283)	47
45.2.1.130.1	Lane 19 aligned (1.283.11)	47
45.2.1.130.2	Lane 18 aligned (1.283.10)	47
45.2.1.130.3	Lane 17 aligned (1.283.9)	47
45.2.1.130.4	Lane 16 aligned (1.283.8)	47
45.2.1.130.5	Lane 15 aligned (1.283.7)	47
45.2.1.130.6	Lane 14 aligned (1.283.6)	47
45.2.1.130.7	Lane 13 aligned (1.283.5)	47
45.2.1.130.8	Lane 12 aligned (1.283.4)	47
45.2.1.130.9	Lane 11 aligned (1.283.3)	48

45.2.1.130.10	Lane 10 aligned (1.283.2)	48
45.2.1.130.11	Lane 9 aligned (1.283.1)	48
45.2.1.130.12	Lane 8 aligned (1.283.0)	48
45.2.1.131a	RS-FEC codeword error bin registers 1 through 15 (Registers 1.340 through 1.369).....	48
45.2.1.146	RS-FEC degraded SER activate threshold register (Register 1.650, 1.651)	49
45.2.1.147	RS-FEC degraded SER deactivate threshold register (Register 1.652, 1.653).....	49
45.2.1.148	RS-FEC degraded SER interval register (Register 1.654, 1.655)	49
45.2.1.161	BASE-R PAM4 PMD training LP control, lane 0 through lane 3 registers (Register 1.1120 through 1.1123).....	49
45.2.1.163	BASE-R PAM4 PMD training LP status, lane 0 through lane 3 registers (Register 1.1220 through 1.1223).....	50
45.2.1.165	BASE-R PAM4 PMD training LD control, lane 0 through lane 3 registers (Register 1.1320 through 1.1323).....	51
45.2.1.167	BASE-R PAM4 PMD training LD status, lane 0 through lane 3 registers (Register 1.420 through 1.423).....	52
45.2.1.169	Test-pattern ability (Register 1.1500).....	53
45.2.1.171a	PRBS9Q testing control (Register 1.1511).....	54
45.2.1.216	IFEC control register (Register 1.2200)	55
45.2.1.216.a	IFEC enable (1.2200.2).....	55
45.2.7	Auto-Negotiation registers	56
45.2.7.12	Backplane Ethernet, BASE-R copper status (Register 7.48).....	56
45.2.7.12.3	Negotiated Port Type (7.48.1, 7.48.2, 7.48.3, 7.48.5, 7.48.6, 7.48.8, 7.48.9, 7.48.10, 7.48.11, 7.48.12, 7.48.13, 7.48.14, 7.48.15, 7.49.0, 7.49.1, 7.49.2, 7.49.3, 7.49.4, 7.49.5)	56
45.2.7.13	Backplane Ethernet, BASE-R copper status 2 (Register 7.49).....	56
45.2.7.13.a	RS-FEC-Int negotiated (7.49.6).....	57
45.2.7.13.1	Negotiated Port Type	57
69.	Introduction to Ethernet operation over electrical backplanes	58
69.1	Overview.....	58
69.1.1	Scope	58
69.1.2	Relationship of Backplane Ethernet to the ISO OSI reference model	59
69.2	Summary of Backplane Ethernet Sublayers	59
69.2.1	Reconciliation sublayer and media independent interfaces	59
69.2.3	Physical Layer signaling systems.....	60
69.2.6	Low-Power Idle.....	60
69.3	Delay constraints.....	60
69.5	Protocol implementation conformance statement (PICS) proforma.....	60
73.	Auto-Negotiation for backplane and copper cable assembly	61
73.2	Relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model	61
73.5	DME transmission	61
73.5.1	DME electrical specifications	61
73.6	Link codeword encoding	62
73.6.4	Technology Ability Field	62
73.6.5	FEC capability.....	63
73.6.5.a	FEC resolution for 100GBASE-P PHYs which include RS-FEC-Int	63
73.6.5.3	FEC control variables	63
73.7	Receive function requirements	64
73.7.6	Priority Resolution function.....	64
73.8	Management register requirements.....	65

73.10	State diagrams and variable definitions	65
73.10.1	State diagram variables	65
73.10.2	State diagram timers	66
73.11	Protocol implementation conformance statement (PICS) proforma for Clause 73, Auto-Negotiation for backplane and copper cable assembly	66
73.11.4	PICS proforma tables for Auto-Negotiation for backplane and copper cable assembly	66
73.11.4.3	Link codeword encoding	66
80.	Introduction to 40 Gb/s and 100 Gb/s networks	67
80.1	Overview	67
80.1.3	Relationship of 40 Gigabit and 100 Gigabit Ethernet to the ISO OSI reference model	67
80.1.4	Nomenclature	68
80.1.5	Physical Layer signaling systems	69
80.2	Summary of 40 Gigabit and 100 Gigabit Ethernet sublayers	72
80.2.3	Forward error correction (FEC) sublayers	72
80.2.5	Physical Medium Dependent (PMD) sublayer	72
80.2.6	Auto-Negotiation	72
80.4	Delay constraints	73
80.5	Skew constraints	74
80.7	Protocol implementation conformance statement (PICS) proforma	76
82.	Physical Coding Sublayer (PCS) for 64B/66B, type 40GBASE-R and 100GBASE-R	77
82.1	Overview	77
82.1.4	Inter-sublayer interfaces	77
82.2	Physical Coding Sublayer (PCS)	77
82.2.13	PCS lane deskew	77
82.6	Auto-Negotiation	77
82.7	Protocol implementation conformance statement (PICS) proforma for Clause 82, Physical Coding Sublayer (PCS) for 64B/66B, type 40GBASE-R and 100GBASE-R	78
82.7.4	PICS proforma tables for PCS, type 40GBASE-R and 100GBASE-R	78
82.7.4.11	Auto-Negotiation for Backplane Ethernet functions	78
83.	Physical Medium Attachment (PMA) sublayer, type 40GBASE-R and 100GBASE-R	79
83.1	Overview	79
83.1.1	Scope	79
91.	Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs	80
91.5	Functions within the RS-FEC sublayer	80
91.5.2	Transmit function	80
91.5.2.7	Reed-Solomon encoder	80
91.5.3	Receive function	80
91.5.3.3	Reed-Solomon decoder	80
91.5.3.3.1	FEC Degraded SER (optional)	81
91.6	RS-FEC MDIO function mapping	81
91.6.3	four_lane_pmd	81
91.6.7a	100G_RS_FEC_enable	81
91.7	Protocol implementation conformance statement (PICS) proforma for Clause 91, Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs	82

91.7.3	Major capabilities/options	82
91.7.4	PICS proforma tables for Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs	83
91.7.4.1	Transmit function	83
91.7.4.2	Receive function	83
116.	Introduction to 200 Gb/s and 400 Gb/s networks	85
116.1	Overview	85
116.1.2	Relationship of 200 Gigabit and 400 Gigabit Ethernet to the ISO OSI reference model	85
116.1.3	Nomenclature	86
116.1.4	Physical Layer signaling systems	87
116.2	Summary of 200 Gigabit and 400 Gigabit Ethernet sublayers	89
116.2.5	Physical Medium Dependent (PMD) sublayer	89
116.2.5a	Auto-Negotiation	89
116.4	Delay constraints	90
116.5	Skew constraints	92
118.	200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)	94
118.1	Overview	94
118.1.3	200GAUI-n/400GAUI-n	94
119.	Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R	95
119.6	Auto-Negotiation	95
119.7	Protocol implementation conformance statement (PICS) proforma for Clause 119, Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R	95
119.7.4	PICS proforma tables for Physical Coding Sublayer (PCS) 64B/66B, type 200GBASE-R and 400GBASE-R	95
119.7.4.12	Auto-Negotiation for Backplane Ethernet functions	95
120.	Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R	96
120.1	Overview	96
120.1.1	Scope	96
120.1.3	Summary of functions	96
120.1.4	PMA sublayer positioning	96
120.3	PMA service interface	97
120.5	Functions within the PMA	97
120.5.1	Per input-lane clock and data recovery	97
120.5.2	Bit-level multiplexing	97
120.5.5	Clocking architecture	97
120.5.6	Signal drivers	98
120.5.7	PAM4 Encoding	98
120.5.7.1	Gray mapping for PAM4 encoded lanes	98
120.5.7.2	Precoding for PAM4 encoded lanes	98
120.5.11	PMA test patterns (optional)	99
120.5.11.2	Test patterns for PAM4 encoded signals	99
120.5.11.2.a	PRBS9Q test pattern	99
120.5.11.2.4	Square wave (quaternary) test pattern	100
120.6	PMA MDIO function mapping	100

120.7	Protocol implementation conformance statement (PICS) proforma for Clause 120, Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R	102
120.7.3	Major capabilities/options	102
121.	Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR4	104
121.1	Overview	104
122.	Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8	105
122.1	Overview	105
123.	Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16	106
123.1	Overview	106
124.	Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4	107
124.1	Overview	107
135.	Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P	108
135.1	Overview	108
135.1.1	Scope	108
135.1.4	PMA sublayer positioning	109
135.5	Functions within the PMA	110
135.5.1	Per input-lane clock and data recovery	110
135.5.2	Bit-level multiplexing	110
135.5.6	Signal drivers	111
135.5.7	PAM4 encoding	111
135.5.7.2	Precoding for PAM4 encoded lanes	111
135.5.10	PMA test patterns (optional)	112
135.5.10.2	Test patterns for PAM4 encoded signals	112
135.5.10.2.4	Square wave (quaternary) test pattern	112
135.7	Protocol implementation conformance statement (PICS) proforma for Clause 135, Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P	113
135.7.3	Major capabilities/options	113
135.7.4	PICS proforma tables for Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P	114
135.7.4.3	Electrical	114
138.	Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8	115
138.1	Overview	115
140.	Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1	116
140.1	Overview	116
150.	Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR4.2	117

150.1	Overview.....	117
151.	Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-FR4 and 400GBASE-LR4-6.....	118
151.1	Overview.....	118
152.	Inverse RS-FEC sublayer.....	119
152.6	Inverse RS-FEC MDIO function mapping	119
152.6.2a	IFEC_enable.....	119
154.	Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-ZR.....	120
154.1	Overview.....	120
161.	Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs	121
161.1	Overview.....	121
161.1.1	Scope	121
161.1.2	Position of RS-FEC-Int in the 100GBASE-P sublayers	121
161.2	FEC service interface.....	121
161.3	PMA compatibility	122
161.4	Delay constraints.....	122
161.5	Functions within the RS-FEC-Int sublayer.....	122
161.5.1	Functional block diagram.....	122
161.5.2	Transmit function	122
161.5.2.1	Lane block synchronization.....	122
161.5.2.2	Alignment lock and deskew	122
161.5.2.3	Lane reorder.....	122
161.5.2.4	Alignment marker removal.....	122
161.5.2.5	64B/66B to 256B/257B transcoder.....	122
161.5.2.6	Alignment marker mapping and insertion	122
161.5.2.6.1	Alignment marker mapping	124
161.5.2.6.2	Alignment marker insertion	125
161.5.2.7	Pre-FEC distribution.....	126
161.5.2.8	Reed-Solomon encoder.....	126
161.5.2.9	Symbol distribution	126
161.5.2.10	Transmit bit ordering	127
161.5.3	Receive function.....	128
161.5.3.1	Alignment lock and deskew	128
161.5.3.2	Lane reorder and de-interleave	128
161.5.3.3	Reed-Solomon decoder.....	128
161.5.3.3.1	Bypass error indication (optional).....	128
161.5.3.3.2	FEC Degraded SER (optional).....	129
161.5.3.4	Post FEC Interleave	129
161.5.3.5	Alignment marker removal.....	129
161.5.3.6	256B/257B to 64B/66B transcoder.....	129
161.5.3.7	Block distribution	129
161.5.3.8	Alignment marker mapping and insertion	129
161.5.4	Detailed functions and state diagrams.....	130
161.5.4.1	State diagram conventions.....	130
161.5.4.2	State variables.....	130

161.5.4.2.1	Variables	130
161.5.4.2.2	Functions	131
161.5.4.2.3	Counters	131
161.5.4.3	State diagrams.....	131
161.6	RS-FEC-Int MDIO function mapping	133
161.6.1	FEC_bypass_indication_enable	134
161.6.2	FEC_degraded_SER_enable	134
161.6.3	100G_RS_FEC_Int_enable.....	134
161.6.4	FEC_degraded_SER_activate_threshold	134
161.6.5	FEC_degraded_SER_deactivate_threshold	135
161.6.6	FEC_degraded_SER_interval	135
161.6.7	FEC_bypass_indication_ability	135
161.6.8	hi_ser.....	135
161.6.9	FEC_degraded_SER_ability	135
161.6.10	FEC_degraded_SER.....	135
161.6.11	amps_lock<x>	135
161.6.12	fec_align_status.....	135
161.6.13	FEC_corrected_cw_counter.....	135
161.6.14	FEC_uncorrected_cw_counter.....	135
161.6.15	FEC_lane_mapping<x>	135
161.6.16	FEC_symbol_error_counter_i.....	135
161.6.17	FEC_codeword_error_bin_i.....	135
161.6.18	align_status.....	136
161.6.19	BIP_error_counter_i.....	136
161.6.20	lane_mapping<x>.....	136
161.6.21	FEC_cw_counter.....	136
161.6.22	block_lock<x>.....	136
161.6.23	am_lock<x>.....	136
161.7	Protocol implementation conformance statement (PICS) proforma for Clause 161, Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs	137
161.7.1	Introduction	137
161.7.2	Identification	137
161.7.2.1	Implementation identification.....	137
161.7.2.2	Protocol summary	137
161.7.3	Major capabilities/options	138
161.7.4	PICS proforma tables for Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs.....	138
161.7.4.1	Transmit function	138
161.7.4.2	Receive function	139
161.7.4.3	State diagrams.....	140
162.	Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4	141
162.1	Overview	141
162.2	Conventions	143
162.3	PMD service interfaces	144
162.4	PCS requirements for Auto-Negotiation (AN) service interface	145
162.5	Delay constraints.....	145
162.6	Skew constraints	145
162.6.1	Skew Constraints for 100GBASE-CR1	145
162.6.2	Skew Constraints for 200GBASE-CR2 and 400GBASE-CR4.....	146
162.7	PMD MDIO function mapping.....	146

162.8	PMD functional specifications.....	148
162.8.1	Link block diagram	148
162.8.2	PMD transmit function.....	150
162.8.3	PMD receive function	150
162.8.4	PMD global signal detect function.....	151
162.8.5	PMD lane-by-lane signal detect function.....	151
162.8.6	PMD global transmit disable function (optional).....	151
162.8.7	PMD lane-by-lane transmit disable function (optional).....	151
162.8.8	PMD fault function.....	151
162.8.9	PMD transmit fault function (optional).....	151
162.8.10	PMD receive fault function (optional)	152
162.8.11	PMD control function.....	152
162.9	PMD electrical characteristics	154
162.9.1	AC-coupling	154
162.9.2	Signal paths	154
162.9.3	Reference impedance	155
162.9.4	Transmitter characteristics	155
162.9.4.1	Transmitter output waveform	156
162.9.4.1.1	Linear fit to the measured waveform	157
162.9.4.1.2	Steady-state voltage and linear fit pulse peak ratio.....	158
162.9.4.1.3	Coefficient initialization	158
162.9.4.1.4	Coefficient step size	158
162.9.4.1.5	Coefficient range	159
162.9.4.2	Transmitter linearity	159
162.9.4.3	Signal-to-residual-intersymbol-interference ratio	160
162.9.4.4	Peak-to-peak AC common-mode voltage.....	160
162.9.4.5	Insertion loss, TP0 to TP2 or TP3 to TP5.....	160
162.9.4.6	Output SNDR	160
162.9.4.7	Output jitter.....	160
162.9.4.8	Transmitter effective return loss (ERL).....	162
162.9.4.9	Common-mode to common-mode return loss	162
162.9.4.10	Common-mode to differential-mode return loss	163
162.9.5	Receiver characteristics.....	164
162.9.5.1	Receiver signaling rate	164
162.9.5.2	Receiver amplitude tolerance	165
162.9.5.3	Receiver interference tolerance	165
162.9.5.3.1	Test setup	165
162.9.5.3.2	Test channel	166
162.9.5.3.3	Test channel calibration	166
162.9.5.3.4	Pattern generator and noise injection	168
162.9.5.3.5	Test procedure.....	169
162.9.5.4	Receiver jitter tolerance.....	169
162.9.5.4.1	Test setup	169
162.9.5.4.2	Test procedure.....	169
162.9.5.5	Receiver ERL	170
162.9.5.6	Differential-mode to common-mode return loss	170
162.10	Channel characteristics	171
162.11	Cable assembly characteristics	171
162.11.1	Characteristic impedance and reference impedance	171
162.11.2	Cable assembly insertion loss	171
162.11.3	Cable assembly ERL	173
162.11.4	Differential-mode to common-mode return loss	173
162.11.5	Differential-mode to common-mode insertion loss	174
162.11.6	Common-mode to common-mode return loss.....	175

162.11.7	Cable assembly Channel Operating Margin.....	176
162.11.7.1	Channel signal and crosstalk path calculations	178
162.11.7.1.1	Channel signal path.....	179
162.11.7.1.2	Channel crosstalk paths.....	179
162.11.7.2	Signal and crosstalk paths used in calculation of COM	180
162.12	MDI specifications.....	181
162.13	Environmental specifications.....	181
162.14	Protocol implementation conformance statement (PICS) proforma for Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4	182
162.14.1	Introduction	182
162.14.2	Identification	182
162.14.2.1	Implementation identification.....	182
162.14.2.2	Protocol summary	182
162.14.3	Major capabilities/options	183
162.14.4	PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4...	184
162.14.4.1	PMD functional specifications	184
162.14.4.2	PMD control function.....	185
162.14.4.3	Transmitter specifications	185
162.14.4.4	Receiver specifications	186
162.14.4.5	Cable assembly specifications	186
162.14.4.6	Environmental specifications	187
163.	Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4.....	188
163.1	Overview.....	188
163.2	Conventions	191
163.3	PMD service interfaces	191
163.4	PCS requirements for Auto-Negotiation (AN) service interface.....	191
163.5	Delay constraints.....	191
163.6	Skew constraints	192
163.6.1	Skew Constraints for 100GBASE-KR1	192
163.6.2	Skew Constraints for 200GBASE-KR2 and 400GBASE-KR4	192
163.7	PMD MDIO function mapping.....	192
163.8	PMD functional specifications.....	193
163.8.1	Link block diagram	193
163.8.2	PMD transmit function.....	193
163.8.3	PMD receive function	193
163.8.4	PMD global signal detect function.....	193
163.8.5	PMD lane-by-lane signal detect function.....	193
163.8.6	PMD global transmit disable function (optional).....	193
163.8.7	PMD lane-by-lane transmit disable function (optional).....	193
163.8.8	PMD fault function.....	194
163.8.9	PMD transmit fault function (optional).....	194
163.8.10	PMD receive fault function (optional)	194
163.8.11	PMD control function.....	194
163.9	Electrical characteristics	194
163.9.1	Reference impedance	194
163.9.2	Transmitter characteristics	194
163.9.2.1	Transmitter test fixture	195
163.9.2.1.1	Test fixture insertion loss.....	195
163.9.2.1.2	Test fixture effective return loss (ERL)	196

163.9.2.1.3	Test fixture common-mode to common-mode return loss.....	196
163.9.2.2	Transmitter difference ERL.....	196
163.9.2.3	Transmitter common-mode to common-mode return loss	197
163.9.2.4	Difference steady-state voltage	197
163.9.2.5	Difference linear fit pulse peak ratio	197
163.9.2.6	Signal to AC common-mode noise ratio	197
163.9.3	Receiver characteristics.....	197
163.9.3.1	Receiver signaling rate	198
163.9.3.2	Receiver test fixture	198
163.9.3.3	Receiver difference ERL	198
163.9.3.4	Receiver differential-mode to common-mode return loss	198
163.9.3.5	Receiver interference tolerance	199
163.9.3.6	Receiver jitter tolerance.....	201
163.10	Channel characteristics	201
163.10.1	Channel operating margin	201
163.10.2	Channel insertion loss (recommended).....	204
163.10.3	Channel ERL.....	204
163.10.4	Channel differential-mode to common-mode return loss	205
163.10.5	Channel differential-mode to common-mode insertion loss	206
163.10.6	Channel common-mode to differential-mode insertion loss	207
163.10.7	AC-coupling	207
163.11	MDI specifications.....	208
163.12	Environmental specifications.....	208
163.13	Protocol implementation conformance statement (PICS) proforma for Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4.....	209
163.13.1	Introduction	209
163.13.2	Identification	209
163.13.2.1	Implementation identification.....	209
163.13.2.2	Protocol summary	209
163.13.3	Major capabilities/options	210
163.13.4	PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 ..	211
163.13.4.1	Functional specifications	211
163.13.4.2	PMD control function.....	212
163.13.4.3	Transmitter characteristics.....	212
163.13.4.4	Receiver characteristics	213
163.13.4.5	Channel characteristics	213
163.13.4.6	Environmental specifications	214
167.	Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4.....	215
167.1	Overview.....	215
Annex 93A	(normative) Specification methods for electrical channels	216
93A.1	Channel Operating Margin	216
93A.1.2	Transmitter and receiver device package models	219
93A.1.2.1	Cascade connection of two-port networks	219
93A.1.2.2	Two-port network for a shunt capacitance.....	219
93A.1.2.2a	Two-port network for a series inductance.....	219
93A.1.2.3	Two-port network for the package transmission line.....	219
93A.1.2.4	Assembly of transmitter and receiver device package models	220

93A.1.6	Determination of variable equalizer parameters	221
93A.1.6.1	Determination of DFE floating tap locations and magnitudes	222
93A.5	Effective Return Loss (ERL)	223
93A.5.1	Pulse time-domain reflection signal.....	223
93A.5.2	Effective reflection waveform	224
Annex 120A	(informative) 200 Gb/s and 400 Gb/s PMA sublayer partitioning examples.....	225
120A.5	Partitioning examples supporting 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4.....	225
Annex 120F	(normative) Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)	226
120F.1	Overview.....	226
120F.2	Compliance points	228
120F.2.1	Reference impedance	228
120F.3	Electrical characteristics	228
120F.3.1	Transmitter electrical characteristics	228
120F.3.1.1	Peak-to-peak AC common-mode voltage	229
120F.3.1.2	Signal to AC common-mode noise ratio	229
120F.3.1.3	Signal-to-residual-intersymbol-interference ratio	230
120F.3.1.4	Transmitter difference effective return loss (ERL)	230
120F.3.1.5	Transmitter equalization settings	230
120F.3.1.6	Output jitter	231
120F.3.1.7	Transmitter control.....	232
120F.3.2	Receiver characteristics.....	232
120F.3.2.1	Receiver signaling rate	232
120F.3.2.2	Receiver difference ERL	233
120F.3.2.3	Receiver differential-mode to common-mode return loss.....	233
120F.3.2.4	Receiver interference tolerance.....	233
120F.3.2.5	Receiver jitter tolerance	235
120F.3.2.6	Transmitter equalization feedback (optional)	235
120F.4	Channel characteristics	236
120F.4.1	Channel operating margin.....	236
120F.4.2	Channel insertion loss (recommended).....	238
120F.4.3	Channel ERL.....	239
120F.4.4	Channel differential-mode to common-mode return loss	239
120F.4.5	Channel AC-Coupling.....	239
120F.5	Protocol implementation conformance statement (PICS) proforma for Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)	240
120F.5.1	Introduction.....	240
120F.5.2	Identification	240
120F.5.2.1	Implementation identification	240
120F.5.2.2	Protocol summary	241
120F.5.3	Major capabilities/options.....	241
120F.5.4	PICS proforma tables for Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)	242
120F.5.4.1	Transmitter	242
120F.5.4.2	Receiver.....	243

120F.5.4.3 Channel.....	243
Annex 120G (normative) Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)	
120G.1 Overview.....	244
120G.1.1 Bit error ratio.....	245
120G.2 Compliance point definitions.....	245
120G.3 Electrical characteristics	246
120G.3.1 Host output characteristics	246
120G.3.1.1 Host output common-mode to differential-mode return loss	247
120G.3.1.2 Host output effective return loss (ERL)	248
120G.3.1.3 Differential termination mismatch	248
120G.3.1.4 Transition time	248
120G.3.1.5 Host output eye height and vertical eye closure (VEC)	249
120G.3.2 Module output characteristics	249
120G.3.2.1 Module output modes.....	250
120G.3.2.2 Module output eye height and VEC.....	251
120G.3.2.2.1 Near-end and far-end eye measurement methodology	252
120G.3.2.3 Module output ERL.....	252
120G.3.2.4 Module output common-mode voltage tolerance.....	252
120G.3.3 Host input characteristics	253
120G.3.3.1 Host input signaling rate	253
120G.3.3.2 Host input AC common-mode voltage tolerance.....	253
120G.3.3.3 Host input differential-mode to common-mode return loss	253
120G.3.3.4 Host input ERL.....	254
120G.3.3.5 Host stressed input tolerance.....	254
120G.3.3.5.1 Host stressed input test setup	254
120G.3.3.5.2 Host stressed input test calibration	256
120G.3.3.5.3 Host stressed input test procedure	257
120G.3.4 Module input characteristics	258
120G.3.4.1 Module input signaling rate.....	258
120G.3.4.2 Module input AC common-mode voltage tolerance	258
120G.3.4.3 Module stressed input tolerance	258
120G.3.4.3.1 Module stressed input test setup	259
120G.3.4.3.2 Module stressed input test calibration	260
120G.3.4.3.3 Module stressed input test procedure.....	262
120G.3.4.4 Module input ERL.....	263
120G.3.4.5 Module input common-mode voltage tolerance.....	263
120G.4 Channel insertion loss (recommended).....	263
120G.5 Measurement methodology.....	264
120G.5.1 Signal levels	264
120G.5.2 Eye opening measurement method	264
120G.5.3 Steady-state voltage	266
120G.5.4 HCB/MCB characteristics.....	266
120G.6 Protocol implementation conformance statement (PICS) proforma for Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M) and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)	268
120G.6.1 Introduction.....	268
120G.6.2 Identification	268
120G.6.2.1 Implementation identification	268
120G.6.2.2 Protocol summary	268

120G.6.3	Major capabilities/options.....	269
120G.6.4	PICS proforma tables for Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M) and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)	269
120G.6.4.1	Host output	269
120G.6.4.2	Module output	270
120G.6.4.3	Host input	270
120G.6.4.4	Module input	271
Annex 135A	(informative) 50 Gb/s and 100 Gb/s PMA sublayer partitioning examples.....	272
135A.2	Partitioning examples of 100GBASE-P PHYs.....	272
Annex 162A	(informative) Transmitter, receiver and channel parameters associated with test points TP0 and TP5 for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4.....	274
162A.1	Overview.....	274
162A.2	Transmitter characteristics at TP0	274
162A.3	Receiver characteristics at TP5	274
162A.4	Transmitter and receiver differential printed circuit board (PCB) trace insertion loss	274
162A.5	Channel insertion loss	276
162A.6	Channel effective return loss (ERL)	279
162A.7	Channel Operating Margin (COM).....	279
Annex 162B	(normative) Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M	280
162B.1	Test fixtures	280
162B.2	TP2 or TP3 test fixture.....	280
162B.2.1	TP2 or TP3 test fixture insertion loss.....	280
162B.3	Cable assembly test fixture	281
162B.3.1	Cable assembly test fixture insertion loss	281
162B.4	Mated test fixtures	282
162B.4.1	Mated test fixtures insertion loss	282
162B.4.2	Mated test fixtures effective return loss (ERL).....	282
162B.4.3	Mated test fixtures common-mode to differential-mode insertion loss	283
162B.4.4	Mated test fixtures common-mode to common-mode return loss	284
162B.4.5	Mated test fixtures common-mode to differential-mode return loss.....	285
162B.4.6	Mated test fixtures integrated crosstalk noise	286
162B.5	Protocol implementation conformance statement (PICS) proforma for Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M.....	288
162B.5.1	Introduction.....	288
162B.5.2	Identification	288
162B.5.2.1	Implementation identification	288
162B.5.2.2	Protocol summary	288
162B.5.3	Major capabilities/options.....	289
162B.5.4	PICS proforma tables for test fixtures for Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M	289
Annex 162C	(normative) MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4	290

162C.1 Overview.....	290
162C.2 MDI connector types	296
162C.2.1 SFP112.....	296
162C.2.2 SFP-DD112.....	297
162C.2.3 DSFP	298
162C.2.4 QSFP112.....	299
162C.2.5 QSFP-DD800.....	300
162C.2.6 OSFP	301
162C.3 Protocol implementation conformance statement (PICS) proforma for Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4.....	303
162C.3.1 Introduction.....	303
162C.3.2 Identification	303
162C.3.2.1 Implementation identification	303
162C.3.2.2 Protocol summary	303
162C.3.3 Major capabilities/options.....	304
162C.3.4 PICS proforma tables for MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4.....	304
162C.3.4.1 Contact Mapping.....	304
 Annex 162D (informative) Cable assemblies and hosts for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4.....	305
162D.1 Overview.....	305
162D.1.1 Cable assembly types.....	305
 Annex 163A (normative) Specification methods for electrical interfaces.....	308
163A.1 Overview.....	308
163A.2 Test fixture.....	309
163A.3 Test methods for transmitters	309
163A.3.1 Reference parameter determination	309
163A.3.1.1 Steady-state voltage and pulse peak reference values.....	309
163A.3.1.2 ERL reference value.....	310
163A.3.1.3 Transition time reference value.....	310
163A.3.2 Difference parameter determination	311
163A.3.2.1 Steady-state voltage and pulse peak ratio	311
163A.3.2.2 Difference ERL	312
163A.4 Test methods for receivers.....	312
163A.4.1 Reference parameter determination	313
163A.4.1.1 ERL reference value.....	313
163A.4.1.2 Difference ERL	313
 Annex 163B (informative) Example test fixture	314
163B.1 Overview.....	314
163B.2 Characteristics.....	314
163B.3 Reference Values	315

IEEE Standard for Ethernet

Amendment 4: Physical Layer Specifications and Management Parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s Electrical Interfaces Based on 100 Gb/s Signaling

(This amendment is based on IEEE Std 802.3™-2022 as amended by IEEE Std 802.3dd™-2022, IEEE Std 802.3cs™-2022, and IEEE Std 802.3db™-2022.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~strike through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.⁶

⁶ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1. Introduction

1.1 Overview

1.1.3 Architectural perspectives

1.1.3.2 Compatibility interfaces

Change list item q) “100 Gb/s Attachment Unit Interface (CAUI-n/100GAUI-n)” in 1.1.3.2 as follows:

- q) *100 Gb/s Attachment Unit Interface (CAUI-n/100GAUI-n).* The CAUI-n/100GAUI-n is a physical instantiation of the PMA service interface to extend the connection between 100 Gb/s capable PMAs. While conformance with implementation of this interface is not necessary to ensure communication, it is recommended, since it allows maximum flexibility in intermixing PHYs and DTEs at 100 Gb/s speeds. The CAUI-n/100GAUI-n is intended for use as a chip-to-chip or a chip-to-module interface. ThreeFor chip-to-chip interfaces and for chip-to-module interfaces, four widths of CAUI-n/100GAUI-n are defined: a ten-lane version (CAUI-10) in Annex 83A and Annex 83B, a four-lane version (CAUI-4/100GAUI-4) in Annex 83D, Annex 83E, Annex 135D, and Annex 135E, ~~and~~ a two-lane version (100GAUI-2) in Annex 135F and Annex 135G, and a one-lane version (100GAUI-1) in Annex 120F and Annex 120G. No mechanical connector is specified for use with the CAUI-n/100GAUI-n. The CAUI-n/100GAUI-n is optional.

Change list item t) “200 Gb/s Attachment Unit Interface (200GAUI-n)” in 1.1.3.2 as follows:

- t) *200 Gb/s Attachment Unit Interface (200GAUI-n).* The 200GAUI-n is a physical instantiation of the PMA service interface to extend the connection between 200 Gb/s capable PMAs. While conformance with implementation of this interface is not necessary to ensure communication, it is recommended, since it allows maximum flexibility in intermixing PHYs and DTEs at 200 Gb/s speeds. The 200GAUI-n is intended for use as a chip-to-chip or a chip-to-module interface. TwoFor chip-to-chip interfaces and for chip-to-module interfaces, three widths of 200GAUI-n are defined: an eight-lane version (200GAUI-8) in Annex 120B and Annex 120C, ~~and~~ a four-lane version (200GAUI-4) in Annex 120D and Annex 120E, and a two-lane version (200GAUI-2) in Annex 120F and Annex 120G. No mechanical connector is specified for use with the 200GAUI-n. The 200GAUI-n is optional.

Change list item v) “400 Gb/s Attachment Unit Interface (400GAUI-n)” in 1.1.3.2 as follows:

- v) *400 Gb/s Attachment Unit Interface (400GAUI-n).* The 400GAUI-n is a physical instantiation of the PMA service interface to extend the connection between 400 Gb/s capable PMAs. While conformance with implementation of this interface is not necessary to ensure communication, it is recommended, since it allows maximum flexibility in intermixing PHYs and DTEs at 400 Gb/s speeds. The 400GAUI-n is intended for use as a chip-to-chip or a chip-to-module interface. TwoFor chip-to-chip interfaces and for chip-to-module interfaces, three widths of 400GAUI-n are defined: a sixteen-lane version (400GAUI-16) in Annex 120B and Annex 120C, ~~and~~ an eight-lane version (400GAUI-8) in Annex 120D and Annex 120E, and a four-lane version (400GAUI-4) in Annex 120F and Annex 120G. No mechanical connector is specified for use with the 400GAUI-n. The 400GAUI-n is optional.

1.3 Normative references

Insert the following new references in alphanumeric order:

DSFP MSA Dual small form factor pluggable module, Rev. 1.0, September 12, 2018.⁷

OSFP MSA Specification for OSFP octal small form factor pluggable module, Rev 4.1, August 2, 2021.⁸

QSFP-DD/QSFP-DD800/QSFP112 Hardware Specification for QSFP Double Density 8X and QSFP 4X Pluggable Transceivers, Revision 6.01, May 28, 2021.⁹

SFP-DD/SFP-DD112/SFP112 Hardware Specification for SFP112 and SFP Double Density Pluggable Transceiver, Revision 5.0, October 1, 2021.¹⁰

Change the entry for SFF-8432 as follows:

SFF-8432, Rev 5.15.2a, August 8, 2012November 30, 2018, Specification for SFP+ Module and Cage.

1.4 Definitions

Insert the following new definition after 1.4.24 “100BASE-X”:

1.4.24a 100GBASE-CR1: IEEE 802.3 Physical Layer specification for 100 Gb/s using 100GBASE-R encoding over one lane of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 162.)

Insert the following new definition after 1.4.31 “100GBASE-KP4”:

1.4.31a 100GBASE-KR1: IEEE 802.3 Physical Layer specification for 100 Gb/s using 100GBASE-R encoding over one lane of an electrical backplane. (See IEEE Std 802.3, Clause 163.)

Change 1.4.44 as follows:

1.4.44 100 Gb/s Attachment Unit Interface (CAUI-n, 100GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 100 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module interconnections. ThreeFor chip-to-module interconnections and for chip-to-chip interconnections, four widths are defined: a ten-lane version (CAUI-10), two four-lane versions (CAUI-4, 100GAUI-4), ~~and~~ a two-lane version (100GAUI-2), and a single-lane version (100GAUI-1). (See IEEE Std 802.3, Annex 83A and Annex 83B for CAUI-10, Annex 83D and Annex 83E for CAUI-4, Clause 135, Annex 135D, and Annex 135E for 100GAUI-4, ~~or~~ Clause 135, Annex 135F, and Annex 135G for 100GAUI-2, or Clause 135, Annex 120F, and Annex 120G for 100GAUI-1.)

Insert the following new definition after 1.4.101 “2.5GSEI”:

1.4.101a 200GBASE-CR2: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding over two lanes of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 162.)

⁷ DSFP specifications are available from DSFP MSA (<https://dsfpmsa.org/>).

⁸ OSFP specifications are available from OSFP MSA (<https://osfpmsa.org/>).

⁹ QSFP-DD, QSFP-DD800, and QSFP112 specifications are available from QSFP-DD MSA (<http://www.qsfp-dd.com/>).

¹⁰ SFP-DD112 and SFP112 specifications are available from SFP-DD MSA (<http://sfp-dd.com/>).

Insert the following new definition after 1.4.105 “200GBASE-FR4”:

1.4.105a 200GBASE-KR2: IEEE 802.3 Physical Layer specification for 200 Gb/s using 200GBASE-R encoding over two lanes of an electrical backplane. (See IEEE Std 802.3, Clause 163.)

Change 1.4.110 as follows:

1.4.110 200 Gb/s Attachment Unit Interface (200GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 200 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module interconnections. ~~Two~~For chip-to-module interconnections and for chip-to-chip interconnections, three widths of 200GAUI-n are defined: an eight-lane version (200GAUI-8), ~~and a four-lane version (200GAUI-4), and a two-lane version (200GAUI-2).~~ (See IEEE Std 802.3, [Annex 120B](#) and [Annex 120C](#) for 200GAUI-8, ~~or Annex 120D~~ and [Annex 120E](#) for 200GAUI-4, ~~or Annex 120F and Annex 120G for 200GAUI-2.~~)

Insert the following new definition after 1.4.134 “2-Event class signature”:

1.4.134a 400GBASE-CR4: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding over four lanes of shielded balanced copper cabling. (See IEEE Std 802.3, Clause 162.)

Insert the following new definition after 1.4.138 “400GBASE-FR8”:

1.4.138a 400GBASE-KR4: IEEE 802.3 Physical Layer specification for 400 Gb/s using 400GBASE-R encoding over four lanes of an electrical backplane. (See IEEE Std 802.3, Clause 163.)

Change 1.4.145 as follows:

1.4.145 400 Gb/s Attachment Unit Interface (400GAUI-n): A physical instantiation of the PMA service interface to extend the connection between 400 Gb/s capable PMAs over n lanes, used for chip-to-chip or chip-to-module interconnections. ~~Two~~For chip-to-module interconnections and for chip-to-chip interconnections, three widths of 400GAUI-n are defined: a sixteen-lane version (400GAUI-16), ~~and an eight-lane version (400GAUI-8), and a four-lane version (400GAUI-4).~~ (See IEEE Std 802.3, [Annex 120B](#) and [Annex 120C](#) for 400GAUI-16, ~~or Annex 120D~~ and [Annex 120E](#) for 400GAUI-8, ~~or Annex 120F and Annex 120G for 400GAUI-4.~~)

1.5 Abbreviations

Change the abbreviation of 100GAUI as follows:

100GAUI-n 100 Gb/s Attachment Unit Interface over n lanes

Insert the following new acronym in alphanumeric order:

VEC vertical eye closure

30. Management

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 50GBASE-SR:

100GBASE-CR1	100GBASE-R PCS/PMA over shielded balanced copper cable PMD as specified in Clause 162
--------------	---

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 100GBASE-KP4:

100GBASE-KR1	100GBASE-R PCS/PMA over an electrical backplane PMD as specified in Clause 163
--------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 100GBASE-ZR:

200GBASE-CR2	200GBASE-R PCS/PMA over 2 lane shielded balanced copper cable PMD as specified in Clause 162
--------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 200GBASE-FR4:

200GBASE-KR2	200GBASE-R PCS/PMA over an electrical backplane PMD as specified in Clause 163
--------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 200GBASE-VR2 (as inserted by IEEE Std 802.3db-2022):

400GBASE-CR4	400GBASE-R PCS/PMA over 4 lane shielded balanced copper cable PMD as specified in Clause 162
--------------	--

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.2 after the entry for 400GBASE-FR8:

400GBASE-KR4	400GBASE-R PCS/PMA over an electrical backplane PMD as specified in Clause 163
--------------	--

30.5.1.1.16 aFECmode

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.5.1.1.16 after the entry for RS-FEC enabled:

RS-FEC-Int enabled	RS-FEC-Int enabled
--------------------	--------------------

Change the first three paragraphs of the “BEHAVIOUR DEFINED AS:” section of 30.5.1.1.16 as follows:

A read-write value for a PHY that supports an optional FEC sublayer that indicates the mode of operation of the FEC sublayer for forward error correction across the MDI (see 65.2, Clause 74, Clause 91, ~~and Clause 108~~, and Clause 161).

A GET operation returns the current mode of operation of the PHY. A SET operation changes the mode of operation of the PHY to the indicated value. The enumerations “BASE-R enabled”, ~~and “RS-FEC enabled”, and “RS-FEC-Int enabled”~~ are only used ~~for by~~ PHYs which support more than one type of FEC operation. For 25GBASE-CR, 25GBASE-CR-S, 25GBASE-KR, and 25GBASE-KR-S PHYs where operation in the no-FEC mode maps to the enumeration “disabled”, operation in the BASE-R FEC mode maps to the enumeration “BASE-R enabled”, and operation in the RS-FEC mode maps to the enumeration “RS-FEC enabled” (see 110.6 and 111.6). For 100GBASE-CR1 and 100GBASE-KR1 PHYs, operation in RS-FEC mode maps to the enumeration “RS-FEC enabled” (see 91.6.7a) and operation in interleaved RS-FEC mode maps to the enumeration “RS-FEC-Int enabled” (see 161.6.3).

When Clause 73 Auto-Negotiation is enabled for a 25GBASE-R PHY, a SET operation is not allowed and a GET operation maps to the variables FEC_enable in Clause 74 and FEC_enable in Clause 108. When Clause 73 Auto-Negotiation is enabled for a non-25GBASE-R PHY supporting Clause 74 FEC a SET operation is not allowed and a GET operation maps to the variable FEC_enable in Clause 74. When Clause 73 Auto-Negotiation is enabled for a 100GBASE-R PHY supporting Clause 161 FEC a SET operation is not allowed and a GET operation maps to the variable 100G_RS_FEC_enable in Clause 91 and 100G_RS_FEC_Int_enable in Clause 161.

30.5.1.1.17 aFECCorrectedBlocks

Change the final paragraph of 30.5.1.1.17 as follows:

If a Clause 45 MDIO Interface is present, then this attribute maps to the FEC corrected blocks counter(s) (see 45.2.10.5 and 45.2.1.109 for 10GBASE-R, 45.2.3.41 for 10GBASE-PR and 10/1GBASE-PRX, 45.2.1.131 for BASE-R, 45.2.1.118 for RS-FEC and RS-FEC-Int, 45.2.3.62 for PCS FEC, and 45.2.1.227 for SC-FEC).;

30.5.1.1.18 aFECUncorrectedBlocks

Change the final paragraph of 30.5.1.1.18 as follows:

If a Clause 45 MDIO Interface is present, then this attribute maps to the FEC uncorrectable blocks counter(s) (see 45.2.10.6 and 45.2.1.110 for 10GBASE-R, 45.2.3.42 for 10GBASE-PR and 10/1GBASE-PRX, 45.2.1.149 for BASE-R, 45.2.1.119 for RS-FEC and RS-FEC-Int, 45.2.3.63 for PCS FEC, and 45.2.1.228 for SC-FEC).;

30.6 Management for link Auto-Negotiation

30.6.1 Auto-Negotiation managed object class

30.6.1.1 Auto-Negotiation attributes

30.6.1.1.5 aAutoNegLocalTechnologyAbility

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.6.1.1.5 after the entry for 100GR2:

100GR1	100GBASE-CR1 as specified in Clause 162 or 100GBASE-KR1 as specified in Clause 163
--------	--

Insert the following new entries into the “APPROPRIATE SYNTAX” section of 30.6.1.1.5 after the entry for 200GR4:

200GR2	200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as specified in Clause 163
400GR4	400GBASE-CR4 as specified in Clause 162 or 400GBASE-KR4 as specified in Clause 163

Insert the following new entry into the “APPROPRIATE SYNTAX” section of 30.6.1.1.5 after the entry for BASE-RFEC25G Req:

RS-FEC-Int Req	RS-FEC-Int requested as specified in 73.6.5 and Clause 161
----------------	--

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface Registers

45.2.1 PMA/PMD registers

Change Table 45–3 as follows (some unchanged rows not shown):

Table 45–3—PMA/PMD registers

Register address	Register name	Subclause
...		
1.207 through 1.209	RS-FEC codeword counter Reserved	<u>45.2.1.120a</u>
...		
1.300 through 1.339	BASE-R FEC corrected blocks counter, lanes 0 through 19	<u>45.2.1.131</u>
<u>1.340 through 1.369</u>	<u>RS-FEC codeword error bin 1 through 15</u>	<u>45.2.1.131a</u>
1.340 <u>1.370</u> through 1.399	Reserved	
...		
1.1511	PRBS9Q testing control Reserved	<u>45.2.1.171a</u>
...		

45.2.1.6 PMA/PMD control 2 register (Register 1.7)

Change the description for bits 1.7.6:0 in Table 45–7 as follows (unchanged table rows and bit description lines not shown):

Table 45–7—PMA/PMD control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.7.6:0	PMA/PMD type selection	6 5 4 3 2 1 0 ... 1 0 1 1 1 1 0 = 400GBASE-CR4 PMA/PMD reserved 1 0 1 1 1 0 1 = 400GBASE-KR4 PMA/PMD reserved ... 1 0 1 0 1 1 x = reserved <u>1 0 1 0 1 1 1 = 200GBASE-CR2 PMA/PMD</u> <u>1 0 1 0 1 1 0 = 200GBASE-KR2 PMA/PMD</u> ... 1 0 0 0 1 1 x = reserved <u>1 0 0 0 1 1 1 = 100GBASE-CR1 PMA/PMD</u> <u>1 0 0 0 1 1 0 = 100GBASE-KR1 PMA/PMD</u> ...	R/W

^aR/W = Read/Write, RO = Read only

45.2.1.7 PMA/PMD status 2 register (Register 1.8)

45.2.1.7.4 Transmit fault (1.8.11)

Change Table 45–9 as follows (some unchanged rows not shown):

Table 45–9—Transmit fault description location

PMA/PMD	Description location
...	
50GBASE-BR10, 50GBASE-BR20, 50GBASE-BR40	160.5.8
<u>100GBASE-KR1, 200GBASE-KR2, 400GBASE-KR4</u>	<u>163.8.9</u>
100GBASE-KP4	94.3.8
100GBASE-KR4	93.7.10
<u>100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4</u>	<u>162.8.9</u>
100GBASE-CR4	92.7.10
...	

45.2.1.7.5 Receive fault (1.8.10)

Change Table 45–10 as follows (some unchanged rows not shown):

Table 45–10—Receive fault description location

PMA/PMD	Description location
...	
50GBASE-BR10, 50GBASE-BR20, 50GBASE-BR40	160.5.9
<u>100GBASE-KR1, 200GBASE-KR2, 400GBASE-KR4</u>	<u>163.8.10</u>
100GBASE-KP4	94.3.9
100GBASE-KR4	93.7.11
<u>100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4</u>	<u>162.8.10</u>
100GBASE-CR4	92.7.11
...	

45.2.1.8 PMD transmit disable register (Register 1.9)

Change Table 45–12 as follows (some unchanged rows not shown):

Table 45–12—Transmit disable description location

PMA/PMD	Description location
...	
50GBASE-BR10, 50GBASE-BR20, and 50GBASE-BR40	160.5.6
<u>100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4</u>	<u>163.8.6</u>
100GBASE-KP4	94.3.6.6
100GBASE-KR4	93.7.6
<u>100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4</u>	<u>162.8.6</u>
100GBASE-CR4	92.7.6
...	

45.2.1.21 200G PMA/PMD extended ability (Register 1.23)

Change the row for bits 1.23.8:7 in Table 45–24 (as modified by IEEE Std 802.3db-2022) as follows (unchanged rows not shown):

Table 45–24—200G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.23.8:7	Reserved	Value always 0	RO
<u>1.23.8</u>	<u>200GBASE-CR2 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-CR2</u> <u>0 = PMA/PMD is not able to perform 200GBASE-CR2</u>	<u>RO</u>
<u>1.23.7</u>	<u>200GBASE-KR2 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-KR2</u> <u>0 = PMA/PMD is not able to perform 200GBASE-KR2</u>	<u>RO</u>
...			

^aRO = Read only

Insert 45.2.1.21.1c and 45.2.1.21.1d after 45.2.1.21.1b (inserted by IEEE Std 802.3db-2022) as follows:

45.2.1.21.1c 200GBASE-CR2 ability (1.23.8)

When read as a one, bit 1.23.8 indicates that the PMA/PMD is able to operate as a 200GBASE-CR2 PMA/PMD type. When read as a zero, bit 1.23.8 indicates that the PMA/PMD is not able to operate as a 200GBASE-CR2 PMA/PMD type.

45.2.1.21.1d 200GBASE-KR2 ability (1.23.7)

When read as a one, bit 1.23.7 indicates that the PMA/PMD is able to operate as a 200GBASE-KR2 PMA/PMD type. When read as a zero, bit 1.23.7 indicates that the PMA/PMD is not able to operate as a 200GBASE-KR2 PMA/PMD type.

45.2.1.22 400G PMA/PMD extended ability register (Register 1.24)

Change the row for bits 1.24.1:0 in Table 45–25 as follows (unchanged rows not shown):

Table 45–25—400G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.24.1:0	Reserved	Value always 0	RO
<u>1.24.1</u>	<u>400GBASE-CR4 ability</u>	<u>1 = PMA/PMD is able to perform 400GBASE-CR4</u> <u>0 = PMA/PMD is not able to perform 400GBASE-CR4</u>	<u>RO</u>
<u>1.24.0</u>	<u>400GBASE-KR4 ability</u>	<u>1 = PMA/PMD is able to perform 400GBASE-KR4</u> <u>0 = PMA/PMD is not able to perform 400GBASE-KR4</u>	<u>RO</u>

^aRO = Read only

Insert 45.2.1.22.11 and 45.2.1.22.12 after 45.2.1.22.10 as follows:

45.2.1.22.11 400GBASE-CR4 ability (1.24.1)

When read as a one, bit 1.24.1 indicates that the PMA/PMD is able to operate as a 400GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.24.1 indicates that the PMA/PMD is not able to operate as a 400GBASE-CR4 PMA/PMD type.

45.2.1.22.12 400GBASE-KR4 ability (1.24.0)

When read as a one, bit 1.24.0 indicates that the PMA/PMD is able to operate as a 400GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.24.0 indicates that the PMA/PMD is not able to operate as a 400GBASE-KR4 PMA/PMD type.

45.2.1.24 40G/100G PMA/PMD extended ability 2 (Register 1.26)

Change the row for bits 1.26.1:0 in Table 45–27 (as modified by IEEE Std 802.3db-2022) as follows (unchanged rows not shown):

Table 45–27—40G/100G PMA/PMD extended ability register 2 bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.26.1:0	Reserved	Value always 0	RO
<u>1.26.1</u>	<u>100GBASE-CR1 ability</u>	<u>1 = PMA/PMD is able to perform 100GBASE-CR1</u> <u>0 = PMA/PMD is not able to perform 100GBASE-CR1</u>	<u>RO</u>
<u>1.26.0</u>	<u>100GBASE-KR1 ability</u>	<u>1 = PMA/PMD is able to perform 100GBASE-KR1</u> <u>0 = PMA/PMD is not able to perform 100GBASE-KR1</u>	<u>RO</u>

^aRO = Read only

Insert 45.2.1.24.9 and 45.2.1.24.10 after 45.2.1.24.8 (as inserted by IEEE Std 802.3db-2022) as follows:

45.2.1.24.9 100GBASE-CR1 ability (1.26.1)

When read as a one, bit 1.26.1 indicates that the PMA/PMD is able to operate as a 100GBASE-CR1 PMA/PMD type. When read as a zero, bit 1.26.1 indicates that the PMA/PMD is not able to operate as a 100GBASE-CR1 PMA/PMD type.

45.2.1.24.10 100GBASE-KR1 ability (1.26.0)

When read as a one, bit 1.26.0 indicates that the PMA/PMD is able to operate as a 100GBASE-KR1 PMA/PMD type. When read as a zero, bit 1.26.0 indicates that the PMA/PMD is not able to operate as a 100GBASE-KR1 PMA/PMD type.

45.2.1.95 BASE-R PMD control register (Register 1.150)

Change the first sentence of 45.2.1.95 as follows:

The BASE-R PMD control register is used for 10GBASE-KR and other PHY types using the PMDs described in [Clause 72](#), [Clause 84](#), [Clause 85](#), [Clause 92](#), [Clause 93](#), [Clause 94](#), [Clause 110](#), [Clause 111](#), [Clause 136](#), ~~or~~ [Clause 137](#), [Clause 162](#), or [Clause 163](#).

45.2.1.96 BASE-R PMD status register (Register 1.151)

Change the first sentence of 45.2.1.96 as follows:

The BASE-R PMD status register is used for 10GBASE-KR and other PHY types using the PMDs described in [Clause 72](#), [Clause 84](#), [Clause 85](#), [Clause 92](#), [Clause 93](#), [Clause 94](#), [Clause 110](#), [Clause 111](#), [Clause 136](#), ~~or~~ [Clause 137](#), [Clause 162](#), or [Clause 163](#).

45.2.1.116 RS-FEC control register (Register 1.200)

Change the row for bits 1.200.15:5 in Table 45–94 as follows (unchanged rows not shown):

Table 45–94—RS-FEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.200.15:7	Reserved	Value always 0	RO
1.200.6	100G RS-FEC enable ^b	1 = Clause 91 RS-FEC is enabled 0 = Clause 91 RS-FEC is disabled	R/W
1.200.5	100G RS-FEC-Int enable	1 = Clause 161 codeword-interleaved RS-FEC is enabled 0 = Clause 161 codeword-interleaved RS-FEC is disabled	R/W
...			

^aR/W = Read/Write, RO = Read only

^bBit 1.200.6 is present only in PHYs that include an alternative FEC sublayer, such as RS-FEC-Int defined in Clause 161

Insert 45.2.1.116.a and 45.2.1.116.b before 45.2.116.1 as follows:

45.2.1.116.a 100G RS-FEC enable (1.200.6)

Bit 1.200.6 enables the Reed-Solomon FEC described in Clause 91 for PHYs that include multiple FEC sublayers.

45.2.1.116.b 100G RS-FEC-Int enable (1.200.5)

Bit 1.200.5 enables the codeword-interleaved Reed-Solomon FEC described in Clause 161.

45.2.1.116.1 FEC degraded SER enable (1.200.4)

Change the first sentence of 45.2.1.116.1 as follows:

This bit enables the RS-FEC decoder to indicate the presence of a degraded SER (see 91.5.3.3.1, ~~and 134.5.3.3.2, and 161.5.3.3.2~~).

45.2.1.116.4 FEC bypass indication enable (1.200.1)

Change the last sentence of 45.2.1.116.4 as follows:

Writes to bit 1.200.1 are ignored and reads return a zero if the RS-FEC does not have the ability to bypass indicating decoding errors to the PCS layer (see 91.5.3.3, ~~108.5.3.2, and 134.5.3.3.1, and 161.5.3.3.1~~).

45.2.1.117 RS-FEC status register (Register 1.201)

45.2.1.117.1 PCS align status (1.201.15)

Change the second sentence of 45.2.1.117.1 as follows:

For the RS-FEC described in Clause 91, ~~or Clause 134,~~ and Clause 161, PCS alignment is defined as block lock, alignment marker lock, and deskew of all transmit PCS lanes.

45.2.1.117.2 RS-FEC align status (1.201.14)

Change the second sentence of 45.2.1.117.2 as follows:

For the RS-FEC described in Clause 91, ~~or Clause 134,~~ and Clause 161, PMA alignment is defined as alignment marker lock and deskew of all lanes on the PMA service interface.

45.2.1.117.3 FEC AM lock 3 (1.201.11)

Change 45.2.1.117.3 as follows:

When read as a one, bit 1.201.11 indicates that the RS-FEC described in Clause 91 and Clause 161 has locked and aligned lane 3 of the PMA service interface. When read as a zero, bit 1.201.11 indicates that the RS-FEC has not locked and aligned lane 3 of the PMA service interface. This bit reflects the state of `amps_lock[3]` (see [91.5.3.1](#) and 161.5.3.1).

45.2.1.117.4 FEC AM lock 2 (1.201.10)

Change 45.2.1.117.4 as follows:

When read as a one, bit 1.201.10 indicates that the RS-FEC described in Clause 91 and Clause 161 has locked and aligned lane 2 of the PMA service interface. When read as a zero, bit 1.201.10 indicates that the RS-FEC has not locked and aligned lane 2 of the PMA service interface. This bit reflects the state of `amps_lock[2]` (see [91.5.3.1](#) and 161.5.3.1).

45.2.1.117.5 FEC AM lock 1 (1.201.9)

Change 45.2.1.117.5 as follows:

When read as a one, bit 1.201.9 indicates that the RS-FEC described in Clause 91, ~~or Clause 134,~~ and Clause 161 has locked and aligned lane 1 of the PMA service interface. When read as a zero, bit 1.201.9 indicates that the RS-FEC has not locked and aligned lane 1 of the PMA service interface. This bit reflects the state of `amps_lock[1]` (see [91.5.3.1](#) and 161.5.3.1).

45.2.1.117.6 FEC AM lock 0 (1.201.8)

Change 45.2.1.117.6 as follows:

When read as a one, bit 1.201.8 indicates that the RS-FEC described in Clause 91, ~~or Clause 134,~~ and Clause 161 has locked and aligned lane 0 of the PMA service interface. When read as a zero, bit 1.201.8 indicates that the RS-FEC has not locked and aligned lane 0 of the PMA service interface. This bit reflects the state of `amps_lock[0]` (see [91.5.3.1](#) and 161.5.3.1).

45.2.1.117.8 FEC degraded SER (1.201.4)

Change the first sentence of 45.2.1.117.8 as follows:

This bit (1.201.4) reflects the state of the variable FEC_degraded_SER (see [91.6.12](#), ~~and [134.6.9](#)~~, and [161.6.10](#)).

45.2.1.117.9 FEC degraded SER ability (1.201.3)

Change the first sentence of 45.2.1.117.9 as follows:

This bit is set to one to indicate that the decoder has the ability to indicate the presence of a degraded SER (see [91.5.3.3.1](#), ~~and [134.5.3.3.2](#)~~, and [161.5.3.3.2](#)).

45.2.1.117.10 RS-FEC high SER (1.201.2)

Change the first sentence of 45.2.1.117.10 as follows:

When FEC_bypass_indication_enable is set to one, bit 1.201.2 is set to one if the number of RS-FEC symbol errors in a window of 8192 codewords exceeds the threshold (see [91.5.3.3](#), [108.5.3.2](#), ~~and [134.5.3.3.1](#)~~, and [161.5.3.3.1](#)) and is set to zero otherwise.

45.2.1.117.11 FEC bypass indication ability (1.201.1)

Change the first sentence of 45.2.1.117.11 as follows:

The Reed-Solomon decoder may have the option to perform error detection without error indication (see [91.5.3.3](#), [108.5.3.2](#), ~~and [134.5.3.3.1](#)~~, and [161.5.3.3.1](#)) to reduce the delay contributed by the RS-FEC sublayer.

45.2.1.118 RS-FEC corrected codewords counter (Register 1.202, 1.203)

Change the second sentence of 45.2.1.118 as follows:

See [91.6.16](#), ~~and [108.6.7](#)~~, and [161.6.13](#) for a definition of this register.

45.2.1.119 RS-FEC uncorrected codewords counter (Register 1.204, 1.205)

Change the second sentence of 45.2.1.119 as follows:

See [91.6.17](#), ~~and [108.6.8](#)~~, and [161.6.14](#) for a definition of this register.

Insert 45.2.1.120a before 45.2.121 as follows:

45.2.1.120a RS-FEC codeword counter register (Register 1.207, 1.208, 1.209)

The assignment of bits in the RS-FEC codeword counter register is shown in Table 45–98a. The RS-FEC codeword counter register applies to the codeword-interleaved RS-FEC defined in Clause 161. See [161.6.21](#) for a definition of this register. It is a 48-bit counter that counts once for each FEC codeword received when RS-FEC align status (1.201.14) is set to one. Its bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Registers 1.207, 1.208, and 1.209 are used to read the value of the 48-bit counter. When using these registers to read the 48-bit counter value, the register 1.207 is read first, the values of the registers 1.208 and 1.209 are

latched when (and only when) register 1.207 is read, and reads of registers 1.208 and 1.209 return their latched values rather than the current value of the counter.

Table 45–98a—RS-FEC codeword counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.207.15:0	RS-FEC codewords lower	FEC_cw_counter[15:0]	RO, NR
1.208.15:0	RS-FEC codewords middle	FEC_cw_counter[31:16]	RO, NR
1.209.15:0	RS-FEC codewords upper	FEC_cw_counter[47:32]	RO, NR

^aRO = Read only, NR = Non Roll-over

45.2.1.121 RS-FEC symbol error counter lane 0 (Register 1.210, 1.211)

Change the third sentence of 45.2.1.121 as follows:

See [91.6.19](#), ~~and 108.6.9~~, and [161.6.16](#) for a definition of this register.

45.2.1.123 RS-FEC BIP error counter lane 0 (Register 1.230)

Change the second sentence of 45.2.1.123 as follows:

The RS-FEC described in Clause 91, ~~or Clause 134~~, and [Clause 161](#) calculates a BIP value for each PCS lane (see [91.5.2.4](#), ~~91.6.8~~[91.6.21](#), [161.5.2.6](#)).

45.2.1.127 RS-FEC PCS alignment status 1 register (Register 1.280)

45.2.1.127.1 Block lock 7 (1.280.7)

Change the last sentence of 45.2.1.127.1 as follows:

This bit reflects the state of block_lock[7] (see [91.5.2.1](#), [161.6.22](#)).

45.2.1.127.2 Block lock 6 (1.280.6)

Change the last sentence of 45.2.1.127.2 as follows:

This bit reflects the state of block_lock[6] (see [91.5.2.1](#), [161.6.22](#)).

45.2.1.127.3 Block lock 5 (1.280.5)

Change the last sentence of 45.2.1.127.3 as follows:

This bit reflects the state of block_lock[5] (see [91.5.2.1](#), [161.6.22](#)).

45.2.1.127.4 Block lock 4 (1.280.4)

Change the last sentence of 45.2.1.127.4 as follows:

This bit reflects the state of block_lock[4] (see [91.5.2.1](#), [161.6.22](#)).

45.2.1.127.5 Block lock 3 (1.280.3)

Change the last sentence of 45.2.1.127.5 as follows:

This bit reflects the state of block_lock[3] (see 91.5.2.1, 161.6.22).

45.2.1.127.6 Block lock 2 (1.280.2)

Change the last sentence of 45.2.1.127.6 as follows:

This bit reflects the state of block_lock[2] (see 91.5.2.1, 161.6.22).

45.2.1.127.7 Block lock 1 (1.280.1)

Change the last sentence of 45.2.1.127.7 as follows:

This bit reflects the state of block_lock[1] (see 91.5.2.1, 161.6.22).

45.2.1.127.8 Block lock 0 (1.280.0)

Change the last sentence of 45.2.1.127.8 as follows:

This bit reflects the state of block_lock[0] (see 91.5.2.1, 161.6.22).

45.2.1.128 RS-FEC PCS alignment status 2 register (Register 1.281)

45.2.1.128.1 Block lock 19 (1.281.11)

Change the last sentence of 45.2.1.128.1 as follows:

This bit reflects the state of block_lock[19] (see 91.5.2.1, 161.6.22).

45.2.1.128.2 Block lock 18 (1.281.10)

Change the last sentence of 45.2.1.128.2 as follows:

This bit reflects the state of block_lock[18] (see 91.5.2.1, 161.6.22).

45.2.1.128.3 Block lock 17 (1.281.9)

Change the last sentence of 45.2.1.128.3 as follows:

This bit reflects the state of block_lock[17] (see 91.5.2.1, 161.6.22).

45.2.1.128.4 Block lock 16 (1.281.8)

Change the last sentence of 45.2.1.128.4 as follows:

This bit reflects the state of block_lock[16] (see 91.5.2.1, 161.6.22).

45.2.1.128.5 Block lock 15 (1.281.7)

Change the last sentence of 45.2.1.128.5 as follows:

This bit reflects the state of block_lock[15] (see 91.5.2.1, 161.6.22).

45.2.1.128.6 Block lock 14 (1.281.6)

Change the last sentence of 45.2.1.128.6 as follows:

This bit reflects the state of block_lock[14] (see 91.5.2.1, 161.6.22).

45.2.1.128.7 Block lock 13 (1.281.5)

Change the last sentence of 45.2.1.128.7 as follows:

This bit reflects the state of block_lock[13] (see 91.5.2.1, 161.6.22).

45.2.1.128.8 Block lock 12 (1.281.4)

Change the last sentence of 45.2.1.128.8 as follows:

This bit reflects the state of block_lock[12] (see 91.5.2.1, 161.6.22).

45.2.1.128.9 Block lock 11 (1.281.3)

Change the last sentence of 45.2.1.128.9 as follows:

This bit reflects the state of block_lock[11] (see 91.5.2.1, 161.6.22).

45.2.1.128.10 Block lock 10 (1.281.2)

Change the last sentence of 45.2.1.128.10 as follows:

This bit reflects the state of block_lock[10] (see 91.5.2.1, 161.6.22).

45.2.1.128.11 Block lock 9 (1.281.1)

Change the last sentence of 45.2.1.128.11 as follows:

This bit reflects the state of block_lock[9] (see 91.5.2.1, 161.6.22).

45.2.1.128.12 Block lock 8 (1.281.0)

Change the last sentence of 45.2.1.128.12 as follows:

This bit reflects the state of block_lock[8] (see 91.5.2.1, 161.6.22).

45.2.1.129 RS-FEC PCS alignment status 3 register (Register 1.282)

45.2.1.129.1 Lane 7 aligned (1.282.7)

Change the last sentence of 45.2.1.129.1 as follows:

This bit reflects the state of am_lock[7] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.2 Lane 6 aligned (1.282.6)

Change the last sentence of 45.2.1.129.2 as follows:

This bit reflects the state of am_lock[6] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.3 Lane 5 aligned (1.282.5)

Change the last sentence of 45.2.1.129.3 as follows:

This bit reflects the state of am_lock[5] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.4 Lane 4 aligned (1.282.4)

Change the last sentence of 45.2.1.129.4 as follows:

This bit reflects the state of am_lock[4] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.5 Lane 3 aligned (1.282.3)

Change the last sentence of 45.2.1.129.5 as follows:

This bit reflects the state of am_lock[3] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.6 Lane 2 aligned (1.282.2)

Change the last sentence of 45.2.1.129.6 as follows:

This bit reflects the state of am_lock[2] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.7 Lane 1 aligned (1.282.1)

Change the last sentence of 45.2.1.129.7 as follows:

This bit reflects the state of am_lock[1] (see 91.5.2.2, 161.5.2.2).

45.2.1.129.8 Lane 0 aligned (1.282.0)

Change the last sentence of 45.2.1.129.8 as follows:

This bit reflects the state of am_lock[0] (see 91.5.2.2, 161.5.2.2).

45.2.1.130 RS-FEC PCS alignment status 4 register (Register 1.283)

45.2.1.130.1 Lane 19 aligned (1.283.11)

Change the last sentence of 45.2.1.130.1 as follows:

This bit reflects the state of am_lock[19] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.2 Lane 18 aligned (1.283.10)

Change the last sentence of 45.2.1.130.2 as follows:

This bit reflects the state of am_lock[18] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.3 Lane 17 aligned (1.283.9)

Change the last sentence of 45.2.1.130.3 as follows:

This bit reflects the state of am_lock[17] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.4 Lane 16 aligned (1.283.8)

Change the last sentence of 45.2.1.130.4 as follows:

This bit reflects the state of am_lock[16] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.5 Lane 15 aligned (1.283.7)

Change the last sentence of 45.2.1.130.5 as follows:

This bit reflects the state of am_lock[15] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.6 Lane 14 aligned (1.283.6)

Change the last sentence of 45.2.1.130.6 as follows:

This bit reflects the state of am_lock[14] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.7 Lane 13 aligned (1.283.5)

Change the last sentence of 45.2.1.130.7 as follows:

This bit reflects the state of am_lock[13] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.8 Lane 12 aligned (1.283.4)

Change the last sentence of 45.2.1.130.8 as follows:

This bit reflects the state of am_lock[12] (see [91.5.2.2](#), [161.5.2.2](#)).

45.2.1.130.9 Lane 11 aligned (1.283.3)

Change the last sentence of 45.2.1.130.9 as follows:

This bit reflects the state of am_lock[11] (see 91.5.2.2, 161.5.2.2).

45.2.1.130.10 Lane 10 aligned (1.283.2)

Change the last sentence of 45.2.1.130.10 as follows:

This bit reflects the state of am_lock[10] (see 91.5.2.2, 161.5.2.2).

45.2.1.130.11 Lane 9 aligned (1.283.1)

Change the last sentence of 45.2.1.130.11 as follows:

This bit reflects the state of am_lock[9] (see 91.5.2.2, 161.5.2.2).

45.2.1.130.12 Lane 8 aligned (1.283.0)

Change the last sentence of 45.2.1.130.12 as follows:

This bit reflects the state of am_lock[8] (see 91.5.2.2, 161.5.2.2).

Insert 45.2.1.131a before 45.2.132 as follows:

45.2.1.131a RS-FEC codeword error bin registers 1 through 15 (Registers 1.340 through 1.369)

The assignment of bits in the RS-FEC codeword error bin 1 register is shown in Table 45–105a. The assignment of bits in the other RS-FEC codeword error bin registers is identical to that of bin 1. The RS-FEC codeword error bin registers increment depending upon the error signature of a corrected codeword (see 161.6.17). Their bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Two registers are used to read the value of each 32-bit counter, the value of the second register is latched when the first register is read. For example when registers 1.340 and 1.341 are used to read the 32-bit counter value of FEC_codeword_error_bin_1, the register 1.340 is read first, the value of the register 1.341 is latched when (and only when) register 1.340 is read, and reads of register 1.341 return the latched value rather than the current value of the counter.

Table 45–105a—RS-FEC codeword error bin 1 bit definitions

Bit(s)	Name	Description	R/W ^a
1.340.15:0	RS-FEC codeword error bin 1 lower	FEC_codeword_error_bin_1[15:0]	RO, NR
1.341.15:0	RS-FEC codeword error bin 1 upper	FEC_codeword_error_bin_1[31:16]	RO, NR

^aRO = Read only, NR = Non Roll-over

45.2.1.146 RS-FEC degraded SER activate threshold register (Register 1.650, 1.651)

Change the second sentence of 45.2.1.146 as follows:

The value controls the threshold used to set the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1, ~~and 134.5.3.3.2, and 161.5.3.3.2.~~

45.2.1.147 RS-FEC degraded SER deactivate threshold register (Register 1.652, 1.653)

Change the second sentence of 45.2.1.147 as follows:

The value controls the threshold used to clear the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1, ~~and 134.5.3.3.2, and 161.5.3.3.2.~~

45.2.1.148 RS-FEC degraded SER interval register (Register 1.654, 1.655)

Change the second sentence of 45.2.1.148 as follows:

The value controls the interval used to set and clear the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1, ~~and 134.5.3.3.2, and 161.5.3.3.2.~~

45.2.1.161 BASE-R PAM4 PMD training LP control, lane 0 through lane 3 registers (Register 1.1120 through 1.1123)

Change Table 45–129 as follows:

Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1120.15:14	Reserved	Value always 0	RO
1.1120.13:12 <u>1.1120.13:11</u>	Initial condition request ^b	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control	R/W
1.1120.11:10 <u>1.1120.10</u>	Reserved	Value always 0	RO
1.1120.9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	R/W
1.1120.7:5	Reserved	Value always 0	RO

Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.1120.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1120.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^aR/W = Read/Write, RO = Read only

^bSupport for a given preset is PHY dependent

^cSupport for a given coefficient is PHY dependent

45.2.1.163 BASE-R PAM4 PMD training LP status, lane 0 through lane 3 registers (Register 1.1220 through 1.1223)

Change Table 45–130 as follows:

Table 45–130—BASE-R PAM4 PMD training LP status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1220.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1220.14:12	Reserved	Value always 0	RO
1.1220.11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	RO
1.1220.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1220.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1220.7	Parity	Even parity bit	RO
1.1220.6	Reserved	Value always 0	RO

Table 45–130—BASE-R PAM4 PMD training LP status, lane 0 register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.1220.5:3	Coefficient select echo ^b	5 4 3 $\underline{1} \ \underline{0} \ \underline{0} = \text{Reserved}$ $\underline{1} \ \underline{0} \ \underline{1} = c(-3)$ $\underline{1} \ \underline{1} \ \underline{0} = c(-2)$ $\underline{1} \ \underline{1} \ \underline{1} = c(-1)$ $\underline{0} \ \underline{0} \ \underline{0} = c(0)$ $\underline{0} \ \underline{0} \ \underline{1} = c(1)$ $\underline{0} \ \underline{1} \ \underline{x} = \text{Reserved}$	RO
1.1220.2:0	Coefficient status	2 1 0 $\underline{1} \ \underline{1} \ \underline{1} = \text{Reserved}$ $\underline{1} \ \underline{1} \ \underline{0} = \text{Coefficient at limit and equalization limit}$ $\underline{1} \ \underline{0} \ \underline{1} = \text{Reserved}$ $\underline{1} \ \underline{0} \ \underline{0} = \text{Equalization limit}$ $\underline{0} \ \underline{1} \ \underline{1} = \text{Coefficient not supported}$ $\underline{0} \ \underline{1} \ \underline{0} = \text{Coefficient at limit}$ $\underline{0} \ \underline{0} \ \underline{1} = \text{Updated}$ $\underline{0} \ \underline{0} \ \underline{0} = \text{Not updated}$	RO

^aRO = Read only

^bSupport for a given coefficient is PHY dependent

45.2.1.165 BASE-R PAM4 PMD training LD control, lane 0 through lane 3 registers (Register 1.1320 through 1.1323)

Change Table 45–131 as follows:

Table 45–131—BASE-R PAM4 PMD training LD control, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1320.15:14	Reserved	Value always 0	RO
1.1320.13:12 1.1320.13:11	Initial condition request ^b	13 12 11 $\underline{1} \ \underline{1} \ \underline{1} = \text{Reserved}$ $\underline{1} \ \underline{0} \ \underline{1} = \text{Reserved}$ $\underline{0} \ \underline{1} \ \underline{1} = \text{Preset 5}$ $\underline{0} \ \underline{0} \ \underline{1} = \text{Preset 4}$ $\underline{1} \ \underline{1} \ \underline{0} = \text{Preset 3}$ $\underline{1} \ \underline{0} \ \underline{0} = \text{Preset 2}$ $\underline{0} \ \underline{1} \ \underline{0} = \text{Preset 1}$ $\underline{0} \ \underline{0} \ \underline{0} = \text{Individual coefficient control}$	R/W
1.1320.11:10 1.1320.10	Reserved	Value always 0	RO
1.1320.9:8	Modulation and precoding request	9 8 $\underline{1} \ \underline{1} = \text{PAM4 with precoding}$ $\underline{1} \ \underline{0} = \text{PAM4}$ $\underline{0} \ \underline{1} = \text{Reserved}$ $\underline{0} \ \underline{0} = \text{PAM2}$	R/W
1.1320.7:5	Reserved	Value always 0	RO

Table 45–131—BASE-R PAM4 PMD training LD control, lane 0 register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.1320.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1320.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^aR/W = Read/Write, RO = Read only

^bSupport for a given preset is PHY dependent

^cSupport for a given coefficient is PHY dependent

45.2.1.167 BASE-R PAM4 PMD training LD status, lane 0 through lane 3 registers (Register 1.420 through 1.423)

Change Table 45–132 as follows:

Table 45–132—BASE-R PAM4 PMD training LD status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1420.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1420.14:12	Reserved	Value always 0	RO
1.1420.11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	RO
1.1420.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1420.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1420.7	Parity	Even parity bit	RO
1.1420.6	Reserved	Value always 0	RO

Table 45–132—BASE-R PAM4 PMD training LD status, lane 0 register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.1420.5:3	Coefficient select echo ^b	5 4 3 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	RO
1.1420.2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated	RO

^aRO = Read only

^bSupport for a given coefficient is PHY dependent

45.2.1.169 Test-pattern ability (Register 1.1500)

Change the row for bits 1.1500.15:14 in Table 45–134 as follows (unchanged rows not shown):

Table 45–134—Test-pattern ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1500.15:14	Reserved PRBS9Q Tx generator ability	Value always 0 1 = PRBS9Q transmit direction pattern generator supported 0 = PRBS9Q transmit direction pattern generator not supported	RO
1.1500.14	PRBS9Q Rx generator ability	1 = PRBS9Q receive direction pattern generator supported 0 = PRBS9Q receive direction pattern generator not supported	RO
...			

^aRO = Read only

Insert new paragraph before the last paragraph of 45.2.1.169 as follows:

If PRBS9Q pattern testing is supported and register 1.1500 is implemented then bit 1.1500.15 shall indicate the generation ability in the transmit direction and bit 1.1500.14 shall indicate the generation ability in the receive direction. The PRBS9Q pattern test is controlled by register 1.1511 (see 45.2.1.171a).

Insert 45.2.1.171a after 45.2.1.171 as follows:

45.2.1.171a PRBS9Q testing control (Register 1.1511)

The PRBS9Q testing control register is used for PHY types that implement PRBS9Q testing in the PMA. This function is described in 120.5.11.2.a. The assignment of bits in the PRBS9Q testing control register is shown in Table 45–136a.

Table 45–136a—PRBS9Q testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1511.15:8	Reserved	Value always 0	RO
1.1511.7	Lane 7 PRBS9Q enable	1 = Enable PRBS9Q on lane 7 0 = Disable PRBS9Q on lane 7	R/W
1.1511.6	Lane 6 PRBS9Q enable	1 = Enable PRBS9Q on lane 6 0 = Disable PRBS9Q on lane 6	R/W
1.1511.5	Lane 5 PRBS9Q enable	1 = Enable PRBS9Q on lane 5 0 = Disable PRBS9Q on lane 5	R/W
1.1511.4	Lane 4 PRBS9Q enable	1 = Enable PRBS9Q on lane 4 0 = Disable PRBS9Q on lane 4	R/W
1.1511.3	Lane 3 PRBS9Q enable	1 = Enable PRBS9Q on lane 3 0 = Disable PRBS9Q on lane 3	R/W
1.1511.2	Lane 2 PRBS9Q enable	1 = Enable PRBS9Q on lane 2 0 = Disable PRBS9Q on lane 2	R/W
1.1511.1	Lane 1 PRBS9Q enable	1 = Enable PRBS9Q on lane 1 0 = Disable PRBS9Q on lane 1	R/W
1.1511.0	Lane 0 PRBS9Q enable	1 = Enable PRBS9Q on lane 0 0 = Disable PRBS9Q on lane 0	R/W

^aR/W = Read/Write, RO = Read only

Register 1.1511, bits 0 through 7 enable PRBS9Q output on PMA lanes 0 through 7 respectively. The assertion of register 1.1511 bits 0 through 7 operates in conjunction with register 1.1501 bits 3 and 1. If neither bit is asserted, then register 1.1511 bits 0 through 7 have no effect. Lanes for which a PRBS9Q pattern is not enabled act as determined by other registers.

45.2.1.216 IFEC control register (Register 1.2200)

Change Table 45–180 as follows:

Table 45–180—RS-FEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2200.15:32	Reserved	Value always 0	RO
1.2200.2	IFEC enable	1 = Inverse RS-FEC is enabled 0 = Inverse RS-FEC is disabled	R/W
1.2200.1	IFEC bypass indication enable	1 = IFEC decoder does not indicate errors 0 = IFEC decoder indicates errors	R/W
1.2200.0	IFEC bypass correction enable	1 = IFEC decoder performs error detection without error correction 0 = IFEC decoder performs error detection and error correction	R/W

^aR/W = Read/Write, RO = Read only

Insert 45.2.1.216.a before 45.2.1.216.1 as follows:

45.2.1.216.a IFEC enable (1.2200.2)

Bit 1.2200.2 enables the Inverse Reed-Solomon FEC described in Clause 152 for PHYs that have the ability to bypass the FEC sublayer below the Inverse RS-FEC sublayer.

45.2.7 Auto-Negotiation registers

45.2.7.12 Backplane Ethernet, BASE-R copper status (Register 7.48)

Change the title of 45.2.7.12.3 as follows:

45.2.7.12.3 Negotiated Port Type (7.48.1, 7.48.2, 7.48.3, 7.48.5, 7.48.6, 7.48.8, 7.48.9, 7.48.10, 7.48.11, 7.48.12, 7.48.13, 7.48.14, 7.48.15, 7.49.0, 7.49.1, 7.49.2, 7.49.3, 7.49.4, 7.49.5)

45.2.7.13 Backplane Ethernet, BASE-R copper status 2 (Register 7.49)

Change Table 45–388 as follows:

**Table 45–388—Backplane Ethernet, BASE-R copper status 2 register (Register 7.49)
bit definitions**

Bit(s)	Name	Description	RO ^a
7.49.15:7 3	Reserved	Value always 0	RO
<u>7.49.6</u>	<u>RS-FEC-Int negotiated</u>	<u>1 = PMA/PMD is negotiated to perform interleaved RS-FEC</u> <u>0 = PMA/PMD is not negotiated to perform interleaved RS-FEC</u>	<u>RO</u>
<u>7.49.5</u>	<u>400GBASE-KR4 or 400GBASE-CR4</u>	<u>1 = PMA/PMD is negotiated to perform 400GBASE-KR4 or 400GBASE-CR4</u> <u>0 = PMA/PMD is not negotiated to perform 400GBASE-KR4 or 400GBASE-CR4</u>	<u>RO</u>
<u>7.49.4</u>	<u>200GBASE-KR2 or 200GBASE-CR2</u>	<u>1 = PMA/PMD is negotiated to perform 200GBASE-KR2 or 200GBASE-CR2</u> <u>0 = PMA/PMD is not negotiated to perform 200GBASE-KR2 or 200GBASE-CR2</u>	<u>RO</u>
<u>7.49.3</u>	<u>100GBASE-KR1 or 100GBASE-CR1</u>	<u>1 = PMA/PMD is negotiated to perform 100GBASE-KR1 or 100GBASE-CR1</u> <u>0 = PMA/PMD is not negotiated to perform 100GBASE-KR1 or 100GBASE-CR1</u>	<u>RO</u>
7.49.2	200GBASE-KR4 or 200GBASE-CR4	1 = PMA/PMD is negotiated to perform 200GBASE-KR4 or 200GBASE-CR4 0 = PMA/PMD is not negotiated to perform 200GBASE-KR4 or 200GBASE-CR4	RO
7.49.1	100GBASE-KR2 or 100GBASE-CR2	1 = PMA/PMD is negotiated to perform 100GBASE-KR2 or 100GBASE-CR2 0 = PMA/PMD is not negotiated to perform 100GBASE-KR2 or 100GBASE-CR2	RO
7.49.0	50GBASE-KR or 50GBASE-CR	1 = PMA/PMD is negotiated to perform 50GBASE-KR or 50GBASE-CR 0 = PMA/PMD is not negotiated to perform 50GBASE-KR or 50GBASE-CR	RO

^aRO = Read only

Insert new subclause 45.2.7.13.a before 45.2.7.13.1 as follows:

45.2.7.13.a RS-FEC-Int negotiated (7.49.6)

When the Auto-Negotiation process has completed as indicated by the AN complete bit (7.1.5), bit 7.49.6 indicates that forward error correction codeword-interleaved (RS-FEC-Int) operation as defined in Clause 161 has been negotiated. This bit is set only if RS-FEC-Int operation has been negotiated for a 100GBASE-P PHY supporting negotiation of RS-FEC-Int operation.

Change 45.2.7.13.1 as follows:

45.2.7.13.1 Negotiated Port Type

See 45.2.7.12.3 for the definition of bits 7.49.5 through 7.49.0.

69. Introduction to Ethernet operation over electrical backplanes

69.1 Overview

69.1.1 Scope

Change the second paragraph of 69.1.1 as follows:

Backplane Ethernet supports the IEEE 802.3 full duplex MAC operating at 1000 Mb/s, 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, 40 Gb/s, 50 Gb/s, 100 Gb/s, ~~or 200 Gb/s~~, or 400 Gb/s. The following Physical Layers are supported:

Change the dashed list after the second paragraph of 69.1.1 as follows:

- 1000BASE-KX for 1 Gb/s operation over a single lane
- 2.5GBASE-KX for 2.5 Gb/s operation over a single lane
- 5GBASE-KR for 5 Gb/s operation over a single lane
- 10GBASE-KX4 for 10 Gb/s operation over four lanes
- 10GBASE-KR for 10 Gb/s operation over a single lane
- 25GBASE-KR and 25GBASE-KR-S for 25 Gb/s operation over a single lane
- 40GBASE-KR4 for 40 Gb/s operation over four lanes
- 50GBASE-KR for 50 Gb/s operation over a single lane
- 100GBASE-KR4 and 100GBASE-KP4 for 100 Gb/s operation over four lanes
- 100GBASE-KR2 for 100 Gb/s operation over two lanes
- 100GBASE-KR1 for 100 Gb/s operation over one lane
- 200GBASE-KR4 for 200 Gb/s operation over four lanes
- 200GBASE-KR2 for 200 Gb/s operation over two lanes
- 400GBASE-KR4 for 400 Gb/s operation over four lanes

69.1.2 Relationship of Backplane Ethernet to the ISO OSI reference model

Replace Figure 69–5 with the following figure:

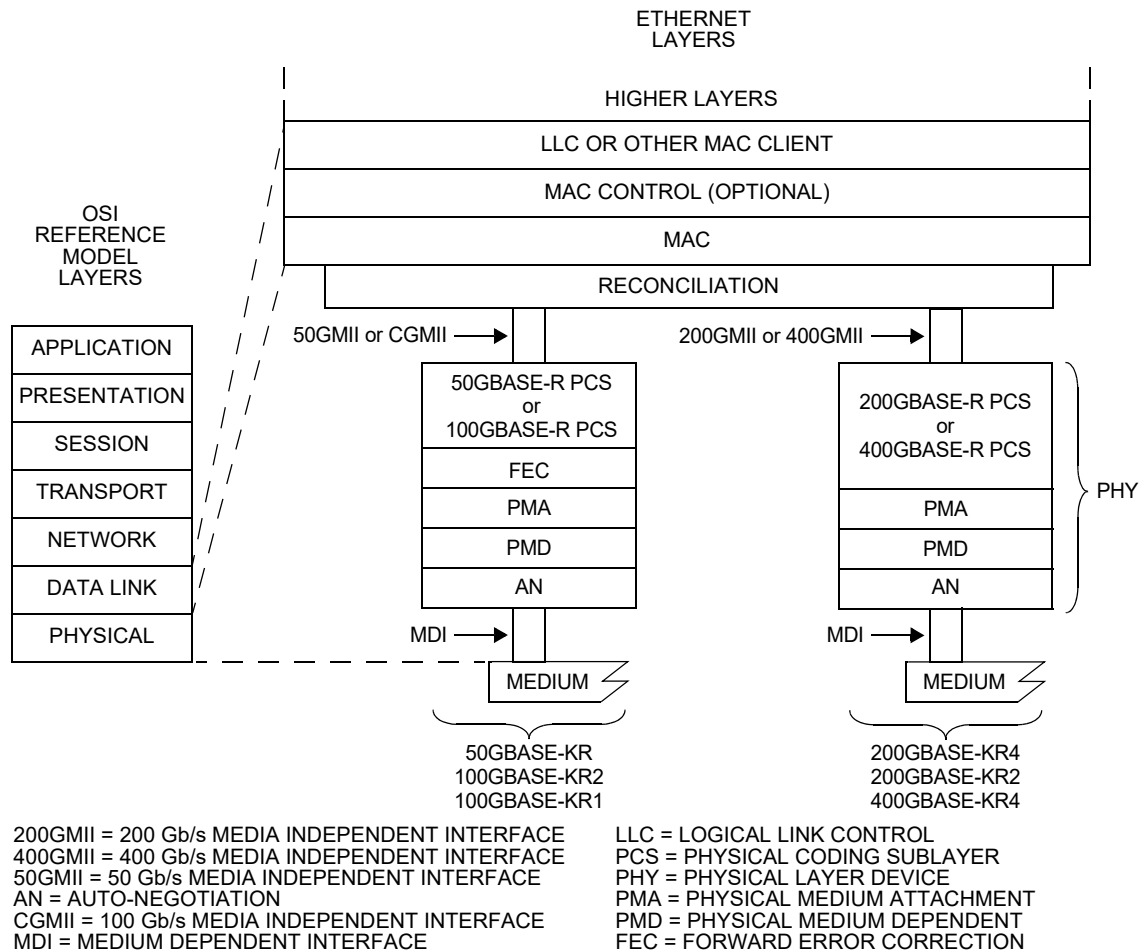


Figure 69–5—Architectural positioning of 50 Gb/s, 100 Gb/s, 200 Gb/s, and 400 Gb/s PAM4 Backplane Ethernet

69.2 Summary of Backplane Ethernet Sublayers

69.2.1 Reconciliation sublayer and media independent interfaces

Change 69.2.1 as follows:

The [Clause 35](#) RS and GMII, the [Clause 46](#) RS and XGMII, the [Clause 106](#) RS and 25GMII, the [Clause 81](#) RS, XLGMII, and CGMII, the [Clause 132](#) RS and 50GMII, and the [Clause 117](#) RS, ~~and 200GMII, and 400GMII~~ are employed for the same purpose in Backplane Ethernet, that being the interconnection between the MAC sublayer and the PHY.

69.2.3 Physical Layer signaling systems

Insert the following new paragraph into 69.2.3 after the tenth paragraph (“Backplane Ethernet also specifies 100GBASE-KP4...”):

Backplane Ethernet also specifies 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4. The 100GBASE-KR1 embodiment employs the PCS defined in Clause 82, the RS-FEC defined in Clause 91, the RS-FEC-Int defined in Clause 161, the PMA defined in Clause 135, and the PMD defined in Clause 163, and specifies 100 Gb/s operation using 4-level PAM over one differential path in each direction. The 200GBASE-KR2 embodiment employs the PCS defined in Clause 119, the PMA defined in Clause 120, and the PMD defined in Clause 163, and specifies 200 Gb/s operation using 4-level PAM over two differential paths in each direction. The 400GBASE-KR4 embodiment employs the PCS defined in Clause 119, the PMA defined in Clause 120, and the PMD defined in Clause 163, and specifies 400 Gb/s operation using 4-level PAM over four differential paths in each direction.

Change the last paragraph in 69.2.3 as follows:

Table 69–1 specifies the correlation between nomenclature and clauses for 1 Gb/s and 10 Gb/s backplane Ethernet. For other backplane PHY types, refer to Table 125–2 (2.5 Gb/s and 5 Gb/s), Table 105–1 (25 Gb/s), Table 80–2 (40 Gb/s), Table 131–2 (50 Gb/s), Table 80–3 (100 Gb/s), and Table 116–3 (200 Gb/s), and Table 116–3a (400 Gb/s).

69.2.6 Low-Power Idle

Change 69.2.6 as follows:

~~With the optional EEE feature, described in Clause 78, Backplane Ethernet PHYs can~~ Some Backplane Ethernet PHYs support the optional EEE feature, described in Clause 78, to achieve lower power consumption during periods of low link utilization. The EEE capabilities are advertised during Auto-Negotiation for Backplane Ethernet. The Backplane Ethernet LPI allows each link direction to enter sleep, refresh, or wake states asymmetric from the other direction.

69.3 Delay constraints

Insert the following paragraph into 69.3 after the tenth paragraph (“For 100GBASE-KR2...”):

For 100GBASE-KR1, normative delay specifications may be found in 81.1.4, 82.5, 135.5.4, and 163.5, and also referenced in 80.4.

Insert the following paragraph into 69.3 after the now twelfth paragraph (“For 200GBASE-KR4...”):

For 200GBASE-KR2 and 400GBASE-KR4, normative delay specifications may be found in 117.1.4, 119.5, 120.5.4, and 163.5, and also referenced in 116.4.

69.5 Protocol implementation conformance statement (PICS) proforma

Change the first paragraph of 69.5 as follows:

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 70 through Clause 74, Clause 84, Clause 91, Clause 93, Clause 94, Clause 108, Clause 111, Clause 119, Clause 128, Clause 130, Clause 134, Clause 137, Clause 161, Clause 163, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

73. Auto-Negotiation for backplane and copper cable assembly

73.2 Relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model

Replace Figure 73–1 with the following figure:

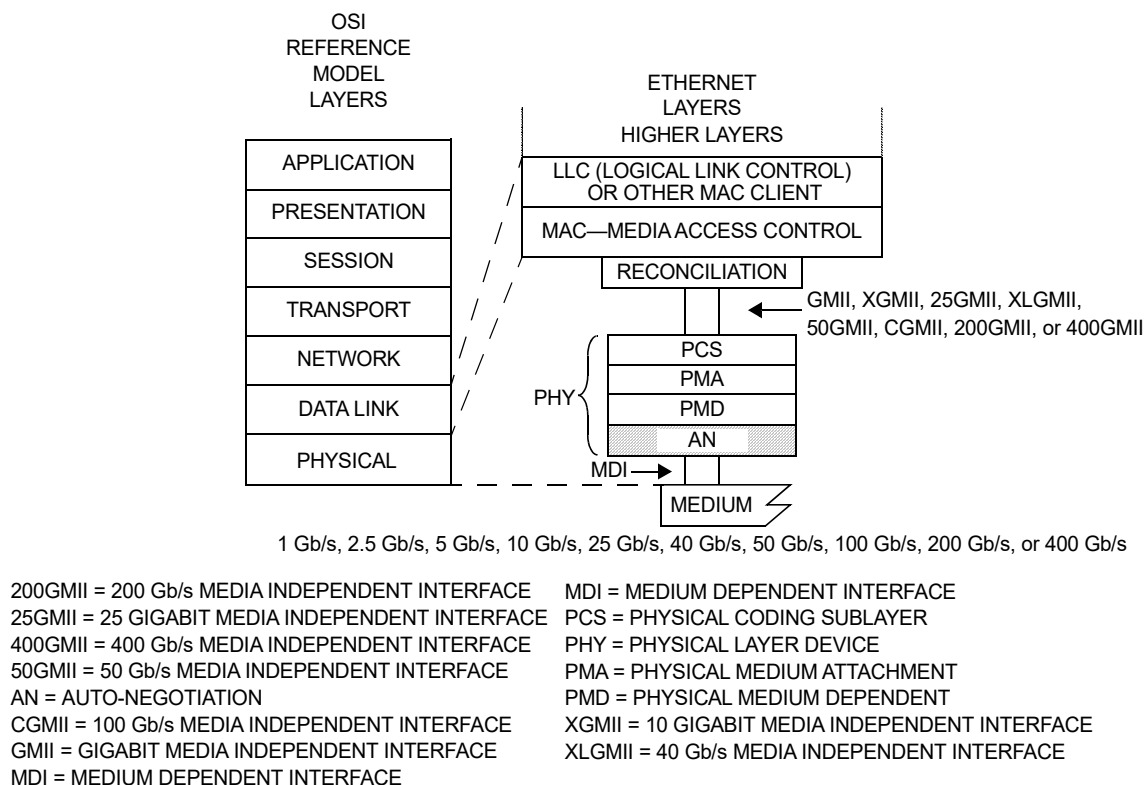


Figure 73–1—Location of Auto-Negotiation function within the ISO/IEC OSI reference model

73.5 DME transmission

73.5.1 DME electrical specifications

Change the second paragraph of 73.5.1 as follows:

For any multi-lane PHY, DME pages shall be transmitted only on lane 0. The transmitters on other lanes should be disabled as specified in 71.6.7, 84.7.7, 85.7.7, 92.7.7, 93.7.7, 94.3.6.7, 136.8.7, ~~or 137.8.7~~, 162.8.7, or 163.8.7.

73.6 Link codeword encoding

Replace Figure 73–6 with the following figure to make D43 indicate F4 rather than A22:

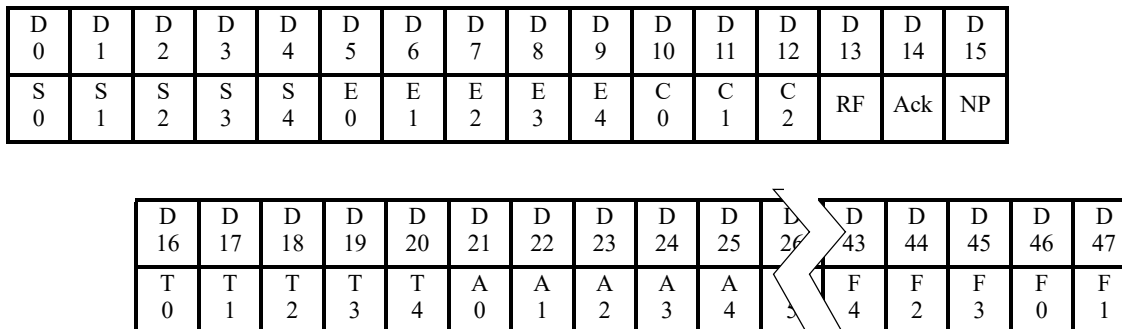


Figure 73–6—Link codeword Base Page

Change the last two sentences of the final paragraph of 73.6 as follows:

D[4342:21] contains the Technology Ability Field. D[47:4344] contains FEC capability (see 73.6.5).

73.6.4 Technology Ability Field

Change the first sentence of 73.6.4 as follows:

Technology Ability Field (A[2221:0]) is a 2322-bit wide field containing information indicating supported technologies specific to the selector field value when used with the Auto-Negotiation for backplane and copper cable assembly.

Change Table 73–4 as follows (some unchanged rows not shown):

Table 73–4—Technology Ability Field encoding

Bit	Technology
...	
A15	200GBASE-KR4 or 200GBASE-CR4
A16	100GBASE-KR1 or 100GBASE-CR1
A17	200GBASE-KR2 or 200GBASE-CR2
A18	400GBASE-KR4 or 400GBASE-CR4
A19 through A22	Reserved

Change the last paragraph of 73.6.4 as follows:

The fields A[2221:1916] are reserved for future use. Reserved fields shall be sent as zero and ignored on receive.

73.6.5 FEC capability

Change 73.6.5 as follows:

FEC (F4, F2, F3, F0, F1) is encoded in bits ~~D44:D43~~D43:D47 of the base link codeword. The ~~four~~ FEC bits are used as follows:

- a) F0 is 10 Gb/s per lane FEC ability
- b) F1 is 10 Gb/s per lane FEC requested
- c) F2 is 25G RS-FEC requested
- d) F3 is 25G BASE-R FEC requested
- e) F4 is 100GBASE-P RS-FEC-Int requested

~~Bits F2 and F3 are used for resolving FEC operation for 25G PHYs, while bits F0 and F1 are used for 10 Gb/s per lane operation. Bits F0 and F1 are not used for 25G PHYs.~~

Bits F0 and F1 are used for 10 Gb/s per lane operation PHYs. F2 and F3 are used for resolving FEC operation for 25G PHYs. F4 is used by 100GBASE-P PHYs where RS-FEC-Int (see Clause 161) is an alternative to the default RS-FEC (see Clause 91).

Insert 73.6.5.a before 73.6.5.1 as follows:

73.6.5.a FEC resolution for 100GBASE-P PHYs that include RS-FEC-Int

For 100GBASE-P PHYs that include RS-FEC-Int (see Clause 161) in addition to RS-FEC (see Clause 91), the F4 field is used to negotiate which FEC sublayer is to be used. If either PHY requests RS-FEC-Int operation then the RS-FEC-Int sublayer is enabled and the RS-FEC sublayer is disabled. Otherwise, the RS-FEC-Int sublayer is disabled and the RS-FEC sublayer is enabled.

73.6.5.3 FEC control variables

Insert the following paragraph after the second paragraph of 73.6.5.3 as follows:

The variable `an_rs_fec_int_negotiated_control` indicates that RS-FEC-Int operation has been negotiated. If the value is false, then RS-FEC-Int has not been negotiated. If the value is true, then RS-FEC-Int has been negotiated.

73.7 Receive function requirements

73.7.6 Priority Resolution function

Change Table 73–5 as follows:

Table 73–5—Priority Resolution

Priority	Technology	Capability
<u>1</u>	<u>400GBASE-KR4 or 400GBASE-CR4</u>	<u>400 Gb/s 4 lane, highest priority</u>
<u>2</u>	<u>200GBASE-KR2 or 200GBASE-CR2</u>	<u>200 Gb/s 2 lane</u>
3 <u>1</u>	200GBASE-KR4 or 200GBASE-CR4	200 Gb/s 4 lane, highest priority
<u>4</u>	<u>100GBASE-KR1 or 100GBASE-CR1</u>	<u>100 Gb/s 1 lane</u>
5 <u>2</u>	100GBASE-KR2 or 100GBASE-CR2	100 Gb/s 2 lane
6 <u>3</u>	100GBASE-CR4	100 Gb/s 4 lane
7 <u>4</u>	100GBASE-KR4	100 Gb/s 4 lane
8 <u>5</u>	100GBASE-KP4	100 Gb/s 4 lane
9 <u>6</u>	100GBASE-CR10	100 Gb/s 10 lane
<u>10</u> 7	50GBASE-KR or 50GBASE-CR	50 Gb/s 1 lane
<u>11</u> 8	40GBASE-CR4	40 Gb/s 4 lane
<u>12</u> 9	40GBASE-KR4	40 Gb/s 4 lane
<u>13</u> 10	25GBASE-KR or 25GBASE-CR	25 Gb/s 1 lane
<u>14</u> 11	25GBASE-KR-S or 25GBASE-CR-S	25 Gb/s 1 lane, short reach
<u>15</u> 12	10GBASE-KR	10 Gb/s 1 lane
<u>16</u> 13	10GBASE-KX4	10 Gb/s 4 lane
<u>17</u> 14	5GBASE-KR	5 Gb/s 1 lane
<u>18</u> 15	2.5GBASE-KX	2.5 Gb/s 1 lane
<u>19</u> 16	1000BASE-KX	1 Gb/s 1 lane, lowest priority

73.8 Management register requirements

Change Table 73–6 as follows (some unchanged rows not shown):

Table 73–6—Backplane Ethernet Auto-Negotiation variable to MDIO register mapping

Variable	Description
...	
link_control_[x]	{7.48.15:8, 7.48.6:5, 7.48.3:1, 7.49.52:0} Negotiated Port Type
an_baser_fec_control	7.48.4 BASE-R FEC negotiated
an_rs_fec_control	7.48.7 RS-FEC negotiated
<u>an_rs_fec_int_negotiated_control</u>	<u>7.49.6 RS-FEC-Int negotiated</u>

73.10 State diagrams and variable definitions

73.10.1 State diagram variables

Insert the following new entry into the list below the first paragraph of 73.10.1 after the 100GR2 entry:

100GR1; represents the 100GBASE-KR1 or 100GBASE-CR1 PMD.

Insert the following new entries into the list below the first paragraph of 73.10.1 after the 200GR4 entry:

200GR2; represents the 200GBASE-KR2 or 200GBASE-CR2 PMD.

400GR4; represents the 400GBASE-KR4 or 400GBASE-CR4 PMD.

Change the single_link_ready entry in the list after the third paragraph (‘Variables of the form “mr_x”, ...’) in 73.10.1 as follows:

single_link_ready

Status indicating that an_receive_idle = true and only one of the following indications is being received:

- 1) link_status_[1GKX] = OK
- 2) link_status_[2.5GKX] = OK
- 3) link_status_[5GKR] = OK
- 4) link_status_[10GKX4] = OK
- 5) link_status_[10GKR] = OK
- 6) link_status_[25GR] = OK
- 7) link_status_[40GKR4] = OK
- 8) link_status_[40GCR4] = OK
- 9) link_status_[50GR] = OK
- 10) link_status_[100GCR10] = OK
- 11) link_status_[100GKP4] = OK
- 12) link_status_[100GKR4] = OK
- 13) link_status_[100GCR4] = OK
- 14) link_status_[100GR2] = OK
- 15) link_status_[100GR1] = OK

~~1645)~~ link_status_[200GR4] = OK

17) link_status_[200GR2] = OK

18) link_status_[400GR4] = OK

73.10.2 State diagram timers

Change Table 73–7 as follows (some unchanged rows not shown):

Table 73–7—Timer min/max value summary

Parameter	Min	Value and tolerance	Max	Units
...				
interval_timer		3.2 ± 0.01%		ns
link_fail_inhibit_timer (when the link is 100GBASE-KR1, 100GBASE-CR1, 200GBASE-KR2, 200GBASE-CR2, 400GBASE-KR4, or 400GBASE-CR4)	<u>12.3</u>		<u>12.4</u>	s
link_fail_inhibit_timer (when the link is 50GBASE-KR, 50GBASE-CR, 100GBASE-KR2, 100GBASE-CR2, 200GBASE-KR4, or 200GBASE-CR4)	3.1		3.2	s
...				

73.11 Protocol implementation conformance statement (PICS) proforma for Clause 73, Auto-Negotiation for backplane and copper cable assembly¹¹

73.11.4 PICS proforma tables for Auto-Negotiation for backplane and copper cable assembly

73.11.4.3 Link codeword encoding

Change the row for item LE10 in the table in 73.11.4.3 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
LE10	FEC capability resolution	73.6.5	Resolve enabling of FEC capability based on F0 and F1 bits	M	Yes []
...					

¹¹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

80. Introduction to 40 Gb/s and 100 Gb/s networks

80.1 Overview

80.1.3 Relationship of 40 Gigabit and 100 Gigabit Ethernet to the ISO OSI reference model

Change the second paragraph in 80.1.3 (as modified by IEEE Std 802.3db-2022) as follows:

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience. The only exceptions are as follows:

- a) The XLGMII and CGMII, which, when implemented as a logical interconnection port between the MAC sublayer and the Physical Layer device (PHY), uses a 64-bit wide data path as specified in [Clause 81](#). Physical instantiations of this interface may use other data-path widths.
- b) The management interface, which, when physically implemented as the MDIO/MDC (Management Data Input/Output and Management Data Clock) at an observable interconnection port, uses a bit-wide data path as specified in [Clause 45](#).
- c) The PMA service interface, which, when physically implemented as XLAUI (40 Gb/s Attachment Unit Interface) or CAUI-4 and 100GAUI-4 (100 Gb/s four-lane Attachment Unit Interface) at an observable interconnection port, uses a 4 lane data path as specified in [Annex 83A](#), [Annex 83B](#), [Annex 83D](#), [Annex 83E](#), [Annex 135D](#), or [Annex 135E](#).
- d) The PMA service interface, which, when physically implemented as CAUI-10 (100 Gb/s ten-lane Attachment Unit Interface) at an observable interconnection port, uses a 10 lane data path as specified in [Annex 83A](#) or [Annex 83B](#).
- e) The PMA service interface, which, when physically implemented as 100GAUI-2 (100 Gb/s 2 lane Attachment Unit Interface) at an observable interconnection port, uses a 2 lane data path as specified in [Annex 135F](#) or [Annex 135G](#).
- f) The PMA service interface, which, when physically implemented as 100GAUI-1 (100 Gb/s single lane Attachment Unit Interface) at an observable interconnection port, uses a single lane data path as specified in [Annex 120F](#) or [Annex 120G](#).
- ~~g)~~ ~~The PMD service interface, which, when physically implemented as XLPPI (40 Gb/s Parallel Physical Interface) at an observable interconnection port, uses a 4 lane data path as specified in [Annex 86A](#).~~
- ~~h)~~ ~~The PMD service interface, which, when physically implemented as CPPI (100 Gb/s Parallel Physical Interface) at an observable interconnection port, uses a 10 lane data path as specified in [Annex 86A](#).~~
- i) ~~h)~~ The MDIs as specified in:
 - [Clause 89](#) for 40GBASE-FR
 - [Clause 140](#) for 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1
 - [Clause 154](#) for 100GBASE-ZR
 - [Clause 162 for 100GBASE-CR1](#)
 - [Clause 163 for 100GBASE-KR1](#)
 - [Clause 167](#) for 100GBASE-VR1 and 100GBASE-SR1use a single lane data path.

- j) ~~h)~~ The MDIs as specified in:
 - Clause 85 for 40GBASE-CR4
 - Clause 86 for 40GBASE-SR4
 - Clause 87 for 40GBASE-LR4 and 40GBASE-ER4
 - Clause 88 for 100GBASE-LR4 and 100GBASE-ER4
 - Clause 92 for 100GBASE-CR4
 - Clause 95 for 100GBASE-SR4all use a 4 lane data path.
- k) ~~j)~~ The MDIs as specified in Clause 85 for 100GBASE-CR10, and in Clause 86 for 100GBASE-SR10 use a 10 lane data path.
- l) ~~k)~~ Although there is no electrical or mechanical specification of the MDI for backplane Physical Layers, the PMDs as specified in:
 - Clause 84 for 40GBASE-KR4
 - Clause 93 for 100GBASE-KR4
 - Clause 94 for 100GBASE-KP4all use a 4 lane data path.
- m) ~~h)~~ The MDI as specified in Clause 113 for 40GBASE-T uses a 4 lane data path.
- n) ~~m)~~ The MDIs as specified in:
 - Clause 136 for 100GBASE-CR2
 - Clause 137 for 100GBASE-KR2
 - Clause 138 for 100GBASE-SR2all use a 2 lane data path.

80.1.4 Nomenclature

Change the sixth paragraph in 80.1.4 as follows:

100GBASE-P represents Physical Layer devices using the Clause 82 Physical Coding Sublayer for 100 Gb/s operation over multiple PCS lanes (see Clause 82) and a PMD implementing more than 2-level pulse amplitude modulation (PAM). 100GBASE-P Physical Layer devices also use the transcoding and FEC of Clause 91 or Clause 161.

Change Table 80–1 as shown (some unchanged rows not shown):

Table 80–1—40 Gb/s and 100 Gb/s PHYs

Name	Description
...	
100GBASE-KP4	100 Gb/s PHY using 100GBASE-R encoding, Clause 91 RS-FEC and 4-level pulse amplitude modulation over four lanes of an electrical backplane, with a total insertion loss up to 33 dB at 7 GHz (see Clause 94)
<u>100GBASE-KR1</u>	<u>100 Gb/s PHY using 100GBASE-R encoding, Clause 91 or Clause 161 FEC, and 4-level pulse amplitude modulation over one lane of an electrical backplane, with a total insertion loss up to 28 dB at 26.56 GHz (see Clause 163)</u>
100GBASE-KR2	100 Gb/s PHY using 100GBASE-R encoding, Clause 91 RS-FEC and 4-level pulse amplitude modulation over two lanes of an electrical backplane, with a total insertion loss up to 30 dB at 13.28 GHz (see Clause 137)
<u>100GBASE-CR1</u>	<u>100 Gb/s PHY using 100GBASE-R encoding and Clause 91 or Clause 161 FEC, and 4-level pulse amplitude modulation over one lane of shielded balanced copper cabling, with reach up to at least 2 m (see Clause 162)</u>
100GBASE-CR2	100 Gb/s PHY using 100GBASE-R encoding and Clause 91 RS-FEC and 4-level pulse amplitude modulation over two lanes of shielded balanced copper cabling, with reach up to at least 3 m (see Clause 136)
...	

80.1.5 Physical Layer signaling systems

Change the first paragraph of 80.1.5 as follows:

This standard specifies a family of Physical Layer implementations. [Table 80–2](#), [Table 80–3](#), [Table 80–3a](#), [Table 80–4](#), [Table 80–5](#), and [Table 80–6](#) specify the correlation between nomenclature and clauses. Implementations conforming to one or more nomenclatures meet the requirements of the corresponding clauses.

Replace Table 80–3 with the following table:

Table 80–3—Nomenclature and clause correlation (100GBASE-R copper)

Nomenclature	Clause ^a												
	73	74	78	81		82	83	83A	83D	85	91	92	93
	Auto-Negotiation	BASE-R FEC	EEE	RS	CGMII	100GBASE-R PCS	100GBASE-R PMA	CAUI-10 C2C	CAUI-4 C2C	100GBASE-CR10 PMD	100GBASE-R RS-FEC	100GBASE-CR4 PMD	100GBASE-KR4 PMD
100GBASE-KR4	M		O	M	O	M	M	O	O		M		M
100GBASE-CR4	M		O	M	O	M	M	O	O		M	M	
100GBASE-CR10	M	O	O	M	O	M	M	O	O	M			

^aO = Optional, M = Mandatory.

Insert new Table 80–3a after Table 80–3 as follows:

Table 80–3a—Nomenclature and clause correlation (100GBASE-P copper)

Nomenclature	Clause ^a																																							
	73		78		81		82		83		83A		83D		91		94		120F		135		135D		135F		136		137		152		161		162		163			
Auto-Negotiation			EEE		RS		CGMII		100GBASE-R PCS		100GBASE-R PMA		CAUI-10 C2C		CAUI-4 C2c		100GBASE-R RS-FEC		100GBASE-KP4 PMD		100GAUI-1 C2C		100GBASE-P PMA		100GAUI-4 C2C		100GAUI-2 C2C		100GBASE-CR2 PMD		100GBASE-KR2 PMD		Inverse RS-FEC		100GBASE-P RS-FEC-Int		100GBASE-CR1 PMD		100GBASE-KR1 PMD	
100GBASE-KR1	M		M	O	M	O	O	O	M		O	M	O	O			O	M	O	O							O	M								M				
100GBASE-KR2	M	O	M	O	M		O	O	M			M	O	O		M												M												
100GBASE-KP4	M	O	M	O	M	O	O	O	M	M																														
100GBASE-CR1	M		M	O	M	O	O	O	M		O	M	O	O													O	M	M											
100GBASE-CR2	M	O	M	O	M		O	O	M				M	O	O	M																								

^aO = Optional, M = Mandatory.

Change Table 80–5 (as modified by IEEE Std 802.3db-2022) as follows:

Table 80–5—Nomenclature and clause correlation (100GBASE-P optical)

Nomenclature	Clause ^a																								
	78		81		82	83	83A	83B	83D	83E	91	120F	120G	135	135D	135E	135F	135G	138	140		167			
	EEE		RS		PCS	100GBASE-R PMA	CAUI-10 C2C	CAUI-10 C2M	CAUI-4 C2C	CAUI-4 C2M	RS-FEC	100GAUI-1 C2C	100GAUI-1 C2M	100GBASE-P PMA	100GAUI-4 C2C	100GAUI-4 C2M	100GAUI-2 C2C	100GAUI-2 C2M	100GBASE-SR2	100GBASE-DR	100GBASE-FR1	100GBASE-LR1	100GBASE-VR1	100GBASE-SR1	
	CGMII																								
100GBASE-VR1	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O					M		
100GBASE-SR2	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O	M						
100GBASE-SR1	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O						M	
100GBASE-DR	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O		M					
100GBASE-FR1	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O			M				
100GBASE-LR1	O	M	O	M	O	O	O	O	O	O	M	<u>O</u>	<u>O</u>	M	O	O	O	O				M			

^aO = Optional, M = Mandatory.

Change Table 80–6 as follows:

Table 80–6—Nomenclature and clause correlation (100GBASE-Z optical)

Nomenclature	Clause ^a																				
	78	81		82	83	83A	83B	83D	83E	91	120F	120G	135	135D	135E	135F	135G	152	153	154	
	EEE	RS	CGMII	100GBASE-R PCS	100GBASE-R PMA	CAUI-10	CAUI-10	CAUI-4	CAUI-4	RS-FEC	100GAUI-1 C2C	100GAUI-1 C2M	100GBASE-P PMA	100GAUI-4 C2C	100GAUI-4 C2M	100GAUI-2 C2C	100GAUI-2 C2M	Inverse RS-FEC	SC-FEC and 100GBASE-ZR PMA	100GBASE-ZR PMD	
100GBASE-ZR	O	M	O	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	C ^b	M	M

^a O = Optional, M = Mandatory, C = Conditional.

^b Clause 152 inverse RS-FEC mandatory when Clause 91 RS-FEC is present.

80.2 Summary of 40 Gigabit and 100 Gigabit Ethernet sublayers

80.2.3 Forward error correction (FEC) sublayers

Change the first paragraph in 80.2.3 (as modified by IEEE Std 802.3db-2022) as follows:

A forward error correction sublayer is optional for 40GBASE-KR4, 40GBASE-CR4, and 100GBASE-CR10 PHYs and mandatory for 100GBASE-CR1, 100GBASE-CR2, 100GBASE-CR4, 100GBASE-KR1, 100GBASE-KR2, 100GBASE-KR4, 100GBASE-KP4, 100GBASE-VR1, 100GBASE-SR1, 100GBASE-SR2, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, 100GBASE-LR1, and 100GBASE-ZR PHYs. The FEC sublayer can be placed in between the PCS and PMA sublayers or between two PMA sublayers.

80.2.5 Physical Medium Dependent (PMD) sublayer

Change the second paragraph of 80.2.5 as follows:

The 40GBASE-R, 100GBASE-R, 100GBASE-P, and 100GBASE-Z PMDs and their corresponding media are specified in [Clause 84](#) through [Clause 89](#), [Clause 92](#) through [Clause 95](#), [Clause 136](#) through [Clause 138](#), [Clause 140](#), ~~and [Clause 154](#)~~, [Clause 162](#), and [Clause 163](#).

80.2.6 Auto-Negotiation

Change the second paragraph of 80.2.6 as follows:

Clause 73 Auto-Negotiation is used by the 40 Gb/s and 100 Gb/s backplane PHYs (40GBASE-KR4, 100GBASE-KR1, 100GBASE-KR2, 100GBASE-KP4, and 100GBASE-KR4) and the 40 Gb/s and 100 Gb/s copper PHYs (40GBASE-CR4, 100GBASE-CR1, 100GBASE-CR2, 100GBASE-CR10, and 100GBASE-CR4).

80.4 Delay constraints

Change Table 80–7 as follows (some unchanged rows not shown):

Table 80–7—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c ,
...				
100GBASE-R RS-FEC	40 960	80	409.60	See 91.4.
<u>100GBASE-P RS-FEC-Int</u>	<u>51 200</u>	<u>100</u>	<u>512.00</u>	<u>See 161.4.</u>
Inverse RS-FEC	40 960	80	409.60	See 152.4.
...				
100GBASE-P PMA	9 216	18	92.16	See 135.5.4.
<u>100GBASE-KR1 PMD</u>	<u>4 096</u>	<u>8</u>	<u>40.96</u>	<u>Includes allocation of 14 ns for one direction through backplane medium. See 163.5.</u>
100GBASE-KR2 PMD	4 096	8	40.96	Includes allocation of 20 ns for one direction through backplane medium. See 137.5.
100GBASE-KR4 PMD	2 048	4	20.48	Includes delay of one direction through backplane medium. See 93.4.
100GBASE-KP4 PMA/PMD	8 192	16	81.92	Includes delay of one direction through backplane medium. See 94.2.5.
<u>100GBASE-CR1 PMD</u>	<u>4 096</u>	<u>8</u>	<u>40.96</u>	<u>Includes allocation for 14 ns for one direction through cable medium. See 162.5.</u>
100GBASE-CR2 PMD	4 096	8	40.96	Includes allocation for 20 ns for one direction through cable medium. See 136.5.
...				

^a For 40GBASE-R, 1 bit time (BT) is equal to 25 ps and for 100GBASE-R, 1 bit time (BT) is equal to 10 ps. (See 1.4.215 for the definition of bit time.)

^b For 40GBASE-R, 1 pause_quantum is equal to 12.8 ns and for 100GBASE-R, 1 pause_quantum is equal to 5.12 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

80.5 Skew constraints

Change Table 80–8 (as modified by IEEE Std 802.3db-2022) as follows:

Table 80–8—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 40GBASE-R PCS lane (UI) ^b	Maximum Skew for 100GBASE-R PCS lane (UI) ^c	Notes ^d
SP0	29	N/A	≈ 150	See 83.5.3.1 or 135.5.3
SP1	29	≈ 299	≈ 150	See 83.5.3.2 or 135.5.3
SP2	43	≈ 443	≈ 222	See 83.5.3.4, 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, 154.3.2, <u>162.6</u> , <u>163.6</u> , or 167.3.2
SP3	54	≈ 557	≈ 278	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, 154.3.2, <u>162.6</u> , <u>163.6</u> , or 167.3.2
SP4	134	≈ 1382	≈ 691	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, 154.3.2, <u>162.6</u> , <u>163.6</u> , or 167.3.2
SP5	145	≈ 1495	≈ 748	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 140.3, 154.3.2, <u>162.6</u> , <u>163.6</u> , or 167.3.2
SP6	160	≈ 1649	≈ 824	See 83.5.3.6 or 135.5.3
SP7	29	N/A	≈ 150	See 83.5.3.8 or 135.5.3
At PCS receive	180	≈ 1856	≈ 928	See 82.2.13
At RS-FEC_or RS-FEC-Int transmit	49	N/A	≈ 253	See 91.5.2.2 or 161.5.2.2
At RS-FEC_or RS-FEC-Int receive	180	N/A	≈ 4641	See 91.5.3.1 or 161.5.3.1
At PCS receive (with RS-FEC)	49	N/A	≈ 253	See 82.2.13

^aThe Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^bThe symbol ≈ indicates approximate equivalent of maximum Skew in UI for 40GBASE-R, based on 1 UI equals 96.969697 ps at PCS lane signaling rate of 10.3125 GBd.

^cThe symbol ≈ indicates approximate equivalent of maximum Skew in UI for 100GBASE-R, based on 1 UI equals 193.939394 ps at PCS lane signaling rate of 5.15625 GBd.

^dShould there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 80–9 (as modified by IEEE Std 802.3db-2022) as follows:

Table 80–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation				Notes ^a
	(ns)	UI for 10.3125 GBd lane ^b	UI for 25.78125 GBd lane ^c	UI for 26.5625 GBd lane ^d	
SP0	0.2	≈ 2	≈ 5	N/A	See 83.5.3.1 or 135.5.3
SP1	0.2	≈ 2	≈ 5	≈ 5	See 83.5.3.2 or 135.5.3
SP2	0.4	≈ 4	≈ 10	≈ 11	See 83.5.3.4, 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 135.5.3, 136.6, 137.6, 138.3, 162.6, 163.6, or 167.3
SP3	0.6	≈ 6	≈ 15	≈ 16	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 136.6, 137.6, 138.3, 162.6, 163.6, or 167.3
SP4	3.4	≈ 35	≈ 88	≈ 90	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 95.3.2, 136.6, 137.6, 138.3, 162.6, 163.6, or 167.3
SP5	3.6	≈ 37	≈ 93	≈ 96	See 84.5, 85.5, 86.3.2, 87.3.2, 88.3.2, 89.3.2, 92.5, 93.5, 94.3.4, 5.3.2, 135.5.3, 136.6, 137.6, 138.3, 162.6, 163.6, or 167.3
SP6	3.8	≈ 39	≈ 98	≈ 101	See 83.5.3.6 or 135.5.3
SP7	0.2	≈ 2	N/A	N/A	See 83.5.3.8 or 135.5.3
At PCS receive	4	≈ 41	N/A	N/A	See 82.2.13
At RS-FEC or RS-FEC-Int transmit	0.4	N/A	≈ 10	≈ 11	See 91.5.2.2 or 161.5.2.2
At RS-FEC or RS-FEC-Int receive ^e	4	N/A	≈ 103	≈ 106	See 91.5.3.1 or 161.5.3.1
At PCS receive (with RS-FEC)	0.4	N/A	≈ 10	N/A	See 82.2.13

^aShould there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

^bThe symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 40GBASE-R, based on 1 UI equals 96.969697 ps at PMD lane signaling rate of 10.3125 GBd.

^cThe symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 100GBASE-R, based on 1 UI equals 38.787879 ps at PMD lane signaling rate of 25.78125 GBd.

^dThe symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI for 100GBASE-R, based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.

^eThe skew at the RS-FEC receive is the skew between RS-FEC lanes.

80.7 Protocol implementation conformance statement (PICS) proforma

Change the first paragraph of 80.7 as follows:

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 45, Clause 73, [Clause 74](#), [Clause 81](#) through [Clause 89](#), Clause 91 through [Clause 95](#), Clause 135 through Clause 138, Clause 140, Clause 152 through [Clause 154](#), ~~Clause 161 through Clause 163~~, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

82. Physical Coding Sublayer (PCS) for 64B/66B, type 40GBASE-R and 100GBASE-R

82.1 Overview

82.1.4 Inter-sublayer interfaces

Change the first paragraph of 82.1.4 as follows:

The upper interface of the PCS may connect to the Reconciliation Sublayer through the XLGMII/CGMII. The lower interface of the PCS connects to the PMA sublayer to support a PMD. If the optional BASE-R FEC sublayer is implemented (see [Clause 74](#)) and an optional physical instantiation, i.e., XLAUI or CAUI-n, is not implemented directly below the PCS sublayer, then the lower interface connects to the BASE-R FEC sublayer. For Physical Layers that use [Clause 91 RS-FEC](#) or [Clause 161 FEC](#), if an optional physical instantiation (i.e., CAUI-n) is not implemented directly below the PCS sublayer, then the lower interface connects to the RS-FEC sublayer. The 40GBASE-R PCS has a nominal rate at the PMA/FEC service interface of 10.3125 Gtransfers/s per PCS lane, which provides capacity for the MAC data rate of 40 Gb/s. The 100GBASE-R PCS has a nominal rate at the PMA/FEC service interface of 5.15625 Gtransfers/s per PCS lane, which provides capacity for the MAC data rate of 100 Gb/s.

82.2 Physical Coding Sublayer (PCS)

82.2.13 PCS lane deskew

Change Table 82–7 as follows:

Table 82–7—Skew tolerance requirements

PCS	Maximum Skew	Maximum Skew Variation
40GBASE-R	180 ns (~1856 bits)	4ns (~41 bits)
100GBASE-R	180 ns (~928 bits)	4ns (~21 bits)
100GBASE-R with RS-FEC or RS-FEC-Int	49 ns (~253 bits)	0.4 ns (~2 bits)

82.6 Auto-Negotiation

Change 82.6 as follows:

The following requirements apply to a PCS used with a 40GBASE-KR4 ~~PMD~~, 40GBASE-CR4 ~~PMD~~, 100GBASE-CR10, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-CR1, 100GBASE-KR4, 100GBASE-KP4, ~~or 100GBASE-KR2~~, or 100GBASE-KR1 PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the primitive AN_LINK.indication(link_status) (see [73.9](#)). The parameter link_status shall take the value FAIL when PCS_status=false and the value OK when PCS_status=true. The primitive shall be generated when the value of link_status changes.

82.7 Protocol implementation conformance statement (PICS) proforma for Clause 82, Physical Coding Sublayer (PCS) for 64B/66B, type 40GBASE-R and 100GBASE-R¹²

82.7.4 PICS proforma tables for PCS, type 40GBASE-R and 100GBASE-R

82.7.4.11 Auto-Negotiation for Backplane Ethernet functions

Change the table in 82.7.4.11 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*AN1	Support for use with a 40GBASE-KR4, 40GBASE-CR4, 100GBASE-CR10, 100GBASE-CR4, 100GBASE-CR2, <u>100GBASE-CR1</u> , 100GBASE-KR4, 100GBASE-KP4, or 100GBASE-KR2, <u>or</u> 100GBASE-KR1 PMD	82.6	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN2	AN_LINK.indication primitive	82.6	Support of the primitive AN_LINK.indication(link_status)	AN1:M	Yes [] N/A []
AN3	link_status parameter	82.6	Takes the value OK or FAIL, as described in 82.6	AN1:M	Yes [] N/A []
AN4	Generation of AN_LINK.indication primitive	82.6	Generated when the value of link_status changes	AN1:M	Yes [] N/A []

¹²Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

83. Physical Medium Attachment (PMA) sublayer, type 40GBASE-R and 100GBASE-R

83.1 Overview

83.1.1 Scope

Change the first paragraph of 83.1.1 as follows:

This clause specifies the Physical Medium Attachment sublayer (PMA) that is common to two families of (40 Gb/s and 100 Gb/s) Physical Layer implementations, known as 40GBASE-R and 100GBASE-R. The PMA allows the PCS (specified in Clause 82) to connect in a media-independent way with a range of physical media. The 40GBASE-R PMA(s) can support any of the 40 Gb/s PMDs in Table 80–2. The 100GBASE-R PMA(s) can support any of the 100 Gb/s PMDs in Table 80–3 or Table 80–4, ~~but does not provide the PMD service interface for 100GBASE-KP4 (Clause 94).~~ The terms 40GBASE-R and 100GBASE-R are used when referring generally to Physical Layers using the PMA defined in this clause. The 100GBASE-R PMA may also be used with some PMDs in Table 80–3a.

91. Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs

91.5 Functions within the RS-FEC sublayer

91.5.2 Transmit function

91.5.2.7 Reed-Solomon encoder

Change the second sentence of the second paragraph of 91.5.2.7 (as modified by IEEE Std 802.3db-2022) as follows:

When used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, ~~or~~ 100GBASE-LR1, 100GBASE-CR1, or 100GBASE-KR1 PHY, the RS-FEC sublayer shall implement RS(544,514).

91.5.3 Receive function

91.5.3.3 Reed-Solomon decoder

Change the second sentence of the second paragraph of 91.5.3.3 (as modified by IEEE Std 802.3db-2022) as follows:

When used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, ~~or~~ 100GBASE-LR1, 100GBASE-CR1, or 100GBASE-KR1 PHY, the RS-FEC sublayer shall be capable of correcting any combination of up to $t=15$ symbol errors in a codeword.

Change the third paragraph of 91.5.3.3 (as modified by IEEE Std 802.3db-2022) as follows:

The Reed-Solomon decoder may provide the option to perform error detection without error correction to reduce the delay contributed by the RS-FEC sublayer. The presence of this option is indicated by the assertion of the FEC_bypass_correction_ability variable (see 91.6.8). When the option is provided, it is enabled by the assertion of the FEC_bypass_correction_enable variable (see 91.6.1). This option shall not be used when the RS-FEC sublayer is used to form part of a 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, ~~or~~ 100GBASE-LR1, 100GBASE-CR1, or 100GBASE-KR1 PHY.

Change the last sentence of the last paragraph of 91.5.3.3 (as modified by IEEE Std 802.3db-2022) as follows:

When the RS-FEC sublayer is used to form a 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, ~~or~~ 100GBASE-LR1, 100GBASE-CR1, or 100GBASE-KR1 PHY, the symbol error threshold shall be $K=6380$.

91.5.3.3.1 FEC Degraded SER (optional)

Change the first paragraph of 91.5.3.3.1 (as modified by IEEE Std 802.3db-2022) as follows:

For 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, ~~and~~ 100GBASE-LR1, 100GBASE-CR1, and 100GBASE-KR1 PHYs an optional FEC degraded symbol error ratio function is available.

91.6 RS-FEC MDIO function mapping

Change Table 91–2 as follows (some unchanged rows not shown):

Table 91–2—MDIO/RS-FEC control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
...			
FEC degraded SER enable	RS-FEC control register	1.200.4	FEC_degraded_SER_enable
<u>100G RS-FEC enable</u>	<u>RS-FEC control register</u>	<u>1.200.6</u>	<u>100G_RS_FEC_enable</u>
FEC degraded SER activate threshold	RS-FEC degraded SER activate threshold register	1.650, 1.651	FEC_degraded_SER_activate_threshold
...			

91.6.3 four_lane_pmd

Change 91.6.3 (as modified by IEEE Std 802.3db-2022) as follows:

When this variable is set to one, the alignment marker mapping function substitutes the fixed bytes of the alignment markers corresponding to PCS lanes 17, 18, and 19 with the fixed bytes for the alignment marker corresponding to PCS lane 16 (see 91.5.2.6). When this variable is set to zero, the alignment markers corresponding to PCS lanes 17, 18, and 19 are passed through unmodified. The default value of the variable is one which is the value required by the 100GBASE-CR4, 100GBASE-KR4, 100GBASE-KP4, and 100GBASE-SR4 PMDs. This variable is set to zero for the 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, ~~and~~ 100GBASE-LR1, 100GBASE-CR1, and 100GBASE-KR1 PMDs. This variable is mapped to the bit defined in 45.2.1.116 (1.200.3).

Insert 91.6.7a after 91.6.7:

91.6.7a 100G_RS_FEC_enable

For PHYs supporting RS-FEC-Int operation (see Clause 161) this sublayer shall have the capability to enable or disable its FEC function. For these PHYs, an MDIO interface or an equivalent management interface shall be provided to access the variable 100G_RS_FEC_Enable for the RS-FEC sublayer. When the 100G_RS_FEC_Enable variable is set to one, the RS-FEC sublayer performs the transmit function as specified in 91.5.2 and the receive function as specified in 91.5.3. When the variable is set to zero, the

RS-FEC transmit and receive functions are disabled, and the 100G RS-FEC sublayer is not used allowing the RS-FEC-Int sublayer (see Clause 161) to be used instead. This variable is mapped to the bit defined in 45.2.1.116 (1.200.6).

91.7 Protocol implementation conformance statement (PICS) proforma for Clause 91, Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs¹³

91.7.3 Major capabilities/options

Change the table in 91.7.3 (as modified by IEEE Std 802.3db-2022) as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*KP1	<u>100GBASE-CR1 or 100GBASE-KR1</u>		<u>Used to form complete 100GBASE-CR1 or 100GBASE-KR1 PHY</u>	<u>O</u>	<u>Yes []</u> <u>No []</u>
*KP4	100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1 <u>100GBASE-CR1, or</u> <u>100GBASE-KR1</u>		Used to form complete 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or</u> <u>100GBASE-KR1 PHY</u>	O	Yes [] No []
*FDDP	100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or</u> <u>100GBASE-KR1</u>	91.5.3.3.1	Used to form complete 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or</u> <u>100GBASE-KR1 PHY</u>	O	Yes [] No []
*FDD	Support for optional FEC degraded SER detection	91.5.3.3.1		FDDP:O	Yes [] No [] N/A []
<u>FE</u>	<u>100G RS-FEC Enable</u>	<u>91.6.7a</u>	<u>Has the capability to enable and disable the RS-FEC function</u>	<u>KP1:M</u>	<u>Yes []</u> <u>N/A []</u>
...					

¹³Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

91.7.4 PICS proforma tables for Reed-Solomon forward error correction (RS-FEC) sublayer for 100GBASE-R PHYs

91.7.4.1 Transmit function

Change the row for item TF11 in the table in 91.7.4.1 (as modified by IEEE Std 802.3db-2022) as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
TF11	Reed-Solomon encoder for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or 100GBASE-KR1</u>	91.5.2.7	RS(544,514)	KP4:M	Yes [] N/A []
...					

91.7.4.2 Receive function

Change the rows for items RF4, RF6, and RF12 in the table in 91.7.4.2 (as modified by IEEE Std 802.3db-2022) as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
RF4	Reed-Solomon decoder for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or 100GBASE-KR1</u>	91.5.3.3	Corrects any combination of up to $t=15$ symbol errors in a codeword unless error correction bypassed	KP4:M	Yes [] N/A []
...					

Item	Feature	Subclause	Value/Comment	Status	Support
RF6	Error correction bypass for 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-SR4, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or 100GBASE-KR1</u>	91.5.3.3	Error correction is not bypassed	SR4:M or FDDP:M	Yes [] N/A []
...					
RF12	Symbol error threshold for 100GBASE-KP4, 100GBASE-CR2, 100GBASE-KR2, 100GBASE-VR1, 100GBASE-SR2, 100GBASE-SR1, 100GBASE-DR, 100GBASE-FR1, or 100GBASE-LR1, <u>100GBASE-CR1, or 100GBASE-KR1</u>	91.5.3.3	$K=6380$	BEI* KP4:M	Yes [] N/A []
...					

116. Introduction to 200 Gb/s and 400 Gb/s networks

116.1 Overview

116.1.2 Relationship of 200 Gigabit and 400 Gigabit Ethernet to the ISO OSI reference model

Change the lettered list in 116.1.2 (as modified by IEEE Std 802.3db-2022) as follows (unchanged items not shown):

- e) The PMA service interface, which, when physically implemented as 200GAUI-4 (200 Gb/s four-lane Attachment Unit Interface) or 400GAUI-4 (400 Gb/s four-lane Attachment Unit Interface) at an observable interconnection port, uses a 4-lane data path as specified in Annex 120D, ~~or~~ Annex 120E, Annex 120F, or Annex 120G.
- f) The PMA service interface, which, when physically implemented as 200GAUI-2 (200 Gb/s two-lane Attachment Unit Interface) at an observable interconnection port, uses a 2-lane data path as specified in Annex 120F or Annex 120G.
- ~~g)~~ ~~The MDI as specified in Clause 123 for 400GBASE-SR16 uses a 16-lane data path.~~
- h) ~~g)~~ The MDIs as specified in:
 - Clause 122 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8
 - Clause 138 for 400GBASE-SR8
 - Clause 150 for 400GBASE-SR4.2all use an 8-lane data path.
- i) ~~h)~~ The MDIs as specified in:
 - Clause 121 for 200GBASE-DR4
 - Clause 122 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4
 - Clause 124 for 400GBASE-DR4
 - Clause 136 for 200GBASE-CR4
 - Clause 137 for 200GBASE-KR4
 - Clause 138 for 200GBASE-SR4
 - Clause 151 for 400GBASE-FR4 and 400GBASE-LR4-6
 - Clause 162 for 400GBASE-CR4
 - Clause 163 for 400GBASE-KR4
 - Clause 167 for 400GBASE-VR4 and 400GBASE-SR4all use a 4-lane data path.
- j) ~~i)~~ The MDIs as specified in:
 - Clause 162 for 200GBASE-CR2
 - Clause 163 for 200GBASE-KR2
 - Clause 167 for 200GBASE-VR2 and 200GBASE-SR2all use a 2-lane data path.

116.1.3 Nomenclature

Change Table 116–1 as follows (some unchanged rows not shown):

Table 116–1—200 Gb/s PHYs

Name	Description
<u>200GBASE-KR2</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over two lanes of an electrical backplane (see Clause 163)</u>
200GBASE-KR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of an electrical backplane (see Clause 137)
<u>200GBASE-CR2</u>	<u>200 Gb/s PHY using 200GBASE-R encoding over two lanes of twinaxial copper cable (see 1.4.559 and Clause 162)</u>
200GBASE-CR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 136)
...	

Change Table 116–2 (as modified by IEEE Std 802.3db-2022) as follows (some unchanged rows not shown):

Table 116–2—400 Gb/s PHYs

Name	Description
<u>400GBASE-KR4</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over four lanes of an electrical backplane (see Clause 163)</u>
<u>400GBASE-CR4</u>	<u>400 Gb/s PHY using 400GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 162)</u>
400GBASE-VR4	400 Gb/s PHY using 400GBASE-R encoding over four lanes of multimode fiber, with reach up to at least 50 m (see Clause 167)
...	

116.1.4 Physical Layer signaling systems

Change Table 116–3 as follows:

Table 116–3—PHY type and clause correlation (200GBASE copper)

PHY type	Clause ^a													
	73	78	117		118	119	120	120B	120D	120F	136	137	162	163
	Auto-Negotiation	EEE	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R PMA	200GAUI-8 C2C	200GAUI-4 C2C	200GAUI-2 C2C	200GBASE-CR4 PMD	200GBASE-KR4 PMD	200GAUI-CR2 PMD	200GAUI-KR2 PMD
	200GBASE-KR2	M		M	O	O	M	M	O	O	O			M
	200GBASE-KR4	M	O	M	O	O	M	M	O	O		M		
200GBASE-CR2	M		M	O	O	M	M	O	O	O			M	
200GBASE-CR4	M	O	M	O	O	M	M	O	O		M			

^aO = Optional, M = Mandatory.

Insert Table 116–3a after Table 116–3 as follows:

Table 116–3a—PHY type and clause correlation (400GBASE copper)

PHY type	Clause ^a										
	73	117		118	119	120	120B	120D	120F	162	163
	Auto-Negotiation	RS	400GMII	400GMII Extender	400GBASE-R PCS	400GBASE-R PMA	400GAUI-16 C2C	400GAUI-8 C2C	400GAUI-4 C2C	400GAUI-CR4 PMD	400GAUI-KR4 PMD
400GBASE-KR4	M	M	O	O	M	M	O	O	O		M
400GBASE-CR4	M	M	O	O	M	M	O	O	O	M	

^aO = Optional, M = Mandatory.

Change Table 116–4 (as modified by IEEE Std 802.3db-2022) as follows:

Table 116–4—PHY type and clause correlation (200GBASE optical)

PHY type	Clause ^a																		
	78	117		118	119	120	120B	120C	120D	120E	120F	120G	121	122			138	167	
	EEE	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R PMA	200GAUI-8 C2C	200GAUI-8 C2M	200GAUI-4 C2C	200GAUI-4 C2M	200GAUI-2 C2C	200GAUI-2 C2M	200GBASE-DR4 PMD	200GBASE-FR4 PMD	200GBASE-LR4 PMD	200GBASE-ER4 PMD	200GBASE-SR4 PMD	200GBASE-VR2	200GBASE-SR2
200GBASE-VR2	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>						M	
200GBASE-SR4	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>					M		
200GBASE-SR2	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>							M
200GBASE-DR4	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>	M						
200GBASE-FR4	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>		M					
200GBASE-LR4	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>			M				
200GBASE-ER4	O	M	O	O	M	M	O	O	O	O	<u>O</u>	<u>O</u>				M			

^a O = Optional, M = Mandatory.

Change Table 116–5 (as modified by IEEE Std 802.3db-2022) as follows:

Table 116–5—PHY type and clause correlation (400GBASE optical)

PHY type	Clause ^a																																										
	78		117		118		119		120		120B		120C		120D		120E		120F		120G		167		123		138		150		124		122		151		122						
	EEE	RS	400GMII		400GMII Extender		400GBASE-R PCS		400GBASE-R PMA		400GAUI-16 C2C		400GAUI-16 C2M		400GAUI-8 C2C		400GAUI-8 C2M		400GAUI-4 C2C		400GAUI-4 C2M		400GBASE-VR4		400GBASE-SR4 PMD		400GBASE-SR16 PMD		400GBASE-SR8 PMD		400GBASE-SR4.2 PMD		400GBASE-DR4 PMD		400GBASE-FR8 PMD		400GBASE-FR4 PMD		400GBASE-LR4 PMD		400GBASE-LR8 PMD		400GBASE-ER8 PMD
400GBASE-VR4	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O	M																				
400GBASE-SR16	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O					M																	
400GBASE-SR8	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O						M																
400GBASE-SR4	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O		M																				
400GBASE-SR4.2	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O							M															
400GBASE-DR4	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O									M													
400GBASE-FR8	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O											M											
400GBASE-FR4	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O														M								
400GBASE-LR4-6	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O															M							
400GBASE-LR8	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O																	M					
400GBASE-ER8	O	M	O	O	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	O																					M	

^a O = Optional, M = Mandatory.

116.2 Summary of 200 Gigabit and 400 Gigabit Ethernet sublayers

116.2.5 Physical Medium Dependent (PMD) sublayer

Change the second paragraph in 116.2.5 (as modified by IEEE Std 802.3db-2022) as follows:

The 200GBASE-R PMDs and their corresponding media are specified in Clause 121, Clause 122, Clause 136 through Clause 138, Clause 162, Clause 163, and Clause 167. The 400GBASE-R PMDs and their corresponding media are specified in Clause 122 through Clause 124, Clause 138, Clause 150, Clause 162, Clause 163, and Clause 167.

Insert new subclause 116.2.5a after 116.2.5 as follows:

116.2.5a Auto-Negotiation

Auto-Negotiation provides a linked device with the capability to detect the abilities (modes of operation) supported by the device at the other end of the link, determine common abilities, and configure for joint operation.

Clause 73 Auto-Negotiation is used by the 200 Gb/s and 400 Gb/s backplane PHYs (200GBASE-KR4, 200GBASE-KR2, and 400GBASE-KR4) and the 200 Gb/s and 400 Gb/s copper PHYs (200GBASE-CR4, 200GBASE-CR2, and 400GBASE-CR4).

116.4 Delay constraints

Change Table 116–6 as follows (some unchanged rows and footnotes not shown):

Table 116–6—Sublayer delay constraints (200GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c
...				
200GBASE-R PMA	18 432	36	92.16	See 120.5.4.
<u>200GBASE-KR2 PMD</u>	<u>8 192</u>	<u>16</u>	<u>40.96</u>	<u>Includes allocation of 14 ns for one direction through backplane medium. See 163.5.</u>
200GBASE-KR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through backplane medium. See 137.5.
<u>200GBASE-CR2 PMD</u>	<u>8 192</u>	<u>16</u>	<u>40.96</u>	<u>Includes allocation of 14 ns for one direction through cable medium. See 162.5.</u>
200GBASE-CR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through cable medium. See 137.5.
...				

^a For 200GBASE-R, 1 bit time (BT) is equal to 5 ps. (See 1.4.215 for the definition of bit time.)

^b For 200GBASE-R, 1 pause_quantum is equal to 2.56 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 116–7 (as modified by IEEE Std 802.3db-2022) as follows (some unchanged rows not shown):

Table 116–7—Sublayer delay constraints (400GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c
...				
400GBASE-R PMA	36 864	72	92.16	See 120.5.4.
<u>400GBASE-KR4 PMD</u>	<u>8 192</u>	<u>16</u>	<u>20.48</u>	<u>Includes allocation of 14 ns for one direction through backplane medium. See 163.5.</u>
<u>400GBASE-CR4 PMD</u>	<u>8 192</u>	<u>16</u>	<u>20.48</u>	<u>Includes allocation for 14 ns for one direction through cable medium. See 162.5.</u>
400GBASE-VR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 167.3.1.
...				

^a For 400GBASE-R, 1 bit time (BT) is equal to 2.5 ps. (See 1.4.215 for the definition of bit time.)

^b For 400GBASE-R, 1 pause_quantum is equal to 1.28 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

116.5 Skew constraints

Change Table 116–8 (as modified by IEEE Std 802.3db-2022) as follows:

Table 116–8—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 200GBASE-R or 400GBASE-R PCS lane (UI) ^b	Notes ^c
SP1	29	≈ 770	See 120.5.3.1
SP2	43	≈ 1142	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, <u>162.6.2</u> , <u>163.6.2</u> , or 167.3.2
SP3	54	≈ 1434	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, <u>162.6.2</u> , <u>163.6.2</u> , or 167.3.2
SP4	134	≈ 3559	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 138.3.2.2, 150.3.2, 151.3.2, <u>162.6.2</u> , <u>163.6.2</u> , or 167.3.2
SP5	145	≈ 3852	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, <u>162.6.2</u> , <u>163.6.2</u> , or 167.3.2
SP6	160	≈ 4250	See 120.5.3.5
At PCS receive	180	≈ 4781	See 119.2.5.1

^a The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^b The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Change Table 116–9 (as modified by IEEE Std 802.3db-2022) as follows:

Table 116–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 26.5625 GBd PMD lane (UI) ^a	Maximum Skew Variation for 53.125 GBd PMD lane (UI) ^b	Notes ^c
SP1	0.2	≈ 5	N/A	See 120.5.3.1
SP2	0.4	≈ 11	≈ 21	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2
SP3	0.6	≈ 16	≈ 32	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2
SP4	3.4	≈ 90	≈ 181	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2
SP5	3.6	≈ 96	≈ 191	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, 151.3.2, 162.6.2, 163.6.2, or 167.3.2
SP6	3.8	≈ 101	N/A	See 120.5.3.5
At PCS receive	4	≈ 106	N/A	See 119.2.5.1

^a The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.

^b The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

118. 200GMII Extender, 400GMII Extender, 200GMII Extender Sublayer (200GXS), and 400GMII Extender Sublayer (400GXS)

118.1 Overview

118.1.3 200GAUI-n/400GAUI-n

Change 118.1.3 as follows:

A 200GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (200GAUI-n), as shown in [Figure 118–1](#):

- 200GAUI-8 chip-to-chip ([Annex 120B](#))
- 200GAUI-8 chip-to-module ([Annex 120C](#))
- 200GAUI-4 chip-to-chip ([Annex 120D](#))
- 200GAUI-4 chip-to-module ([Annex 120E](#))
- 200GAUI-2 chip-to-chip ([Annex 120F](#))
- 200GAUI-2 chip-to-module ([Annex 120G](#))

A 400GMII Extender may use any of the following electrical interfaces for the connection between its PMA sublayers (400GAUI-n), as shown in [Figure 118–1](#):

- 400GAUI-16 chip-to-chip ([Annex 120B](#))
- 400GAUI-16 chip-to-module ([Annex 120C](#))
- 400GAUI-8 chip-to-chip ([Annex 120D](#))
- 400GAUI-8 chip-to-module ([Annex 120E](#))
- 400GAUI-4 chip-to-chip ([Annex 120F](#))
- 400GAUI-4 chip-to-module ([Annex 120G](#))

119. Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R

119.6 Auto-Negotiation

Change 119.6 as follows:

The following requirements apply to a PCS used with a 200GBASE-CR4, ~~200GBASE-CR2, or~~ 200GBASE-KR4, ~~200GBASE-KR2,~~ 400GBASE-CR4, or 400GBASE-KR4 PMD where support for the Auto-Negotiation process defined in Clause 73 is mandatory. The PCS shall support the AN_LINK.indication(link_status) primitive (see 73.9). The parameter link_status shall take the value FAIL when PCS_status=false and the value OK when PCS_status=true. The primitive shall be generated when the value of link_status changes.

119.7 Protocol implementation conformance statement (PICS) proforma for Clause 119, Physical Coding Sublayer (PCS) for 64B/66B, type 200GBASE-R and 400GBASE-R¹⁴

119.7.4 PICS proforma tables for Physical Coding Sublayer (PCS) 64B/66B, type 200GBASE-R and 400GBASE-R

119.7.4.12 Auto-Negotiation for Backplane Ethernet functions

Change the table in 119.7.4.12 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
*AN	Support for use with a 200GBASE-CR4, or 200GBASE-CR2, 200GBASE-KR4, 200GBASE-KR2, 400GBASE-CR4, or 400GBASE-KR4 PMD	119.6	AN technology dependent interface described in Clause 73	O	Yes [] No []
AN1	AN_LINK.indication primitive	119.6	Support of the AN_LINK.indication(link_status) primitive	AN:M	Yes [] N/A []
AN2	link_status parameter	119.6	Takes the value OK or FAIL, as described in 119.6	AN:M	Yes [] N/A []
AN3	Generation of AN_LINK.indication primitive	119.6	Generated when the value of link_status changes	AN:M	Yes [] N/A []

¹⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120. Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R

120.1 Overview

120.1.1 Scope

Change the third paragraph of 120.1.1 as follows:

The interfaces for the inputs of the 200GBASE-R and 400GBASE-R PMAs are defined in an abstract manner and do not imply any particular implementation. For 200GBASE-R PMAs, electrical interfaces connecting PMA sublayers, known as 200GAUI-n, are defined in [Annex 120B](#), [Annex 120C](#), [Annex 120D](#), ~~and [Annex 120E](#)~~, [Annex 120F](#), or [Annex 120G](#). For 400GBASE-R PMAs, electrical interfaces connecting PMA sublayers, known as 400GAUI-n, are defined in [Annex 120B](#), [Annex 120C](#), [Annex 120D](#), ~~and [Annex 120E](#)~~, [Annex 120F](#), or [Annex 120G](#).

120.1.3 Summary of functions

Change item j) in the lettered list below the first paragraph of 120.1.3 as follows:

- j) Perform PAM4 encoding and decoding for 200GBASE-R PMAs where the number of physical lanes is 2 or 4, and for 400GBASE-R PMAs where the number of physical lanes is 4 or 8.

120.1.4 PMA sublayer positioning

Change item b) in the lettered list below the fourth paragraph of 120.1.4) as follows:

- b) 200GAUI-n is a physical instantiation of the connection between two adjacent 200GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface. 400GAUI-n is a physical instantiation of the connection between two adjacent 400GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface.
 - 1) As physical instantiations, these define electrical and timing specification as well as requiring a receive re-timing function.
 - 2) 200GAUI-8 is a 26.5625 GBd by 8 lane NRZ physical instantiation of the 200 Gb/s connection. 400GAUI-16 is a 26.5625 GBd by 16 lane NRZ physical instantiation of the 400 Gb/s connection.
 - 3) 200GAUI-4 is a 26.5625 GBd by 4 lane PAM4 physical instantiation of the 200 Gb/s connection. 400GAUI-8 is a 26.5625 GBd by 8 lane PAM4 physical instantiation of the 400 Gb/s connection.
 - 4) 200GAUI-2 is a 53.125 GBd by 2 lane PAM4 physical instantiation of the 200 Gb/s connection. 400GAUI-4 is a 53.125 GBd by 4 lane PAM4 physical instantiation of the 400 Gb/s connection.
 - 5) For a PHY that includes a 200GAUI-2 interface, it is recommended that the signaling rate range for a 200GAUI-8 or 200GAUI-4 PMA output that is in the same package as the PCS be limited to 26.5625 GBd $\pm$ 50 ppm.
 - 6) For a PHY that includes a 400GAUI-4 interface, it is recommended that the signaling rate range for a 400GAUI-16 or 400GAUI-8 PMA output that is in the same package as the PCS be limited to 26.5625 GBd $\pm$ 50 ppm.

120.3 PMA service interface

Change the fourth paragraph of 120.3 as follows:

If a PMA client is another PMA, a 200GBASE-R PMA with $p=8$ or a 400GBASE-R PMA with $p=16$ physical input lanes receives bits from one PCSL on each of its input lanes. A 200GBASE-R PMA with $p=2$ or $p=4$ physical input lanes receives PAM4 symbols on each of its input lanes at the PCSL rate, each symbol formed from two bits. A 400GBASE-R PMA with $p=4$ or $p=8$ physical input lanes receives PAM4 symbols on each of its input lanes at $8/p$ times the PCSL rate, each symbol formed from two bits. The bit stream represented by the input symbols carries z/p bit-multiplexed PCSLs on each physical input lane, where $z=8$ for 200GBASE-R and $z=16$ for 400GBASE-R. Skew may exist between different PCSLs received on the same physical input lane even though all PCSLs originate from the same synchronous source, so there is independence of arrival of bits from each PCSL on each physical input lane.

120.5 Functions within the PMA

Change the last sentence of 120.5 as follows:

The symbols received from input lanes or sent on output lanes are PAM4 symbols for 200GBASE-R interfaces where the number of lanes is 4 or 2, or 400GBASE-R interfaces where the number of lanes is 8 or 4.

120.5.1 Per input-lane clock and data recovery

Change the first paragraph of 120.5.1 as follows:

If the interface between the PMA client and the PMA is physically instantiated as 200GAUI-n or 400GAUI-n, the PMA shall meet the electrical and timing specifications in [Annex 120B](#), [Annex 120C](#), [Annex 120D](#), ~~or Annex 120E~~, [Annex 120F](#), or [Annex 120G](#), as appropriate. If the interface between the sublayer below the PMA and the PMA is physically instantiated as 200GAUI-n or 400GAUI-n, the PMA shall meet the electrical and timing specifications at the service interface below the PMA as specified in [Annex 120B](#), [Annex 120C](#), [Annex 120D](#), ~~or Annex 120E~~, [Annex 120F](#), or [Annex 120G](#), as appropriate.

120.5.2 Bit-level multiplexing

Change the third paragraph in 120.5.2 as follows:

For a PMA with m input lanes (Tx or Rx direction), each input lane carries, bit multiplexed, z/m PCSLs. Each input lane has a nominal bit rate of $26.5625 \times z/m$ Gb/s. Note that the signaling (Baud) rate is equal to the bit rate when the number of physical lanes is 8 for 200GBASE-R or 16 for 400GBASE-R (bits are sent or received on the lanes). The Baud rate is equal to half of the bit rate when the number of physical lanes is 4 or 2 for 200GBASE-R or the number of physical lanes is 8 or 4 for 400GBASE-R (PAM4 symbols are sent or received on the lanes). If necessary, PAM4 symbols are converted to pairs of bits on the input lanes and/or pairs of bits are converted to PAM4 symbols on the output lanes. If bit x received on an input lane belongs to a particular PCSL, the next bit of that same PCSL is received on the same input lane at bit position $x+(z/m)$. The z/m PCSLs may arrive in any sequence on a given input lane.

120.5.5 Clocking architecture

Change the first paragraph of 120.5.5 as follows:

A PMA with m input lanes and n output lanes shall clock the output lanes such that the bit rate of the output lanes is m/n times the bit rate of the input lanes. For input or output lanes encoded as PAM4 (for

200GBASE-R, where the number of input or output lanes is 2 or 4, or for 400GBASE-R, where the number of input or output lanes is 4 or 8), the symbol rate is half of the bit rate. This applies in both the Tx and Rx directions of transmission. In the case where the interfaces between the PMA client and the PMA and/or the PMA and the sublayer below the PMA are physically instantiated, the PMA may derive its input clock(s) from the electrical interface on one or more of the input lanes, and generate the output clock(s) with an appropriate PLL multiplier/divider circuit.

120.5.6 Signal drivers

Change 120.5.6 as follows:

For cases where the interface between the PMA client and the PMA, or between the PMA and the sublayer below the PMA represent a physically instantiated interface, the PMA provides electrical signal drivers for that interface. The electrical and jitter/timing specifications for these interfaces appear in

- Annex 120B, which specifies the 200GAUI-8 and 400GAUI-16 interfaces for chip-to-chip applications.
- Annex 120C, which specifies the 200GAUI-8 and 400GAUI-16 interfaces for chip-to-module applications.
- Annex 120D, which specifies the 200GAUI-4 and 400GAUI-8 interfaces for chip-to-chip applications.
- Annex 120E, which specifies the 200GAUI-4 and 400GAUI-8 interfaces for chip-to-module applications.
- Annex 120F, which specifies the 200GAUI-2 and 400GAUI-4 interfaces for chip-to-chip applications.
- Annex 120G, which specifies the 200GAUI-2 and 400GAUI-4 interfaces for chip-to-module applications.

For 200GAUI-8 or 400GAUI-16, the modulation format is NRZ. For 200GAUI-4, 200GAUI-2, ~~or~~ 400GAUI-8, or 400GAUI-4 the modulation format is PAM4.

120.5.7 PAM4 Encoding

120.5.7.1 Gray mapping for PAM4 encoded lanes

Change the first paragraph of 120.5.7.1 as follows:

For output lanes encoded as PAM4 (for 200GBASE-R, where the number of output lanes is 2 or 4, or for 400GBASE-R, where the number of output lanes is 4 or 8), the PMA transmit process shall map consecutive pairs of bits {A, B}, where A is the bit arriving first, to a Gray-coded symbol as follows:

Change the second paragraph of 120.5.7.1 as follows:

For input lanes encoded as PAM4 (for 200GBASE-R, where the number of input lanes is 2 or 4, or for 400GBASE-R, where the number of input lanes is 4 or 8), the PMA receive process shall map Gray-coded PAM4 symbols to pairs of bits {A, B} where A is considered to be the first bit as follows:

120.5.7.2 Precoding for PAM4 encoded lanes

Change the first paragraph of 120.5.7.2 as follows:

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of a 200GBASE-R or 400GBASE-R PMD that includes the PMD control

function defined in [136.8.11](#) ([200GBASE-CR2](#), [200GBASE-CR4](#), ~~or~~ [200GBASE-KR2](#), [200GBASE-KR4](#), [400GBASE-CR4](#), or [400GBASE-KR4](#)).

120.5.11 PMA test patterns (optional)

120.5.11.2 Test patterns for PAM4 encoded signals

Change the first paragraph of 120.5.11.2 as follows:

For a 200GBASE-R PMA with 2 or 4 output lanes or a 400GBASE-R PMA with 4 or 8 output lanes using PAM4 encoding, the test patterns described in this clause may optionally be supported.

Insert new subclause 120.5.11.2.a before 120.5.11.2.1 as follows:

120.5.11.2.a PRBS9Q test pattern

A PMA may optionally include a PRBS9Q pattern generator as specified in this subclause. The ability to perform this function is indicated by the PRBS9Q_gen_Tx_ability and PRBS9Q_gen_Rx_ability status variables, reflecting the ability to send this test pattern in the transmit direction (towards the PMD) and receive direction (towards the MAC), respectively. The PRBS9Q test pattern is enabled in the transmit direction by the Tx_gen_enable and PRBS9Q_enable_0 through PRBS9Q_enable_7 control variables and in the receive direction by the Rx_gen_enable and PRBS9Q_enable_0 through PRBS9Q_enable_7 control variables.

If a Clause 45 MDIO is implemented, the variables are mapped to register bits as follows:

- The PRBS9Q_gen_Tx_ability status variable is accessible through the PRBS9Q Tx generator ability bit 1.1500.15 (45.2.1.169).
- The PRBS9Q_gen_Rx_ability status variable is accessible through the PRBS9Q Rx generator ability bit 1.1500.14 (see 45.2.1.169).
- The Tx_gen_enable control variable is accessible through the Tx generator enable bit 1.1501.3 (see [45.2.1.170](#)).
- The Rx_gen_enable control variable is accessible through the Rx generator enable bit 1.1501.1 (see [45.2.1.170](#)).
- The PRBS9_enable_0 through PRBS9_enable_7 control variables are accessible through bits 1.1511.0 through 1.1511.7 (see 45.2.1.171a).

When the PRBS9Q test pattern is enabled, it replaces the signal on the output lane(s) for which it is enabled. The PRBS9Q test pattern is a repeating 511-symbol sequence formed by Gray coding pairs of bits from two repetitions of the PRBS9 pattern into PAM4 symbols as described in 120.5.7. The PRBS9 pattern generator produces the same result as the implementation shown in Figure 120–7, which implements the generator polynomial shown in Equation (120–1). Since the PRBS9 pattern is an odd number of bits in length, bits that are mapped as the first bit of a PAM4 symbol during one repetition of the PRBS9 sequence are mapped as the second bit of a PAM4 symbol during the next repetition of the PRBS9 sequence, and bits that are mapped as the second bit of a PAM4 symbol are mapped as the first bit of the following symbol in the next repetition of the PRBS9 sequence. For example, if the PRBS9 generator used to create the PRBS9Q sequence is initialized to a seed value of 11111111 (with the leftmost bit in S0 and the rightmost in S8), the PRBS9Q sequence is the following sequence of Gray coded PAM4 symbols, transmitted left to right:

```
0012322303231310010331213302202231320111030230213332303130303000
1003020031203332002123313231011003321022213103113222031333131300
0201311013311222101130233203202201221210013321323200113322333330
0110332203232300120233102211211010301312003221320210023220022223
```

0022122011202030031102321012312202130333101201321112010201010000
3010130102311113013221021203033011133122320310321223102110202000
1302033021032223303201211311312302232330021132121300321122111100
033111231121200023121031233233303100202301123213133012123012222

$$G(x) = 1 + x^5 + x^9 \tag{120-1}$$

NOTE—The PRBS9 sequence defined in this subclause is the same as the PRBS9 sequence defined in [Table 68-6](#).

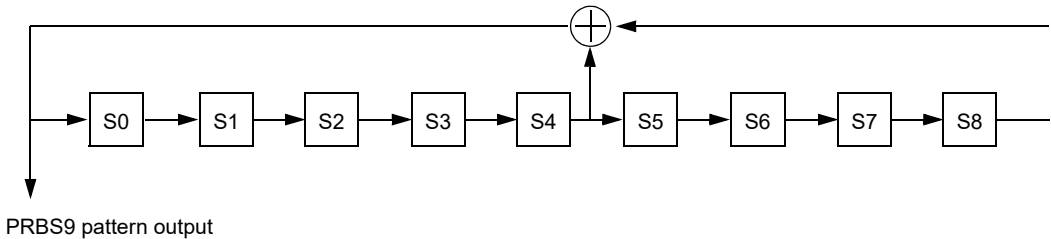


Figure 120-7—PRBS9 pattern generator

120.5.11.2.4 Square wave (quaternary) test pattern

Change the first paragraph of 120.5.11.2.4 as follows:

Transmit square wave (quaternary) test-pattern mode optionally applies to each lane of the Tx direction PMA towards a physically instantiated 200GAUI-4, 200GAUI-2, 400GAUI-8, or 400GAUI-4, or towards a PAM4 PMD service interface whether or not it is physically instantiated.

Change the note at the end of 120.5.11.2.4 as follows:

NOTE—A square wave transmitted over a 200GAUI-4, 200GAUI-2, ~~or 400GAUI-8, or 400GAUI-4~~ may not be correctly forwarded to the PMD transmitter output.

120.6 PMA MDIO function mapping

Change Table 120-3 as follows (some unchanged rows not shown):

Table 120-3—MDIO/PMA control variable mapping

MDIO variable	PMA/PMD register name	Register/ bit number	PMA control variable
...			
Lane 15 SW enable	Square wave testing control	1.1510.15	Square_wave_enable_15
<u>Lane 0 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.0</u>	<u>PRBS9Q_enable_0</u>
<u>Lane 1 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.1</u>	<u>PRBS9Q_enable_1</u>
<u>Lane 2 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.2</u>	<u>PRBS9Q_enable_2</u>
<u>Lane 3 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.3</u>	<u>PRBS9Q_enable_3</u>

Table 120–3—MDIO/PMA control variable mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/ bit number	PMA control variable
<u>Lane 4 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.4</u>	<u>PRBS9Q_enable_4</u>
<u>Lane 5 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.5</u>	<u>PRBS9Q_enable_5</u>
<u>Lane 6 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.6</u>	<u>PRBS9Q_enable_6</u>
<u>Lane 7 PRBS9Q enable</u>	<u>PRBS9Q testing control</u>	<u>1.1511.7</u>	<u>PRBS9Q_enable_7</u>
<u>Lane 0 PRBS13Q enable</u>	<u>PRBS13Q testing control</u>	<u>1.1512.0</u>	<u>PRBS13Q_enable_0</u>
...			

Change Table 120–4 as follows (some unchanged rows not shown):

Table 120–4—MDIO/PMA status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMA status variable
...			
SSPRQ Tx generator ability	Test-pattern ability register	1.1500.13	SSPRQ_gen_Tx_ability
<u>PRBS9Q Rx generator ability</u>	<u>Test-pattern ability register</u>	<u>1.1500.14</u>	<u>PRBS9Q_gen_Rx_ability</u>
<u>PRBS9Q Tx generator ability</u>	<u>Test-pattern ability register</u>	<u>1.1500.15</u>	<u>PRBS9Q_gen_Tx_ability</u>

120.7 Protocol implementation conformance statement (PICS) proforma for Clause 120, Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R¹⁵

120.7.3 Major capabilities/options

Change the table in 120.7.3 as follows (unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*LNS_UPSTR M	Number of lanes in direction of MAC	120.1.4	Divisor of number of PCS lanes	PMA200: M	8 [] 4 [] <u>2</u> []
				PMA400:M	16 [] 8 [] <u>4</u> []
...					
UP_PAM4	PAM4 modulation used for PMA service interface	120.1.4		(PMA200 LNS_UPSTRM=4 <u>LNS_UPSTRM<8</u> or PMA400* LNS_UPSTRM=8 <u>LNS_UPSTRM<16</u>) :M	Yes [] N/A []
*LNS_DNSTR M	Number of lanes in the direction of PMD	120.1.4	Divisor of number of PCS lanes	PMA200: M	8 [] 4 [] <u>2</u> []
				PMA400:M	16 [] 8 [] <u>4</u> []
...					
DN_PAM4	PAM4 modulation used for service interface below the PMA	120.1.4		(PMA200 LNS_DNSTRM=4 <u>LNS_DNSTRM<8</u> or PMA400* LNS_DNSTRM=16 <u>LNS_DNSTRM<16</u>) :M	Yes [] N/A []
RX_CLOCK	Signaling rate of output lanes in Rx direction	120.5.5	26.5625 GBd <u>LNS_DNSTRM=</u> 4 or 8: <u>26.5625 GBd</u> <u>LNS_DNSTRM=</u> 2: <u>53.125 GBd</u>	<u>PMA200:M</u>	Yes [] <u>N/A</u> []
			<u>LNS_DNSTRM=</u> 8 or 16: <u>26.5625 GBd</u> <u>LNS_DNSTRM=</u> 4: <u>53.125 GBd</u>	<u>PMA400:M</u>	<u>Yes</u> [] <u>N/A</u> []

¹⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Item	Feature	Subclause	Value/Comment	Status	Support
TX_CLOCK	Signaling rate of output lanes in Tx direction	120.5.5	26.5625 GBd <u>LNS_DNSTRM=</u> <u>4 or 8:</u> 26.5625 GBd <u>LNS_DNSTRM=</u> <u>2: 53.125 GBd</u>	PMA200:M	Yes [] N/A []
			LNS_DNSTRM= 8 or 16: 26.5625 GBd LNS_DNSTRM= 4: 53.125 GBd	PMA400:M	Yes [] N/A []
...					
UNAU1	Electrical and timing requirements of Annex 120B, Annex 120C, Annex 120D, or Annex 120E through <u>Annex 120G</u> as appropriate met by 200GAUI-n or 400GAUI-n of the PMA service interface	120.5.1, 120.5.5		USP1SP6:M	Yes [] N/A []
DNAUI	Electrical and timing requirements of Annex 120B, Annex 120C, Annex 120D, or Annex 120E through <u>Annex 120G</u> as appropriate met by 200GAUI-n or 400GAUI-n of the service interface below the PMA	120.5.1, 120.5.5		DSP1SP6:M	Yes [] N/A []
...					

121. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-DR4

121.1 Overview

Change Table 121–1 as follows (some unchanged rows not shown):

Table 121–1—Physical Layer clauses associated with the 200GBASE-DR4 PMD

Associated clause	200GBASE-DR4
...	
120E—Chip-to-module 200GAUI-4	Optional
120F—Chip-to-chip 200GAUI-2 (200GAUI-2 C2C)	<u>Optional</u>
120G—Chip-to-module 200GAUI-2 (200GAUI-2 C2M)	<u>Optional</u>
78—Energy Efficient Ethernet	Optional

122. Physical Medium Dependent (PMD) sublayer and medium, type 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8

122.1 Overview

Change Table 122-1 as follows (some unchanged rows not shown):

Table 122–1—Physical Layer clauses associated with the 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 PMDs

Associated clause	200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4	400GBASE-FR8, 400GBASE-LR8, 400GBASE-ER8
...		
120E—Chip-to-module 400GAUI-8	Not applicable	Optional
120F—Chip-to-chip 200GAUI-2 (200GAUI-2 C2C)	Optional	Not applicable
120F—Chip-to-chip 400GAUI-4 (400GAUI-4 C2C)	Not applicable	Optional
120G—Chip-to-module 200GAUI-2 (200GAUI-2 C2M)	Optional	Not applicable
120G—Chip-to-module 400GAUI-4 (400GAUI-4 C2M)	Not applicable	Optional
78—Energy Efficient Ethernet	Optional	Optional

123. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR16

123.1 Overview

Change Table 123–1 as follows (some unchanged rows not shown):

Table 123–1—Physical Layer clauses associated with the 400GBASE-SR16 PMD

Associated clause	400GBASE-SR16
...	
120E—Chip-to-module 400GAUI-8	Optional
120F—Chip-to-chip 400GAUI-4 (400GAUI-4 C2C)	<u>Optional</u>
120G—Chip-to-module 400GAUI-4 (400GAUI-4 C2M)	<u>Optional</u>
78—Energy Efficient Ethernet	Optional

124. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-DR4

124.1 Overview

Change Table 124–1 as follows (some unchanged rows not shown):

Table 124–1—Physical Layer clauses associated with the 400GBASE-DR4 PMD

Associated clause	400GBASE-DR4
...	
120E—Chip-to-module 400GAUI-8	Optional
120F—Chip-to-chip 400GAUI-4 (400GAUI-4 C2C)	<u>Optional</u>
120G—Chip-to-module 400GAUI-4 (400GAUI-4 C2M)	<u>Optional</u>
78—Energy Efficient Ethernet	Optional

135. Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P

135.1 Overview

135.1.1 Scope

Change the first paragraph of 135.1.1 as follows:

This clause specifies the Physical Medium Attachment sublayer (PMA) for the families of 50 Gb/s and 100 Gb/s PAM4 Physical Layer implementations known as 50GBASE-R (see [131.1.3](#)) and 100GBASE-P (see [80.1.4](#)). The PMA allows the PCS (see [Clause 133](#) and [Clause 82](#)) and FEC (see [Clause 134](#), ~~and Clause 91~~, and [Clause 161](#)) to connect in a media-independent way with a range of physical media. The 50GBASE-R PMA can support any of the 50 Gb/s PMDs in [Table 131-1](#), ~~and~~ The 100GBASE-P PMA can support any of the 100 Gb/s PAM4 PMDs ~~in according to Table 80-3~~ [Table 80-3a](#) and [Table 80-5](#), ~~with the exception of the 100GBASE-KP4 PMD ([Clause 94](#)).~~

Change the third paragraph of 135.1.1 as follows:

The interfaces for the inputs of the 50GBASE-R and 100GBASE-P PMAs are defined in an abstract manner and do not imply any particular implementation. For 50GBASE-R PMAs, electrical interfaces connecting PMA sublayers, known as LAUI-2 and 50GAUI-n, are defined in [Annex 135B](#) through [Annex 135G](#). For 100GBASE-P PMAs, electrical interfaces connecting PMA sublayers, known as 100GAUI-n, are defined in [Annex 120F](#), [Annex 120G](#), and [Annex 135D](#) through [Annex 135G](#).

135.1.4 PMA sublayer positioning

Replace Figure 135–2 with the following figure:

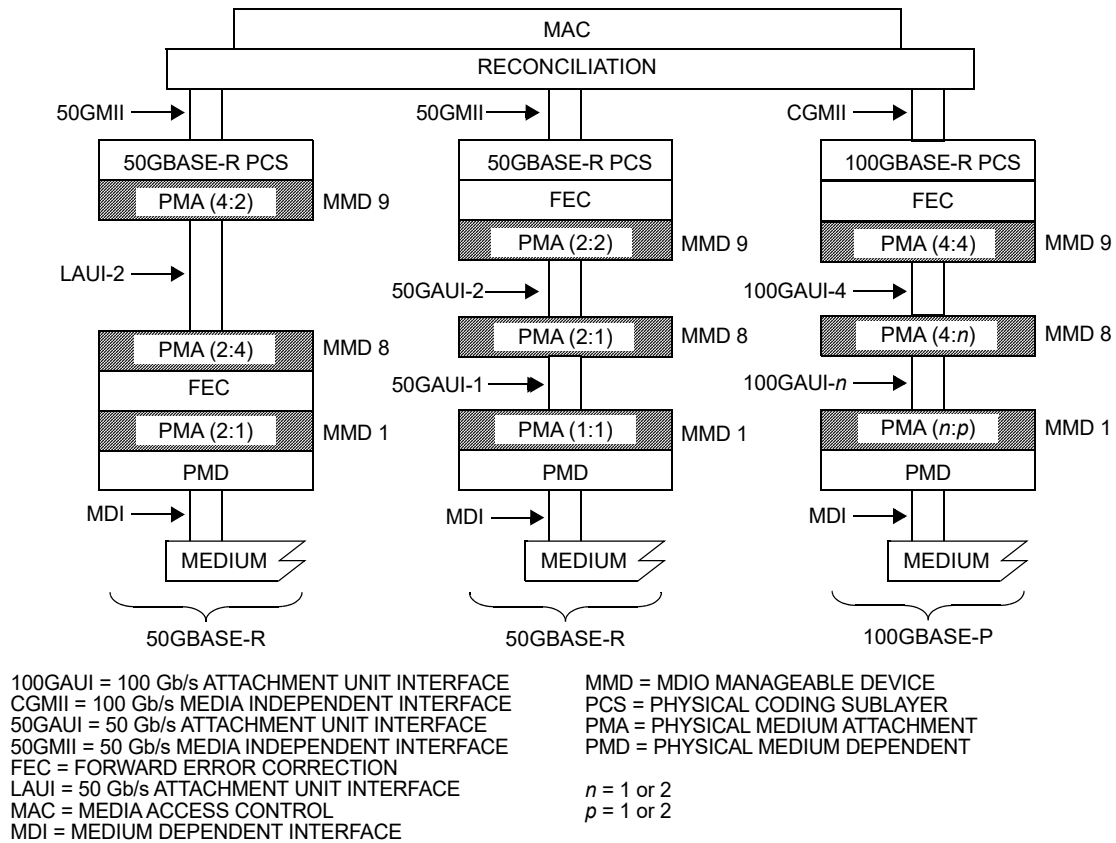


Figure 135–2—Example 50GBASE-R and 100GBASE-P PMA layering

In the fourth paragraph of 135.1.4, change list items g) and h) as follows:

- g) CAUI-n is a physical instantiation of the PMA service interface between two adjacent 100GBASE-R PMA (see Clause 83) sublayers above the FEC for a 100GBASE-P PHY or between adjacent sublayers below the PCS for a 100GBASE-R PHY. CAUI-n is not specified in this clause, but is listed here for completeness.
 - 1) CAUI-n provides bit-multiplexing of 20 PCSLs.
 - 2) CAUI-4 is a 25.78125 GBd by four-lane NRZ physical instantiation of the 100 Gb/s connection. CAUI-4 is specified in Clause 83 and associated annexes.
 - 3) CAUI-10 is a 10.3125 GBd by ten-lane NRZ physical instantiation of the 100 Gb/s connection. CAUI-10 is specified in Clause 83 and associated annexes.
 - 4) For a PHY that includes a 100GAUI-1 interface, it is recommended that the signaling rate range for a CAUI-4 or CAUI-10 PMA output that is in the same package as the PCS be limited to 25.78125 GBd $\pm$ 50 ppm or 10.3125 GBd $\pm$ 50 ppm, respectively.

- h) 100GAUI-n is a physical instantiation of the PMA service interface between two adjacent 100GBASE-P PMA sublayers below the FEC for a 100GBASE-P PHY.
 - 1) 100GAUI-n provides bit-multiplexing of 4 FECLs.
 - 2) 100GAUI-4 is a 26.5625 GBd by four-lane NRZ physical instantiation of the 100 Gb/s connection.
 - 3) 100GAUI-2 is a 26.5625 GBd by two-lane PAM4 physical instantiation of the 100 Gb/s connection.
 - 4) 100GAUI-1 is a 53.125 GBd by one-lane PAM4 physical instantiation of the 100 Gb/s connection.
 - 5) For a PHY that includes a 100GAUI-1 interface, it is recommended that the signaling rate range for a 100GAUI-4 or 100GAUI-2 PMA output that is in the same package as the PCS be limited to 26.5625 GBd $\pm$ 50 ppm.

135.5 Functions within the PMA

135.5.1 Per input-lane clock and data recovery

Change 135.5.1 as follows:

If the interface between the PMA client and the PMA is physically instantiated as LAUI-2, 50GAUI-n, or 100GAUI-n, the PMA shall meet the electrical and timing specifications above the PMA in Annex 120F, Annex 120G, or Annex 135B through Annex 135G, as appropriate.

If the interface between the sublayer below the PMA and the PMA is physically instantiated as LAUI-2, 50GAUI-n, or 100GAUI-n, the PMA shall meet the electrical and timing specifications below the PMA at the service interface as specified in Annex 120F, Annex 120G, or Annex 135B through Annex 135G, as appropriate.

135.5.2 Bit-level multiplexing

Change the fifth paragraph in 135.5.2 as follows:

Each PCSL/FECL received in any temporal position on an input lane is transferred into a temporal position on an output lane. As the PCS (see Clause 133 and Clause 82) and FEC (see Clause 91, and Clause 134, and Clause 161) have fully flexible receive logic, an implementation is free to perform the mapping of PCSLs/FECLs from input lanes to output lanes without constraint. Figure 135–6 illustrates one possible bit ordering for a 100GBASE-P 4:2 PMA bit mux. Other bit orderings are also valid.

135.5.6 Signal drivers

Change 135.5.6 as follows:

For cases where the interface between the PMA client and the PMA or between the PMA and the sublayer below the PMA represent a physically instantiated interface, the PMA provides electrical signal drivers for that interface. The electrical and jitter/timing specifications for these interfaces appear in

- [Annex 135B](#) for LAUI-2 C2C applications
- [Annex 135C](#) for LAUI-2 C2M applications
- [Annex 135D](#) for 50GAUI-2 and 100GAUI-4 C2C applications
- [Annex 135E](#) for 50GAUI-2 and 100GAUI-4 C2M applications
- [Annex 135F](#) for 50GAUI-1 and 100GAUI-2 C2C applications
- [Annex 135G](#) for 50GAUI-1 and 100GAUI-2 C2M applications
- [Annex 120F](#) for 100GAUI-1 C2C applications
- [Annex 120G](#) for 100GAUI-1 C2M applications

The modulation format is NRZ for LAUI-2, 50GAUI-2, and 100GAUI-4 and the modulation format is PAM4 for 50GAUI-1, ~~and~~ 100GAUI-2, and 100GAUI-1.

135.5.7 PAM4 encoding

135.5.7.2 Precoding for PAM4 encoded lanes

Change the first paragraph of 135.5.7.2 as follows:

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of a 50GBASE-R or 100GBASE-R PMD that includes the PMD control function defined in [136.8.11](#) (50GBASE-CR, 50GBASE-KR, 100GBASE-CR1, 100GBASE-CR2, 100GBASE-KR1, or 100GBASE-KR2), or are part of a 50GAUI-1 C2C or 100GAUI-2 C2C link.

Change the seventh paragraph of 135.5.7.2 as follows:

If the PMA is connected to the service interface of a PMD that supports the PMD control function and training is disabled by the management variable `mr_training_enable`, or if the PMA is part of a 50GAUI-1, ~~C2C or a~~ 100GAUI-2, or 100GAUI-1 C2C link, then `precoder_tx_out_enable_i`, `precoder_rx_in_enable_i`, `precoder_tx_in_enable_i`, and `precoder_rx_out_enable_i` are set as required by the implementation. The method described in 135F.3.2.1 may be used for 50GAUI-1, ~~C2C or~~ 100GAUI-2, or 100GAUI-1 C2C.

Replace Figure 135–7 with the following figure:

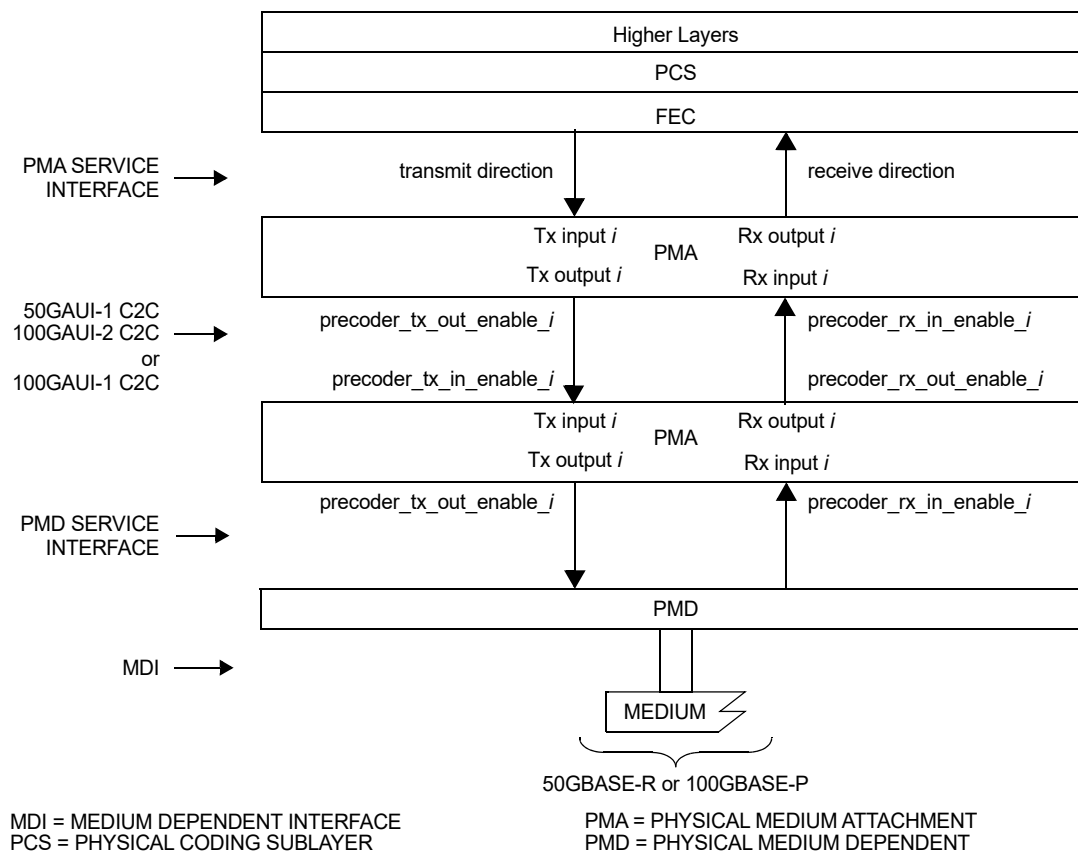


Figure 135–7—Example relating PMA precoder variables to inputs and outputs

135.5.10 PMA test patterns (optional)

135.5.10.2 Test patterns for PAM4 encoded signals

135.5.10.2.4 Square wave (quaternary) test pattern

Change 135.5.10.2.4 as follows:

Transmit square wave (quaternary) test-pattern generator optionally applies to each lane of the Tx direction PMA towards a physically instantiated 50GAUI-1, 100GAUI-2, or 100GAUI-1, or towards a PAM4 PMD service interface whether or not it is physically instantiated as specified in 120.5.11.2.4.

135.7 Protocol implementation conformance statement (PICS) proforma for Clause 135, Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P¹⁶

135.7.3 Major capabilities/options

Change the table in 135.7.3 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/ Comment	Status	Support
...					
*NLA	Number of lanes on physically instantiated PMA service interface	135.1.4		50G*PSIA:M	1 [] 2 [] N/A []
				100G*PSIA:M	1 [] 2 [] 4 [] N/A []
...					
50GA2	Physical instantiation of the PMA service interface is 50GAUI-2	135.1.4		50G(BFEC+APMD)*PSIA*NLA=2:M	Yes [] N/A []
<u>*100GA1</u>	<u>Physical instantiation of the PMA service interface is 100GAUI-1</u>	<u>135.1.4</u>		<u>100G*(BFEC+APMD)*PSIA*NLA=1:M</u>	<u>Yes []</u> <u>N/A []</u>
100GA2	Physical instantiation of the PMA service interface is 100GAUI-2	135.1.4		100G(BFEC+APMD)*PSIA*NLA=2:M	Yes [] N/A []
...					
*PMDE	PMD is 50GBASE-CR, 50GBASE-KR, 100GBASE-CR2, <u>100GBASE-CR1</u> , 100GBASE-KR2, <u>100GBASE-KR1</u>	135.5.7.2		APMD:O	Yes [] N/A []
...					

¹⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

135.7.4 PICS proforma tables for Physical Medium Attachment (PMA) sublayer, type 50GBASE-R and 100GBASE-P

135.7.4.3 Electrical

Change the table in 135.7.4.3 as follows (some unchanged rows not shown):

Item	Feature	Subclause	Value/Comment	Status	Support
...					
E10	Electrical and timing requirements of Annex 135G met by 50GAUI-1 or 100GAUI-2	135.5.1, 135.5.6		(50GA1+100GA2)* (AT=C2M):M	Yes [] N/A []
<u>E11</u>	<u>Electrical and timing requirements of Annex 120F met by 100GAUI-1 C2C</u>	<u>135.5.1, 135.5.6</u>		<u>100GA1*</u> <u>(AT=C2C):M</u>	<u>Yes []</u> <u>N/A []</u>
<u>E12</u>	<u>Electrical and timing requirements of Annex 120G met by 100GAUI-1 C2M</u>	<u>135.5.1, 135.5.6</u>		<u>100GA1*</u> <u>(AT=C2M):M</u>	<u>Yes []</u> <u>N/A []</u>

138. Physical Medium Dependent (PMD) sublayer and medium, type 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8

138.1 Overview

Change Table 138–2 and Table 138–3 as follows (some unchanged rows not shown):

Table 138–2—Physical Layer clauses associated with the 100GBASE-SR2 PMD

Associated clause	100GBASE-SR2
...	
83E—CAUI-4 C2M	Optional
120F—100GAUI-1 C2C	<u>Optional</u>
120G—100GAUI-1 C2M	<u>Optional</u>
135—PMA for 100GBASE-P	Required
...	

Table 138–3—Physical Layer clauses associated with the 200GBASE-SR4 and 400GBASE-SR8 PMDs

Associated clause	200GBASE-SR4	400GBASE-SR8
...		
120E—400GAUI-8 C2M	Not applicable	Optional
120F—200GAUI-2 C2C	<u>Optional</u>	<u>Not applicable</u>
120F—400GAUI-4 C2C	<u>Not applicable</u>	<u>Optional</u>
120G—200GAUI-2 C2M	<u>Optional</u>	<u>Not applicable</u>
120G—400GAUI-4 C2M	<u>Not applicable</u>	<u>Optional</u>
78—Energy-Efficient Ethernet	Optional	Optional

140. Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1

140.1 Overview

Change Table 140–1 as follows (some unchanged rows not shown):

Table 140–1—Physical Layer clauses associated with the 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1 PMDs

Associated clause	100GBASE-DR, 100GBASE-FR1, 100GBASE-LR1
...	
83E—CAUI-4 C2M	Optional
120F—100GAUI-1 C2C	<u>Optional</u>
120G—100GAUI-1 C2M	<u>Optional</u>
135—100GBASE-P PMA	Required
...	

150. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-SR4.2

150.1 Overview

Change Table 150–1 as follows (some unchanged rows not shown):

Table 150–1—Physical Layer clauses associated with the 400GBASE-SR4.2 PMD

Associated clause	400GBASE-SR4.2
...	
120E —400GAUI-8 C2M	Optional
120F —400GAUI-4 C2C	<u>Optional</u>
120G —400GAUI-4 C2M	<u>Optional</u>
78 —Energy-Efficient Ethernet	Optional

151. Physical Medium Dependent (PMD) sublayer and medium, type 400GBASE-FR4 and 400GBASE-LR4-6

151.1 Overview

Change Table 151–1 as follows (some unchanged rows not shown):

Table 151–1—Physical Layer clauses associated with the 400GBASE-FR4 and 400GBASE-LR4-6 PMDs

Associated clause	400GBASE-FR4, 400GBASE-LR4-6
...	
120E —400GAUI-8 C2M	Optional
120F —400GAUI-4 C2C	<u>Optional</u>
120G —400GAUI-4 C2M	<u>Optional</u>
78 —Energy-Efficient Ethernet	Optional

152. Inverse RS-FEC sublayer

152.6 Inverse RS-FEC MDIO function mapping

Change Table 152–1 as follows:

Table 152–1—MDIO/Inverse RS-FEC control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
IFEC bypass correction enable	IFEC control	1.2200.0	IFEC_bypass_correction_enable
IFEC bypass indication enable	IFEC control	1.2200.1	IFEC_bypass_indication_enable
<u>IFEC enable</u>	<u>IFEC control</u>	<u>1.2200.2</u>	<u>IFEC_enable</u>

Insert 152.6.2a after 152.6.2 as follows:

152.6.2a IFEC_enable

For PHYs that support bypassing the FEC operation of the FEC sublayer below the Inverse RS-FEC sublayer, the Inverse RS-FEC sublayer has the capability to enable or disable its FEC operation. An MDIO interface or an equivalent management interface controls the variable IFEC_enable for the Inverse RS-FEC sublayer. When the IFEC_enable variable is set to one, the Inverse RS-FEC sublayer performs the transmit function as specified in 152.5.2 and the receive function as specified in 152.5.3. When the variable is set to zero, the Inverse RS-FEC transmit and receive functions are disabled, and the Inverse RS-FEC sublayer is bypassed, effectively connecting its service interface to the service interface of its underlying sublayer. This variable is mapped to the bit defined in 45.2.1.216.a (1.2200.2).

154. Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-ZR

154.1 Overview

Change Table 151–1 as follows (some unchanged rows not shown):

Table 154–1—Physical Layer clauses associated with the 100GBASE-ZR PMD

Associated clause	100GBASE-ZR
...	
83E—CAUI-4 C2M	Optional
120F—100GAUI-1 C2C	<u>Optional</u>
120G—100GAUI-1 C2M	<u>Optional</u>
152—Inverse RS-FEC	Conditional ^a
...	

^aClause 152 inverse RS-FEC mandatory when Clause 91 RS-FEC is present.

Insert new clauses Clause 161 through Clause 163 after Clause 160 as follows:

161. Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs

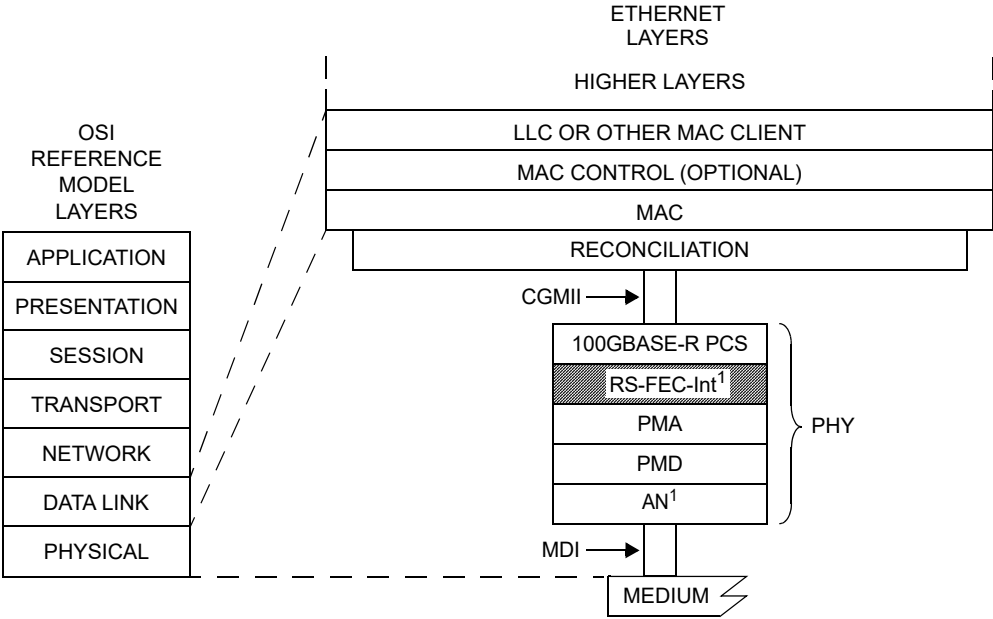
161.1 Overview

161.1.1 Scope

This clause specifies a Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for some 100GBASE-P PHYs.

161.1.2 Position of RS-FEC-Int in the 100GBASE-P sublayers

Figure 161–1 shows the relationship of the RS-FEC-Int sublayer to the ISO/IEC Open System Interconnection (OSI) reference model.



AN = AUTO-NEGOTIATION
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
RS-FEC-Int = REED-SOLOMON FORWARD ERROR
CORRECTION CODEWORD-INTERLEAVED

NOTE 1—CONDITIONAL BASED ON PHY TYPE

Figure 161–1—RS-FEC-Int relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

161.2 FEC service interface

The FEC service interface is specified identically as in 91.2, with the exception that EEE is not supported.

161.3 PMA compatibility

PMA compatibility requirements are identical to those in 91.3.

161.4 Delay constraints

The maximum delay contributed by the RS-FEC-Int sublayer (sum of transmit and receive delays at one end of the link) shall be no more than 51 200 bit times (100 pause_quanta or 512 ns). A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 80.4 and its references.

161.5 Functions within the RS-FEC-Int sublayer

When operating in the RS-FEC-Int mode, all functions operate as described in 161.5.1 through 161.5.4.

161.5.1 Functional block diagram

A functional block diagram of the RS-FEC-Int sublayer is shown in Figure 161–2.

161.5.2 Transmit function

161.5.2.1 Lane block synchronization

Lane block synchronization is specified identically as in 91.5.2.1.

161.5.2.2 Alignment lock and deskew

Alignment lock and deskew is specified identically as in 91.5.2.2.

161.5.2.3 Lane reorder

Lane reorder is specified identically as in 91.5.2.3.

161.5.2.4 Alignment marker removal

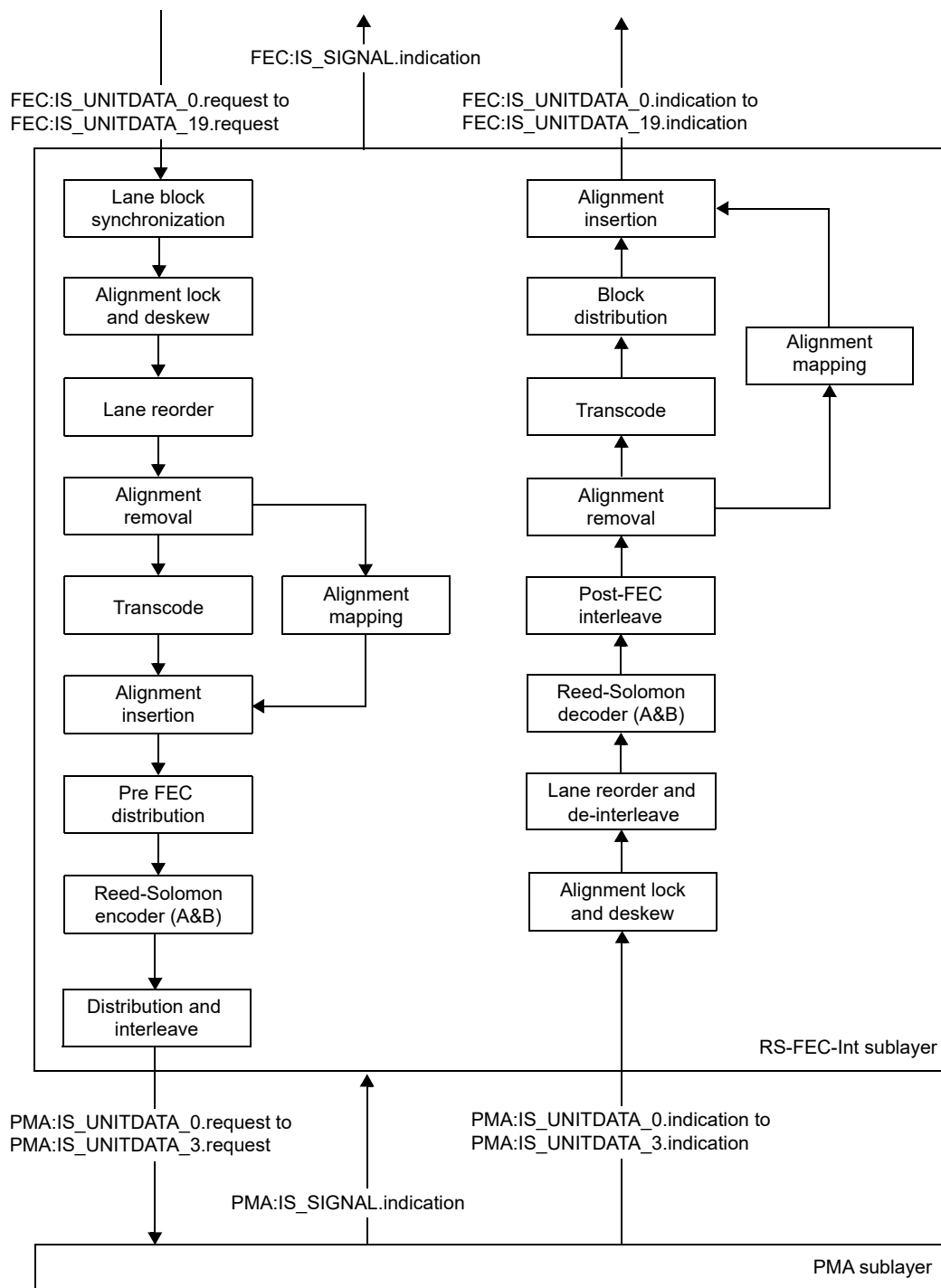
Alignment marker removal is specified identically as in 91.5.2.4, with the exception that EEE is not supported.

161.5.2.5 64B/66B to 256B/257B transcoder

The transcoder is specified identically as in 91.5.2.5.

161.5.2.6 Alignment marker mapping and insertion

The alignment markers that were removed per 161.5.2.4 are re-inserted (see 161.5.2.6.2) after being processed by the alignment marker mapping function (see 161.5.2.6.1). The alignment marker mapping function compensates for the operation of the symbol distribution function defined in 161.5.2.9 and rearranges the alignment marker bits so that they appear on the FEC lanes intact and in the desired sequence. This preserves the properties of the alignment markers (e.g., DC balance, transition density) and provides a deterministic pattern for the purpose of synchronization. The RS-FEC-Int receive function uses knowledge of this mapping to determine the FEC lane that is received on a given lane of the PMA service interface, to compensate for skew between FEC lanes, and to identify FEC codeword boundaries.



161.5.2.6.1 Alignment marker mapping

The alignment marker mapping function operates on a group of 20 aligned and reordered alignment markers. Let $\text{am_tx_x}<65:0>$ be the alignment marker for PCS lane x , $x=0$ to 19, where bit 0 is the first bit transmitted. The alignment markers shall be mapped to $\text{am_txmapped}<1284:0>$ in a manner that yields the same result as the process described in the remainder of this subclause.

For $x=0$ to 19, $\text{amp_tx_x}<63:0>$ is constructed as follows:

- a) If $x \leq 3$, $\text{amp_tx_x}<23:0>$ is set to M_0 , M_1 , and M_2 as shown in Figure 82–9 (bits 25 to 2) using the values in Table 82–2 for PCS lane number 0. If $x \geq 4$, $\text{amp_tx_x}<23:0>$ is set to M_4 , M_5 , and M_6 as shown in Figure 82–9 (bits 57 to 34) using the values in Table 82–2 for PCS lane number x .
- b) $\text{amp_tx_x}<31:24> = \text{amp_tx_x}<33:26>$
- c) If $x \leq 3$, $\text{amp_tx_x}<55:32>$ is set to M_4 , M_5 , and M_6 as shown in Figure 82–9 (bits 57 to 34) using the values in Table 82–2 for PCS lane number 0. If $x \geq 4$, $\text{amp_tx_x}<55:32>$ is set to M_0 , M_1 , and M_2 as shown in Figure 82–9 (bits 25 to 2) using the values in Table 82–2 for PCS lane number x .
- d) $\text{amp_tx_x}<63:56> = \text{amp_tx_x}<65:58>$

This process replaces the fixed bytes of the alignment markers received, possibly with errors, with the values from Table 82–2. In addition it substitutes the fixed bytes of the alignment markers corresponding to PCS lanes 1, 2, and 3 with the fixed bytes for the alignment marker corresponding to PCS lane 0. The variable bytes BIP are unchanged. This process simplifies receiver synchronization since the receiver only needs to search for the fixed bytes corresponding to PCS lane 0 on each FEC lane.

Construct a matrix of 4 rows and 320 columns, am_txpayloads , as shown in Figure 161–3. Given $i=0$ to 3, $j=0$ to 4, and $x=i+4j$, the matrix is derived per the following expression:

$$\text{am_txpayloads}<i, (64j+63):64j> = \text{amp_tx_x}<63:0>$$

The variable am_txmapped is derived from am_txpayloads per the following expressions:

```

for all  $k=0$  to 31
  for all  $j=0$  to 1
    if even( $k$ )
       $\text{am\_txmapped}<40k+20j+9:40k+20j> = \text{am\_txpayloads}<2j, 10k+9:10k>$ 
       $\text{am\_txmapped}<40k+20j+19:40k+20j+10> = \text{am\_txpayloads}<2j+1, 10k+9:10k>$ 
    else
       $\text{am\_txmapped}<40k+20j+9:40k+20j> = \text{am\_txpayloads}<2j+1, 10k+9:10k>$ 
       $\text{am\_txmapped}<40k+20j+19:40k+20j+10> = \text{am\_txpayloads}<2j, 10k+9:10k>$ 

```

A 5-bit pad is appended to the mapped alignment markers to yield the equivalent of five 257-bit blocks. The pad bits, $\text{am_txmapped}<1284:1280>$, shall be set to the binary values 00101 and 11010 (the leftmost bit is assigned to the highest bit index) in an alternating pattern. In other words, if a pad value of 00101 is used for the current iteration of the mapping function, a value of 11010 is used in the next iteration and vice versa.

The result of the alignment marker mapping function is a deterministic mapping between alignment marker payloads and FEC lanes (see Figure 161–3).

FEC lane, <i>i</i>	Reed-Solomon symbol index, <i>k</i> (10-bit symbols)																																	
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33
0	A B A B A B						A B A B A B A						B A B A B A						B A B A B A B						A B A B A B						A B			
	0 amp_tx_0 63						0 amp_tx_4 63						0 amp_tx_8 63						0 amp_tx_12 63						0 amp_tx_16 63									
1	B A B A B A						B A B A B A B						A B A B A B						A B A B A B A						B A B A B A						B A			
	0 amp_tx_1 63						0 amp_tx_5 63						0 amp_tx_9 63						0 amp_tx_13 63						0 amp_tx_17 63									
2	A B A B A B						A B A B A B A						B A B A B A						B A B A B A B						A B A B A B						A B			
	0 amp_tx_2 63						0 amp_tx_6 63						0 amp_tx_10 63						0 amp_tx_14 63						0 amp_tx_18 63									
3	B A B A B A						B A B A B A B						A B A B A B						A B A B A B A						B A B A B A						B A			
	0 amp_tx_3 63						0 amp_tx_7 63						0 amp_tx_11 63						0 amp_tx_15 63						0 amp_tx_19 63									

= 5-bit pad

= Resumption of 257-bit blocks

A = from FEC codeword A

B = from FEC codeword B

Figure 161–3—Alignment marker mapping to FEC lanes

As a result of this process, the BIP₃ and BIP₇ fields from normal alignment markers are carried across the link protected by FEC. These fields cannot be used to monitor errors on the link protected by FEC as 64B/66B to 256B/257B transcoding and Reed-Solomon encoding alters the bit sequence. However, these fields may again be used to monitor errors after the original bit sequence is restored, i.e., following Reed-Solomon decoding and 256B/257B to 64B/66B transcoding.

One group of aligned and reordered alignment markers are mapped every $20 \times 16\,384\,66$ -bit blocks. This group of aligned and reordered alignment markers is called the “alignment marker group” and is labeled am_txmapped<1284:0>.

161.5.2.6.2 Alignment marker insertion

An alignment marker group shall be inserted so it appears in the output stream every $81\,920 \times 257$ -bit blocks.

The variable tx_scrambled_am<10279:0> is constructed in one of two ways. Let the set of vectors tx_scrambled_*i*<256:0> represent consecutive values of the transcoder output tx_scrambled<256:0> (see 161.5.2.5 for a definition of the transcoder).

For a 10 280-bit block with an alignment marker group inserted:

tx_scrambled_am<1284:0> = am_txmapped<1284:0>
For all *i*=0 to 34
tx_scrambled_am<257*i*+1541:257*i*+1285> = tx_scrambled_*i*<256:0>

For each 10 280-bit block with an alignment marker group inserted, the first 257-bit block inserted after am_txmapped shall correspond to the four 66-bit blocks received on PCS lanes 0, 1, 2, and 3 that immediately followed the alignment marker on each respective lane.

For a 10 280-bit block without an alignment marker group:

For all *i*=0 to 39
tx_scrambled_am<257*i*+256:257*i*> = tx_scrambled_*i*<256:0>

Alignment marker repetition rate is shown in Figure 161–4.

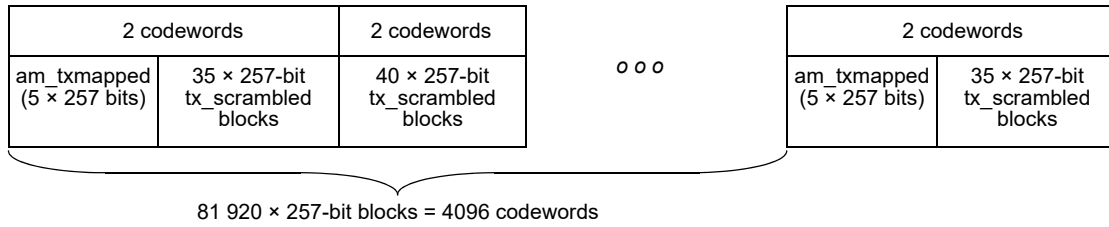


Figure 161–4—Alignment marker insertion period

The contents of tx_scrambled_am<10 279:0> are an input to the Pre-FEC distribution (see 161.5.2.7 for the definition of the Pre-FEC distribution).

161.5.2.7 Pre-FEC distribution

The Pre-FEC distribution is specified identically as in 119.2.4.5.

161.5.2.8 Reed-Solomon encoder

The Reed-Solomon encoder is specified identically as in 119.2.4.6, with the exception that the term “PCS sublayer” is replaced with “RS-FEC-Int sublayer”.

161.5.2.9 Symbol distribution

Once the data has been Reed-Solomon encoded, two resulting FEC codewords ($c_A<543:0>$ and $c_B<543:0>$) are interleaved on a 10-bit basis before the data is distributed to each FEC lane. The interleaving of two codewords shall follow this procedure:

```

For all  $k=0$  to 271
  For all  $j=0$  to 1
    if even( $k$ )
      tx_out< $4k+2j$ > =  $c_A<543-2k-j>$ 
      tx_out< $4k+2j+1$ > =  $c_B<543-2k-j>$ 
    else
      tx_out< $4k+2j$ > =  $c_B<543-2k-j>$ 
      tx_out< $4k+2j+1$ > =  $c_A<543-2k-j>$ 

```

tx_out<1087:0> shall be distributed to four FEC lanes, one 10-bit symbol at a time, from the lowest to the highest FEC lane. The first bit transmitted from each 10-bit symbol is bit 0.

161.5.2.10 Transmit bit ordering

The transmit bit ordering is illustrated in Figure 161–5.

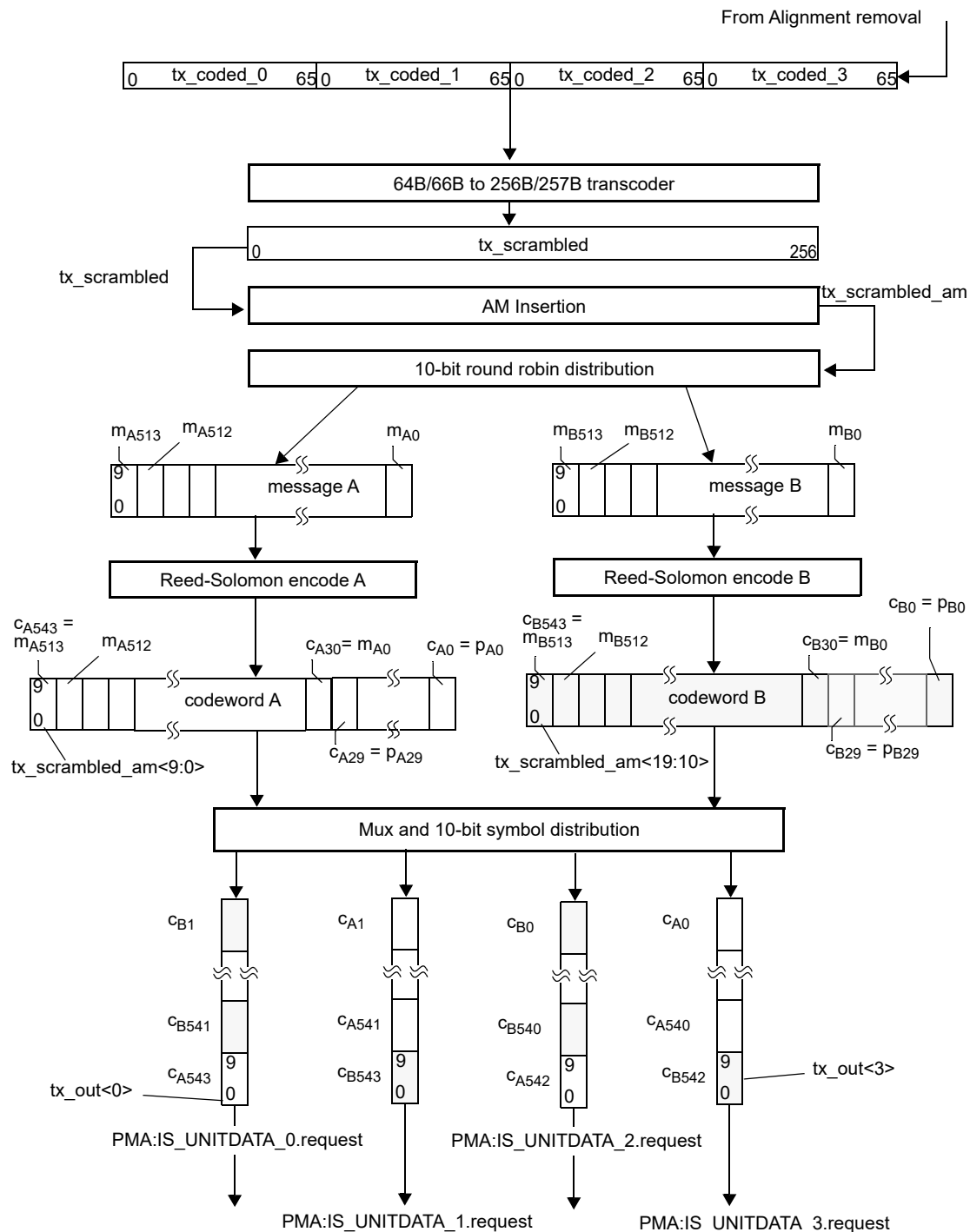


Figure 161–5—Transmit bit ordering

161.5.3 Receive function

161.5.3.1 Alignment lock and deskew

The RS-FEC-Int receive function forms four bit streams by concatenating the bits from each of the four PMA:IS_UNITDATA_*i*.indication primitives in the order they are received. It obtains lock to the alignment markers as specified by the FEC synchronization state diagram shown in Figure 91–8. Note that alignment marker lock is achieved before FEC codewords are processed and therefore the alignment markers are processed in a high error probability environment.

After alignment marker lock is achieved on all four lanes (bit streams), all inter-lane Skew is removed as specified by the FEC alignment state diagram shown in Figure 161–6. The FEC receive function shall support a maximum Skew of 180 ns between FEC lanes and a maximum Skew Variation of 4 ns.

161.5.3.2 Lane reorder and de-interleave

FEC lanes can be received on different lanes of the service interface from which they were originally transmitted. The FEC receive function shall order the FEC lanes according to the FEC lane number. The FEC lane number is defined by the sequence of alignment markers that are mapped to each FEC lane.

After all FEC lanes are aligned, deskewed, and reordered, the two FEC codewords are de-interleaved to reconstruct the original stream of two FEC codewords.

161.5.3.3 Reed-Solomon decoder

The Reed-Solomon decoder extracts the message symbols from the codeword, corrects them as necessary, and discards the parity symbols.

The Reed-Solomon decoder shall be capable of correcting any combination of up to $t=15$ symbol errors in a codeword. The Reed-Solomon decoder shall also be capable of indicating when an errored codeword was not corrected. The probability that the Reed-Solomon decoder fails to indicate a codeword as uncorrected is not expected to exceed 10^{-16} .

If bypass error indication (see 161.5.3.3.1) is not supported or not enabled, when the Reed-Solomon decoder determines that a codeword contains errors that were not corrected, it shall ensure that, for every other 257-bit block within the two associated codewords starting with the first (1st, 3rd, 5th, etc.), the synchronization header for the first 66-bit block at the output of the 256B/257B to 64B/66B transcoder, rx_coded_0<1:0>, is set to 11. In addition, it shall ensure rx_coded_0<1:0> corresponding to the 6th 257-bit block and rx_coded_3<1:0> corresponding to the last (40th) 257-bit block in the two associated codewords are set to 11. This causes the PCS to discard all frames 64 bytes and larger that are fully or partially within the codeword.

161.5.3.3.1 Bypass error indication (optional)

The Reed-Solomon decoder may optionally provide the ability to bypass the error indication feature to reduce the delay contributed by the FEC function. The presence of this option is indicated by the assertion of the FEC_bypass_indication_ability variable (see 161.6). When the option is provided it is enabled by the assertion of the FEC_bypass_indication_enable variable (see 161.6).

When FEC_bypass_indication_enable is asserted, additional error monitoring is performed by the RS-FEC-Int sublayer to reduce the likelihood that errors in a packet are not detected. The Reed-Solomon decoder counts the number of symbol errors detected on all FEC lanes in consecutive non-overlapping blocks of 8192 codewords. When the number of symbol errors in a block of 8192 codewords exceeds 5560, the RS-FEC-Int sublayer shall cause the synchronization header rx_coded<1:0> of each subsequent 66-bit

block that is delivered to the PCS to be assigned a value of 00 or 11 for a period of 60 ms to 75 ms. As a result, the PCS sets `hi_ber=true`, which inhibits the processing of received packets. When Auto-Negotiation is supported and enabled, assertion of `hi_ber` causes Auto-Negotiation to restart.

161.5.3.3.2 FEC Degraded SER (optional)

The FEC degraded SER is specified identically as in 91.5.3.3.1.

161.5.3.3.4 Post FEC Interleave

The post FEC interleave is specified identically as in 119.2.5.4.

161.5.3.3.5 Alignment marker removal

Every $81\,920 \times 257$ -bit blocks (corresponds to 4096 codewords) the first 1285 bits of `rx_scrambled_am` blocks is the vector `am_rxmapped<1284:0>` where bit 0 is the first bit received. The specific codewords that include this vector are indicated by the alignment lock and deskew function (refer to 161.5.3.1).

The vector `am_rxmapped` shall be removed from `rx_scrambled_am` prior to transcoding.

161.5.3.3.6 256B/257B to 64B/66B transcoder

The transcoder is specified identically as in 91.5.3.5, with the exception that `rx_scrambled` is replaced with `rx_scrambled_am`.

161.5.3.3.7 Block distribution

Block distribution is specified identically as in 91.5.3.6.

161.5.3.3.8 Alignment marker mapping and insertion

The alignment marker mapping function compensates for operation of the lane reorder function (refer to 161.5.3.2) to derive the PCS lane alignment markers, `am_rx_x<65:0>` for $x=0$ to 19, from `am_rxmapped<1284:0>` (refer to 161.5.3.5).

The alignment markers shall be derived from `am_rxmapped<1284:0>` in a manner that yields the same result as the following process.

Given $i=0$ to 3, $k=0$ to 31, and $y=i+4k$, `am_rxpayloads` may be derived from `am_rxmapped` per the following expression:

$$\text{am_rxpayloads}\langle i, (10k+9):10k \rangle = \text{am_rxmapped}\langle (10y+9):10y \rangle$$

The 5-bit pad `am_rxmapped<1284:1280>` is ignored. Given $i=0$ to 3, $j=0$ to 4, and $x=i+4j$, `amp_rx_x` may be derived from `am_rxpayloads` by the following expression:

$$\text{amp_rx_x}\langle 63:0 \rangle = \text{am_rxpayloads}\langle i, (64j+63):64j \rangle$$

For $x=0$ to 19, `am_rx_x<65:0>` is constructed as follows:

- a) `am_rx_x<0>=1` and `am_rx_x<1>=0`.
- b) `am_rx_x<25:2>` is set to M_0 , M_1 , and M_2 as shown in Figure 82–9 using the values in Table 82–2 for PCS lane number x .
- c) `am_rx_x<33:26> = amp_rx_x<31:24>`.

- d) `am_rx_x<57:34>` is set to M_4 , M_5 , and M_6 as shown in Figure 82–9 using the values in Table 82–2 for PCS lane number x .
- e) `am_rx_x<65:58> = amp_rx_x<63:56>`.

One vector is mapped to 20 alignment markers every 4096 Reed-Solomon codewords (see 161.5.3.5). The alignment markers are simultaneously transmitted on the 20 PCS lanes after every 16 383rd column of 20 66-bit blocks.

The alignment markers `am_rx_0` to `am_rx_3` shall be inserted so that they are immediately followed by `rx_coded_0` to `rx_coded_3`, respectively, as derived from the first 257-bit block following `am_rxmapped`. Similarly `am_rx_4` to `am_rx_7` are followed by the 66-bit blocks corresponding to the second 257-bit block following `am_rxmapped`, and so on.

161.5.4 Detailed functions and state diagrams

161.5.4.1 State diagram conventions

The body of this subclause is composed of state diagrams, including the associated definitions of variables, functions, and counters. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

The notation used in the state diagrams follows the conventions of 21.5. The notation ++ after a counter or integer variable indicates that its value is to be incremented.

161.5.4.2 State variables

161.5.4.2.1 Variables

The variables are specified identically as in 91.5.4.2.1, with the following exceptions:

- `amp_valid`
Identical to the definition in 91.5.4.2.1 except that 161.5.2.6 defines the alignment marker payload and the optional EEE capability is not supported.
- `cw_bad`
This variable is not defined.
- `cw_A_bad`
A Boolean variable that is set to true if the Reed-Solomon decoder (see 161.5.3.3) fails to correct the current FEC codeword A and is set to false otherwise.
- `cw_B_bad`
A Boolean variable that is set to true if the Reed-Solomon decoder (see 161.5.3.3) fails to correct the current FEC codeword B and is set to false otherwise.
- `fec_alignment_valid`
Identical to the definition in 91.5.4.2.1 except that 161.5.2.6 defines the alignment marker payload.
- `fec_lane`
Identical to the definition of `fec_lane` in 91.5.4.2.1 except that 161.5.2.6 defines the alignment marker payload.
- `fec_lane_mapping<x>`
Identical to the definition of `fec_lane_mapping` in 91.5.4.2.1 except that 161.6.15 defines the FEC lane mapping.
- `fec_optional_states`
Identical to the definition in 91.5.4.2.1, but always set to true to implement the optional states in Figure 91–8.

reset

Identical to the definition in 91.5.4.2.1, with the exception that it refers to the RS-FEC-Int sublayer.

rx_align_status

Boolean variable that is set by the alignment lock and deskew function (see 161.5.2.2).

161.5.4.2.2 Functions

AMP_COMPARE

This function compares the values of first_pcs1 and current_pcs1 to determine if a valid alignment marker payload sequence has been detected and returns the result of the comparison using the variable amp_match. If current_pcs1 and first_pcs1 both found a match and indicate the same PCS lane number, amp_match is set to true. Otherwise, amp_match is set to false.

SLIP

Identical to the definition in 91.5.4.2.2.

161.5.4.2.3 Counters

amp_bad_count

Identical to the definition in 91.5.4.2.3.

amp_counter

Counts the interval of 4096 FEC codewords containing normal alignment marker payload sequences.

cw_A_bad_count

Identical to the definition in 119.2.6.2.4.

cw_B_bad_count

Identical to the definition in 119.2.6.2.4.

161.5.4.3 State diagrams

The FEC shall implement four synchronization processes as shown in Figure 91–8. The synchronization process operates independently on each lane. The synchronization state diagram determines when the FEC has obtained alignment marker lock to the received bit stream for a given lane of the service interface.

Each synchronization process looks for two valid alignment markers $557\,056 \times 10$ -bit Reed-Solomon symbols apart, on a per FEC lane basis, to gain alignment marker lock. When the synchronization process achieves lock for a lane, and if a Clause 45 MDIO is implemented, the FEC shall record the FEC lane number received on that lane of the service interface in the appropriate lane mapping register (1.206).

The FEC shall implement the alignment process as shown in Figure 161–6.

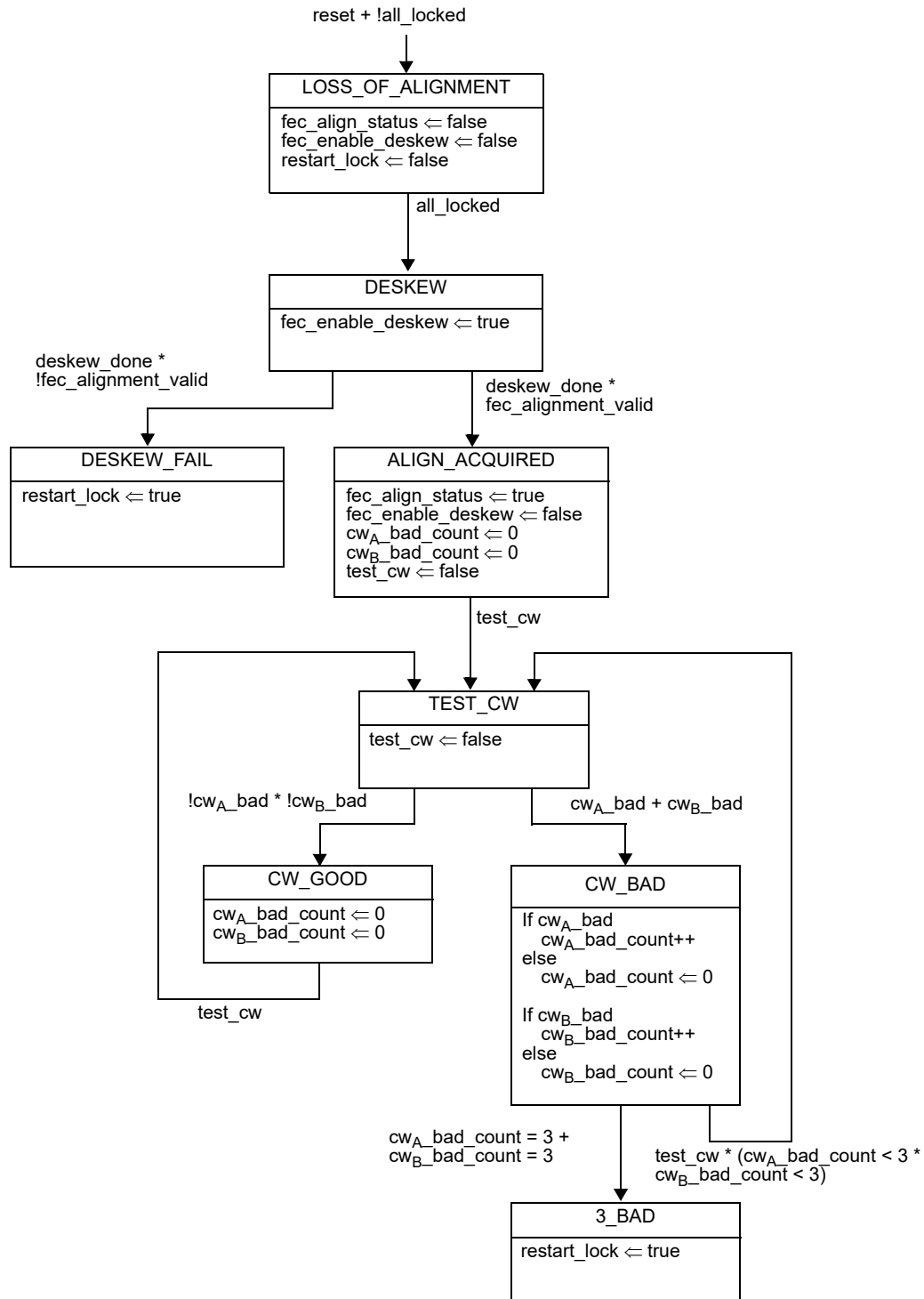


Figure 161–6—FEC alignment state diagram

161.6 RS-FEC-Int MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the RS-FEC-Int. If MDIO is implemented, it shall map MDIO control bits to RS-FEC-Int control variables as shown in Table 161–1, and MDIO status bits to RS-FEC-Int status variables as shown in Table 161–2, and if a separated PMA (see 45.2.1) is connected to the FEC service interface, it shall map additional MDIO status bits to additional RS-FEC-Int status variables as shown in Table 161–3.

Table 161–1—MDIO/RS-FEC-Int control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
FEC bypass indication enable	RS-FEC control register	1.200.1	FEC_bypass_indication_enable
FEC degraded SER enable	RS-FEC control register	1.200.4	FEC_degraded_SER_enable
100G RS-FEC-Int enable	RS-FEC control register	1.200.5	100G_RS_FEC_Int_enable
FEC degraded SER activate threshold	RS-FEC degraded SER activate threshold register	1.650, 1.651	FEC_degraded_SER_activate_threshold
FEC degraded SER deactivate threshold	RS-FEC degraded SER deactivate threshold register	1.652, 1.653	FEC_degraded_SER_deactivate_threshold
FEC degraded SER interval	RS-FEC degraded SER interval register	1.654, 1.655	FEC_degraded_SER_interval

Table 161–2—MDIO/RS-FEC-Int status variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
FEC bypass indication ability	RS-FEC status register	1.201.1	FEC_bypass_indication_ability
RS-FEC high SER	RS-FEC status register	1.201.2	hi_ser
FEC degraded SER ability	RS-FEC status register	1.201.3	FEC_degraded_SER_ability
FEC degraded SER	RS-FEC status register	1.201.4	FEC_degraded_SER
FEC AM lock x , $x=0$ to 3	RS-FEC status register	1.201.8:11	amps_lock< x >
RS-FEC align status	RS-FEC status register	1.201.14	fec_align_status
FEC corrected codewords	RS-FEC corrected codewords counter register	1.202, 1.203	FEC_corrected_cw_counter
FEC uncorrected codewords	RS-FEC uncorrected codewords counter register	1.204, 1.205	FEC_uncorrected_cw_counter
FEC lane x mapping	RS-FEC lane mapping register	1.206	FEC_lane_mapping< x >

Table 161–2—MDIO/RS-FEC-Int status variable mapping (*continued*)

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
RS-FEC codeword counter	RS-FEC codeword counter register	1.207 to 1.209	FEC_cw_counter
FEC symbol errors, FEC lanes 0 to 3	RS-FEC symbol error counter register, FEC lanes 0 to 3	1.210 to 1.217	FEC_symbol_error_counter_i
FEC codeword error bin counter, 1 to 15	RS-FEC codeword error bin counter register	1.340 to 1.369	FEC_codeword_error_bin_i

Table 161–3—MDIO/RS-FEC-Int status variable mapping for separated PMA

MDIO control variable	PMA/PMD register name	Register/bit number	FEC variable
PCS align status	RS-FEC status register	1.201.15	align_status
BIP errors, PCS lanes 0 to 19	RS-FEC BIP error counter register, PCS lanes 0 to 19	1.230 to 1.249	BIP_error_counter_i
PCS lane <i>x</i> mapping	PCS lane <i>x</i> mapping register	1.250 to 1.269	lane_mapping< <i>x</i> >
Block lock <i>x</i>	RS-FEC PCS alignment status 1 and 2 registers	1.280 to 1.281	block_lock< <i>x</i> >
Lane <i>x</i> aligned	RS-FEC PCS alignment status 3 and 4 registers	1.282 to 1.283	am_lock< <i>x</i> >

The following subclauses define variables that are not otherwise defined, e.g., for use by state diagrams.

161.6.1 FEC_bypass_indication_enable

Identical to the definition in 91.6.2 except the reference becomes 161.5.3.3.1.

161.6.2 FEC_degraded_SER_enable

Identical to the definition in 91.6.4 except the reference becomes 161.5.3.3.2.

161.6.3 100G_RS_FEC_Int_enable

The RS-FEC-Int sublayer shall have the capability to enable or disable the FEC function. An MDIO interface or an equivalent management interface shall be provided to access the variable 100G_RS_FEC_Int_enable for the RS-FEC-Int sublayer. When the 100G_RS_FEC_Int_enable variable is set to one, the RS-FEC-Int sublayer performs the transmit function as specified in 161.5.2 and the receive function as specified in 161.5.3. When the variable is set to zero, the transmit and receive functions are disabled, and the RS-FEC-Int sublayer is not used allowing the 100G RS-FEC sublayer (see Clause 91) to be used instead. This variable is mapped to the bit defined in 45.2.1.116.

161.6.4 FEC_degraded_SER_activate_threshold

Identical to the definition in 91.6.5 except the reference becomes 161.5.3.3.2.

161.6.5 FEC_degraded_SER_deactivate_threshold

Identical to the definition in 91.6.6 except the reference becomes 161.5.3.3.2.

161.6.6 FEC_degraded_SER_interval

Identical to the definition in 91.6.7 except the reference becomes 161.5.3.3.2.

161.6.7 FEC_bypass_indication_ability

Identical to the definition in 91.6.9 except the reference becomes 161.5.3.3.1.

161.6.8 hi_ser

Identical to the definition in 91.6.10 except the reference becomes 161.5.3.3.1.

161.6.9 FEC_degraded_SER_ability

Identical to the definition in 91.6.11 except the reference becomes 161.5.3.3.2.

161.6.10 FEC_degraded_SER

Identical to the definition in 91.6.12 except the reference becomes 161.5.3.3.2.

161.6.11 amps_lock<x>

Identical to the definition in 91.6.14 except the reference becomes 161.5.4.3.

161.6.12 fec_align_status

Identical to the definition in 91.6.15 except the reference becomes 161.5.4.3.

161.6.13 FEC_corrected_cw_counter

Identical to the definition in 91.6.16.

161.6.14 FEC_uncorrected_cw_counter

Identical to the definition in 91.6.17.

161.6.15 FEC_lane_mapping<x>

Identical to the definition in 91.6.18, with the exception that it refers to the RS-FEC-Int sublayer.

161.6.16 FEC_symbol_error_counter_{*i*}

Identical to the definition in 91.6.19.

161.6.17 FEC_codeword_error_bin_{*i*}

FEC_codeword_error_bin_{*i*}, where *i*=1 to 15, are optional 32-bit counters. While fec_align_status is true, for each codeword received with exactly *i* correctable 10-bit symbol errors, FEC_codeword_error_bin_{*i*} is incremented. For example, if a codeword has exactly 5 errored 10-bit symbols, then

fec_codeword_error_bin_5 is incremented. These variables are mapped to the registers defined in 45.2.1.131a (1.340 to 1.369).

161.6.18 align_status

Identical to the definition in 91.6.20.

161.6.19 BIP_error_counter_i

Identical to the definition in 91.6.21, with the exception that it refers to the RS-FEC-Int transmit function.

161.6.20 lane_mapping<x>

Identical to the definition in 91.6.22, with the exception that it refers to the RS-FEC-Int transmit function.

161.6.21 FEC_cw_counter

FEC_cw_counter is an optional 48-bit counter that counts once for each FEC codeword received when fec_align_status is true. This variable is mapped to the registers defined in 45.2.1.120a (1.207 to 1.209).

161.6.22 block_lock<x>

Identical to the definition in 91.6.23.

161.6.23 am_lock<x>

Identical to the definition in 91.6.24.

161.7 Protocol implementation conformance statement (PICS) proforma for Clause 161, Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs¹⁷

161.7.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 161, Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

161.7.2 Identification

161.7.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1— Required for all implementations. NOTE 2— May be completed as appropriate in meeting the requirements for the identification. NOTE 3— The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

161.7.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 161, Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

¹⁷*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

161.7.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
DC	Delay constraints	161.4	Conforms to delay constraints specified in 161.4	M	Yes []
*MD	MDIO capability	161.6	Registers and interface supported	O	Yes [] No []
*BEI	Bypass error indication	161.5.3.3	Capability is supported	O	Yes [] No []
*FDD	Support for optional FEC degraded SER detection	161.5.3.3.2		O	Yes [] No []
EF	100G RS-FEC-Int enable	161.6.18	Has the capability to enable and disable the RS-FEC-Int function	M	Yes []

161.7.4 PICS proforma tables for Reed-Solomon forward error correction codeword-interleaved (RS-FEC-Int) sublayer for 100GBASE-P PHYs

161.7.4.1 Transmit function

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Skew tolerance	161.5.2.2	Maximum Skew of 49 ns between PCS lanes and a maximum Skew Variation of 400 ps	M	Yes []
TF2	Lane reorder	161.5.2.3	Order the PCS lanes according to the PCS lane number	M	Yes []
TF3	64B/66B to 256B/257B transcoder	161.5.2.5	tx_xcoded<256:0> constructed per 161.5.2.5	M	Yes []
TF4	257-bit block transmission order	161.5.2.5	First bit transmitted is bit 0	M	Yes []
TF5	Alignment maker mapping	161.5.2.6	Map to am_txmapped<1284:0> per 161.5.2.6	M	Yes []
TF6	Pad value	161.5.2.6	Binary values 00101 and 11010 (the leftmost bit is assigned to the highest bit index) in an alternating pattern per 161.5.2.6	M	Yes []
TF7	Alignment marker insertion	161.5.2.6	First 1285 message bits to be transmitted from every 4096th codeword	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TF8	Alignment marker insertion point	161.5.2.6	First 257-bit block inserted after <code>am_txmapped</code> corresponds to the four 66-bit blocks received on PCS lanes 0, 1, 2, and 3 that immediately followed the alignment marker on each respective lane	M	Yes []
TF9	Pre-FEC Distribution	161.5.2.7	10-bit round robin distribution as described in 161.5.2.7	M	Yes []
TF10	Reed-Solomon encoder	161.5.2.8	RS(544,514)	M	Yes []
TF11	Symbol distribution	161.5.2.9	Distributed to four FEC lanes, one 10-bit symbol at a time in a round robin distribution from the lowest to the highest numbered FEC lane	M	Yes []

161.7.4.2 Receive function

Item	Feature	Subclause	Value/Comment	Status	Support
RF1	Skew tolerance	161.5.3.1	Maximum Skew of 180 ns between FEC lanes and a maximum Skew Variation of 4 ns	M	Yes []
RF2	Lane reorder and de-interleave	161.5.3.2	Order the FEC lanes according to the FEC lane number and de-interleave the FEC codewords	M	Yes []
RF3	Reed-Solomon decoder	161.5.3.3	Corrects any combination of up to $t=15$ symbol errors in a codeword	M	Yes []
RF4	Error indication function	161.5.3.3	Capable of indicating when a codeword was not corrected.	M	Yes []
RF5	Error monitoring while error indication is bypassed	161.5.3.3	When the number of symbol errors in a block of 8192 codewords exceeds 5560 assert <code>hi_ser</code> for 60 ms to 75 ms	BEI:M	Yes [] N/A []
RF7	Alignment marker removal	161.5.3.5	<code>am_rxmapped</code> removed prior to transcoding	M	Yes []
RF8	256B/257B to 64B/66B transcoder	161.5.3.6	<code>rx_coded_j<65:0></code> , $j=0$ to 3 constructed per 161.5.3.6	M	Yes []
RF9	Block distribution	161.5.3.7	One 66-bit block at a time in a round robin fashion from the lowest to the highest numbered PCS lane	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
RF10	Alignment marker mapping	161.5.3.8	Map to am_rx_x, x=0 to 19 per 161.5.3.8	M	Yes []
RF11	Alignment marker insertion point	161.5.3.8	Alignment markers immediately followed by the 66-bit blocks derived from the 257-blocks immediately following am_rxmapped	M	Yes []
RF12	FEC decoder detects FEC degraded SER at a programmable threshold	161.5.3.3.2		FDD:M	Yes [] N/A []

161.7.4.3 State diagrams

Item	Feature	Subclause	Value/Comment	Status	Support
SD1	SLIP function	161.5.4.2.2	Ensure that all possible block positions are evaluated	M	Yes []
SD2	Synchronization process	161.5.4.3	One instance per FEC lane per Figure 161–6	M	Yes []
SD3	Alignment process	161.5.4.3	Per Figure 161–6	M	Yes []

162. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162.1 Overview

This clause specifies the 100GBASE-CR1 PMD, the 200GBASE-CR2 PMD, the 400GBASE-CR4 PMD, and the baseband medium. The specifications for the three PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are four associated annexes. Annex 162A provides information on parameters with test points that may not be testable in an implemented system. Annex 162B specifies test fixtures. Annex 162C specifies MDIs. Annex 162D describes host and cable assembly types.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 162–1, to the appropriate sublayers (as specified in Table 162–1, Table 162–2, and Table 162–3), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 162–1—Physical Layer clauses associated with the 100GBASE-CR1 PMD

Associated clause	100GBASE-CR1
81—RS	Required
81—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC	Required
152—Inverse RS-FEC	Optional ^b
161—RS-FEC-Int	Required
83—PMA for 100GBASE-R	Optional ^c
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135F—100GAUI-2 C2C	Optional
120F—100GAUI-1 C2C	Optional
73—Auto-Negotiation	Required

^a The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^b Inverse RS-FEC is required to convert between RS-FEC and RS-FEC-Int (see 152.1.2).

^c PMA for 100GBASE-R is required when either CAUI-10 or CAUI-4 is used.

Table 162–2—Physical Layer clauses associated with the 200GBASE-CR2 PMD

Associated clause	200GBASE-CR2
117—RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—PCS for 200GBASE-R	Required
120—PMA for 200GBASE-R	Required
120B—200GAUI-8 C2C	Optional
120D—200GAUI-4 C2C	Optional
120F—200GAUI-2 C2C	Optional
73—Auto-Negotiation	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

Table 162–3—Physical Layer clauses associated with the 400GBASE-CR4 PMD

Associated clause	400GBASE-CR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—PCS for 400GBASE-R	Required
120—PMA for 400GBASE-R	Required
120B—400GAUI-16 C2C	Optional
120D—400GAUI-8 C2C	Optional
120F—400GAUI-4 C2C	Optional
73—Auto-Negotiation	Required

^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

For the 100GBASE-CR1 PHY, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-10} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 10^{-10} after processing by the FEC (Clause 91 or Clause 161) and the PCS (Clause 82) if there are negligible errors due to other electrical interfaces (100GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 10^{-10} .

For the 200GBASE-CR2 and 400GBASE-CR4 PHYs, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the

adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 9.2×10^{-13} after processing by the PCS (Clause 119) if there are negligible errors due to other electrical interfaces (200GAUI-n or 400GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 9.2×10^{-13} .

A compliant input signal is a transmitter output of a compliant PHY that has passed through a compliant cable assembly.

Figure 162–1 shows the relationship of the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMD sublayers and MDIs to the ISO/IEC Open System Interconnection (OSI) reference model.

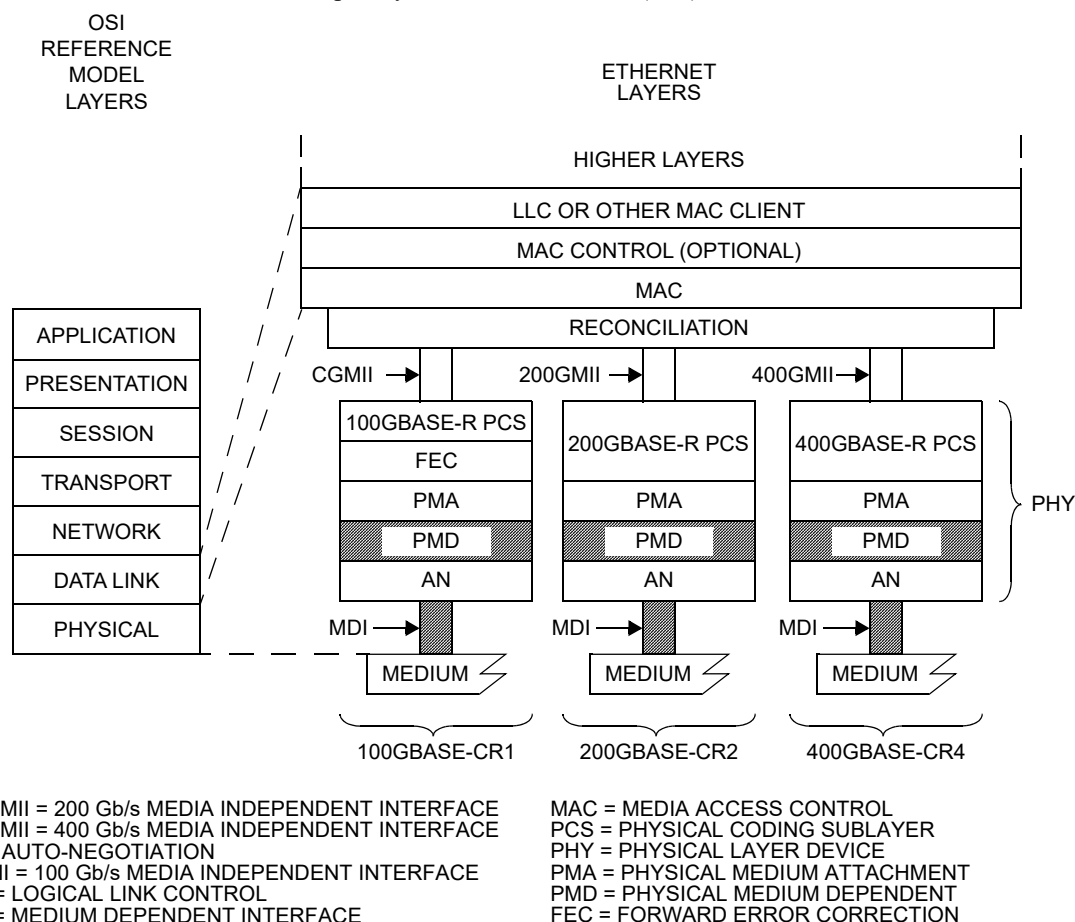


Figure 162–1—100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

162.2 Conventions

Clause 162 describes three PMDs, 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, which have one, two, and four lanes, respectively. For efficient description, the parameter n is used to describe the

number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-CR1, $n = 2$ for 200GBASE-CR2, and $n = 4$ for 400GBASE-CR4.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-CR1 PMD, 200GBASE-CR2 PMD, or 400GBASE-CR4 PMD.

Within this clause, the term FEC refers to either Clause 91 RS-FEC or the Clause 161 RS-FEC-Int for 100GBASE-CR1 or the RS-FEC within the Clause 119 PCS for 200GBASE-CR2 and 400GBASE-CR4.

162.3 PMD service interfaces

This subclause specifies the services provided by the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

The 100GBASE-CR1 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with the exception that there is a single symbol stream ($n = 1$).

The 200GBASE-CR2 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with two parallel symbol streams ($n = 2$).

The 400GBASE-CR4 PMD service interface is an instance of the inter-sublayer service interface defined in 116.3, with four parallel symbol streams ($n = 4$).

The service interface primitives are summarized as follows:

PMD:IS_UNITDATA_ i .request
PMD:IS_UNITDATA_ i .indication
PMD:IS_SIGNAL.indication

The 100GBASE-CR1 PMD has a single symbol stream, hence $i = 0$.

The 200GBASE-CR2 PMD has two parallel symbol streams, hence $i = 0$ to 1.

The 400GBASE-CR4 PMD has four parallel symbol streams, hence $i = 0$ to 3.

In the transmit direction, the PMA continuously sends n streams of PAM4 symbols to the PMD, one per lane, using the PMD:IS_UNITDATA_ i .request primitive, at a nominal signaling rate of 53.125 GBd. The PMD converts these streams of symbols into appropriate signals on the MDI.

In the receive direction, the PMD continuously sends n streams of PAM4 symbols to the PMA, corresponding to the signals received from the MDI, one per lane, using the PMD:IS_UNITDATA_ i .indication primitive, at a nominal signaling rate of 53.125 GBd.

The SIGNAL_OK parameter of the PMD:IS_SIGNAL.indication primitive corresponds to the variable Global_PMD_signal_detect as defined in 162.8.4. When Global_PMD_signal_detect is one, SIGNAL_OK shall be assigned the value OK. When Global_PMD_signal_detect is zero, SIGNAL_OK shall be assigned the value FAIL. When SIGNAL_OK is FAIL, the PMD:IS_UNITDATA_ i .indication parameter is undefined.

162.4 PCS requirements for Auto-Negotiation (AN) service interface

The PCSs associated with the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs are required to support the AN service interface primitive AN_LINK.indication defined in 73.9. (See 82.6 and 119.6.)

100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PHYs may be extended using CAUI-n/100GAUI-n C2C, 200GAUI-n C2C, and 400GAUI-n C2C, respectively, as physical instantiations of the inter-sublayer service interface between devices. If such extension is used, then the AN_LINK(link_status).indication is relayed from the device with the PCS sublayer to the device with the AN sublayer by means at the discretion of the implementer. As examples, the implementer may employ use of pervasive management or employ a dedicated electrical signal to relay the state of link_status as indicated by the PCS sublayer on one device to the AN sublayer on the other device.

162.5 Delay constraints

The sum of the transmit and the receive delays at one end of the link contributed by the PMD and AN sublayers including the medium in one direction shall be no more than the maximum delays listed in Table 162–4. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 162–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-CR1	4 096	8	40.96
200GBASE-CR2	8 192	16	40.96
400GBASE-CR4	16 384	32	40.96

^a One bit time is equal to 10 ps for 100GBASE-CR1, 5 ps for 200GBASE-CR2, and 2.5 ps for 400GBASE-CR4. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-CR1, 2.56 ns for 200GBASE-CR2, and 1.28 ns for 400GBASE-CR4. (See 31B.2 for the definition of pause_quanta.)

Descriptions of overall system delay constraints can be found in 80.4 for 100GBASE-CR1 and in 116.4 for 200GBASE-CR2 and 400GBASE-CR4.

162.6 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

162.6.1 Skew Constraints for 100GBASE-CR1

Skew and Skew Variation are defined in 80.5 and specified at the points SP0 to SP7 shown in Figure 80–9.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by 135.5.3.5. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 80.5. The measurements of Skew and Skew Variation are defined in 89.7.2 with the exception that the measurement clock and data recovery unit high-frequency corner bandwidth is 4 MHz.

162.6.2 Skew Constraints for 200GBASE-CR2 and 400GBASE-CR4

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

162.7 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several registers that provide control and status information for and about the PMD. If MDIO is implemented, it shall map MDIO variables and registers to PMD variables as shown in Table 162–5 through Table 162–7.

Table 162–5—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable	1.9.0	Global_PMD_transmit_disable
PMD transmit disable 3 to PMD transmit disable 2 ^a	PMD transmit disable	1.9.4:3	PMD_transmit_disable_3, PMD_transmit_disable_2
PMD transmit disable 1 ^b	PMD transmit disable	1.9.2	PMD_transmit_disable_1
PMD transmit disable 0	PMD transmit disable	1.9.1	PMD_transmit_disable_0

^a Available only in 400GBASE-CR4.

^b Available only in 200GBASE-CR2 and 400GBASE-CR4.

Table 162–6—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect	1.10.0	Global_PMD_signal_detect
PMD receive signal detect 3 to PMD receive signal detect 2 ^a	PMD receive signal detect	1.10.4:3	PMD_signal_detect_3, PMD_signal_detect_2
PMD receive signal detect 1 ^b	PMD receive signal detect	1.10.2	PMD_signal_detect_1
PMD receive signal detect 0	PMD receive signal detect	1.10.1	PMD_signal_detect_0

^a Available only in 400GBASE-CR4.

^b Available only in 200GBASE-CR2 and 400GBASE-CR4.

Table 162–7—MDIO/PMD control function mapping

MDIO variable	PMA/PMD register name	Register/bit number	PMD variable
Restart training	BASE-R PMD control	1.150.0	mr_restart_training
Training enable	BASE-R PMD control	1.150.1	mr_training_enable
Receiver status	BASE-R PMD status	1.151.0 ^a	local_trained
Frame lock	BASE-R PMD status	1.151.1 ^a	local_tf_lock
Start-up protocol status	BASE-R PMD status	1.151.2 ^a	training
Training failure	BASE-R PMD status	1.151.3 ^a	training_failure
Polynomial identifier	PMD training pattern, lanes 0 to 3	1.1450.12:11 ^b	identifier_i

Table 162–7—MDIO/PMD control function mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/bit number	PMD variable
Seed	PMD training pattern, lanes 0 to 3	1.1450.15:14 1.1450.10:0 ^b	seed_i
Initial condition request	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.13:12 ^b	ic_req
Coefficient select	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.4:2 ^b	coef_sel
Coefficient request	BASE-R PAM4 PMD training LP control, lanes 0 to 3	1.1120.1:0 ^b	coef_req
Receiver ready	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.15 ^b	remote_rx_ready
Modulation and precoding status	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.11:10 ^b	remote_tp_mode
Receiver frame lock	BASE-R PAM4 PMD training LP status, lanes 0 to 3	1.1220.9 ^b	remote_tf_lock
Modulation and precoding request	BASE-R PAM4 PMD training LD control, lanes 0 to 3	1.1320.11:10 ^b	local_tp_mode
Local receiver ready	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.15 ^b	local_rx_ready
Initial condition status	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.8 ^b	ic_sts
Coefficient select echo	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.5:3 ^b	k
Coefficient status	BASE-R PAM4 PMD training LD status, lanes 0 to 3	1.1420.2:0 ^b	coef_sts

^a Bit reference is provided for lane 0. Status for lanes 1 to 3 are located in the same register.

^b Address reference is provided for lane 0. Register for lanes 1 to 3 are located at an offset from the lane 0 register.

162.8 PMD functional specifications

162.8.1 Link block diagram

One direction of a 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 link is shown in Figure 162–2. The PMDs on both ends of the link have connected ground references. Each differential pair in the cable is separately shielded as illustrated in Figure 162–2. The signal shields are connected to ground contacts in the MDI plug connectors on both ends of the cable assembly.

Note that the source lane (SL) signals $SL_i<p>$ and $SL_i<n>$ are the positive and negative sides of the transmitter's differential signal pair on lane i and the destination lane (DL) signals $DL_i<p>$ and $DL_i<n>$ are the positive and negative sides of the receiver's differential signal pair on lane i .

For purposes of system conformance, the PMD sublayer is standardized at the test points described in this subclause.



NOTE—One direction is illustrated.

PMD:IS UNITDATA *i*.indication

Figure 162-2—100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 link

The electrical transmit signal is defined at TP2. Unless specified otherwise, all transmitter measurements and tests defined in 162.9.4 are made at TP2 utilizing the test fixture specified in Annex 162B.

The electrical receive signal is defined at TP3. Unless specified otherwise, all receiver measurements and tests defined in 162.9.5 are performed at TP3 utilizing the test fixture specified in Annex 162B.

A mated connector pair has been included in both the transmitter and receiver specifications defined in 162.9.4 and 162.9.5. The recommended maximum insertion loss from TP0 to TP2 or TP3 to TP5 including the test fixture is provided in 162.9.4.5. Annex 162A provides information on parameters associated with test points TP0 and TP5, which might not be testable in an implemented system.

The channel (see 162.11) is defined between the transmitter (TP0) and receiver (TP5) blocks to include the transmitter and receiver differential controlled impedance printed circuit boards (PCBs) and the cable assembly, as illustrated in Figure 162–2. All cable assembly measurements are to be made between TP1 and TP4 as illustrated in Figure 162–2. The cable assembly test fixture of Annex 162B.1, or its equivalent, is required for measuring the cable assembly specifications in 162.11 at TP1 and TP4. Two cable assembly test fixtures have been included in the cable assembly specifications defined in 162.11. Transmitter and receiver differential controlled impedance PCB insertion losses defined between TP0 and the MDI and between the MDI and TP5, respectively, are provided informatively in 162A.4.

Table 162–8 describes the defined test points illustrated in Figure 162–2.

Table 162–8—Test points

Test points	Description
TP0 to TP5	The channel including the transmitter and receiver differential controlled impedance PCB insertion loss and the cable assembly insertion loss.
TP1 to TP4	All cable assembly measurements are made between TP1 and TP4 as illustrated in Figure 162–2. The cable assembly test fixture of Annex 162B, or its equivalent, is required for measuring the cable assembly specifications in 162.10 at TP1 and TP4.
TP0 to TP2 TP3 to TP5	A mated connector pair has been included in both the transmitter and receiver specifications defined in 162.9.4 and 162.9.5. The recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture is provided in 162.9.4.5.
TP2	Unless specified otherwise, all transmitter measurements defined in 162.9.4 are made at TP2 utilizing the test fixture specified in Annex 162B.
TP3	Unless specified otherwise, all receiver measurements and tests defined in 162.9.5 are made at TP3 utilizing the test fixture specified in Annex 162B.

162.8.2 PMD transmit function

The PMD transmit function has three operating modes: DATA, TRAINING, and QUIET. The operating mode is controlled by the PMD control state diagram (Figure 136–7).

When operating in DATA mode, the PMD transmit function shall convert the symbol stream requested by the PMD service interface message `PMD:IS_UNITDATA_i.request(tx_symbol)` of each lane into an electrical signal, and deliver the electrical signals to the MDI, according to the transmit electrical specifications in 162.9.4. The differential output voltage ($SL_{i<p>} - SL_{i<n>}$) meets the specifications in 162.9.4.1.1 where the PAM4 symbol values 0, 1, 2, and 3 correspond to the `tx_symbol` values zero, one, two, and three, respectively, with the highest differential output voltage corresponding to `tx_symbol` = three and the lowest differential output voltage corresponding to `tx_symbol` = zero.

When operating in TRAINING mode, the PMD transmit function shall convert the symbol stream generated by the PMD control function of each lane into an electrical signal, and deliver the electrical signals to the MDI, according to the transmit electrical specifications in 162.9.4. The differential output voltage ($SL_{i<p>} - SL_{i<n>}$) meets the specifications in 162.9.4.1.1, with the highest differential output voltage corresponding to the PAM4 symbol 3 and the lowest differential output voltage corresponding to the PAM4 symbol 0.

When operating in QUIET mode the PMD transmit function shall disable the transmitter such that the transmitter drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.

162.8.3 PMD receive function

The PMD receive function shall convert the electrical signal from each of the lanes on the MDI into a symbol stream for delivery to the PMD service interface using the messages `PMD:IS_UNITDATA_i.indication(rx_symbol)`. The highest differential input voltage ($DL_{i<p>} - DL_{i<n>}$) shall correspond to `rx_symbol` = three and the lowest differential input voltage shall correspond to `rx_symbol` = zero.

162.8.4 PMD global signal detect function

For the 100GBASE-CR1 PMD, the variable `Global_PMD_signal_detect` is set to `PMD_signal_detect_0` (see 162.8.5). For the 200GBASE-CR2 and 400GBASE-CR4 PMDs, the variable `Global_PMD_signal_detect` is set to the logical AND of the values of `PMD_signal_detect_i` from each of the PMD lanes.

162.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function is used by the PMD to indicate the successful completion of the start-up protocol by the PMD control function (see 162.8.11). `PMD_signal_detect_i` shall be set to zero when `signal_detect` on lane *i* is set to false, and to one when `signal_detect` on lane *i* is set to true. The `signal_detect` variables are set independently on each lane by the PMD control state diagram (Figure 136–7).

If training is disabled by the management variable `mr_training_enable` (see 162.7), `PMD_signal_detect_i` shall be set to one for all lanes.

When the MDIO is implemented, this function maps the variables to registers and bits as defined in 162.7.

162.8.6 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional. When implemented, it allows the transmitters on all lanes to be disabled with a single variable.

When `Global_PMD_transmit_disable` variable is set to one, this function shall turn off all the transmitters such that each transmitter drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.

If a PMD fault (162.8.8) is detected, then the PMD may set `Global_PMD_transmit_disable` to one.

162.8.7 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional and allows the transmitter in each lane to be selectively disabled. When this function is supported, it shall meet the following requirements:

- When a `PMD_transmit_disable_i` variable is set to one, this function shall turn off the transmitter in lane *i* so that it drives a constant level (i.e., no transitions) and does not exceed the differential peak-to-peak output voltage (max) with Tx disabled in Table 162–11.
- If a PMD fault (162.8.8) is detected, then the PMD may set each `PMD_transmit_disable_i` to one, turning off the transmitter in each lane.

162.8.8 PMD fault function

`PMD_fault` is the logical-OR of `PMD_receive_fault`, `PMD_transmit_fault`, and any other implementation specific fault. If the MDIO interface is implemented, then `PMD_fault` shall be mapped to the fault bit as specified in 45.2.1.2.3.

162.8.9 PMD transmit fault function (optional)

The PMD transmit fault function is optional. The faults detected by this function are implementation specific, but the assertion of `Global_PMD_transmit_disable` is not considered a transmit fault.

If `PMD_transmit_fault` is set to one, then `Global_PMD_transmit_disable` should also be set to one.

If the MDIO interface is implemented, then PMD_transmit_fault shall be mapped to the Transmit fault bit as specified in 45.2.1.7.4.

162.8.10 PMD receive fault function (optional)

The PMD receive fault function is optional. The faults detected by this function are implementation specific. A fault is indicated by setting the variable PMD_receive_fault to one.

If the MDIO interface is implemented, then PMD_receive_fault shall be mapped to the Receive fault bit as specified in 45.2.1.7.5.

162.8.11 PMD control function

The PMD control function performs the PMD start-up protocol. This protocol facilitates timing recovery and equalization while providing a mechanism through which the receiver can configure the transmitter to optimize performance. The protocol supports these functions through the continuous exchange of fixed-length training frames.

The PMD shall implement one instance of the PMD control function described in 136.8.11 for each lane with the following exceptions:

- a) The control field structure is specified in Table 162–9 and the status field structure is specified in Table 162–10.
- b) For k_list as specified in 136.8.11.4.4, the set of valid transmitter equalizer coefficient indices is $\{-3, -2, -1, 0, +1\}$.
- c) For the initial condition request as described in 136.8.11.2.1, five predefined transmitter equalizer settings are specified in 162.9.4.1.3.
- d) The “No equalization” value (see 136.8.11.2.4) of $c(-3)$ is 0.
- e) The terminal count of max_wait_timer as specified in 136.8.11.7.3 is 12 s.
- f) A receiver is expected to assert local_tf_lock within 275 ms from entry into the AN_GOOD_CHECK state in Figure 73–11 provided that there is a compliant signal containing valid training frames at the PMD input.
- g) The value of use_quiet_in_training (see 136.8.11.7.1) is TRUE.

Table 162–9—Control field structure

Bit(s)	Name	Description
15:14	Reserved	Transmit as 0, ignore on receipt
13:11	Initial condition request	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control
10	Reserved	Transmit as 0, ignore on receipt
9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2
7:5	Reserved	Transmit as 0, ignore on receipt
4:2	Coefficient select	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved
1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold

Table 162–10—Status field structure

Bit(s)	Name	Description
15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue
14:12	Reserved	Transmit as 0, ignore on receipt
11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2
9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified
8	Initial condition status	1 = Updated 0 = Not updated
7	Parity	Even parity bit
6	Reserved	Transmit as 0, ignore on receipt
5:3	Coefficient select echo	5 4 3 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$
2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated

162.9 PMD electrical characteristics

162.9.1 AC-coupling

Interoperability between PMD components operating from different supply voltages is facilitated by AC-coupling within the cable assembly as specified in 162.11.

162.9.2 Signal paths

The MDI transmit and receive paths are point-to-point connections. Each path corresponds to one MDI lane and comprises two complementary signals, which form a balanced differential pair.

For 100GBASE-CR1, there is one differential lane in each direction for a total of two pairs, or four connections. For 200GBASE-CR2, there are two differential lanes in each direction for a total of four pairs, or eight connections. For 400GBASE-CR4, there are four differential lanes in each direction for a total of eight pairs, or sixteen connections.

162.9.3 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

162.9.4 Transmitter characteristics

The transmitter on each lane shall meet the specifications given in Table 162–11 and detailed in the referenced subclauses. Unless specified otherwise, all transmitter signal measurements are made for each lane separately, at TP2, utilizing the test fixtures specified in Annex 162B and a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth. The connection from TP2 to the test equipment is AC-coupled.

Measurement of the DC common-mode voltage is made with a high-impedance connection to TP2 where each conductor of the differential pair is AC-coupled to a 50 Ω termination.

Recommended transmitter characteristics at TP0 are provided in 162A.2.

Table 162–11—Summary of transmitter specifications at TP2

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)		53.125 $\pm$ 50 ppm ^a	GBd
Differential pk-pk voltage with Tx disabled (max) ^b	93.8.1.3	30	mV
DC common-mode voltage (max) ^b	93.8.1.3	1.9	V
AC common-mode peak-to-peak voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	162.9.4.4	30 80	mV mV
Differential pk-pk voltage, v_{di} (max) ^b	93.8.1.3	1200	mV
Effective return loss, ERL (min)	162.9.4.8	7.3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	162.9.4.9	See Equation (162–6)	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	162.9.4.10	See Equation (162–7)	dB
Transmitter steady-state voltage, v_f (min) Transmitter steady-state voltage, v_f (max)	162.9.4.1.2	0.387 0.6	V
Linear fit pulse peak ratio, R_{peak} (min)	162.9.4.1.2	0.397	—
Level separation mismatch ratio R_{LM} (min)	162.9.4.2	0.95	—
Transmitter output waveform absolute value of step size for all taps (min) absolute value of step size for all taps (max) value at minimum state for $c(-3)$ (max) value at maximum state for $c(-2)$ (min) value at minimum state for $c(-1)$ (max) value at minimum state for $c(0)$ (max) value at minimum state for $c(1)$ (max)	162.9.4.1.4 162.9.4.1.4 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5 162.9.4.1.5	0.005 0.025 –0.06 0.12 –0.34 0.5 –0.2	— — — — — — —

Table 162–11—Summary of transmitter specifications at TP2 (*continued*)

Parameter	Subclause reference	Value	Units
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	31.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	162.9.4.3	26.7	dB
Output jitter (max)			
J_{RMS}	162.9.4.7	0.023	UI
J_{3u03}	162.9.4.7	0.115	UI
J_{3u}	162.9.4.7	0.125	UI
Even-odd jitter, pk-pk	162.9.4.7	0.025	UI

^a For a PMD in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

162.9.4.1 Transmitter output waveform

The transmit function includes programmable equalization to compensate for the frequency-dependent loss of the channel and facilitate data recovery at the receiver. The functional model for the transmit equalizer on each lane is the five-tap transversal filter shown in Figure 162–3.

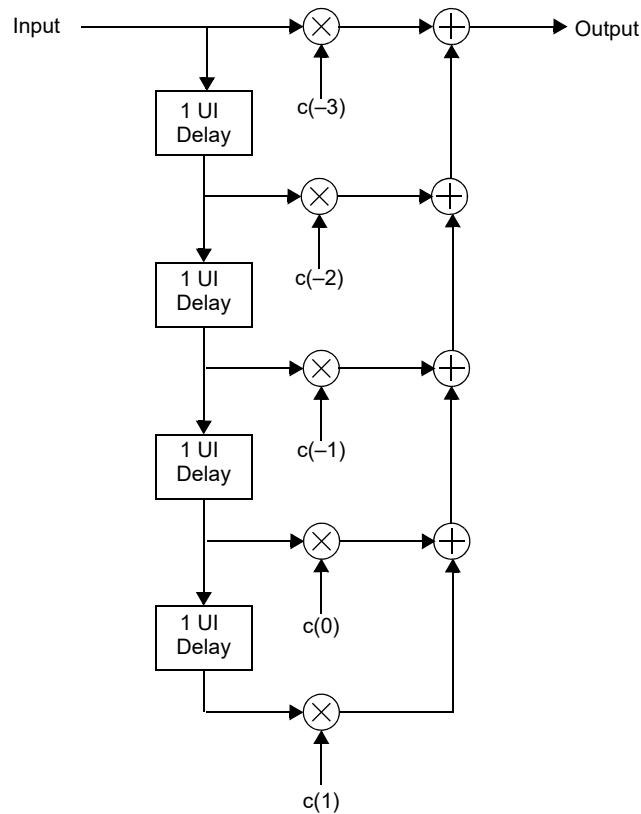


Figure 162–3—Transmit equalizer functional model

The state of the transmit equalizer and hence the transmitted output waveform may be manipulated via the PMD control function defined in 162.8.11 or via the management interface. The transmit function responds

to a set of commands issued by the link partner's receive function and conveyed by a back-channel communications path. This command set includes instructions to increment, decrement, hold, or set to zero a selected coefficient $c(k)$, where $k = -3$ to 1. In addition, it includes commands to set all coefficients to one of five initial conditions.

In response, the transmit function relays status information to the link partner's receive function. The status messages indicate the selected coefficient index k , the status of the coefficient $c(k)$ (updated, not updated, or at limit), and the status of the initial condition (updated or not updated).

162.9.4.1.1 Linear fit to the measured waveform

The following procedure is used to determine the linear fit pulse response, linear fit error, and normalized transmitter coefficient values.

Set the transmitter under test to transmit the PRBS13Q test pattern (defined in 120.5.11.2.1). For each configuration of the transmit equalizer, capture at least one complete cycle of the test pattern at TP2, as specified in 85.8.3.3.4. The reference clock recovery unit (CRU) used in the measurement acts as a high-pass jitter filter with a corner frequency of 4 MHz and a slope of 20 dB/decade.

In the following calculations, M should be an integer greater than or equal to 32. Interpolation of the captured waveform may be used to achieve this.

Compute the linear fit pulse response $p(k)$ and linear fit error $e(k)$, $k=1$ to $M \times N_p$, from the captured waveform, as specified in 85.8.3.3.5, with $N_p = 200$ and $D_p = 4$, where the aligned symbols $x(n)$ are assigned normalized amplitudes -1 , $-ES$, ES , and 1 to represent the PAM4 symbol values 0, 1, 2, and 3 respectively. ES is defined as $(|ES1| + |ES2|)/2$ where $ES1$ and $ES2$ are calculated according to 120D.3.1.2.

Define $r(k)$ to be the linear fit pulse response when transmit equalizer coefficients have been set to the "preset 1" values. The normalized coefficients for any configuration of the transmit equalizer are computed using the following method.

For each value of m in the range $-M/2$ to $M/2 - 1$ when M is even and $-(M-1)/2$ to $(M-1)/2$ when M is odd:

- a) Define an $M \times N_p$ -by-5 matrix R_m . The elements of R_m are assigned values per Equation (162-1) where $i = -3$ to 1 and $j = 1$ to $M \times N_p$.

$$R_m(j, i+4) = \begin{cases} r(m+j-M \times i) & \text{if } 1 \leq m+j-M \times i \leq M \times N_p \\ 0 & \text{otherwise} \end{cases} \quad (162-1)$$

- b) The normalized coefficients of the transmit equalizer are computed using Equation (162-2).

$$\begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} = (R_m^T R_m)^{-1} R_m^T \begin{bmatrix} p(1) \\ \dots \\ p(M \times N_p) \end{bmatrix} \quad (162-2)$$

- c) The linear fit pulse response is reconstructed from the matrix R_m and the normalized coefficients using Equation (162-3).

$$\begin{bmatrix} p_m(1) \\ \dots \\ p_m(M \times N_p) \end{bmatrix} = R_m \begin{bmatrix} c_m(-3) \\ c_m(-2) \\ c_m(-1) \\ c_m(0) \\ c_m(1) \end{bmatrix} \quad (162-3)$$

d) $\varepsilon^2(m)$ is computed using Equation (162-4).

$$\varepsilon^2(m) = \sum_{k=1}^{M \times N_p} (p(k) - p_m(k))^2 \quad (162-4)$$

The normalized transmit equalizer coefficients $c(i)$ for a given linear fit pulse $p(k)$ are the values $c_m(i)$ for the value of m that minimizes $\varepsilon^2(m)$.

162.9.4.1.2 Steady-state voltage and linear fit pulse peak ratio

The linear fit pulse peak v_{peak} and steady-state voltage v_f are defined using the linear fit pulse response $p(1)$ through $p(M \times N_p)$, measured with transmit equalizer set to preset 1 (no equalization). N_p is set equal to 200. The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 162.9.4.1.1.

v_{peak} is defined as the maximum value of $p(k)$.

v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M .

The linear fit pulse peak ratio R_{peak} is defined by Equation (162-5).

$$R_{peak} = \frac{v_{peak}}{v_f} \quad (162-5)$$

162.9.4.1.3 Coefficient initialization

When the Coefficient update state diagram (Figure 136-9) is in either the OUT_OF_SYNC state or the NEW_IC state, the coefficients of the transmit equalizer shall be configured to values within the ranges specified in Table 162-12 (according to the value of the variable `ic_req`, which is defined in 136.8.11.7.1). These requirements apply upon the assertion of initial condition status of “Updated”.

162.9.4.1.4 Coefficient step size

The change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “increment” shall be between 0.005 and 0.025, and the change in the normalized transmit equalizer coefficient $c(\text{coef_sel})$ corresponding to a request to “decrement” shall be between −0.025 and −0.005. It is recommended that the same step size is used for all coefficients.

The coefficients other than $c(\text{coef_sel})$ are not expected to change. The absolute change in any coefficient other than $c(\text{coef_sel})$ shall be less than 0.005.

The change in the normalized transmit equalizer coefficient is defined to be the difference in the value measured prior to the assertion of the “increment” or “decrement” request (i.e., `coef_req` is “Hold”) and the value upon the assertion of a coefficient status of “updated”.

Table 162–12—Coefficient initial conditions

Coefficient update state	ic_req	$c(-3)$	$c(-2)$	$c(-1)$	$c(0)$	$c(1)$
OUT_OF_SYNC ^a	N/A	0	0	0	1	0
NEW_IC	preset 1 ^a	0	0	0	1	0
	preset 2	0 ± 0.0125	0 ± 0.0125	0 ± 0.0125	0.5 ± 0.0125	0 ± 0.0125
	preset 3	0 ± 0.0125	0 ± 0.0125	−0.075 ± 0.0125	0.75 ± 0.0125	0 ± 0.0125
	preset 4	0 ± 0.0125	0.05 ± 0.0125	−0.2 ± 0.0125	0.75 ± 0.0125	0 ± 0.0125
	preset 5	−0.025 ± 0.0125	0.075 ± 0.0125	−0.25 ± 0.0125	0.65 ± 0.0125	0 ± 0.0125

^a PRESET1 is the reference for the calculation of the normalized coefficients of the transmit equalizer (see 162.9.4.1.1). As a result the normalized coefficients for PRESET1 and OUT_OF_SYNC do not include any tolerances.

162.9.4.1.5 Coefficient range

When sufficient “increment” or “decrement” requests have been received for a given coefficient, the coefficient reaches a lower or upper bound based on the range of that coefficient or the combination of coefficients.

With $c(-3)$, $c(-2)$, and $c(-1)$ set to zero and both $c(0)$ and $c(1)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(1)$ shall be less than or equal to −0.2.

With $c(-3)$, $c(-2)$, $c(-1)$, and $c(1)$ set to zero and having received sufficient “decrement” requests so that it is at its minimum value, $c(0)$ shall be less than or equal to 0.5.

With $c(-3)$, $c(-2)$, and $c(1)$ set to zero and both $c(-1)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-1)$ shall be less than or equal to −0.34.

With $c(-3)$, $c(-1)$, and $c(1)$ set to zero, $c(0)$ having received sufficient “decrement” requests so that it is at its minimum value, and $c(-2)$ having received sufficient “increment” requests so that it is at its maximum value, $c(-2)$ shall be greater than or equal to 0.12.

With $c(-2)$, $c(-1)$, and $c(1)$ set to zero and both $c(-3)$ and $c(0)$ having received sufficient “decrement” requests so that they are at their respective minimum values, $c(-3)$ shall be less than or equal to −0.06.

NOTE—Any of the coefficients $c(-3)$, $c(-2)$, $c(-1)$, or $c(1)$ may be set to zero by asserting a coefficient request of “no equalization” for that coefficient, using the control function specified in 162.8.11, or by implementation specific means.

162.9.4.2 Transmitter linearity

Transmitter linearity is defined using the method in 120D.3.1.2.

The transmitter linearity shall meet the requirement when the transmitter equalization is set to each of the initial conditions defined in Table 162–12.

162.9.4.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is determined using the method in 120D.3.1.7 with the following exceptions:

- The linear fit pulse response $p(k)$ is determined using the linear fit procedure in 162.9.4.1.1.
- The continuous time filter parameters are provided in Table 162–20.
- For calculation of SNR_{ISI} using Equation (120D–8) a value of 6 is used for N_b .
- A time offset is added to t_p whose value is swept from -0.5 UI to 0.5 UI when calculating ISI_{cursors} . SNR_{ISI} is defined as the lowest value found across the time offset sweep.
- The transmitter equalizer setting is chosen, within the required coefficient ranges (see 162.9.4.1.5), to result in the highest SNR_{ISI} value.

162.9.4.4 Peak-to-peak AC common-mode voltage

Full-band peak-to-peak AC common-mode voltage, $VC_{M_{FB}}$, is defined as the AC common-mode voltage (see 93.8.1.3) range that includes all but 10^{-4} of the measured distribution, from 0.00005 to 0.99995 of the cumulative distribution. The transmitter equalization is turned off (preset 1 condition).

Low-frequency peak-to-peak AC common-mode voltage, $VC_{M_{LF}}$, is defined in the same way as $VC_{M_{FB}}$, except that it is measured with a low-pass filter defined by Equation (93A–20) with f_c set to 100 MHz.

The low-frequency and full-band peak-to-peak AC common-mode voltages shall meet the specifications for $VC_{M_{LF}}$ (max) and $VC_{M_{FB}}$ (max) in Table 162–11.

NOTE 1—The common-mode noise measurement may be sensitive to mismatches between the single-ended paths in the test fixture and the test setup. Careful design and calibration of the test system are recommended.

NOTE 2—The common-mode noise measurement should take into consideration frequencies down to the specified AC-coupling frequency limit (e.g., the 50 kHz limit specified in 162.11 for the cable assembly).

162.9.4.5 Insertion loss, TP0 to TP2 or TP3 to TP5

The recommended insertion loss for TP0 to TP2 and for TP3 to TP5 is provided in 162A.4.

162.9.4.6 Output SNDR

The transmitter SNDR is defined by the measurement method described in 120D.3.1.6, with the exceptions that the test system response is specified in 162.9.4 and the linear fit procedure in 162.9.4.1.1 is used.

The transmitter SNDR shall meet the requirement when the transmitter equalization is set to each of the initial conditions defined in Table 162–12.

162.9.4.7 Output jitter

Output jitter is characterized by four parameters: J_{RMS} , J_{3u} , $J_{3u_{03}}$, and even-odd jitter. These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the transmitter package and host channel. The equalizer setting is chosen to minimize any or all of the jitter parameters.

J_{RMS} is calculated using the measurement method specified in 120D.3.1.8.1. J_{3u} is calculated using the measurement method for J_{4u} in 120D.3.1.8.1, except that J_{3u} is defined as the time interval that includes all but 10^{-3} of $f_j(t)$, from the 0.05th to the 99.95th percentile of $f_j(t)$. $J_{3u_{03}}$ is calculated the same way as J_{3u} except that the jitter calculation uses only transitions R03 and F30 in Table 162–13.

Even-odd jitter is calculated using the measurement method specified in 120D.3.1.8.2. with the following exceptions:

- a) The test pattern is either PRBS13Q or alternatively PRBS9Q. PRBS9Q is defined in 120.5.11.2.a. Meeting the even-odd jitter requirement with only one pattern is sufficient.
- b) If the test pattern is PRBS13Q, the corner frequency of the clock recovery unit (CRU) is set to 4 MHz or to 1 MHz. Meeting the even-odd jitter requirement with only one CRU bandwidth is sufficient.

NOTE 1—If the measuring instrument is triggered by a clock based on the signaling rate divided by an even number, the even-odd jitter might not be correctly observed. As a result, the observation of J_{3u} and J_{RMS} might also be affected.

NOTE 2— J_{3u} is sensitive to measurement noise being converted to timing errors. Hence, accounting for measurement noise effects is recommended.

The pattern symbols and thresholds used to define each transition for PRBS13Q are defined in Table 120D-4. The pattern symbols and thresholds used to define each transition for PRBS9Q are defined in Table 162-13.

Table 162-13—PRBS9Q pattern symbols used for jitter measurement

Label	Description	Gray coded PAM4 symbols	Index of first symbol	Index transition begins	Index transition ends	Index of last symbol	Threshold level
REF	Reference for symbol index	33333	1	—	—	5	—
R03	0 to 3 rise	1000 331	260	263	264	266	$(V_0+V_3)/2^a$
F30	3 to 0 fall	233333 001	511	5	6	8	
R12	1 to 2 rise	311 2221	464	466	467	470	$(V_1+V_2)/2$
F21	2 to 1 fall	122 11110	254	256	257	261	
R01	0 to 1 rise	200 113	503	505	506	508	$(V_0+V_1)/2$
F10	1 to 0 fall	21111 0003	256	260	261	264	
R23	2 to 3 rise	3222 330	210	213	214	216	$(V_2+V_3)/2$
F32	3 to 2 fall	133 223	507	509	510	1	
R02	0 to 2 rise	200 22223	63	65	66	70	$(V_0+V_2)/2$
F20	2 to 0 fall	12222 001	321	325	326	328	
R13	1 to 3 rise	0111 331	166	169	170	172	$(V_1+V_3)/2$
F31	3 to 1 fall	033 1112	263	265	266	269	

^a V_0 , V_1 , V_2 , and V_3 are defined in 120D.3.1.2.

162.9.4.8 Transmitter effective return loss (ERL)

The ERL of the transmitter at TP2 is defined by the procedure in 93A.5 using the values in Table 162–14 and Table 162–20, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Table 162–14—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

Transmitter ERL at TP2 shall meet the requirement in Table 162–11.

162.9.4.9 Common-mode to common-mode return loss

The transmitter common-mode to common-mode return loss shall meet Equation (162–6).

$$RL_{cc}(f) \geq \begin{cases} 2 & 0.2 \leq f < 4 \\ 1.6 + 0.1f & 4 \leq f \leq 30 \\ 8.5 - 0.13f & 30 < f \leq 40 \end{cases} \quad (162-6)$$

where

$RL_{cc}(f)$ is the transmitter common-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

The transmitter common-mode to common-mode return loss limit is illustrated in Figure 162–4.

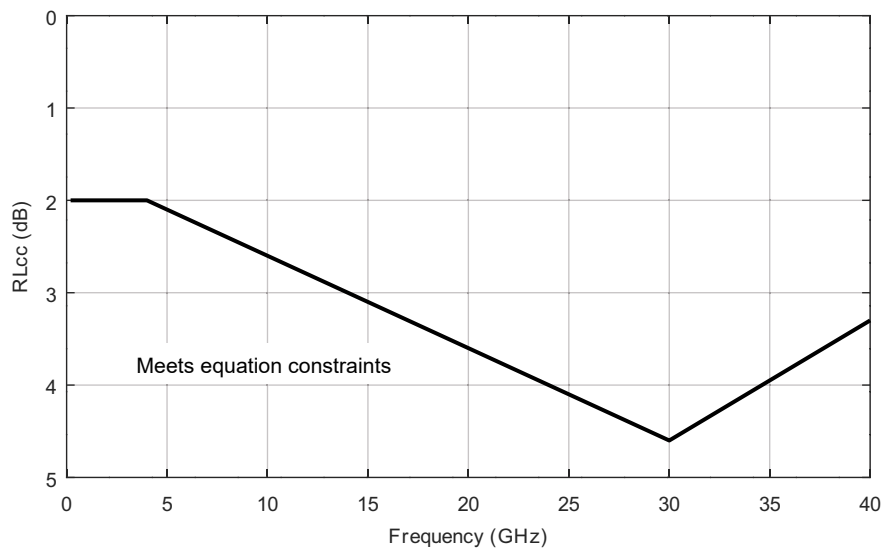


Figure 162-4—Transmitter common-mode to common-mode return loss

162.9.4.10 Common-mode to differential-mode return loss

The transmitter common-mode to differential-mode return loss shall meet Equation (162-7).

$$RL_{dc}(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (162-7)$$

where

$RL_{dc}(f)$ is the transmitter common-mode to differential-mode return loss at frequency f in dB
 f is the frequency in GHz

The transmitter common-mode to differential-mode return loss limit is illustrated in Figure 162-5.

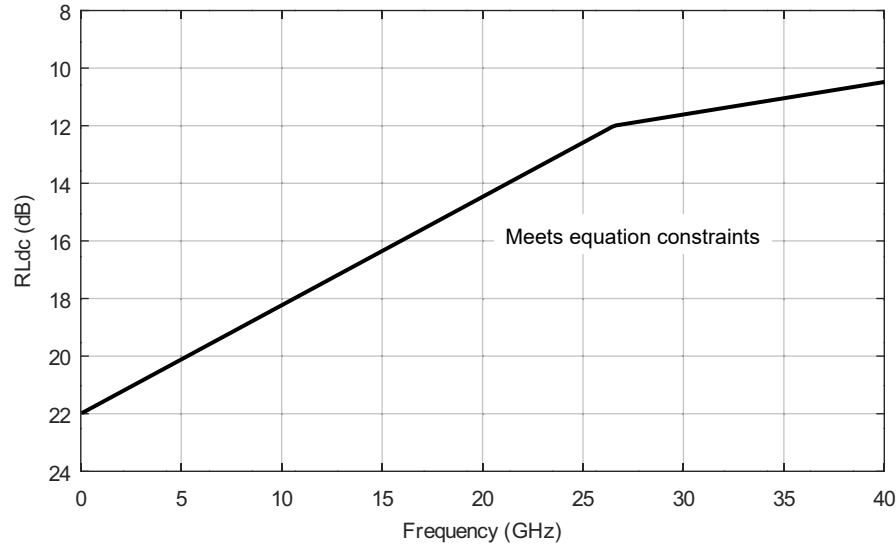


Figure 162-5—Transmitter common-mode to differential-mode return loss

162.9.5 Receiver characteristics

The receiver on each lane shall meet the specifications given in Table 162-15 and detailed in the referenced subclauses. Unless specified otherwise, all receiver measurements are made for each lane separately, at TP3, utilizing the test fixtures specified in Annex 162B.

Recommended receiver characteristics at TP5 are provided in 162A.3.

Table 162-15—Summary of receiver specifications at TP3

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)	162.9.5.1	53.125 ± 100 ppm	GBd
Amplitude tolerance	162.9.5.2	1200 ^a	mV
Interference tolerance	162.9.5.3	Table 162-16	—
Jitter tolerance	162.9.5.4	Table 162-17	—
Effective return loss, ERL (min)	162.9.5.5	7.3	dB
Differential-mode to common-mode return loss, <i>RL_{cd}</i> (min)	162.9.5.6	Equation (162-18)	dB

^a Amplitude is measured at TP2.

162.9.5.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 162.9.5.2, 162.9.5.3, and 162.9.5.4 for any signaling rate in the range 53.125 GBd ± 100 ppm.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ± 50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ± 100 ppm, when derived from an intermediate interface (e.g., 100GAUI-4).

162.9.5.2 Receiver amplitude tolerance

When a PMD receiver is connected to a compliant transmitter whose peak-to-peak differential output voltage (see Table 162–11 footnote b) measured at the preset 1 equalizer setting is 1200 mV, using a compliant cable assembly with the minimum insertion loss specified in 162.11.2, the PMD receiver operation shall enable an FEC symbol error ratio better than 10^{-3} assuming errors are sufficiently uncorrelated to support the required frame loss ratio (see 162.1). If errors are not sufficiently uncorrelated, the FEC symbol error ratio shall be lower as appropriate to support the required frame loss ratio.

The receiver is allowed to control the transmitter equalizer coefficients, using the PMD control function defined in 162.8.11 or an equivalent process, to meet these requirements.

162.9.5.3 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure described in 162.9.5.3.1 through 162.9.5.3.5. Receiver interference tolerance test requirements are specified in Table 162–16.

Two tests are defined. Test 1 includes a low-loss channel. Test 2 includes a high-loss channel. The cable assembly used in the test channel (see 110.8.4.2.2) shall meet the cable assembly Channel Operating Margin (COM) as specified in 162.11.7.

Table 162–16—Interference tolerance test parameters

Parameter	Test 1 (low loss)		Test 2 (high loss)		Units
	Min	Max	Min	Max	
Test pattern	Scrambled idle encoded by FEC				
FEC symbol error ratio required ^a	$< 10^{-3}$				
Test channel insertion loss at 26.56 GHz ^b	10.5	11.5	23.625	24.625	dB
Cable assembly insertion loss at 26.56 GHz	10.5	11.5	17.75	19.75	dB
COM ^c	3		3		dB

^a See 162.9.5.3.5 for definition of FEC symbol error ratio.

^b Insertion loss between the two test references (see Figure 110–3b).

^c The COM value is the target value for the SNR_{TX} calibration defined in 162.9.5.3.3 item h). The SNR_{TX} value measured at the Tx test reference should be as close as practical to the value needed to produce the target COM. If lower SNR_{TX} values are used, this would demonstrate margin to the specification but this is not required for compliance.

162.9.5.3.1 Test setup

The interference tolerance test is performed with the setup similar to the one described in 110.8.4.2.1. The test setup includes noise injection for all lanes of the pattern generator.

162.9.5.3.2 Test channel

The test channel is the same as the one defined in 110.8.4.2.2, except that the cable assembly meets the requirements of 162.11, the test channel loss meets the requirements of Table 162–16 and the cable assembly test fixture meets the requirements of 162B.3.

162.9.5.3.3 Test channel calibration

The scattering parameters of the test channel are measured at the test references as illustrated in Figure 110–3b using the cable assembly test fixtures specified in Annex 162B.1.

The insertion loss at 26.56 GHz of the signal path between the test references in Figure 110–3b is within the limits in Table 162–16.

The COM is calculated using the method and parameters of 162.11.7 with the following considerations:

- a) The channel signal path is $SCHS_p = \text{cascade}(S^{(CTSP)}, S^{(HOSPR)})$, where $\text{cascade}()$ is defined in 93A.1.2.1, $S^{(HOSPR)}$ is defined in 162.11.7.1.1, and $S^{(CTSP)}$ is the measured channel between the test references in Figure 110–3b.
- b) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 162–20 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- c) The augmented signal path in 93A.1.2 is replaced by S_p determined from Equation (162–8) (effectively omitting the transmitter device package model $S^{(tp)}$). The filtered voltage transfer function $H^{(k)}(f)$ calculated in Equation (93A–19) uses T_r equal to the transition time at the Tx test reference. T_r is measured using the method in 120E.3.1.5 with the transmit equalizer turned off (i.e., coefficients set to the preset 1 values, see 162.9.4.1.3) with an exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 40 GHz.
- d) Even-odd jitter, J3u, and J_{RMS} without noise injection (see 162.9.5.3.4) are measured at the Tx test reference and comply with the specification in Table 162–11. In the calculation of COM, A_{DD} , and σ_{RJ} are calculated from the measured values of J3u and J_{RMS} using Equation (162–9) and Equation (162–10), replacing the values in Table 162–20. It is recommended to adjust the pattern generator jitter such that J3u and J_{RMS} are as close as practical to their limits in Table 162–11.
- e) In the calculation of COM, SNR_{TX} is set to the SNDR value measured at the Tx test reference using the procedure in 120D.3.1 with the exception that the linear fit in 120D.3.1.3 is performed with a pulse length (N_p) of 29 UI and with pattern generator noise disabled.
- f) For the CTLE and FFE settings used in the calculation of COM in items a) to e), find the value of σ_{TX} required to achieve the COM value specified in Table 162–16. From σ_{TX} , determine the value of σ_{ne} using Equation (162–11).
- g) From σ_{ne} , the voltage transfer function $H_{21}^{(0)}(f)$ of the test channel, the transfer function $H_{ctf}(f)$ of the CTLE used in COM, and $H_{hp}(f)$ defined in Equation (162–13), find σ_{bn} , using Equation (162–12). From σ_{bn} and $H_{hp}(f)$ find σ_{hp} using Equation (162–14).
- h) Set the RMS voltage of the noise added to the signal before the Tx test reference to σ_{hp} .

$$S_p = \text{cascade}(SCHS_p, S^{(rp)}) \quad (162-8)$$

where

$\text{cascade}()$	is defined in 93A.1.2.1
$SCHS_p$	is defined in item a) above
$S^{(rp)}$	is defined in 93A.1.2.4

$$A_{DD} = \frac{\frac{J3u}{2} + Q3d \sqrt{(Q3d^2 + 1) \times J_{RMS}^2 - \left(\frac{J3u}{2}\right)^2}}{Q3d^2 + 1} \quad (162-9)$$

$$\sigma_{RJ} = \frac{\frac{J3u}{2} - A_{DD}}{Q3d} \quad (162-10)$$

where

$$Q3d = 3.0902$$

NOTE 1— $Q3d$ is an approximated solution of $Q(Q3d) = 10^{-3}$, where the Q function is defined in Equation (95-1).

NOTE 2—The calculation of A_{DD} may, under certain conditions, pose a negative discriminant. If this condition occurs, the recommended solution is to increase sinusoidal jitter or bounded uncorrelated jitter such that the discriminant is positive.

$$\sigma_{TX}^2 = [h^{(0)}(t_s)]^2 10^{-SNR_{TX}/10} + \sigma_{ne}^2 \quad (162-11)$$

$$\sigma_{ne}^2 = \frac{\sigma_{bn}^2}{(f_b/2)} \int_0^{f_b/2} |H_{hp}(f)H_{21}^{(0)}(f)H_r(f)H_{ctf}(f)|^2 df \quad (162-12)$$

$$H_{hp}(f) = \frac{j \frac{f}{f_{hp}}}{1 + j \frac{f}{f_{hp}}} \quad (162-13)$$

$$\sigma_{hp}^2 = \frac{\sigma_{bn}^2}{(f_b/2)} \left(\int_0^{f_b/2} |H_{hp}(f)|^2 df \right) \approx 0.6954 \sigma_{bn}^2 \quad (162-14)$$

where

σ_{TX}	is the RMS transmitter noise
SNR_{TX}	is the transmitter signal-to-noise ratio
$H_{hp}(f)$	is the high-pass spectral shaping of the noise
f_{hp}	is the 3 dB cutoff frequency of the high-pass spectral shaping filter with a value of 6 GHz
σ_{bn}	is the RMS broadband noise amplitude
σ_{ne}	is the RMS filtered noise amplitude
σ_{hp}	is the RMS noise amplitude added at the transmitter test reference
$h^{(0)}$	is defined in 93A.1.5.
$H_{21}^{(0)}(f)$	is defined in 93A.1.3
$H_r(f)$	is defined in 93A.1.4.1
$H_{ctf}(f)$	is defined in 93A.1.4.3

162.9.5.3.4 Pattern generator and noise injection

The pattern generator transmits data to all lanes of the device under test. At the start of transmitter training, the pattern generator output amplitude on all lanes shall be 800 mV peak-to-peak differential when measured on an alternating zero-three sequence. The output amplitude, measured on an alternating zero-three sequence, is not permitted to exceed 800 mV peak-to-peak differential during transmitter training. The output waveform and the ERL of the pattern generator comply with 162.9.4.

Noise is added to the signal before the Tx test reference, with noise level set according to step h) in 162.9.5.3.3.

The noise required for each lane is calibrated. The noise may be added either to one lane at a time or using multiple noise sources to all lanes at the same time.

To avoid excessive low-frequency weighting to the receiver input noise, the noise added to the signal is bounded by the normalized spectral density mask defined in Equation (162–15), Equation (162–16), and Equation (162–17) with $f_1 = 8$ GHz and $f_2 = 5$ GHz, and illustrated in Figure 162–6.

$$NSD_{avg} = \frac{1}{(f_b/2 - f_1)} \int_{f_1}^{f_b/2} NSD(f) df \quad (162-15)$$

$$10\log_{10}\left(\frac{NSD(f)}{NSD_{avg}}\right) \geq -3 + 3.6(f/f_b) \quad f_1 \leq f \leq f_b/2 \quad (162-16)$$

$$10\log_{10}\left(\frac{NSD(f)}{NSD_{avg}}\right) \leq \begin{cases} -12 + 15(f/f_2) & 0 \leq f < f_2 \\ 3 & f_2 \leq f \leq f_b/2 \end{cases} \quad (162-17)$$

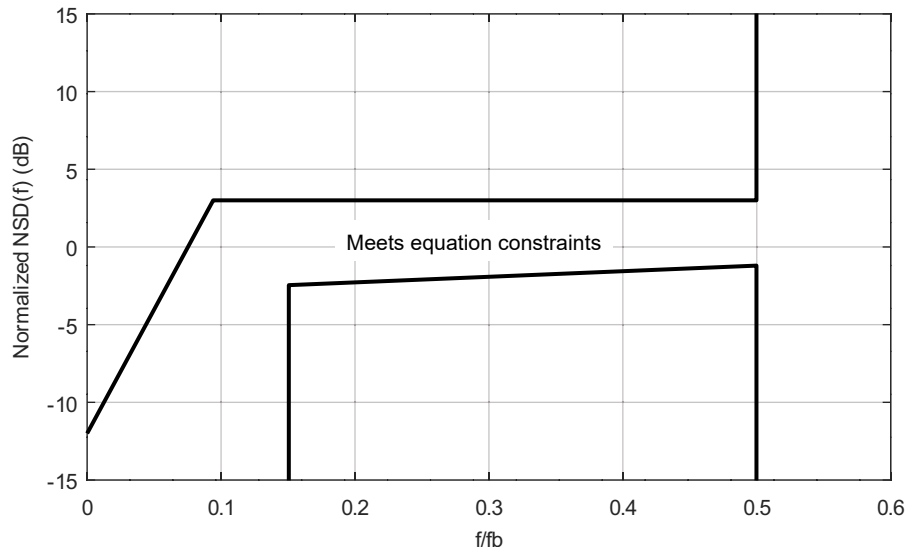


Figure 162–6—NSD(f) constraints

162.9.5.3.5 Test procedure

The pattern generator is first configured to transmit the training pattern defined in 162.8.11. During this initialization period, the device under test (DUT) configures the pattern generator transmit equalizer to the coefficient settings it would select using the protocol described in 162.8.11 and the receiver is tuned using its optimization method. The coefficient settings may be communicated via the start-up protocol or by other means.

After the pattern generator equalizer has been configured and the receiver tuned, the pattern generator is set to generate the test pattern specified in Table 162–16. During the test, the transmitters in the device under test transmit the same pattern type specified for the test, with equalization turned off (preset 1 condition).

Symbol error ratio is measured using the per-lane FEC symbol error counters (see 91.6 or 161.6) in the adjacent FEC sublayer, or the per-lane PCS symbol error counters (see 119.3.1) in the adjacent PCS, as appropriate.

The FEC symbol error ratio is defined as the sum of the symbol error counters on all lanes divided by the number of symbols transmitted on all lanes, which may be estimated from the test time.

A PHY shall meet the FEC symbol error ratio requirement in all tests defined in Table 162–16 with broadband noise added to all lanes (see 162.9.5.3.4). The FEC symbol error ratio requirement assumes that errors are sufficiently uncorrelated to support the required frame loss ratio (see 162.1). If errors are not sufficiently uncorrelated, the FEC symbol error ratio shall be lower as appropriate to support the required frame loss ratio.

NOTE—If noise is applied to each of the n lanes one at a time, results of the n measurements are summed to yield the FEC symbol error ratio. The result may need to be corrected based on the FEC symbol error ratio with no noise added on any lane.

162.9.5.4 Receiver jitter tolerance

Receiver jitter tolerance is defined by the procedure described in 162.9.5.4.1 and 162.9.5.4.2

162.9.5.4.1 Test setup

Jitter tolerance is measured with a channel meeting the insertion loss of Test 2 as specified in Table 162–16. The pattern generator includes sinusoidal jitter injection.

162.9.5.4.2 Test procedure

The jitter tolerance test procedure is similar to that of 162.9.5.3, with the exception that no noise is injected (i.e., step h) in 162.9.5.3.3 is not performed). Instead, jitter with the specified frequency and amplitude is applied to the pattern generator and the jitter amplitude is adjusted to obtain the peak-to-peak jitter specified for that frequency in Table 162–17 at the Tx test reference (see Figure 110–3a). The test channel COM, calculated per 162.9.5.3.3 with the jitter-stressed transmitter output, shall not be lower than the value in Table 162–16.

A PHY shall meet the FEC symbol error ratio requirement defined in Table 162–16 for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17 with jitter added to all lanes.

NOTE 1—If jitter is applied to each of the n lanes one at a time, results of the n measurements are summed to yield the FEC symbol error ratio. The result may need to be corrected based on the FEC symbol error ratio with no jitter applied on any lane.

NOTE 2—The A_{DD} (Equation (162–9)) and σ_{RJ} (Equation (162–10)) calculated from transmitter measurements in this test may be higher than the values in Table 162–20. A suitable channel should be chosen in order to meet the COM requirement with these values.

Table 162–17—Receiver jitter tolerance parameters

Parameter	Case A	Case B	Case C	Case D	Case E	Case F	Units
Jitter frequency	0.04	0.4	1.333	4	12	40	MHz
Jitter amplitude (pk-pk)	5	0.5	0.15	0.05	0.05	0.05	UI

162.9.5.5 Receiver ERL

The ERL of the receiver at TP3 is computed using the procedure in 93A.5 with the values in Table 162–14 and Table 162–20, and with the value of T_{jx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Receiver ERL at TP3 shall meet the requirement specified in Table 162–15.

162.9.5.6 Differential-mode to common-mode return loss

The receiver differential-mode to common-mode return loss shall meet Equation (162–18) as illustrated in Figure 162–7.

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (162-18)$$

where

$RLcd(f)$ is the receiver differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

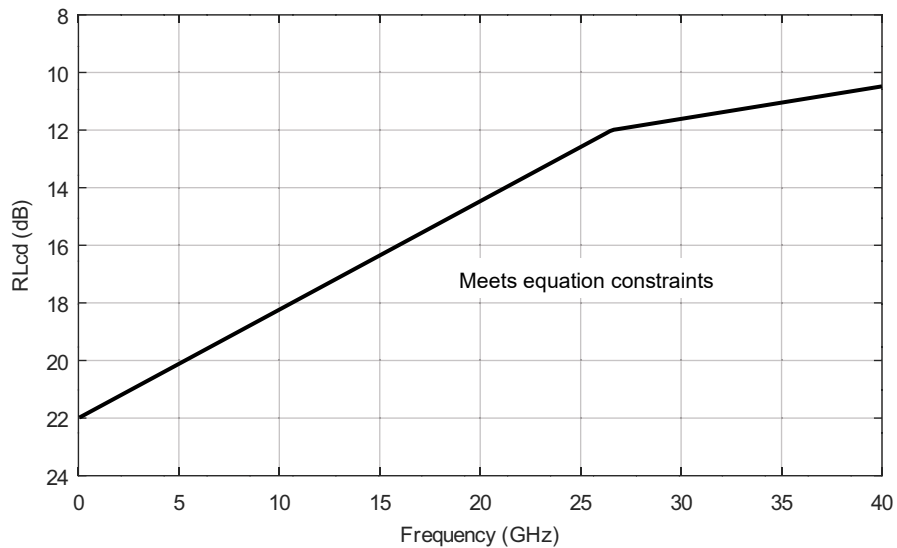


Figure 162–7—Receiver differential-mode to common-mode return loss

162.10 Channel characteristics

The channel is defined between TP0 and TP5 to include the transmitter and receiver differential controlled impedance PCB and the cable assembly as illustrated in Figure 162–2. The channel insertion loss, return loss, COM, and the transmitter and receiver differential controlled impedance PCB parameters are provided informatively in 162A.4 through 162A.7.

Channel definitions apply for links between two PHYs of the same type: 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4.

162.11 Cable assembly characteristics

Cable assemblies defined in this subclause contain insulated conductors terminated in a connector at each end for use as link segments between MDIs. Cable assemblies are primarily intended as point-to-point links between 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 PHYs using controlled impedance cables.

Three cable assembly types are specified:

- a) 100GBASE-CR1: Cable assembly that supports single-lane links between two 100GBASE-CR1 PHYs with achievable cable length of at least 2 m.
- b) 200GBASE-CR2: Cable assembly that supports two-lane links between two 200GBASE-CR2 PHYs with achievable cable length of at least 2 m.
- c) 400GBASE-CR4: Cable assembly that supports four-lane links between two 400GBASE-CR4 PHYs with achievable cable length of at least 2 m.

NOTE—It may be possible to construct compliant cable assemblies longer than indicated. Length of a cable assembly does not imply compliance to specifications.

100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 cable assembly types are described in Annex 162D. The MDIs are defined in Annex 162C.

For 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, the lanes are AC-coupled. The AC-coupling shall be within the cable assembly. It is recommended that it is within the plug connectors. It should be noted that there may be various methods for AC-coupling in actual implementations. The low-frequency 3 dB cutoff of the AC-coupling shall be less than 50 kHz. It is recommended that the value of the coupling capacitors be 100 nF. The capacitor limits the inrush charge and baseline wander.

All cable assembly measurements are to be made between TP1 and TP4 with cable assembly test fixtures as specified in Annex 162B. These cable assembly specifications are based upon twinaxial cable characteristics, but other cable types are acceptable if the specifications of this subclause are met.

Table 162–18 provides a summary of the cable assembly characteristics for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, and references to the subclauses addressing each parameter.

The specifications for the three cable assembly types are identical except the number of lanes.

162.11.1 Characteristic impedance and reference impedance

The nominal differential characteristic impedance of the cable assembly is 100 Ω . The differential reference impedance for cable assembly specifications shall be 100 Ω .

162.11.2 Cable assembly insertion loss

The measured insertion loss at 26.56 GHz of a cable assembly shall be less than or equal to 19.75 dB.

Table 162–18—Cable assembly characteristics summary

Description	Reference	Value	Unit
Insertion loss at 26.56 GHz, IL_{dd} (max)	162.11.2	19.75	dB
Insertion loss at 26.56 GHz, IL_{dd} (min)	162.11.2	11	dB
Minimum cable assembly ERL ^a	162.11.3	8.25	dB
Differential-mode to common-mode return loss, RL_{cd}	162.11.4	Equation (162–20)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	162.11.5	Equation (162–21)	dB
Common-mode to common-mode return loss, RL_{cc}	162.11.6	Equation (162–22)	dB
Minimum COM	162.11.7	3	dB

^a Cable assemblies with a COM greater than 4 dB are not required to meet minimum ERL.

The measured insertion loss of a cable assembly shall be greater than or equal to the minimum cable assembly insertion loss given in Equation (162–19) and illustrated in Figure 162–8.

$$IL_{dd}(f) \geq IL_{dd_{min}}(f) = \left\{ \begin{array}{ll} 0.418\sqrt{f} + 0.177f + 0.0059f^2 & 0.05 \leq f < 26.56 \\ 1.222\sqrt{f} + 0.138f + 0.0015f^2 & 26.56 \leq f \leq 40 \end{array} \right\} \quad (162-19)$$

where

$IL_{dd}(f)$ is the measured cable assembly insertion loss in dB
 $IL_{dd_{min}}(f)$ is the minimum cable assembly insertion loss in dB
 f is the frequency in GHz

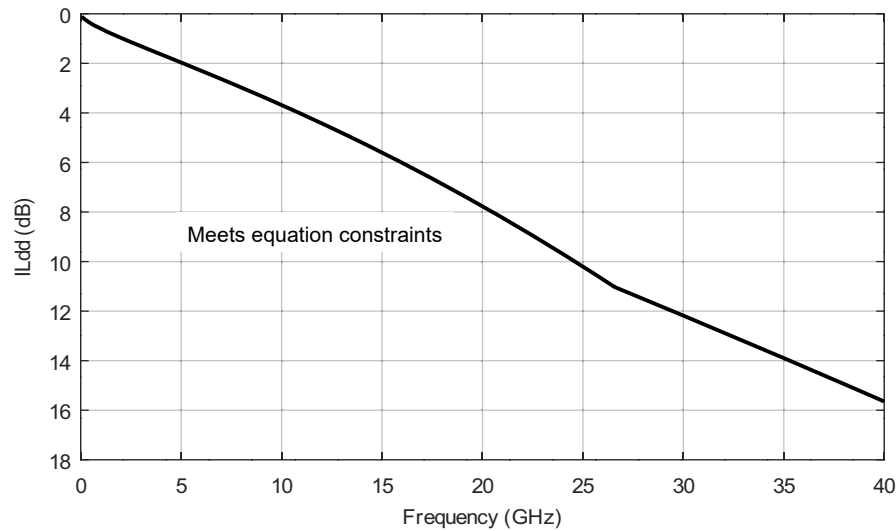


Figure 162–8—Minimum cable assembly insertion loss

162.11.3 Cable assembly ERL

The cable assembly ERL at TP1 and at TP4 is defined by the procedure in 93A.5 with the values in Table 162–19 and Table 162–20, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture cable-facing connection minus 0.2 ns.

The values of cable assembly ERL at TP1 and at TP4 shall meet the requirement specified in Table 162–18 for cable assemblies that have a COM less than 4 dB.

Table 162–19—Cable assembly ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	4500	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

162.11.4 Differential-mode to common-mode return loss

The cable assembly differential-mode to common-mode return loss shall meet Equation (162–20).

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (162-20)$$

where

$RLcd(f)$ is the cable assembly differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The cable assembly differential-mode to common-mode return loss limit is illustrated in Figure 162–9.

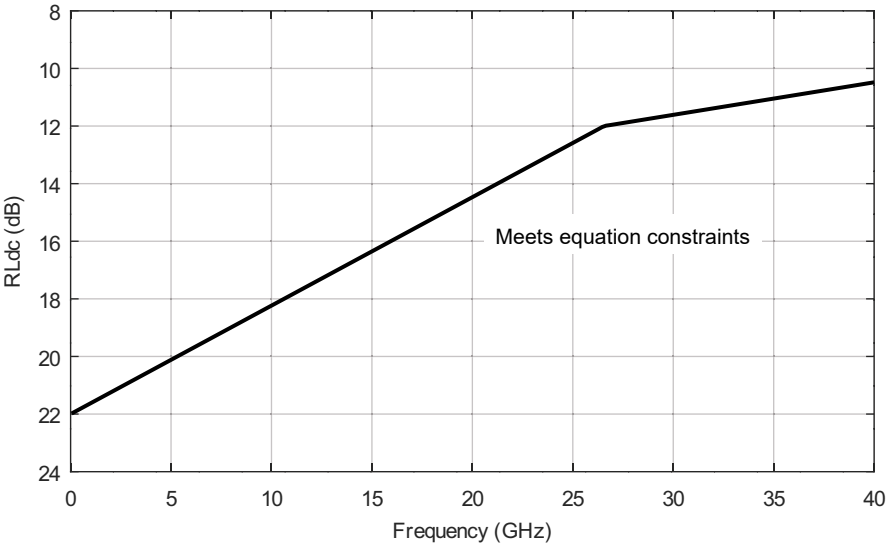


Figure 162-9—Cable assembly differential-mode to common-mode return loss

162.11.5 Differential-mode to common-mode insertion loss

The cable assembly differential-mode to common-mode insertion loss is specified relative to the insertion loss. The difference between the cable assembly differential-mode to common-mode insertion loss and the cable assembly insertion loss shall meet Equation (162-21) illustrated in Figure 162-10.

$$IL_{cd}(f) - IL_{dd}(f) \geq \left\{ \begin{array}{ll} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{array} \right\} \tag{162-21}$$

where

$IL_{cd}(f)$	is the cable assembly differential-mode to common-mode insertion loss at frequency f in dB
$IL_{dd}(f)$	is the cable assembly insertion loss at frequency f in dB
f	is the frequency in GHz

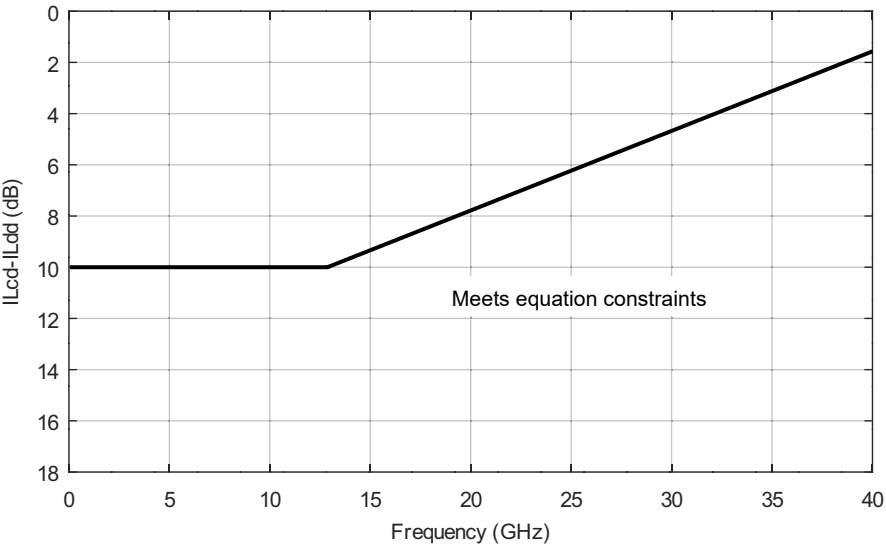


Figure 162-10—Cable assembly differential-mode to common-mode insertion loss to insertion loss difference

162.11.6 Common-mode to common-mode return loss

The cable assembly common-mode to common-mode return loss shall meet Equation (162-22).

$$RL_{cc}(f) \geq \left\{ \begin{array}{ll} 1.4 & 0.05 \leq f \leq 6 \\ 0.68 + 0.12f & 6 < f \leq 30 \\ 10.28 - 0.2f & 30 < f \leq 40 \end{array} \right\} \tag{162-22}$$

where

$RL_{cc}(f)$ is the cable assembly common-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The cable assembly common-mode to common-mode return loss limit is illustrated in Figure 162-11.

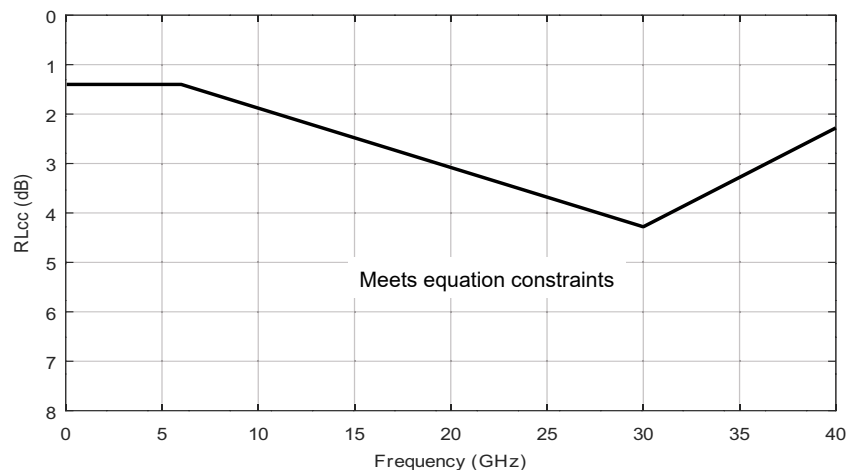


Figure 162–11—Cable assembly common-mode to common-mode return loss

162.11.7 Cable assembly Channel Operating Margin

The cable assembly Channel Operating Margin (COM) for each lane is derived from measurements of the cable assembly signal, near-end crosstalk, and far-end crosstalk paths. COM is computed using the path calculations defined in 162.11.7.1 and the procedure in 93A.1, where T_r is 7.5 ps for $H_t(f)$ as used in Equation (93A–19). The specific paths used depend on cable assembly types (see Annex 162D), as described in 162.11.7.2.

COM parameter values for the three cable assembly types are provided in Table 162–20.

COM shall be computed twice, Test 1 and Test 2, which differ in the value of the device package model transmission line length z_p . COM for any lane within the cable assembly shall be greater than or equal to 3 dB for both Test 1 and Test 2.

Table 162–20—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	1.2×10^{-4}	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1	z_p	12	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Transmission line characteristic impedance	Z_c	87.5	Ω
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line 2 characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient	$c(-3)$		
Minimum value		−0.06	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, 2 nd pre-cursor coefficient	$c(-2)$		
Minimum value		0	
Maximum value		0.12	—
Step size		0.02	
Transmitter equalizer, 1 st pre-cursor coefficient	$c(-1)$		
Minimum value		−0.34	
Maximum value		0	—
Step size		0.02	
Transmitter equalizer, post-cursor coefficient	$c(1)$		
Minimum value		−0.2	
Maximum value		0	—
Step size		0.02	
Continuous time filter, DC gain	g_{DC}		
Minimum value		−20	dB
Maximum value		0	dB
Step size		1	dB
Continuous time filter, DC gain 2	g_{DC2}		
Minimum value		−6	dB
Maximum value		0	dB
Step size		1	dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz

Table 162–20—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage			
Victim	A_v	0.413	V
Far-end aggressor	A_{fe}	0.413	V
Near-end aggressor	A_{ne}	0.608	V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	32.5	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	12	UI
Normalized DFE coefficient maximum limit	$bb_{\max}(n)$		
for $n = 1$		0.85	—
for $n = 2$		0.3	—
for $n = 3$ to 6		0.2	—
for $n = 7$ to N_b		0.1	—
Normalized DFE coefficient minimum limit	$bb_{\min}(n)$		
for $n = 1$		0.3	—
for $n = 2$		0.05	—
for $n = 3$ to N_b		−0.03	—
Number of DFE floating tap banks	N_{bg}	3	—
Number of DFE floating taps per bank	N_{bf}	3	—
DFE maximum span including floating taps	N_f	40	UI
Normalized coefficient magnitude limit for DFE floating taps	b_{gmax}	0.03	—
DFE floating tap tail root-sum-of-squares limit	σ_{tmax}	0.02	—
DFE floating tap tail starting position	N_{ts}	25	UI
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	9×10^{-9}	V ² /GHz
Target detector error ratio	DER_0	10^{-4}	—

162.11.7.1 Channel signal and crosstalk path calculations

The channel paths between TP0 and TP5 used for calculation of the cable assembly COM consist of measured cable assembly signal and crosstalk paths, representative transmitter PCB signal paths, and representative receiver PCB signal paths.

The scattering parameters for a PCB transmission line are calculated using the method defined in 93A.1.2.3 with the parameter values given in Table 162–21. The PCB trace length parameter z_p has a different value for each specific signal path, as specified in 162.11.7.1.1 and 162.11.7.1.2.

The channel path calculations use the function `cascade()` defined in 93A.1.2.1.

Table 162–21—PCB model parameters and values

Parameter	Value	Units
γ_0	0	1/mm
a_1	3.8206×10^{-4}	ns ^{1/2} /mm
a_2	9.5909×10^{-5}	ns/mm
τ	5.79×10^{-3}	ns/mm
C_0	2.9×10^{-5}	nF
C_1	1.9×10^{-5}	nF
Z_c	100	Ω
R_0	50	Ω

162.11.7.1.1 Channel signal path

The scattering parameters of the channel signal path from TP0 to TP5 are calculated using Equation (162–25). The transmitter and receiver PCB signal paths are denoted as $S^{(HOSPT)}$ and $S^{(HOSPR)}$, and are calculated using Equation (162–23) and Equation (162–24), respectively. The PCB transmission line scattering parameters are denoted as $S^{(l)}$ and are calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 110.3$ mm in length and the parameter values given in Table 162–21, representing an insertion loss of 4.33 dB at 26.56 GHz on each PCB. The scattering parameters for the PCB via capacitances, $S^{(c0)}$ and $S^{(c1)}$, are calculated using Equation (93A–8) and the values in Table 162–21.

$$S^{(HOSPT)} = \text{cascade}(S^{(c0)}, S^{(l)}, S^{(c1)}) \quad (162-23)$$

$$S^{(HOSPR)} = \text{cascade}(S^{(c1)}, S^{(l)}, S^{(c0)}) \quad (162-24)$$

$$SCHS_p^{(k)} = \text{cascade}(S^{(HOSPT)}, S^{(CASP)}, S^{(HOSPR)}) \quad (162-25)$$

where

$SCHS_p^{(k)}$	is the channel signal path
$S^{(HOSPT)}$	is the host transmitter PCB signal path
$S^{(HOSPR)}$	is the host receiver PCB signal path
$S^{(CASP)}$	is the cable assembly signal path (TP1 to TP4)
k	is equal to zero

162.11.7.1.2 Channel crosstalk paths

The MDI is a significant contributor to crosstalk and is included in and characterized by the cable assembly crosstalk measurements. Crosstalk includes a near-end path where the aggressor is the PMD transmitter, and in some cases, additional near-end, far-end, and alien far-end crosstalk paths where the aggressors are other PMD transmitters that are connected to the same cable assembly.

For the channel crosstalk paths, the receiver PCB model is $S^{(HOSPR)}$ as defined in 162.11.7.1.1. The aggressor transmitter host PCB model is denoted as $S^{(HOTxSP)}$ and is calculated from Equation (93A-13) and Equation (93A-14) using $z_p = 110.3$ mm in length and the parameter values given in Table 162-21, representing an insertion loss of 4.33 dB at 26.56 GHz.

The scattering parameters of the channel near-end crosstalk paths are calculated using Equation (162-26). The scattering parameters of the channel alien far-end crosstalk paths are calculated using Equation (162-27).

$$SCHNXT_p^{(k)} = \text{cascade}(S^{(HOTxSP)}, S^{(CANXTk)}, S^{(HOSPR)}) \quad (162-26)$$

where

$SCHNXT_p^{(k)}$	is the near-end crosstalk path
$S^{(HOSPR)}$	is the host receiver PCB signal path defined in 162.11.7.1.1
$S^{(HOTxSP)}$	is the aggressor transmitter PCB signal path
$S^{(CANXTk)}$	is the cable assembly near-end crosstalk path k (TP1 to TP4)
k	is the index of the near-end crosstalk path

$$SCHFXT_p^{(k)} = \text{cascade}(S^{(HOTxSP)}, S^{(CAFXTk)}, S^{(HOSPR)}) \quad (162-27)$$

where

$SCHFXT_p^{(k)}$	is the alien far-end crosstalk path
$S^{(HOSPR)}$	is the host receiver PCB signal path defined in 162.11.7.1.1
$S^{(HOTxSP)}$	is the aggressor transmitter PCB signal path
$S^{(CAFXTk)}$	is the cable assembly far-end crosstalk path k (TP1 to TP4)
k	is the index of the alien far-end crosstalk path

162.11.7.2 Signal and crosstalk paths used in calculation of COM

Cable assemblies have several types, as described in Annex 162D. The choice of signal and crosstalk paths for calculation of COM is specific to each cable assembly type, as specified in Table 162-22.

The signal path is calculated using Equation (162-25).

The near-end crosstalk paths are calculated using Equation (162-26), with k values from 1 to n ; where n is the number of near-end crosstalk paths given in Table 162-22 for the cable assembly type considered.

The far-end crosstalk paths are calculated using Equation (162-27), with k values from 1 to n ; where n is the number of far-end crosstalk paths given in Table 162-22 for the cable assembly type considered.

The specifications for the cable assembly types are identical except the number of lanes.

The number of crosstalk paths based on MDI type are given in Table 162–22.

Table 162–22—Number of crosstalk paths used in COM

Victim (one end)	NEXT	FEXT (other end)			
		SFP112	SFP-DD112 or DSFP	QSFP112	QSFP-DD800 or OSFP
SFP112	1	0	1	3	7
SFP-DD112 or DSFP	2	1	1	3	7
QSFP112	4	3	3	3	7
QSFP-DD800 or OSFP	8	7	7	7	7

162.12 MDI specifications

The MDI couples the PMD (specified in 162.8 and 162.9) to the cable assembly (specified in 162.11).

100GBASE-CR1 has six specified MDI connectors: SFP112, SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

200GBASE-CR2 has five specified MDI connectors: SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

400GBASE-CR4 has three specified MDI connectors: QSFP112, QSFP-DD800, and OSFP.

Annex 162C specifies the MDIs for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4.

162.13 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 14.7.

162.14 Protocol implementation conformance statement (PICS) proforma for Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4¹⁸

162.14.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

162.14.2 Identification

162.14.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

162.14.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

¹⁸Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

162.14.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CR1	100GBASE-CR1 PMD	162.1	Can operate as a 100GBASE-CR1 PMD	O.1	Yes [] No []
*CR2	200GBASE-CR2 PMD	162.1	Can operate as a 200GBASE-CR2 PMD	O.1	Yes [] No []
*CR4	400GBASE-CR4 PMD	162.1	Can operate as a 400GBASE-CR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-P PMA	162.1	Device implements Clause 135 PMA for 100GBASE-P	CR1:M	Yes [] N/A []
PMA200	200GBASE-R PMA	162.1	Device implements Clause 120 PMA for 200GBASE-R	CR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	162.1	Device implements Clause 120 PMA for 400GBASE-R	CR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	162.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	162.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!CR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	162.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	162.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC	162.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	CR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	162.1	Device implements Clause 82 100GBASE-R PCS	CR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	162.1	Device implements Clause 119 200GBASE-R PCS	CR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	162.1	Device implements Clause 119 400GBASE-R PCS	CR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	162.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	162.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	162.6	Conforms to constraints	M	Yes []
*MD	MDIO capability	162.7	Device implements Clause 45 MDIO with the variable mapping in 162.7.	O	Yes [] No []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CBL	Cable assembly	162.11	Item marked with CBL are cable assembly specifications, these items are the only ones applicable to cable assemblies and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	162.9.4		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

162.14.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162.14.4.1 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

162.14.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed_ <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	162.9.4.1.3	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []
PC9	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC10	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []

162.14.4.3 Transmitter specifications

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	162.9.4	Per Table 162–11	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD is not in same package as PCS	162.9.4	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	Output voltage specifications	93.8.1.3	Per Table 162–11	M	Yes []
TC4	ERL	162.9.4.8	Per Table 162–11	M	Yes []
TC5	Common-mode to differential-mode return loss	162.9.4.10	Per Table 162–11	M	Yes []
TC6	Common-mode to common-mode return loss	162.9.4.9	Per Table 162–11	M	Yes []
TC7	Steady-state voltage, v_f	162.9.4.1.2	Per Table 162–11	M	Yes []
TC8	Linear fit pulse peak (min)	162.9.4.1.2	Per Table 162–11	M	Yes []
TC9	Level separation mismatch ratio	162.9.4	Per Table 162–11	M	Yes []
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 162–11	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Absolute change in coefficients other than $c(\text{coef_sel})$	162.9.4.1.4	Per Table 162–11	M	Yes []
TC12	Coefficient range	162.9.4.1.5	Per Table 162–11	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120D.3.1.6	Per Table 162–11	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	162.9.4	Per Table 162–11	M	Yes []
TC15	Output jitter	162.9.4.7	Per Table 162–11	M	Yes []
TC16	Low-frequency AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 162–11	M	Yes []
TC17	Full-band AC common-mode peak-to-peak voltage	162.9.4.4	Per Table 162–11	M	Yes []

162.14.4.4 Receiver specifications

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Signaling rate range	162.9.5.1	Per Table 162–15	M	Yes []
RC2	Differential-mode to common-mode return loss at TP3	162.9.5.6	Per Table 162–15	M	Yes []
RC3	Input amplitude tolerance	162.9.5.2	Meets FEC symbol error ratio requirement under described conditions	M	Yes []
RC4	Interference tolerance	162.9.5.3	PHY meets FEC symbol error ratio requirement in all tests defined in Table 162–16 with the specified test setup and procedure	M	Yes []
RC5	Jitter tolerance	162.9.5.4	PHY meets FEC symbol error ratio requirement with the specified test setup and procedure	M	Yes []
RC6	ERL	162.9.5.5	Per Table 162–15	M	Yes []

162.14.4.5 Cable assembly specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CA1	Differential reference impedance	162.11.1	100 Ω	CBL:M	Yes [] N/A []
CA2	Minimum insertion loss	162.11.2	Per Equation (162–19)	CBL:M	Yes [] N/A []
CA3	Maximum insertion loss at 26.56 GHz	162.11.2	19.75 dB	CBL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CA4	ERL	162.11.3	Greater than or equal to 8.25 dB for cable assemblies that have a COM less than 4 dB	CBL:M	Yes [] N/A []
CA5	Differential-mode to common-mode input and output return loss	162.11.4	Per Equation (162–20)	CBL:M	Yes [] N/A []
CA6	Differential-mode to common-mode insertion loss	162.11.5	Per Equation (162–21)	CBL:M	Yes [] N/A []
CA7	Common-mode to common-mode return loss	162.11.6	Per Equation (162–22)	CBL:M	Yes [] N/A []
CA8	Cable assembly Channel Operating Margin (COM)	162.11.7	Greater than or equal to 3 dB for both Test 1 and Test 2, for any lane within the cable assembly	CBL:M	Yes [] N/A []
CA9	AC-coupling	162.11	3 dB cutoff frequency less than 50 kHz	CBL:M	Yes [] N/A []

162.14.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Environmental specifications	162.13	Conform to applicable requirements of 14.7	M	Yes []

163. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.1 Overview

This clause specifies the 100GBASE-KR1 PMD, the 200GBASE-KR2 PMD, the 400GBASE-KR4 PMD, and the baseband medium. The specifications for the three PMDs are similar, except for the number of lanes and associated parameters and the MDI.

There are two associated annexes. Annex 163A provides measurement methods and test points for backplane and chip-to-chip interfaces. Annex 163B provides information on an example test fixture.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 163–1, to the appropriate sublayers (as specified in Table 163–1, Table 163–2, and Table 163–3), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 163–1—Physical Layer clauses associated with the 100GBASE-KR1 PMD

Associated clause	100GBASE-KR1
81—RS	Required
81—CGMII ^a	Optional
82—PCS for 100GBASE-R	Required
91—RS-FEC	Required
152—Inverse RS-FEC	Optional ^b
161—RS-FEC-Int	Required
83—PMA for 100GBASE-R	Optional ^c
83A—CAUI-10	Optional
83D—CAUI-4 C2C	Optional
135—PMA for 100GBASE-P	Required
135D—100GAUI-4 C2C	Optional
135F—100GAUI-2 C2C	Optional
120F—100GAUI-1 C2C	Optional
73—Auto-Negotiation	Required

^a The CGMII is an optional interface. However, if the CGMII is not implemented, a conforming implementation behaves functionally as though the RS and CGMII were present.

^b Inverse RS-FEC is required to convert between RS-FEC and RS-FEC-Int (see 152.1.2).

^c PMA for 100GBASE-R is required when either CAUI-10 or CAUI-4 is used.

Table 163–2—Physical Layer clauses associated with the 200GBASE-KR2 PMD

Associated clause	200GBASE-KR2
117—RS	Required
117—200GMII ^a	Optional
118—200GMII extender	Optional
119—PCS for 200GBASE-R	Required
120—PMA for 200GBASE-R	Required
120B—200GAUI-8 C2C	Optional
120D—200GAUI-4 C2C	Optional
120F—200GAUI-2 C2C	Optional
73—Auto-Negotiation	Required

^a The 200GMII is an optional interface. However, if the 200GMII is not implemented, a conforming implementation behaves functionally as though the RS and 200GMII were present.

Table 163–3—Physical Layer clauses associated with the 400GBASE-KR4 PMD

Associated clause	400GBASE-KR4
117—RS	Required
117—400GMII ^a	Optional
118—400GMII extender	Optional
119—PCS for 400GBASE-R	Required
120—PMA for 400GBASE-R	Required
120B—400GAUI-16 C2C	Optional
120D—400GAUI-8 C2C	Optional
120F—400GAUI-4 C2C	Optional
73—Auto-Negotiation	Required

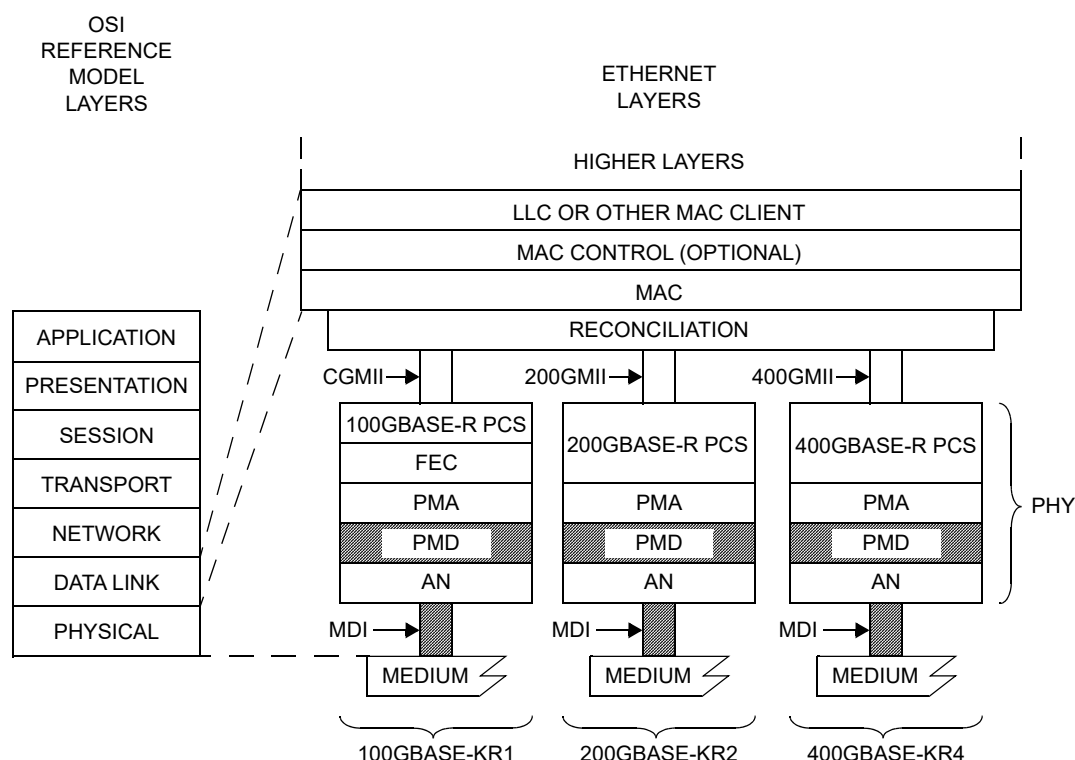
^a The 400GMII is an optional interface. However, if the 400GMII is not implemented, a conforming implementation behaves functionally as though the RS and 400GMII were present.

For the 100GBASE-KR1 PHY, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-10} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 10^{-10} after processing by the FEC (Clause 91 or Clause 161) and the PCS (Clause 82) if there are negligible errors due to other electrical interfaces (100GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 10^{-10} .

For the 200GBASE-KR2 and 400GBASE-KR4 PHYs, in order to support the required frame loss ratio (see 1.4.344) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 9.2×10^{-13} after processing by the PCS (Clause 119) if there are negligible errors due to other electrical interfaces (200GAUI-n or 400GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 9.2×10^{-13} .

A compliant input signal is a transmitter output of a compliant PHY that has passed through a compliant channel.

Figure 163–1 shows the relationship of the PMD and MDI to the ISO/IEC Open System Interconnection (OSI) reference model.



200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
AN = AUTO-NEGOTIATION
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL

MDI = MEDIUM DEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 163–1—100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

163.2 Conventions

Clause 163 describes three PMDs, 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, which have one, two, and four lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-KR1, $n = 2$ for 200GBASE-KR2, and $n = 4$ for 400GBASE-KR4.

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-KR1 PMD, 200GBASE-KR2 PMD, or 400GBASE-KR4 PMD.

Within this clause, the term FEC refers to either Clause 91 RS-FEC or the Clause 161 RS-FEC-Int for 100GBASE-KR1, or the RS-FEC within the Clause 119 PCS for 200GBASE-KR2 and 400GBASE-KR4.

163.3 PMD service interfaces

The service interfaces of the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PMDs are identical to those of the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 PMDs, respectively (see 162.3).

163.4 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with the PMD is required to support the AN service interface primitive AN_LINK.indication defined in 73.9. The requirements are the same as those specified in 162.4.

163.5 Delay constraints

The sum of the transmit and receive delays at one end of the link contributed by the PMD and AN including the medium in one direction shall be no more than the maximum delays listed in Table 163–4. It is assumed that the one-way delay through the medium is no more than 14 ns.

Table 163–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-KR1	4 096	8	40.96
200GBASE-KR2	8 192	16	40.96
400GBASE-KR4	16 384	32	40.96

^a One bit time is equal to 10 ps for 100GBASE-KR1, 5 ps for 200GBASE-KR2, and 2.5 ps for 400GBASE-KR4. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-KR1, 2.56 ns for 200GBASE-KR2, and 1.28 ns for 400GBASE-KR4. (See 31B.2 for the definition of pause_quanta.)

Descriptions of overall system delay constraints can be found in 80.4 for 100GBASE-KR1 and in 116.4 for 200GBASE-KR2 and 400GBASE-KR4.

163.6 Skew constraints

The Skew (relative delay) between the PCS or FEC lanes is kept within limits so that the information on the PCS or FEC lanes can be reassembled by the PCS or FEC. The Skew Variation is also limited to ensure that a given PCS or FEC lane always traverses the same physical lane.

163.6.1 Skew Constraints for 100GBASE-KR1

Skew and Skew Variation are defined in 80.5 and specified at the points SP1 to SP7 shown in [Figure 80–9](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns as defined by [135.5.3.5](#). Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns. Since the signal at the MDI represents a serial bit stream, there is no Skew Variation at this point.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns. Since the signal at the PMD service interface represents a serial bit stream, there is no Skew Variation at this point.

For more information on Skew and Skew Variation, see 80.5. The measurements of Skew and Skew Variation are defined in [89.7.2](#).

163.6.2 Skew Constraints for 200GBASE-KR2 and 400GBASE-KR4

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in [Figure 116–5](#).

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 43 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 54 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 134 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 145 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 116.5.

163.7 PMD MDIO function mapping

The PMD MDIO function mapping requirements for the PMDs defined in this clause are identical to those of 162.7.

163.8 PMD functional specifications

163.8.1 Link block diagram

One direction of a 100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link is shown in Figure 163–2. The PMDs on both ends of the link have connected ground references.

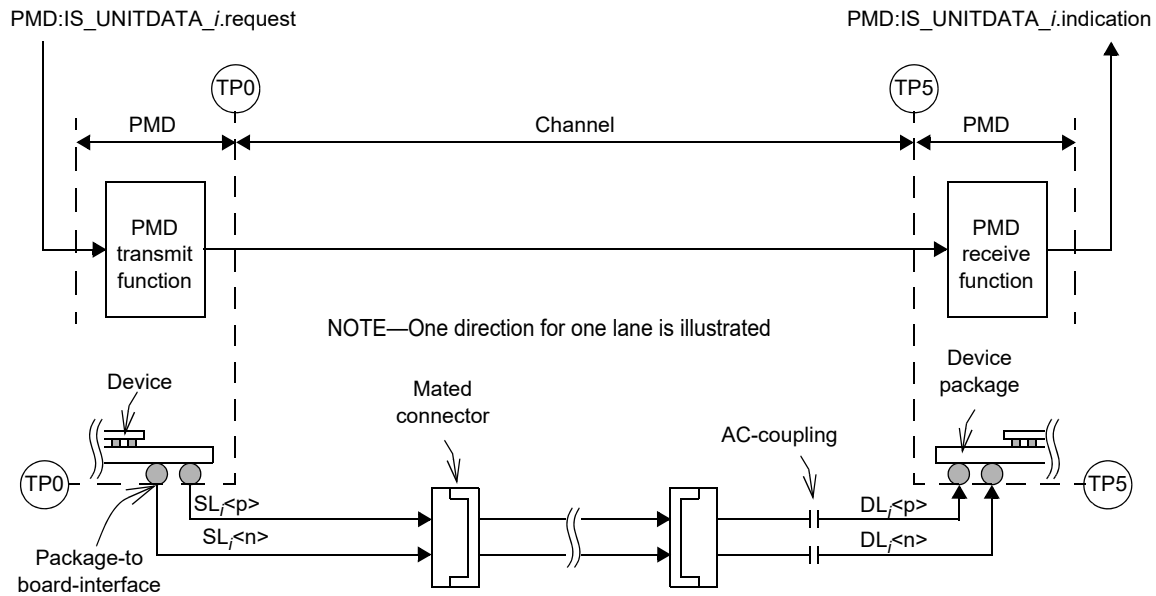


Figure 163–2—100GBASE-KR1, 200GBASE-KR2, or 400GBASE-KR4 link

163.8.2 PMD transmit function

The PMD transmit function specification is identical to that of 162.8.2 with the exception that electrical signals are delivered to the MDI, according to the transmit electrical specifications in 163.9.2.

163.8.3 PMD receive function

The PMD receive function specification is identical to that of 162.8.3 with the exception that electrical signals are received from the MDI, according to the receive electrical specifications in 163.9.3.

163.8.4 PMD global signal detect function

The PMD global signal detect function specification is identical to that of 162.8.4.

163.8.5 PMD lane-by-lane signal detect function

The PMD lane-by-lane signal detect function specification is identical to that of 162.8.5.

163.8.6 PMD global transmit disable function (optional)

The PMD global transmit disable function is optional. Its specification is identical to that of 162.8.6.

163.8.7 PMD lane-by-lane transmit disable function (optional)

The PMD lane-by-lane transmit disable function is optional. Its specification is identical to that of 162.8.7.

163.8.8 PMD fault function

The PMD fault function specification is identical to that of 162.8.8.

163.8.9 PMD transmit fault function (optional)

The PMD transmit fault function is optional. Its specification is identical to that of 162.8.9.

163.8.10 PMD receive fault function (optional)

The PMD receive fault function is optional. Its specification is identical to that of 162.8.10.

163.8.11 PMD control function

The PMD control function specification is identical to that of 162.8.11.

163.9 Electrical characteristics

163.9.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

163.9.2 Transmitter characteristics

The transmitter shall meet the specifications given in Table 163–5. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

Table 163–5—Summary of transmitter specifications at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
Differential pk-pk voltage (max) ^b Transmitter disabled Transmitter enabled	93.8.1.3	30 1200	mV mV
DC common-mode voltage (max) ^b	93.8.1.3	1	V
DC common-mode voltage (min) ^b	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	162.9.4.4	30	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	163.9.2.6	15	dB
Difference effective return loss, $dERL$ (min)	163.9.2.2	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	163.9.2.3	3.25	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163.9.2.5	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 163–5—Summary of transmitter specifications at TP0v (*continued*)

Parameter	Reference	Value	Units
Transmitter waveform			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at minimum state for $c(-3)$ (max)	162.9.4.1.5	−0.06	—
value at maximum state for $c(-2)$ (min)	162.9.4.1.5	0.12	—
value at minimum state for $c(-1)$ (max)	162.9.4.1.5	−0.34	—
value at minimum state for $c(0)$ (max)	162.9.4.1.5	0.5	—
value at minimum state for $c(1)$ (max)	162.9.4.1.5	−0.2	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	162.9.4.3	28	dB
Jitter (max) ^c			
J_{RMS}	162.9.4.7	0.023	UI
J_{3u03}	162.9.4.7	0.106	UI
J_{3u}	162.9.4.7	0.115	UI
Even-odd jitter, pk-pk	162.9.4.7	0.025	UI

^a For a PMD in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c J_{3u} , J_{RMS} , and even-odd jitter measurements are made with a single transmit equalizer setting selected to compensate for the loss of the transmitter package and TP0 to TP0v test fixture.

163.9.2.1 Transmitter test fixture

Unless otherwise noted, measurements of the transmitter are made at the output of a test fixture (TP0v) as shown in Figure 163–3 and described in Annex 163A. An example test fixture is described in Annex 163B.

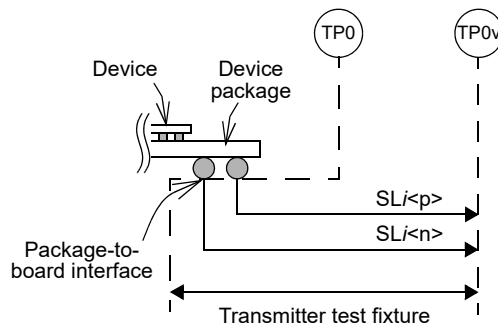


Figure 163–3—Transmitter test fixture and test points

163.9.2.1.1 Test fixture insertion loss

The insertion loss of the test fixture shall be between 1.7 dB and 5 dB at 26.56 GHz. The magnitude of the insertion loss deviation of the test fixture shall be less than or equal to 0.2 dB from 0.05 GHz to 26.56 GHz. Insertion loss deviation is calculated as specified in 93A.4, where T_i is 0.01 ns, and f_b and f_r values are taken from Table 163–11.

163.9.2.1.2 Test fixture effective return loss (ERL)

The ERL of the test fixture at TP0v is computed using the procedure in 93A.5 with the values in Table 163–6. Parameters that do not appear in Table 163–6 take values from Table 163–11.

Table 163–6—Test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

The ERL at TP0v shall be greater than or equal to 15 dB.

163.9.2.1.3 Test fixture common-mode to common-mode return loss

The common-mode to common-mode return loss of the test fixture shall be greater than or equal to 6 dB at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.2 Transmitter difference ERL

The difference ERL of the transmitter at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) in Table 163–5.

Table 163–7—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Tukey window flag	tw	1	—

163.9.2.3 Transmitter common-mode to common-mode return loss

The common-mode to common-mode return loss shall be greater than or equal to RL_{cc} (min) in Table 163–5 at all frequencies between 0.2 GHz and 40 GHz.

163.9.2.4 Difference steady-state voltage

The difference steady-state voltage of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with $N_v = 200$ and other parameter values specified in Table 163–11.

The difference steady-state voltage at TP0v shall meet the specification dV_f (min) in Table 163–5.

163.9.2.5 Difference linear fit pulse peak ratio

The difference linear fit pulse peak ratio of the transmitter at TP0v is computed using the procedure in 163A.3.2.1 with parameters specified in Table 163–11.

The difference linear fit pulse peak ratio at TP0v shall meet the specification dR_{peak} (min) in Table 163–5.

163.9.2.6 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is calculated using Equation (163–1).

$$SCMR = 20\log_{10}\left(\frac{V_{peak}}{VCM_{FB}}\right) \quad (163-1)$$

where

$SCMR$	is the signal to AC common-mode noise ratio in dB
V_{peak}	is defined in 162.9.4.1.2
VCM_{FB}	is the full-band peak-to-peak AC common-mode voltage defined by the method specified in 162.9.4.4 and measured with the transmitter equalization set to “no equalization”

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 163–5.

163.9.3 Receiver characteristics

Receiver electrical characteristics are specified at TP5v. The receiver shall meet the specifications given in Table 163–8.

Table 163–8—Summary of receiver specifications at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	163.9.3.1	53.125 ± 100 ppm	GBd
Difference effective return loss, $dERL$ (min)	163.9.3.3	–3	dB
Differential-mode to common-mode return loss, RL_{cd}	163.9.3.4	Equation (163–2)	dB
Interference tolerance	163.9.3.5	Table 163–9	—
Jitter tolerance	163.9.3.6	Table 162–17	—

163.9.3.1 Receiver signaling rate

A PHY shall comply with the receiver requirements of 163.9.3.5 and 163.9.3.6 for any signaling rate in the range specified in Table 163–8.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ± 50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ± 100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).

163.9.3.2 Receiver test fixture

Unless otherwise noted, measurements of the receiver are made at the input of a test fixture (TP5v) as shown in Figure 163–4.

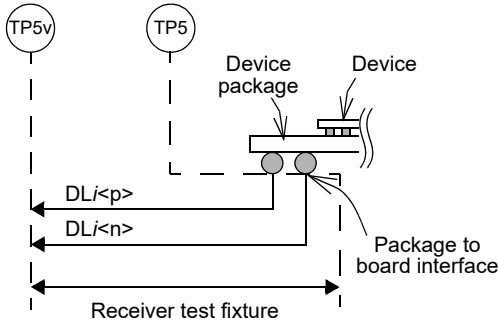


Figure 163–4—Receiver test fixture and test points

Receiver test fixture characteristics are the same as specified in 163.9.2.1.

163.9.3.3 Receiver difference ERL

The difference ERL of the receiver at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 163–7 and Table 163–11, and with the value of T_{fx} equal to twice the delay from TP5v to TP5.

The receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 163–8.

163.9.3.4 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (163–2). The differential-mode to common-mode return loss limit is illustrated by Figure 163–5.

$$RL_{cd}(f) \geq \begin{cases} 25 - 20(f/53.125) & 0.05 \leq f \leq 26.5625 \\ 15 & 26.5625 < f \leq 53.125 \end{cases} \quad (163-2)$$

where

$RL_{cd}(f)$ is the differential-mode to common-mode return loss at frequency f in dB
 f is the frequency in GHz

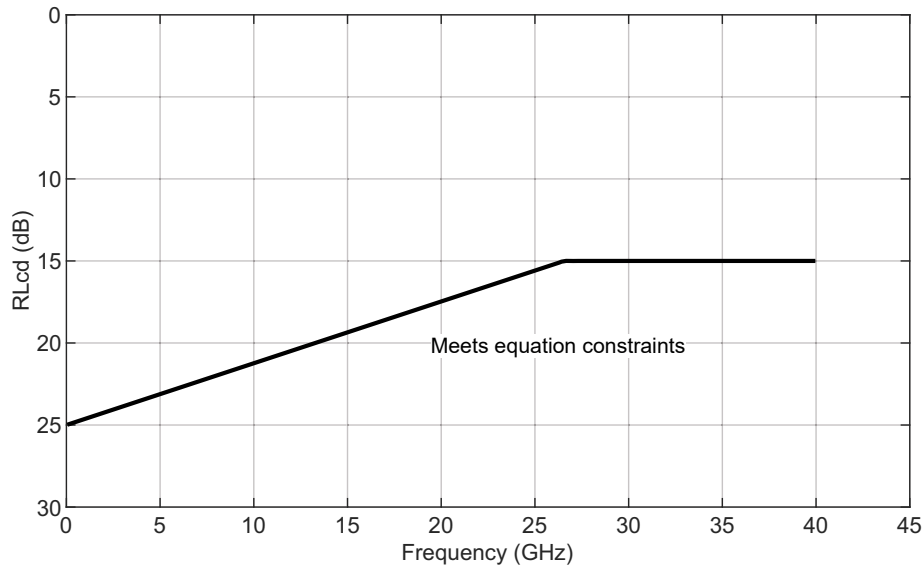


Figure 163-5—Receiver differential-mode to common-mode return loss limit

163.9.3.5 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#). The receiver on each lane shall meet the FEC symbol error ratio requirement with channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 163-9. The following additional considerations apply to the interference tolerance test.

- a) TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A and [Annex 93C](#).
- b) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see [93.8.1.3](#)) is less than or equal to 800 mV.
- c) The ERL of the test setup in [Figure 93C-4](#) measured at TP5 replica towards TPt meets the requirements in [163.10.3](#).
- d) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- e) For the calculation of test channel COM, the transmitter model is determined in one of the following ways.
 - If the transmitter is a device with known S-parameters and transition time T_r , these parameters should be used instead of the transmitter package model in [93A.1.2](#). T_r should be provided as the value at the input of the device S-parameters network, as defined in [120G.3.1.4](#) but with no observation filter.
 - If a calibrated instrument-grade transmitter is used, the TP0 to TP0a trace in [Figure 93C-2](#) and [Figure 93C-3](#) and TP0 to TP0a replica trace in [Figure 93C-4](#) are omitted and the transmitter device package model $S^{(tp)}$ is omitted from [Equation \(93A-3\)](#). The filtered voltage transfer function $H^{(k)}(f)$ calculated in [Equation \(93A-19\)](#) uses the filter $H_f(f)$ defined by [Equation \(93A-46\)](#), where T_r is the measured transmitter transition time (see [120G.3.1.4](#)).
 - If the transmitter is composed of a device with unknown S-parameters or unknown transition time, then the transmitter device package model $S^{(tp)}$ in [93A.1.2](#) is used, and T_r is determined from measurement at TP0v and the TP0 to TP0v S-parameters. The transmitter transition time (see [120G.3.1.4](#)) is measured at TP0v with transmit equalization turned off by setting coefficients to preset 1 values (see [162.9.4.1.3](#)). T_r is set as the value in [Equation \(93A-46\)](#) that

would result in the reference transition time $T_r^{(ref)}$, determined according to 163A.3.1.3 with f_b and A_v equal to values in Table 163–11, being equal to the measured transition time.

- f) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p = 29$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of $J3u$ and J_{RMS} using Equation (163–3) and Equation (163–4) respectively.
- g) Other COM parameters are set according to the values in Table 163–11.
- h) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 163–11 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- i) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
- j) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
- k) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

$$A_{DD} = \left(\frac{J3u}{2} + Q3d \sqrt{\left((Q3d^2 + 1) \times J_{RMS}^2 - \left(\frac{J3u}{2} \right)^2 \right)} \right) / (Q3d^2 + 1) \quad (163-3)$$

$$\sigma_{RJ} = \left(\frac{J3u}{2} - A_{DD} \right) / (Q3d) \quad (163-4)$$

where

$$Q3d = 3.0902$$

NOTE 1— $Q3d$ is an approximated solution of $Q(Q3d) = 1 \times 10^{-3}$, where the Q function is defined in Equation (95–1).

NOTE 2—Calculation of A_{DD} requires that $(Q3d^2 + 1) \times J_{RMS}^2 \geq \left(\frac{J3u}{2} \right)^2$. If this does not hold, a different transmitter should be used in the test setup.

Table 163–9—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-3}	—	—	10^{-3}	—	—
Insertion loss, IL_{dd} , at 26.5625 GHz ^b	13.5	14.5	—	27.5	28.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

163.9.3.6 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. TP0v (TP5v) replaces TP0a (TP5a) in Annex 93A, Annex 93C, and Annex 120D. The test channel meets the insertion loss requirement for Test 2 in Table 163–9. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 163.9.3.5, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items 3) through 7) in 93C.2, is at least 3 dB.
- For the COM parameter calibration described in 93C.2 item 7), the test channel transmitter J_{RMS} and J_{3u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 163–9 for each case in Table 162–17.

163.10 Channel characteristics

Table 163–10 provides a summary of the channel characteristics for 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, and references to the subclauses addressing each parameter.

163.10.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1 with the values in Table 163–11, where T_r is 7.5 ps for $H_t(f)$ as used in Equation (93A–19). COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2.

Table 163–10—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	163.10.1	3	dB
Maximum insertion loss, IL_{dd} , at 26.56 GHz (recommended)	163.10.2	28	dB
Minimum channel ERL	163.10.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	163.10.4	Equation (163–6)	dB
Differential-mode to common-mode insertion loss, IL_{cd}	163.10.5	Equation (163–7)	dB
Common-mode to differential-mode insertion loss, IL_{dc}	163.10.6	Equation (163–8)	dB
Maximum AC-coupling 3 dB corner frequency	163.10.7	50	kHz

Table 163–11—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	1.2×10^{-4}	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1	z_p	12	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Transmission line characteristic impedance	Z_c	87.5	Ω
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line 2 characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient	$c(-3)$	—	—
Minimum value		–0.06	
Maximum value		0	
Step size		0.02	
Transmitter equalizer, 2 nd pre-cursor coefficient	$c(-2)$	—	—
Minimum value		0	
Maximum value		0.12	
Step size		0.02	

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	–0.34 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	–0.2 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	–20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	–6 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	12	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to 6 $n = 7$ to N_b	$bb_{\max}(n)$	0.85 0.3 0.2 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 –0.03	—
Number of DFE floating tap banks	N_{bg}	3	—
Number of DFE floating taps per bank	N_{bf}	3	—
DFE maximum span including floating taps	N_f	40	UI
Normalized coefficient magnitude limit for DFE floating taps	$b_{g\max}$	0.05	—
DFE floating tap tail root-sum-of-squares limit	$\sigma_{t\max}$	0.02	—

Table 163–11—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
DFE floating tap tail starting position	N_{ts}	25	—
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	8.2×10^{-9}	V ² /GHz
Target detector error ratio	DER_0	10^{-4}	—

163.10.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (163–5).

$$ILdd(f) \leq \begin{cases} 0.693 + 2.161\sqrt{f} + 0.607f & 0.01 \leq f \leq 26.5625 \\ -19.12 + 1.773f & 26.5625 < f \leq 40 \end{cases} \quad (163-5)$$

where

$ILdd(f)$ is the insertion loss in dB at frequency f
 f is the frequency in GHz

The insertion loss limit is illustrated by Figure 163–6.

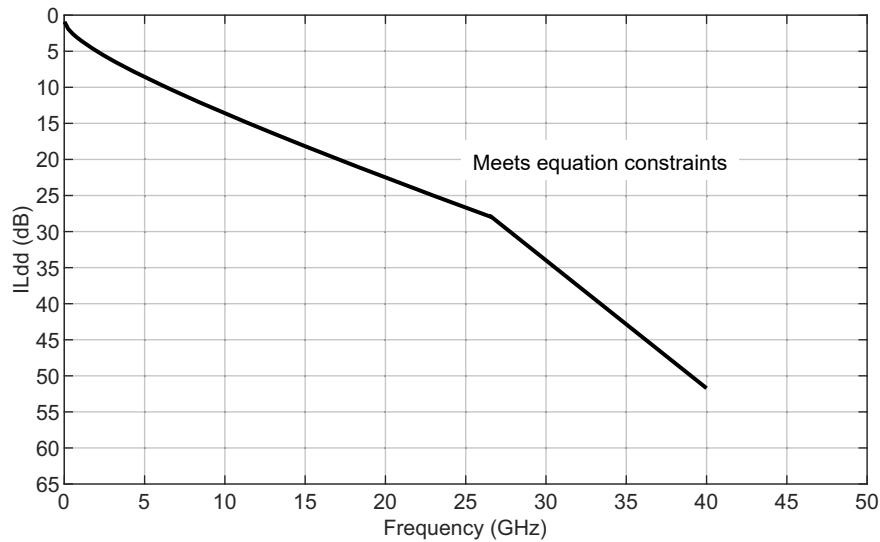


Figure 163–6—Channel insertion loss limit

163.10.3 Channel ERL

The ERL of the channel at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 163–12 and Table 163–11.

Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 163–12—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	3500	UI
Equalizer length associated with reflection signal	N_{bx}	21	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

163.10.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6).

$$RLcd(f) \geq \begin{cases} 22 - 10(f/26.56) & 0.05 \leq f < 26.56 \\ 15 - 3(f/26.56) & 26.56 \leq f \leq 40 \end{cases} \quad (163-6)$$

where

$RLcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode return loss at frequency f in dB

f is the frequency in GHz

The TP0 to TP5 channel differential-mode to common-mode return loss limit is illustrated in Figure 163–7.

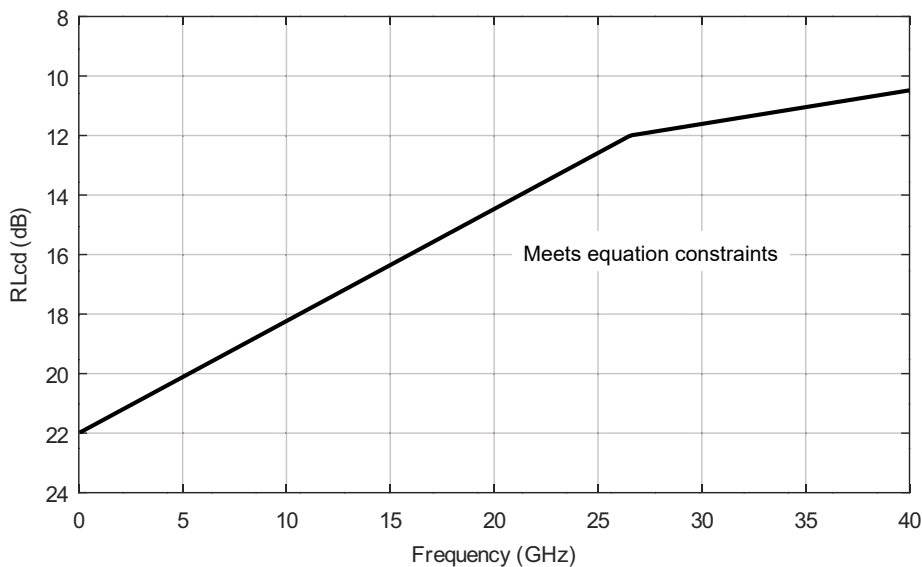


Figure 163-7— Channel differential-mode to common-mode return loss

163.10.5 Channel differential-mode to common-mode insertion loss

The difference between the TP0 to TP5 channel differential-mode to common-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163-7) as illustrated in Figure 163-8.

$$ILcd(f) - ILdd(f) \geq \left\{ \begin{array}{ll} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{array} \right\} \quad (163-7)$$

where

- $ILcd(f)$ is the TP0 to TP5 channel differential-mode to common-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB
- f is the frequency in GHz

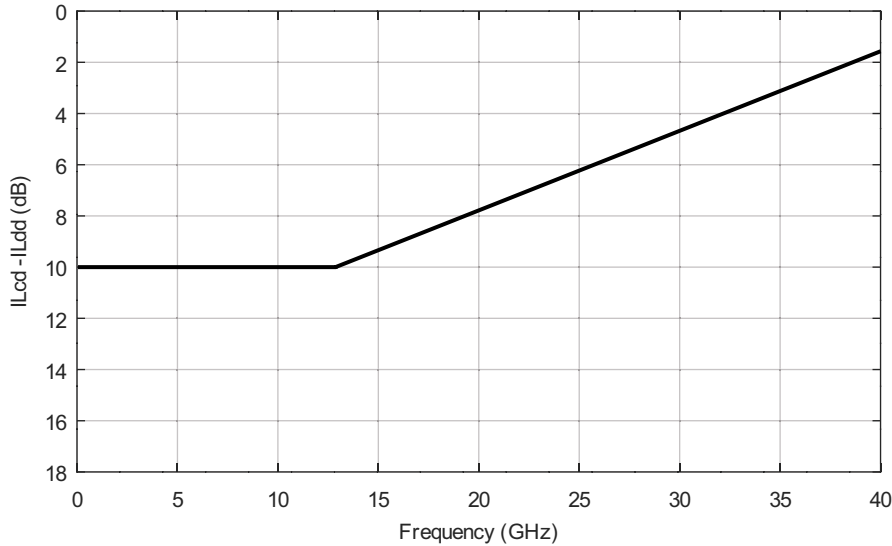


Figure 163–8—Channel differential-mode to common-mode insertion loss to insertion loss difference

163.10.6 Channel common-mode to differential-mode insertion loss

The difference between the TP0 to TP5 channel common-mode to differential-mode insertion loss and the TP0 to TP5 channel insertion loss shall meet Equation (163–8) illustrated in Figure 163–9.

$$ILdc(f) - ILdd(f) \geq \begin{cases} 10 & 0.05 \leq f < 12.89 \\ 14 - 0.3108f & 12.89 \leq f \leq 40 \end{cases} \quad (163-8)$$

where

- $ILdc(f)$ is the TP0 to TP5 channel common-mode to differential-mode insertion loss at frequency f in dB
- $ILdd(f)$ is the TP0 to TP5 insertion loss at frequency f in dB
- f is the frequency in GHz

163.10.7 AC-coupling

AC-coupling shall be implemented within the channel (between TP0 and TP5) using DC blocking capacitors. The low-frequency 3 dB cutoff of the channel shall be less than 50 kHz.

Systems with no AC-coupling within the channel are considered engineered links. It is the system integrator’s responsibility to verify that the transmitter and the receiver are compatible with the common-mode voltage differences that may exist in this configuration.

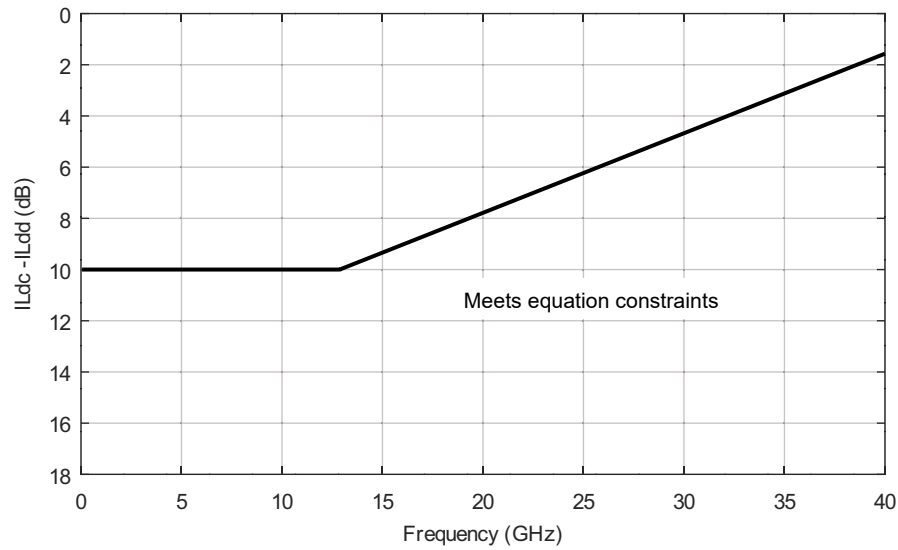


Figure 163-9—Channel common-mode to differential-mode insertion loss to insertion loss difference

163.11 MDI specifications

The MDI for the 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4 PHYs is an implementation-dependent direct electrical connection between the PMD and the medium. The MDI is composed of $2 \times n$ differential pairs, one pair for the transmit function and one pair for the receive function on each lane, marked by TP0 and TP5 in Figure 163-2.

163.12 Environmental specifications

All equipment subject to this clause shall conform to the applicable requirements of 93.10.

163.13 Protocol implementation conformance statement (PICS) proforma for Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4¹⁹

163.13.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

163.13.2 Identification

163.13.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

163.13.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Clause 163, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

¹⁹Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

163.13.3 Major capabilities/options

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*KR1	100GBASE-KR1 PMD	163.1	Can operate as a 100GBASE-KR1 PMD	O.1	Yes [] No []
*KR2	200GBASE-KR2 PMD	163.1	Can operate as a 200GBASE-KR2 PMD	O.1	Yes [] No []
*KR4	400GBASE-KR4 PMD	163.1	Can operate as a 400GBASE-KR4 PMD	O.1	Yes [] No []
PMA100	100GBASE-R PMA	163.1	Device implements Clause 135 PMA for 100GBASE-P	KR1:M	Yes [] N/A []
PMA200	200GBASE-P PMA	163.1	Device implements Clause 120 PMA for 200GBASE-R	KR2:M	Yes [] N/A []
PMA400	400GBASE-R PMA	163.1	Device implements Clause 120 PMA for 400GBASE-R	KR4:M	Yes [] N/A []
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C	163.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	163.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	!KR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII and RS	163.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
AN	Auto-Negotiation	163.1	Device implements Auto-Negotiation	M	Yes []
FEC100	100GBASE-R RS-FEC 100GBASE-P RS-FEC	163.1	Device implements Clause 91 and Clause 161 RS-FEC with RS(544,514)	KR1* !AUIPMD :M	Yes [] N/A []
PCS100	100GBASE-R PCS	163.1	Device implements Clause 82 100GBASE-R PCS	KR1* nGMII:M	Yes [] N/A []
PCS200	200GBASE-R PCS	163.1	Device implements Clause 119 200GBASE-R PCS	KR2* nGMII:M	Yes [] N/A []
PCS400	400GBASE-R PCS	163.1	Device implements Clause 119 400GBASE-R PCS	KR4* nGMII:M	Yes [] N/A []
PCSAN	PCS requirements for AN service interface	163.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
DC	Delay constraints	163.5	Conforms to constraints	M	Yes []
SC	Skew and Skew Variation constraints	163.6	Conforms to constraints	M	Yes []
*MD	MDIO capability	163.7	Device implements Clause 45 MDIO	O	Yes [] No []

Item ^a	Feature	Subclause	Value/Comment	Status	Support
*CHNL	Channel	163.10	Item marked with CHNL are channel specifications, these items are the only ones applicable to channels and are not applicable to PHYs.	O	Yes [] No []
*PCSPKG	PMD in same package as PCS	163.9.2		O	Yes [] No []

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

163.13.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-KR1, 200GBASE-KR2, and 400GBASE-KR4

163.13.4.1 Functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
PF1	Transmit function in DATA mode	162.8.2	Converts each symbol stream from the PMD service interface into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF2	Transmit function in TRAINING mode	162.8.2	Converts each symbol stream generated by the PMD control function into an electrical signal as specified, and delivers it to the MDI, with the electrical specifications in 163.9.2	M	Yes []
PF4	Receive function	162.8.3	Converts each electrical signal from the MDI into a symbol stream as specified, and delivers it to the PMD service interface	M	Yes []
PF5	Lane-by-lane signal detect	162.8.5	Set to one if training is disabled, otherwise set according to signal_detect	M	Yes []
PF6	Global PMD transmit disable	162.8.6	Function is implemented as specified	O	Yes [] No []
PF7	Lane-by-lane PMD transmit disable	162.8.7	Function is implemented as specified	O	Yes [] No []
PF8	PMD fault	162.8.8	PMD_fault variable mapped to the fault bit as specified in 45.2.1.2.3	MD:M	Yes [] N/A []
PF9	PMD transmit fault	162.8.9	Implemented, PMD_transmit_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.4	MD:O	Yes [] No [] N/A []
PF10	PMD receive fault	162.8.10	Implemented, PMD_receive_fault variable mapped to the Transmit fault bit as specified in 45.2.1.7.5	MD:O	Yes [] No [] N/A []

163.13.4.2 PMD control function

Item	Feature	Subclause	Value/Comment	Status	Support
PC1	PMD control function	162.8.11	Implemented as specified, one instance for each lane, operating independently	M	Yes []
PC2	Training pattern	136.8.11.1.3	Each lane implements four generator polynomials defined in Table 136–8	M	Yes []
PC3	Training pattern	136.8.11.1.3	State set to the value of seed <i>i</i> at the start of the training pattern	M	Yes []
PC4	Control field structure	162.8.11	As shown in 162.8.11	M	Yes []
PC5	Receiver frame lock bit	136.8.11.3.3	Initially set to zero, not set to 1 until local_tf_lock is true	M	Yes []
PC6	Initial condition setting	136.8.11.4.1	When requested, set according to the request, with values per Table 162–12	M	Yes []
PC7	Handshake timing	136.8.11.6	When the transmitted frame lock bit is 1, acknowledge requests within less than 2 ms	M	Yes []
PC8	Transmit precoded data	136.8.11.7.5	PMD causes adjacent PMA to use or not use precoding on transmitted data according to modulation and precoding status bit	M	Yes []
PC9	Receive precoded data	136.8.11.7.5	PMD informs adjacent PMA about precoding of received data according to modulation and precoding request bit	M	Yes []
PC10	Training	136.8.11.7.1	The value of use_quiet_in_training is TRUE.	M	Yes []

163.13.4.3 Transmitter characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMD in same package as PCS	163.9.2	Per Table 163–5	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMD not in same package as PCS	163.9.2	Derived from input to PMD transmit function	!PCSP KG:M	Yes [] N/A []
TC3	TP0 to TP0v test fixture	163.9.2.1	Meets specifications	M	Yes []
TC4	Output voltage specifications	93.8.1.3	Per Table 163–5	M	Yes []
TC5	Difference ERL at TP0v	163.9.2.2	Per Table 163–5	M	Yes []
TC6	Common-mode to common-mode return loss	93.8.1.4	Per Table 163–5	M	Yes []
TC7	Difference steady-state voltage	163.9.2.4	Per Table 163–5	M	Yes []
TC8	Difference linear fit pulse peak ratio	163.9.2.5	Per Table 163–5	M	Yes []
TC9	Level separation mismatch ratio	163.9.2	Per Table 163–5	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC10	Absolute coefficient step size	162.9.4.1.4	Per Table 163–5	M	Yes []
TC11	Coefficient range	162.9.4.1.5	Per Table 163–5	M	Yes []
TC12	Signal-to-noise-and-distortion ratio	163.9.2	Per Table 163–5	M	Yes []
TC13	Output jitter	162.9.4.7	Per Table 163–5	M	Yes []
TC14	Transmitter signaling rate	163.9.2	Per Table 163–5	M	Yes []
TC15	Low-frequency peak-to-peak AC common-mode voltage	163.9.2	Per Table 163–5	M	Yes []
TC16	Signal to AC common-mode noise ratio	163.9.2	Per Table 163–5	M	Yes []
TC17	Signal-to-residual-intersymbol-interference ratio	163.9.2	Per Table 163–5	M	Yes []

163.13.4.4 Receiver characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	TP5 to TP5v test fixture	163.9.3.2	Meets specifications	M	Yes []
RC2	Difference ERL at TP5v	163.9.3.3	Greater than or equal to –3 dB	M	Yes []
RC3	Input differential-mode to common-mode return loss	163.9.3.4	Meets constraints of Equation (163–2)	M	Yes []
RC4	Interference tolerance	163.9.3.5	PHY meets FEC symbol error ratio requirement in all tests defined in Table 163–9	M	Yes []
RC5	Jitter tolerance	163.9.3.6	PHY meets FEC symbol error ratio requirement in all cases defined in Table 162–17	M	Yes []

163.13.4.5 Channel characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	ERL	163.10.3	Greater than or equal to 9.7 dB	CHNL:M	Yes [] N/A []
CC2	Channel Operating Margin (COM)	163.10	Greater than or equal to 3 dB	CHNL:M	Yes [] N/A []
CC3	AC-coupling	93.9.4	Between TP0 and TP5, 3 dB cutoff frequency less than 50 kHz	CHNL:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
CC4	Channel differential-mode to common-mode return loss	163.10.4	Per Equation (163–6)	CHNL:M	Yes [] N/A []
CC5	Channel differential-mode to common-mode insertion loss	163.10.5	Per Equation (163–7)	CHNL:M	Yes [] N/A []
CC6	Channel common-mode to differential-mode insertion loss	163.10.6	Per Equation (163–8)	CHNL:M	Yes [] N/A []

163.13.4.6 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	Environmental specifications	163.12	Conform to applicable requirements of 93.10	M	Yes []

Clause 167 was added to IEEE Std 802.3-2022 by IEEE Std 802.3db-2022.

167. Physical Medium Dependent (PMD) sublayer and medium, type 100GBASE-VR1, 200GBASE-VR2, 400GBASE-VR4, 100GBASE-SR1, 200GBASE-SR2, and 400GBASE-SR4

167.1 Overview

Change Table 167–1 and Table 167–2 as follows (some unchanged rows not shown):

Table 167–1—Physical Layer clauses associated with the 100GBASE-VR1 and 100GBASE-SR1 PMDs

Associated clause	100GBASE-VR1, 100GBASE-SR1
...	
83E—CAUI-4 C2M	Optional
120F—100GAUI-1 C2C	<u>Optional</u>
120G—100GAUI-1 C2M	<u>Optional</u>
135—PMA for 100GBASE-P	Required
...	

Table 167–2—Physical Layer clauses associated with the 200GBASE-VR2, 400GBASE-VR4, 200GBASE-SR2, and 400GBASE-SR4 PMDs

Associated clause	200GBASE-VR2, 200GBASE-SR2	400GBASE-VR4, 400GBASE-SR4	Status
...			
120E	200GAUI-4 C2M	400GAUI-8 C2M	Optional
120F	<u>200GAUI-2 C2C</u>	<u>400GAUI-4 C2C</u>	<u>Optional</u>
120G	<u>200GAUI-2 C2M</u>	<u>400GAUI-4 C2M</u>	<u>Optional</u>
78	Energy-Efficient Ethernet		Optional

Annex 93A

(normative)

Specification methods for electrical channels

93A.1 Channel Operating Margin

Change Table 93A–1 as follows (some unchanged rows not shown):

Table 93A–1—COM parameters

Parameter	Reference	Symbol	Units
Signaling rate	93A.1.1	f_b	GBd
Maximum start frequency	93A.1.1	$f_{\min}$	GHz
Maximum frequency step	93A.1.1	Δf	GHz
Device package model Single-ended device capacitance ^a Single-ended device series inductance ^a Single-ended bump capacitance ^a Transmission line length ^a Single-ended package capacitance at package-to-board interface ^a Transmission line characteristic impedance ^a Transmission line 2 length ^a Transmission line parameter, a_1 ^b Transmission line parameter, a_2 ^b Transmission line 2 characteristic impedance ^a	93A.1.2	C_d L_s C_b z_p C_p Z_c z_{p2} a_1 a_2 Z_{c2}	nF nH nF mm nF Ω mm ns ^{1/2} /mm ns/mm Ω
Single-ended reference resistance	93A.1.2	R_0	Ω
Single-ended termination resistance	93A.1.3	R_d	Ω
Receiver 3 dB bandwidth	93A.1.4.1	f_r	GHz
Transmitter equalizer, minimum cursor coefficient	93A.1.4.2	$c(0)$	—
Transmitter equalizer, 3 rd pre-cursor coefficient ^c Minimum value Maximum value Step size	93A.1.4.2	$c(-3)$	— — —
Transmitter equalizer, 2 nd pre-cursor coefficient ^c Minimum value Maximum value Step size	93A.1.4.2	$c(-2)$	— — —
Transmitter equalizer, pre-cursor coefficient Minimum value Maximum value Step size	93A.1.4.2	$c(-1)$	— — —

Table 93A-1—COM parameters (*continued*)

Parameter	Reference	Symbol	Units
Transmitter equalizer, post-cursor coefficient	93A.1.4.2	$c(1)$	—
Minimum value			—
Maximum value			—
Step size			—
...			
Decision feedback equalizer (DFE) length	93A.1.6	N_b	UI
Normalized DFE coefficient magnitude limit ^d	93A.1.6	$b_{\max}(n)$	—
Normalized DFE coefficient maximum limit ^d	93A.1.6	$bb_{\max}(n)$	==
Normalized DFE coefficient minimum limit ^d	93A.1.6	$bb_{\min}(n)$	==
Number of DFE floating tap banks ^d	93A.1.6	N_{bg}	==
Number of DFE floating taps per bank ^d	93A.1.6	N_{bf}	==
DFE maximum span including floating taps ^d	93A.1.6	N_f	UI
Normalized coefficient magnitude limit for DFE floating taps ^d	93A.1.6	$b_{g\max}$	==
DFE floating tap tail root-sum-of-squares limit ^d	93A.1.6.1	$\sigma_{\max}$	==
DFE floating tap tail starting position ^d	93A.1.6.1	N_{ts}	==
Random jitter, RMS	93A.1.6	σ_{RJ}	UI
...			

^aSome clauses that invoke this method do not provide a value for C_d , L_s , C_{b1} , z_{p1} , C_{p2} , Z_{c2} , z_{p2} , or Z_{c2} . See 93A.1.2.

^bSome clauses that invoke this method do not provide a value for a_1 or a_2 . See 93A.1.2.3.

^cSome clauses that invoke this method do not provide a value for $c(-2)$ or $c(-3)$. See 93A.1.6.

^dSome clauses that invoke this method do not provide a value for $b_{\max}(n)$, $bb_{\max}(n)$, $bb_{\min}(n)$, N_{bg} , N_{bf} , N_f , $b_{g\max}$, $\sigma_{\max}$, or N_{ts} . See 93A.1.6.

Change Table 93A–2 as follows (some unmodified rows are not shown):

Table 93A–2—Physical Layer specifications that employ COM

Physical Layer	Parameter values
...	
100GBASE-CR2 (Clause 136)	Table 136–18
100GBASE-CR1 (Clause 162)	Table 162–20
100GBASE-KR4 (Clause 93)	Table 93–8
100GBASE-KP4 (Clause 94)	Table 94–17
100GBASE-KR2 (Clause 137)	Table 137–6
100GBASE-KR1 (Clause 163)	Table 163–11
CAUI-4 C2C (Annex 83D)	Table 83D–6
100GAUI-4 C2C (Annex 135D)	Table 83D–6
100GAUI-2 C2C (Annex 135F)	Table 120D–8
100GAUI-1 C2C (Annex 120F)	Table 120F–8
200GBASE-CR4 (Clause 136)	Table 136–18
200GBASE-CR2 (Clause 162)	Table 162–20
200GBASE-KR4 (Clause 137)	Table 137–6
200GBASE-KR2 (Clause 163)	Table 163–11
200GAUI-8 C2C (Annex 120B)	Table 83D–6 ^a
200GAUI-4 (Annex 120D)	Table 120D–8
200GAUI-2 C2C (Annex 120F)	Table 120F–8
400GBASE-CR4 (Clause 162)	Table 162–20
400GBASE-KR4 (Clause 163)	Table 163–11
400GAUI-16 (Annex 120B)	Table 83D–6 ^a
400GAUI-8 (Annex 120D)	Table 120D–8
400GAUI-4 C2C (Annex 120F)	Table 120F–8

^aWith the exceptions given in 120B.4.

93A.1.2 Transmitter and receiver device package models

93A.1.2.1 Cascade connection of two-port networks

Change the last sentence in 93A.1.2.1 as follows:

For the purpose of this annex, this set of operations is referred to using the shorthand notation $S^{(z)} = \text{cascade}(S^{(x)}, S^{(y)})$. The shorthand notation $S^{(z)} = \text{cascade}(S^{(w)}, S^{(x)}, S^{(y)})$ is equivalent to $S^{(z)} = \text{cascade}(S^{(w)}, \text{cascade}(S^{(x)}, S^{(y)}))$. Following this approach, the notation can be extended to an arbitrary number of components.

93A.1.2.2 Two-port network for a shunt capacitance

Change the second paragraph in 93A.1.2.2 as follows:

The scattering parameters for the device capacitance C_d are denoted as $S^{(d)}=S(C_d)$. the scattering parameters for the device-to-package interface capacitance are denoted as $S^{(b)}=S(C_b)$, and the scattering parameters for the board capacitance C_p are denoted as $S^{(p)}=S(C_p)$. When a value for C_b is not provided by the clause that invokes this method, C_b is set to 0.

Insert 93A.1.2.2a after 93A.1.2.2:

93A.1.2.2a Two-port network for a series inductance

The scattering parameters for a series inductance with value L are defined by Equation (93A–9a).

$$S(L) = \frac{1}{2 + (j\omega L)/R_0} \begin{bmatrix} (j\omega L)/R_0 & 2 \\ 2 & (j\omega L)/R_0 \end{bmatrix} \quad (93A-9a)$$

The scattering parameters for the series inductance L_s are denoted as $S^{(s)}(L_s)$. When a value for L_s is not provided by the clause that invokes this method, L_s is set to 0.

93A.1.2.3 Two-port network for the package transmission line

Change the first paragraph of 93A.1.2.3 as follows:

The scattering parameters for the package transmission line model are a function of the complex propagation coefficient defined by Equation (93A–9), Equation (93A–10), and Equation (93A–11) and the reflection coefficient defined by Equation (93A–12). The values of the parameters that appear in these equations are defined in Table 93A–3 unless alternative values are provided by the clause that invokes this method. Where a value for Z_c is not provided by the clause that invokes this method, it takes the value 78.2 Ω . The units of f are GHz.

Insert the following at the end of 93A.1.2.3:

For clauses that use a second package transmission line segment described by parameters z_{p2} and Z_{c2} , the scattering parameters for the second transmission line are defined by Equation (93A–12), Equation (93A–13), and Equation (93A–14) where values for z_p and Z_c in the equations are provided by z_{p2} and Z_{c2} .

Change the title of Table 93A–3 as follows:

Table 93A–3—Transmission line model parameters and default values

93A.1.2.4 Assembly of transmitter and receiver device package models

Change 93A.1.2.4 as follows:

The scattering parameters for the transmitter device package model $S^{(tp)}$ are the result of the cascade connection of the ~~device capacitance model~~, package transmission line, and board capacitance as shown in Figure 93A–2 and defined by Equation (93A–15) and Equation (93A–15a).

$$\cancel{S^{(tp)} = \text{cascade}(\text{cascade}(S^{(d)}, S^{(l)}), S^{(p)})} \quad S^{(td)} = \text{cascade}(S^{(d)}, S^{(s)}, S^{(b)}) \quad (93A-15)$$

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(p)})} \quad (93A-15a)$$

Similarly, the scattering parameters for the receiver device package model $S^{(rp)}$ are the result of the cascade connection of the board capacitance, package transmission line, and ~~device capacitance model~~ as shown in Figure 93A–2 and defined by Equation (93A–16) and Equation (93A–16a).

$$\cancel{S^{(rp)} = \text{cascade}(\text{cascade}(S^{(p)}, S^{(l)}), S^{(d)})} \quad S^{(rd)} = \text{cascade}(S^{(b)}, S^{(s)}, S^{(d)}) \quad (93A-16)$$

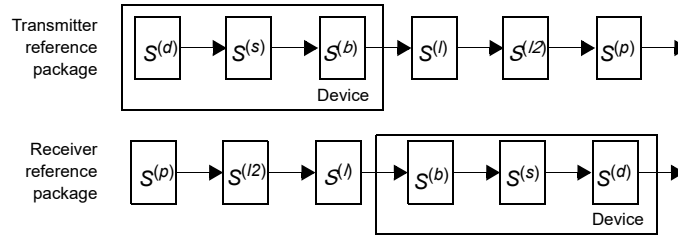
$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l)}, S^{(rd)})} \quad (93A-16a)$$

The transmitter and receiver device package model scattering parameters for clauses that include a second transmission line segment are computed using Equation (93A–16b) and Equation (93A–16c), respectively.

$$\underline{S^{(tp)} = \text{cascade}(S^{(td)}, S^{(l)}, S^{(l2)}, S^{(p)})} \quad (93A-16b)$$

$$\underline{S^{(rp)} = \text{cascade}(S^{(p)}, S^{(l2)}, S^{(l)}, S^{(rd)})} \quad (93A-16c)$$

Insert Figure 93A–2 at the end of 93A.1.2.4 as follows:



NOTE— For some PMDs, $S^{(l/2)}$ is not included in the package model.

$S^{(d)}$ = device capacitance S-parameter
 $S^{(s)}$ = device series inductance S-parameter
 $S^{(b)}$ = bump capacitance S-parameter
 $S^{(l)}$ = package transmission line S-parameter
 $S^{(l/2)}$ = package transmission line 2 S-parameter
 $S^{(p)}$ = package capacitance S-parameter

Figure 93A–2—Reference package models

93A.1.6 Determination of variable equalizer parameters

Change the first paragraph of 93A.1.6 as follows:

COM is a function of the variables $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} . The following procedure is used to determine the values of these variables that are used to calculate COM.

- Compute the pulse response $h^{(k)}(t)$ of each signal path k for a given $c(-3)$, $c(-2)$, $c(-1)$, $c(1)$, g_{DC} , and g_{DC2} using the procedure defined in 93A.1.5.
- Define t_s to be the time that satisfies Equation (93A–25). If there are multiple values of t_s that satisfy the equation, then the first value prior to the peak of $h^{(0)}(t)$ is selected. The ~~fixed-tap~~ coefficients of the ~~decision feedback equalizer~~ DFE $b(n)$ are computed as shown in Equation (93A–26). If N_b is 0, then the $b(n)$ is considered to be zero for all n . If $bb_{\min}(n)$ and $bb_{\max}(n)$ are not provided by the clause that invokes this method then $bb_{\min}(n)$ is set to $-b_{\max}(n)$ and $bb_{\max}(n)$ is set to $b_{\max}(n)$.
- Compute the location and values of floating-tap coefficients of the DFE to minimize the residual ISI using the method described in 93A.1.6.1. If N_{bg} , N_{bf} , N_f , and $b_{g\max}$ are not specified then no floating taps are used and N_f takes the value of N_b from the referring clause.
- ~~d)~~ Define A_s to be $R_{LM}h^{(0)}(t_s)/(L-1)$.
- ~~e)~~ Compute σ_{TX}^2 per Equation (93A–30) and Equation (93A–29). This represents the noise output from the transmitter.
- ~~f)~~ Compute $h_{ISI}(n)$ per Equation (93A–27). This represents the residual intersymbol interference (ISI) after decision feedback equalization. The corresponding ISI amplitude variance σ_{ISI}^2 is computed per Equation (93A–31) and Equation (93A–29).
- ~~g)~~ Compute the slope of the pulse response of the victim path $h_j(n)$ as shown in Equation (93A–28). The variance of the amplitude error due to timing jitter σ_j^2 is computed per Equation (93A–32) and Equation (93A–29).
- ~~h)~~ The variance of the amplitude for path k is given by Equation (93A–33) where the phase index m can assume any integer value from 0 to $M-1$. Denote the value of m that maximizes the variance for path k as i . The variance of the amplitude for the combination of all crosstalk paths σ_{XT}^2 is then computed using Equation (93A–34), which is the sum of the maximum variances for the individual paths $k=1$ to $K-1$.

- i) ~~h)~~ Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 referred to the receiver noise filter input per Equation (93A–35).
- j) ~~i)~~ Compute the figure of merit (FOM) per Equation (93A–36).

Replace Equation (93A–26) in 93A.1.6 with the following equation:

$$b(n) = \begin{cases} bb_{\min}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) < bb_{\min}(n) \\ bb_{\max}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) > bb_{\max}(n) \\ h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) & \text{otherwise} \end{cases} \quad (93A-26)$$

Replace Equation (93A–27) in 93A.1.6 with the following equation:

$$h_{ISI}(n) = \begin{cases} 0 & n = 0 \\ h^{(0)}(t_s + nT_b) - h^{(0)}(t_s)b(n) & 1 \leq n \leq N_f \\ h^{(0)}(t_s + nT_b) & \text{otherwise} \end{cases} \quad (93A-27)$$

Insert 93A.1.6.1 at the end of 93A.1.6 as follows:

93A.1.6.1 Determination of DFE floating tap locations and magnitudes

The following procedure is used to determine the bank location and tap coefficient magnitudes for each bank of DFE floating taps.

First, the locations of the floating banks are determined using the following steps:

- a) Potential locations for the first tap of a floating bank start at the first post-cursor position after the fixed taps, $N_b + 1$, and end at the specified maximum location minus the number of taps in the bank plus one, $N_f - N_{bf} + 1$. Set the initial values of the provisional coefficients, defined as $b'(n)$, to zero for $N_b + 1 \leq n \leq N_f$.
- b) Define σ_{DFE}^2 to be the sum-of-squares of the tap coefficients for a bank, according to Equation (93A–37a). Calculate σ_{DFE}^2 for each potential bank location, where the coefficient magnitudes are determined using Equation (93A–26). When performing the calculation, if any tap of the floating bank that is being placed overlaps with one or more taps of a previously placed bank, then remove that location from consideration.
- c) Place the floating bank at the location that has the maximum σ_{DFE}^2 . The coefficient magnitudes calculated in b) become the provisional coefficients, $b'(n)$, for the bank, subject to the tail constraints in the steps that follow.
- d) Repeat steps a) through c) for each bank.

After placing the floating banks, the coefficient magnitudes for all floating taps are computed as follows:

- e) For all floating taps that are located between $N_b + 1$ and the start of the ISI tail, $N_{ts} - 1$, set $b(n) = b'(n)$.
- f) For all floating taps that are located between N_{ts} and N_f , compute the tail coefficient root-sum-of-squares, σ_t , which is defined in Equation (93A–37b). Determine the final tap coefficient magnitudes, $b(n)$, by applying the tail weight constraint defined in Equation (93A–37c).

$$\sigma_{DFE}^2 = \sum_{k=1}^{N_{bf}} [b'(N_b + k)]^2 \quad (93A-37a)$$

$$\sigma_t^2 = \sum_{k=N_{ts}}^{N_f} [b'(k)]^2 \quad (93A-37b)$$

$$b(n) = \begin{cases} b'(n) & \sigma_t \leq \sigma_{tmax} \\ b'(n) \frac{\sigma_{tmax}}{\sigma_t} & \sigma_t > \sigma_{tmax} \end{cases} \quad (93A-37c)$$

93A.5 Effective Return Loss (ERL)

Change Table 93A-4 as follows (some unchanged rows not shown):.

Table 93A-4—ERL parameters

Parameter	Reference	Symbol	Units
...			
Target detector error ratio	93A.1.7	DER_0	—
Time-gated propagation delay	93A.5.2	T_{fx}	ns
Tukey window flag	93A.5.1	tw	—

93A.5.1 Pulse time-domain reflection signal

Change the second paragraph of 93A.5.1 as follows:

The filtered return loss, $H_{ii}(f)$, is defined by Equation (93A-58). When the parameter tw is equal to 1, $H_{tw}(f)$ is defined by Equation (93A-58a). When the value of tw is 0 or is not provided by the clause that invokes this method, $H_{tw}(f) = 1$.

Replace Equation (93A-58) with the following equation:

$$H_{ii}(f) = H_i(f) s_{ii}(f) H_r(f) H_{tw}(f) \quad (93A-58)$$

Insert new Equation (93A-58a) and Equation (93A-58b) after Equation (93A-58):

$$H_{tw}(f) = \begin{cases} 1 & f \leq f_r \\ 0.5 \left(1 - \cos \left(\frac{2\pi(f-f_b)}{f_{per}} \right) \right) & f_r < f \leq f_b \\ 0 & f_b < f \end{cases} \quad (93A-58a)$$

$$f_{per} = 2(f_b - f_r) \quad (93A-58b)$$

Change the variable list after Equation (93A-58), now after Equation (93A-58b), in 93A.5.1 as follows:

where

f	is the frequency in GHz
$H_r(f)$	is defined by Equation (93A-20)
$H_t(f)$	is defined by Equation (93A-46)
f_{per}	<u>is defined by Equation (93A-58b)</u>
i	is the port index of the scattering parameters, 1 or 2
f_r	<u>is the receiver 3 dB bandwidth in GHz</u>
$H_{tw}(f)$	<u>is the Tukey window function</u>

93A.5.2 Effective reflection waveform

Change the definition of T_{fx} in the variable list after Equation (93A-62) as follows:

T_{fx}	is twice the propagation delay in ns associated with the test fixture, obtained by measurement or inspection, <u>unless its value is specified by the clause that invokes this method</u>
----------	---

Annex 120A

(informative)

200 Gb/s and 400 Gb/s PMA sublayer partitioning examples

Insert new subclause 120A.5 at the end of Annex 120A as follows:

120A.5 Partitioning examples supporting 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4

Figure 120A–8 depicts an example of 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4 PMA layering with a single 200GAUI-*n* or 400GAUI-*n* interface.

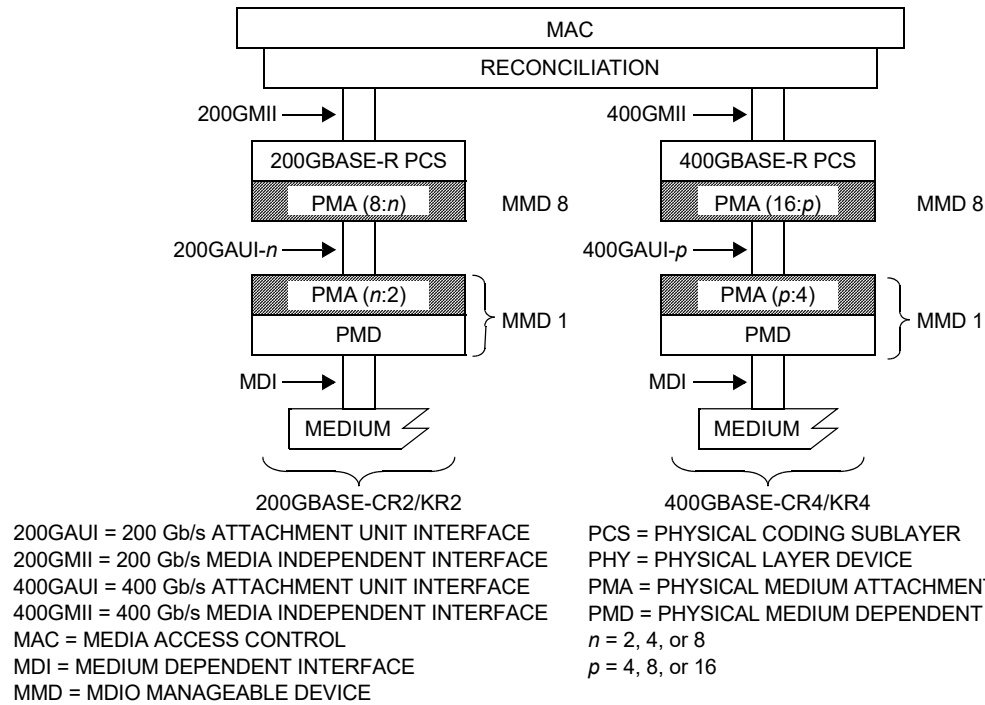


Figure 120A–8—Example 200GBASE-CR2/KR2 and 400GBASE-CR4/KR4 PMA layering with single 200GAUI-2/4/8 or 400GAUI-4/8/16 C2C interface

Insert new annexes Annex 120F and Annex 120G corresponding to Clause 120 as follows:

Annex 120F

(normative)

Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C). The 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces provide electrical characteristics and associated compliance points, which can optionally be used when designing systems with electrical interconnect of approximately 25 cm in length.

100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C are physical instantiations of the PMA service interface between the FEC (for 100GBASE-P PHYs) or PCS (for 200GBASE-R and 400GBASE-R PHYs) and the PMD, as described in 135.1.4 and 120.1.4. Figure 120F–1 shows example relationships of the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces to the ISO/IEC Open System Interconnection (OSI) reference model.

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2C bidirectional link is described in terms of a C2C transmitter, a C2C channel, and a C2C receiver. Figure 120F–2 depicts a typical C2C application. The C2C components at both ends of the link have connected ground references.

The C2C interface is composed of independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, or 400GAUI-4 data path contains one, two, or four, respectively, differential lanes, which are AC-coupled.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C use PAM4 signaling. The nominal signaling rate for each lane is 53.125 GBd.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C channel is specified in 120F.4.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter on each end of the link is adjusted to an appropriate setting, which may be based on channel knowledge. If implemented, the transmitter equalization feedback mechanism described in 120F.3.2.6 may be used to identify an appropriate setting of the link partner transmitter. The adaptive or adjustable receiver performs the remainder of the equalization.

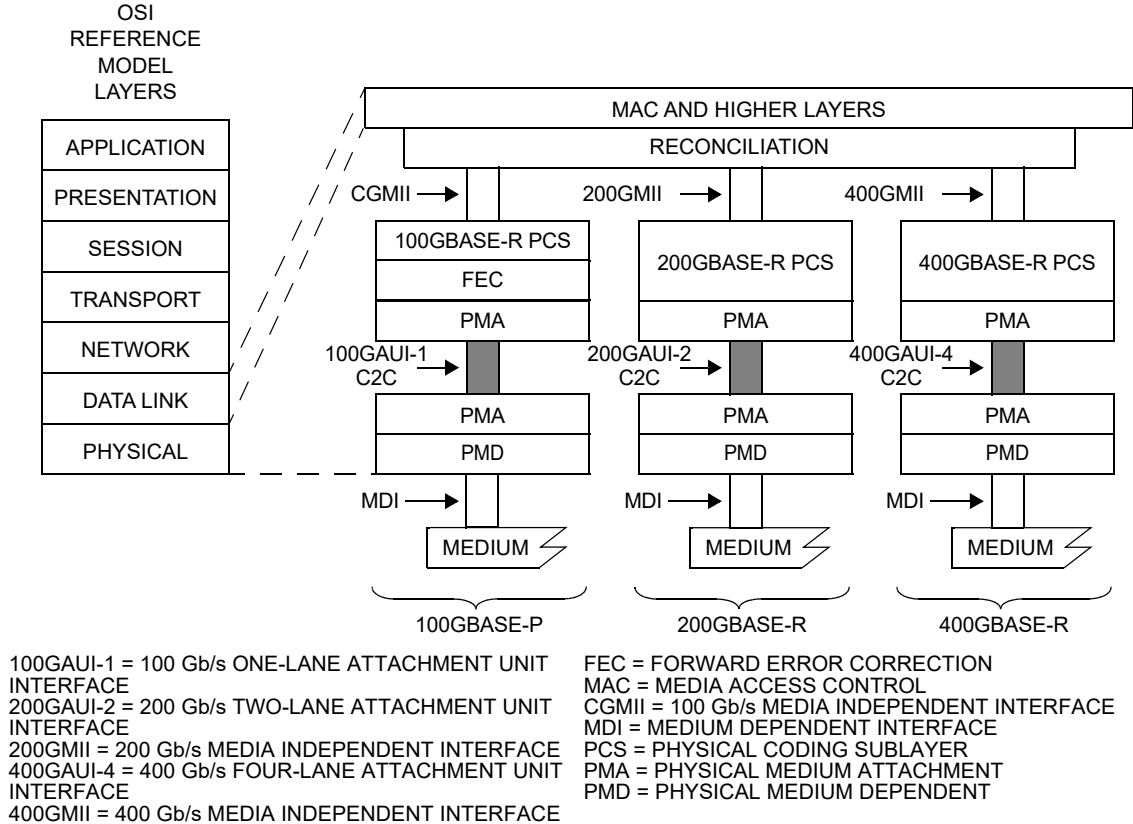


Figure 120F-1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

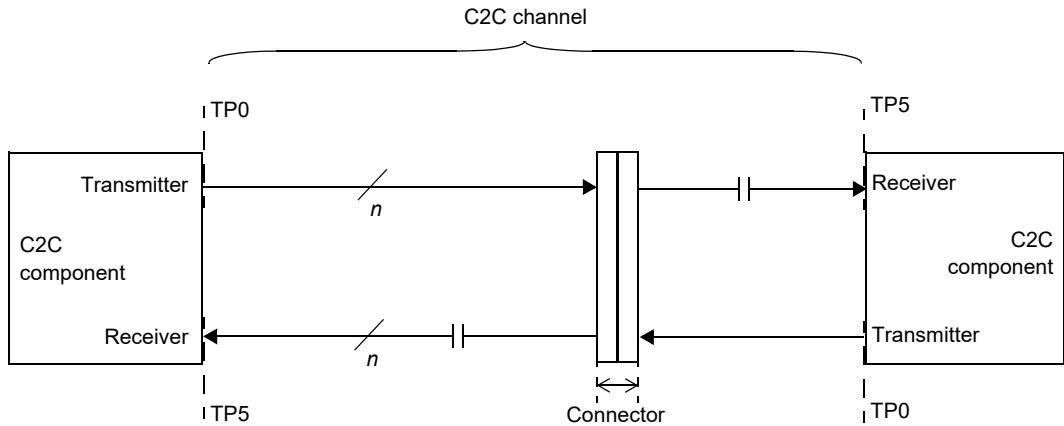


Figure 120F-2—Typical 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C application

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter supports $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2, that may be enabled or disabled as required. The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C receiver may support $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2. Precoding may be enabled and disabled using the precoder request mechanism specified in 135F.3.2.1.

120F.2 Compliance points

The electrical characteristics for the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces are defined at test points as described in 163.9.2.1 and 163.9.3.2.

120F.2.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

120F.3 Electrical characteristics

120F.3.1 Transmitter electrical characteristics

A 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C transmitter shall meet the specifications given in Table 120F–1.

Table 120F–1—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
Differential peak-to-peak output voltage ^b (max) Transmitter disabled Transmitter enabled	93.8.1.3	35 1200	mV mV
Common-mode voltage ^b (max)	93.8.1.3	1	V
Common-mode voltage ^b (min)	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	120F.3.1.1	32	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	120F.3.1.2	15	dB
Difference effective return loss, $dERL$ (min)	120F.3.1.4	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	162.9.4.9	2	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163A.3.2.1	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 120F–1—Transmitter electrical characteristics at TP0v (*continued*)

Parameter	Reference	Value	Units
Output waveform ^c			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at min state for c(−3) (max)	162.9.4.1.5	−0.05	—
value at max state for c(−3) (min)	162.9.4.1.5	0	—
value at min state for c(−2) (max)	162.9.4.1.5	0	—
value at max state for c(−2) (min)	162.9.4.1.5	0.1	—
value at min state for c(−1) (max)	162.9.4.1.5	−0.3	—
value at max state for c(−1) (min)	162.9.4.1.5	0	—
value at min state for c(0) (max)	162.9.4.1.5	0.5	—
value at min state for c(1) (max)	162.9.4.1.5	−0.1	—
value at max state for c(1) (min)	162.9.4.1.5	0	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	120F.3.1.3	28	dB
Output jitter			
J_{RMS} (max)	120F.3.1.6	0.023	UI
J_{4u} (max)	120F.3.1.6	0.128	UI
$J_{4u_{03}}$ (max)	120F.3.1.6	0.118	UI
Even-odd jitter (max)	120F.3.1.6	0.025	UI

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMD, PMA, or FEC sublayers.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c The state of the transmit equalizer is controlled by management interface.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

120F.3.1.1 Peak-to-peak AC common-mode voltage

The low-frequency and full-band peak-to-peak AC common-mode voltage, VC_{MF} and VC_{MB} , respectively, are defined by the method specified in 162.9.4.4 with the exception that the peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP0v that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.

The low-frequency peak-to-peak AC common-mode voltages shall meet the specification for VC_{MF} (max) in Table 120F–1.

120F.3.1.2 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is defined by the method specified in 163.9.2.6 with the exception that the full-band peak-to-peak AC common-mode voltage is defined in 120F.3.1.1.

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 120F–1.

120F.3.1.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is defined by the method specified in 162.9.4.3 with the exception that the continuous time filter settings are provided in Table 120F–8.

Signal-to-residual-intersymbol-interference ratio shall meet the specification for SNR_{ISI} (min) in Table 120F–1.

120F.3.1.4 Transmitter difference effective return loss (ERL)

The transmitter difference ERL at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

Table 120F–2—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Tukey window flag	tw	1	—

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) specified in Table 120F–1.

120F.3.1.5 Transmitter equalization settings

Each transmitter includes programmable equalization to compensate for the frequency-dependent loss of the channel and to facilitate data recovery at the receiver. The functional model for the transmit equalizer is the five-tap transversal filter shown in Figure 120F–3. Transmitter output waveform specifications are provided in Table 120F–1. The state of the transmitter equalizer may be configured via the transmitter control interface described in 120F.3.1.7.

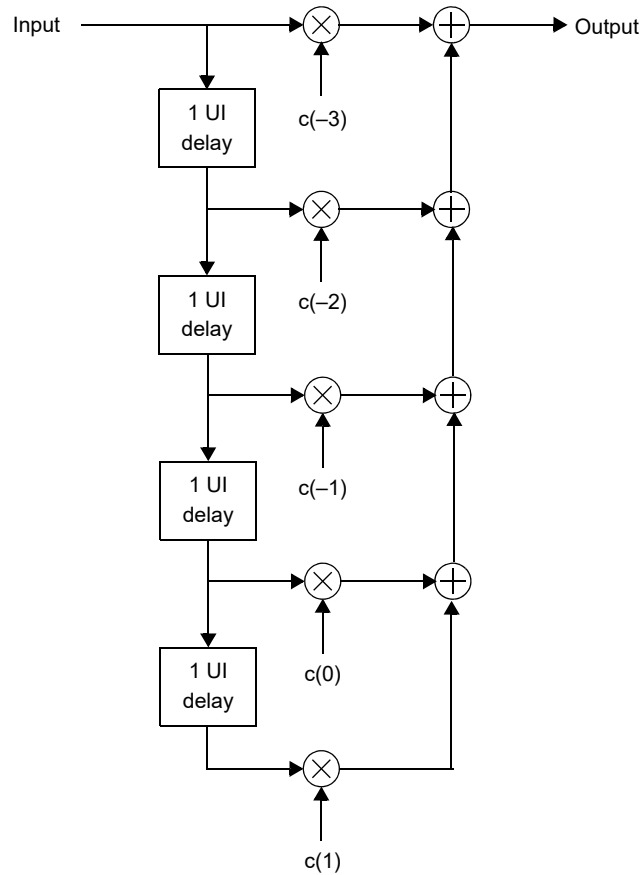


Figure 120F-3—Transmitter equalizer functional model

120F.3.1.6 Output jitter

Output jitter is characterized by four parameters: J_{4u} , $J_{4u_{03}}$, J_{RMS} , and even-odd jitter. These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the transmitter package and test fixture. The equalizer setting is chosen to minimize any or all of the jitter parameters.

J_{4u} and J_{RMS} are calculated using the jitter measurement method specified in 120D.3.1.8.1. $J_{4u_{03}}$ is calculated the same way as J_{4u} except that the calculation uses only transitions R03 and F30 in Table 120D-4.

Even-odd jitter is calculated using the jitter measurement method specified in 162.9.4.7.

NOTE— J_{4u} is sensitive to measurement noise being converted to timing errors. Hence, accounting for measurement noise effects is recommended.

120F.3.1.7 Transmitter control

The transmitter output equalization is characterized using the linear fit method described in 162.9.4.1.1.

Transmitter equalization shall be configurable by management using a set of control and status variables listed in Table 120F–3, or an equivalent interface. The variables that control transmitter equalization are specific for each lane.

Table 120F–3—Transmitter control variables

Variable name	Description	Management access ^a
local_ic_req	Requested initial condition in the local transmitter.	RW
local_coef_sel	Coefficient selection in the local transmitter.	RW
local_coef_req	Update to the selected coefficient in the local transmitter.	RW
local_ic_sts	Status of initial condition request.	RO
local_k	Coefficient select echo.	RO
local_coef_sts	Coefficient update acknowledge.	RO

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–3 are the same as the corresponding ones without the “local_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter control is identical to the “LD” variable mapping in Table 162–7.

120F.3.2 Receiver characteristics

A receiver shall meet the specifications given in Table 120F–4, with precoding enabled on the test transmitter if required by the receiver.

Table 120F–4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	120F.3.2.1	53.125 ± 100 ppm	GBd
Difference effective return loss, <i>dERL</i> (min)	120F.3.2.2	–3	dB
Differential-mode to common-mode return loss, <i>RLcd</i>	120F.3.2.3	Equation (120F–1)	dB
Interference tolerance	120F.3.2.4	Table 120F–5	—
Jitter tolerance	120F.3.2.5	Table 162–17	—

120F.3.2.1 Receiver signaling rate

The receiver shall comply with the requirements of 120F.3.2.4 and 120F.3.2.5 for any signaling rate in the range $53.125 \text{ GBd} \pm 100 \text{ ppm}$.

120F.3.2.2 Receiver difference ERL

The receiver difference ERL at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP5 to TP5v.

Receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 120F–4.

120F.3.2.3 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (120F–1). The differential-mode to common-mode return loss limit is illustrated by Figure 120F–4.

$$RL_{cd}(f) \geq \begin{cases} 25 - 0.72f & 0.05 \leq f \leq 13.9 \\ 15 & 13.9 < f \leq 38 \end{cases} \quad (120F-1)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

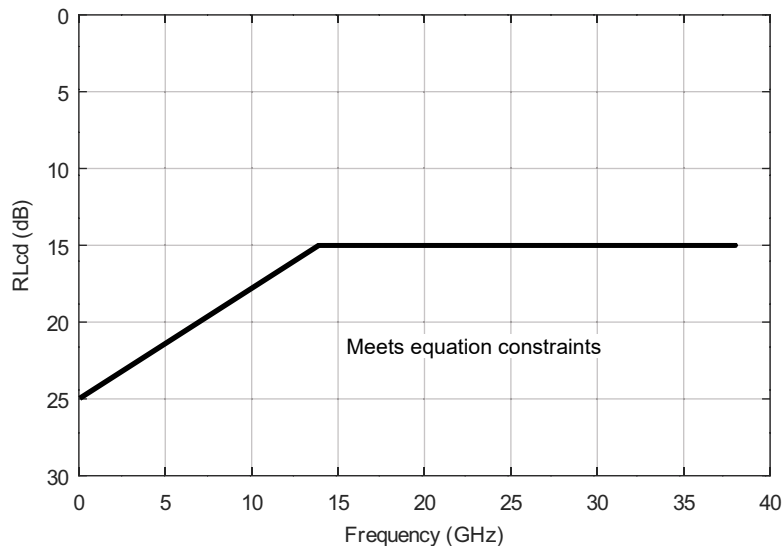


Figure 120F–4—Receiver differential-mode to common-mode return loss limit

120F.3.2.4 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#) with the exception that transmitter equalization is configured by management (see [120D.3.2.3](#)) to the settings that provide the lowest FEC symbol error ratio. The receiver on each lane shall meet the FEC symbol error ratio requirement with

channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 120F–5. The following additional considerations apply to the interference tolerance test.

- a) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see 93.8.1.3) is less than or equal to 800 mV.
- b) The ERL of the test channel measured at TP5 replica towards TPt meets the requirements in 120F.4.3.
- c) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- d) For the calculation of COM, the transmitter model is determined according to step e in 163.9.3.5.
- e) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p=11$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{4u} and J_{RMS} using Equation (120D–10), and Equation (120D–11), respectively, with the exceptions that $Q4$ is replaced with $Q4d$ and $Q4d$ is equal to 3.719.
- f) Other COM parameters are set according to the values in Table 120F–8.
- g) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 120F–8 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- h) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
- i) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
- j) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

Table 120F–5—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-4}	—	—	10^{-4}	—	—
Insertion loss at 26.5625 GHz ^b	9.5	10.5	—	19.5	20.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

120F.3.2.5 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. The test channel meets the insertion loss requirement for Test 2 in Table 120F–5. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 120F.3.2.4, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items d) through g) in 120F.3.2.4, is at least 3 dB.
- For the COM parameter calibration described in 120F.3.2.4 item e), the test channel transmitter J_{RMS} and J_{4u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 120F–5 for each case in Table 162–17.

120F.3.2.6 Transmitter equalization feedback (optional)

Transmitter equalization feedback is an optional capability for a receiver. If implemented, it shall operate as described in this subclause.

A receiver may generate a request to change the transmit equalization coefficients of the remote transmitter using the variables in Table 120F–6, or an equivalent interface.

Table 120F–6—Transmitter equalization feedback variables

Variable name	Description	Management access ^a
remote_ic_req	Initial condition request for the remote transmitter.	RO
remote_coef_sel	Coefficient selection for the remote transmitter.	RO
remote_coef_req	Requested update to the selected coefficient for the remote transmitter.	RO
remote_ic_sts	Status of initial condition request from the remote transmitter.	RW
remote_k	Coefficient select echo from the remote transmitter.	RW
remote_coef_sts	Coefficient update acknowledge from the remote transmitter.	RW

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–6 are the same as the corresponding ones without the “remote_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

The requested setting is generated from information internal to the receiver in an implementation specific manner.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter equalization feedback is identical to the “LP” variable mapping in Table 162–7.

120F.4 Channel characteristics

Table 120F–7 provides a summary of the channel characteristics for 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces, and references to the subclauses addressing each parameter.

Table 120F–7—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	120F.4.1	3	dB
Maximum insertion loss at 26.56 GHz (recommended)	120F.4.2	20	dB
Minimum ERL	120F.4.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	120F.4.4	Equation (163–6)	dB
Maximum AC-coupling 3 dB corner frequency	120F.4.5	50	kHz

120F.4.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1, where T_r is 7.5 ps for $H_f(f)$ in Equation (93A–19), and the parameters in Table 120F–8. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2. This minimum value allocates margin for practical limitations on the receiver implementation as well as the allowed transmitter equalization coefficients.

Table 120F–8—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	$1.2e-4$	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1 Tx	z_p	13	mm
Transmission line length, Test 1 Rx	z_p	11	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Package transmission line nominal characteristic impedance	Z_c	87.5	Ω
Package transmission line 2 nominal characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz

Table 120F-8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient Minimum value Maximum value Step size	$c(-3)$	−0.04 0 0.02	—
Transmitter equalizer, 2 nd pre-cursor coefficient Minimum value Maximum value Step size	$c(-2)$	0 0.1 0.02	—
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	−0.28 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	−0.1 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	−20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	−4 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	6	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\max}(n)$	0.65 0.15 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 −0.04	—

Table 120F-8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	2×10^{-8}	V ² /GHz
Target detector error ratio	DER_0	10^{-5}	—

120F.4.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (120F-2). Actual channel insertion loss could be higher or lower than that given by Equation (120F-2) due to the channel insertion loss deviation (ILD), return loss, and crosstalk. Note that for this equation the channel insertion loss at the Nyquist frequency is less than or equal to 20 dB.

$$ILdd(f) \leq 1.083 + 1.444\sqrt{f} + 0.432f \quad (120F-2)$$

for $0.01 \leq f \leq 53.125$

where

f is the frequency in GHz
 $ILdd$ is the channel insertion loss in dB

The insertion loss limit is illustrated by Figure 120F-5.

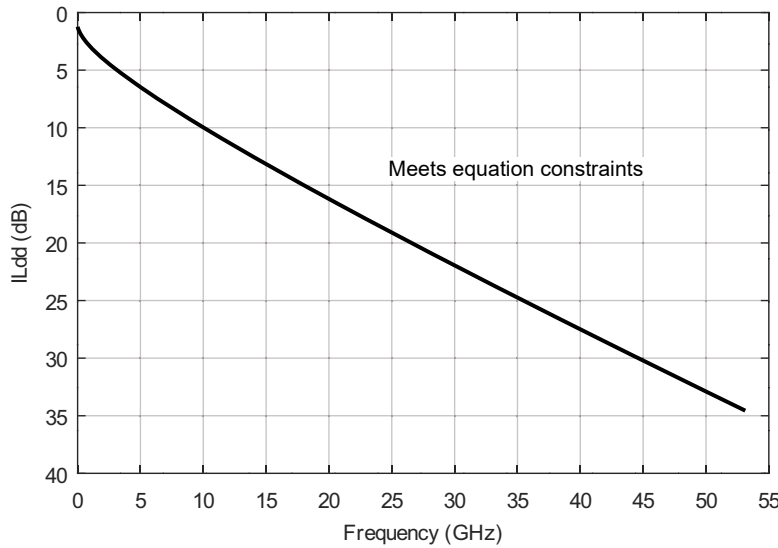


Figure 120F-5—Channel insertion loss limit

120F.4.3 Channel ERL

The channel ERL at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 120F–9. Parameters that do not appear in Table 120F–9 take values from Table 120F–8. Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 120F–9—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	2000	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

120F.4.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6) illustrated in Figure 163–7.

120F.4.5 Channel AC-Coupling

Channel AC-coupling is specified in 163.10.7.

120F.5 Protocol implementation conformance statement (PICS) proforma for Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)²⁰

120F.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120F.5.2 Identification

120F.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120F.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

²⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120F.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120F.1	One independent data path in each direction for 100GAUI-1 C2C, two independent data paths in each direction for 200GAUI-2 C2C, and four independent data paths in each direction for 400GAUI-4 C2C	M	Yes []
*CHAN	Channel	120F.4	Items marked with CHAN include channel specifications not applicable to a PHY manufacturer	O	Yes [] No []
*PCSP KG	PMA in same package as PCS.			O	Yes [] No []

120F.5.4 PICS proforma tables for Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.5.4.1 Transmitter

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMA is in same package as PCS	120F.3.1	Per Table 120F–1	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMA is not in same package as PCS	120F.3.1	Derived from PMA input lanes	!PCSP KG:M	Yes [] N/A []
TC3	Peak-to-peak differential output voltage, transmitter enabled	120F.3.1	Per Table 120F–1	M	Yes []
TC4	Peak-to-peak differential output voltage, transmitter disabled	120F.3.1	Per Table 120F–1	M	Yes []
TC5	DC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC6	Low-frequency peak-to-peak AC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC7	Signal to AC common-mode noise ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC8	Difference ERL	120F.3.1	Per Table 120F–1	M	Yes []
TC9	Common-mode to common-mode return loss	120F.3.1	Per Table 120F–1	M	Yes []
TC10	Output waveform coefficient limits	120F.3.1	Per Table 120F–1	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Output waveform coefficient step size	120F.3.1	Per Table 120F–1	M	Yes []
TC12	Output jitter	120F.3.1	Per Table 120F–1	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC15	Transmitter precoder request	120F.1		O	Yes [] No []
TC16	Transmit equalizer controlled by management interface	120F.3.1.7		M	Yes []
TC17	Difference steady-state voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC18	Difference linear fit pulse peak ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC19	Level separation mismatch ratio	120F.3.1	Per Table 120F–1	M	Yes []

120F.5.4.2 Receiver

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Difference ERL	120F.3.2	Per Table 120F–4	M	Yes []
RC2	Differential-mode to common-mode return loss	120F.3.2	Per Table 120F–4	M	Yes []
RC3	Interference tolerance	120F.3.2.4		M	Yes []
RC4	Jitter tolerance	120F.3.2.5		M	Yes []
RC5	Transmitter equalization feedback	120F.3.2.6		O	Yes [] No []

Channel

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Channel operating margin (COM)	120F.4.1	≥ 3 dB	CHAN :M	Yes [] N/A []
CC2	Channel insertion loss	120F.4.2	Meets Equation (120F–2) constraints	CHAN :O	Yes [] No [] N/A []
CC2	Channel ERL	120F.4.3	≥ 9.7 dB	CHAN :M	Yes [] N/A []
CC4	Channel AC-coupling	120F.4.5	Per 163.10.7	CHAN :M	Yes [] N/A []

Annex 120G

(normative)

Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-module (C2M) 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M). Figure 120G–1 shows the relationship of the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces to the ISO/IEC Open System Interconnection (OSI) reference model. The C2M interface provides electrical characteristics and associated compliance points, which can optionally be used when designing systems with pluggable module interfaces.

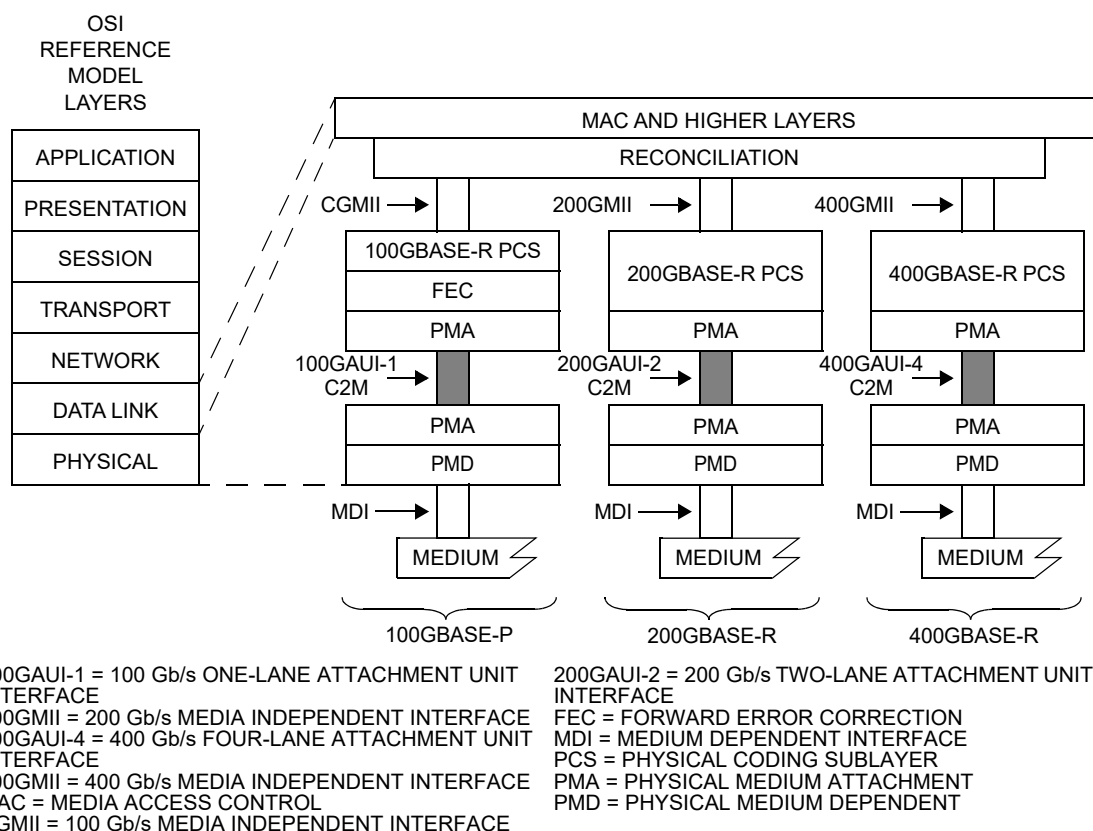
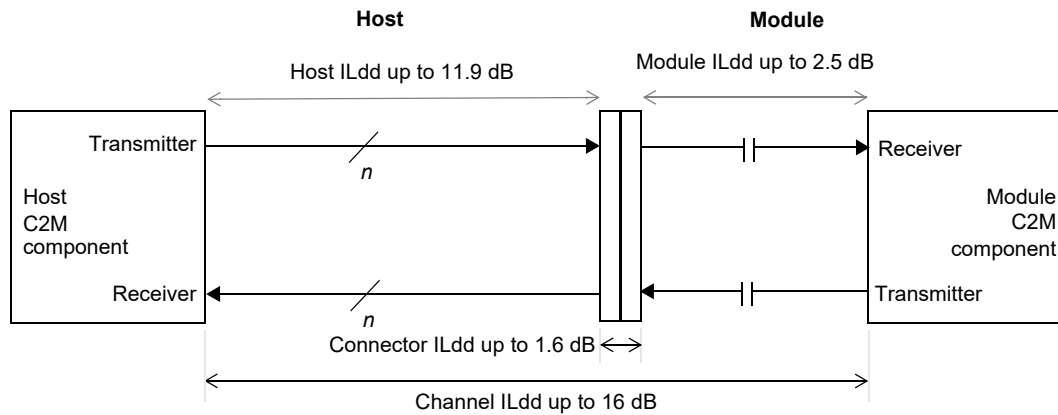


Figure 120G–1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2M, 200GAUI-2 C2M, or 400GAUI-4 C2M relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The C2M link is described in terms of a host C2M component, a C2M channel with associated insertion loss (ILdd), and a module C2M component. The host C2M component and module C2M component have connected ground references. Figure 120G–2 depicts a typical C2M application and summarizes the ILdd budget associated with the C2M application. The ILdd budget reflects the highest ILdd intended to be supported and is characterized by Equation (120G–5) and illustrated in Figure 120G–12. The C2M interface comprises independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M data path contains one, two, or four differential lanes, respectively, using PAM4 signaling, where the highest differential level corresponds to the symbol three and the lowest level corresponds to the symbol zero. Each lane is AC-coupled within the module.



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, and 4 for 400GAUI-4.

Figure 120G–2—100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M insertion loss budget at 26.56 GHz

The nominal signaling rate for each lane is 53.125 GBd.

120G.1.1 Bit error ratio

The bit error ratio (BER) when processed by the PMA according to Clause 135 for 100GAUI-1 C2M or Clause 120 for 200GAUI-2 or 400GAUI-4 C2M shall be less than 10^{-5} .

120G.2 Compliance point definitions

The electrical characteristics for the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M interfaces are defined at compliance points for the host and module. Reference test fixtures, called compliance boards, are used to access the electrical specification parameters.

Figure 120G–3 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M host compliance. The output of the Host Compliance Board (HCB) is used to verify the host electrical output signal at TP1a. The input of the HCB at TP4a is used to verify the host input compliance.

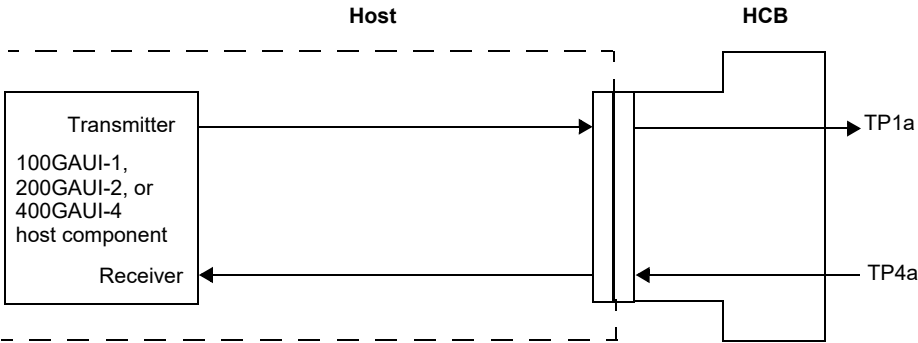


Figure 120G–3—Host compliance points

Figure 120G–4 depicts the location of compliance points when measuring 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2M module compliance. The output of the Module Compliance Board (MCB) is used to verify the module electrical output signal at TP4. The input of the MCB at TP1 is used to verify the module input compliance.

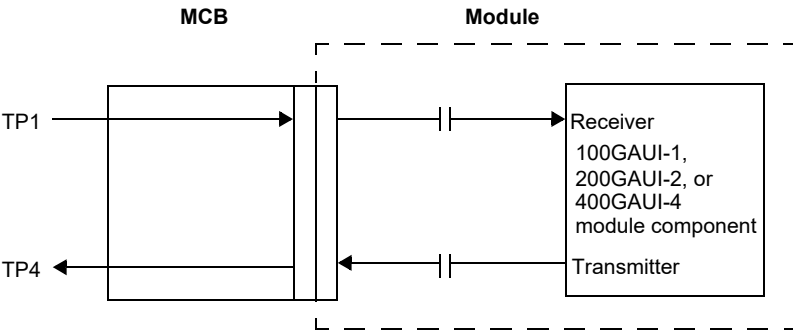


Figure 120G–4—Module compliance points

Additional details on the requirements for the MCB and HCB are given in 120G.5.4.

120G.3 Electrical characteristics

120G.3.1 Host output characteristics

The host output shall meet the specifications given in Table 120G–1.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

Table 120G–1—Host output characteristics at TP1a

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
DC common-mode output voltage (max)	120G.5.1	2.8	V
DC common-mode output voltage (min)	120G.5.1	−0.3	V
Single-ended output voltage (max)	120G.5.1	3.3	V
Single-ended output voltage (min)	120G.5.1	−0.4	V
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Transmitter disabled Transmitter enabled	120G.5.1	35 750	mV
Steady-state voltage, v_f (max)	120G.5.3	375	mV
Eye height (min)	120G.3.1.5	10	mV
Vertical eye closure, VEC (max)	120G.3.1.5	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.1.2	7.3	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min) Host is requesting short mode ^b Host is requesting long mode ^b	120G.3.1.4	10 15	ps ps

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the PMA input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMA or FEC sublayers.

^b Short and long modes are defined in 120G.3.2.1.

120G.3.1.1 Host output common-mode to differential-mode return loss

The host output common-mode to differential-mode return loss shall meet Equation (120G–1) as illustrated in Figure 120G–5.

$$RL_{dc}(f) \geq \begin{cases} 25 - 22(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-1)$$

where

RL_{dc} is the common-mode to differential-mode return loss in dB
 f is the frequency in GHz

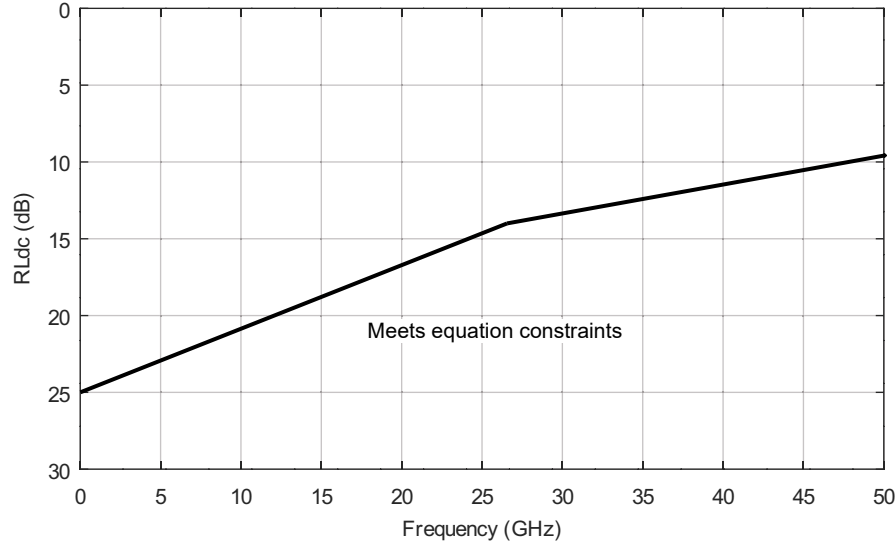


Figure 120G-5—Host output common-mode to differential-mode return loss

120G.3.1.2 Host output effective return loss (ERL)

The ERL of the host output at TP1a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture host-facing connection minus 0.2 ns.

Host output ERL at TP1a shall be greater than or equal to ERL (min) specified in Table 120G-1.

Table 120G-2—Host output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	800	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.1.3 Differential termination mismatch

Differential termination mismatch is defined in 86A.5.3.2.

120G.3.1.4 Transition time

Transition time is defined in 120E.3.1.5, with the exception that the waveform is observed through a fourth-order Bessel-Thomson low-pass response with a 3 dB bandwidth of 40 GHz.

120G.3.1.5 Host output eye height and vertical eye closure (VEC)

Figure 120G–6 depicts an example host output eye height and vertical eye closure (VEC) test configuration. Host output eye height and VEC are measured at TP1a using compliance boards defined in 120G.5.4. Eye height and VEC are measured according to the method described in 120G.5.2.

All counter-propagating signals are asynchronous to the co-propagating signals using the PRBS13Q²¹ (see 120.5.11.2.1) or PRBS31Q²² (see 120.5.11.2.2) pattern, or a valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal. For the case where PRBS13Q or PRBS31Q are used with a common clock, there is at least 31 UI delay between the patterns on one lane and any other lane, so that the symbols on each lane are not correlated. The crosstalk generator is calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, with target differential peak-to-peak voltage of 845 mV and transition time (see 120G.3.1.4) of 8.5 ps.

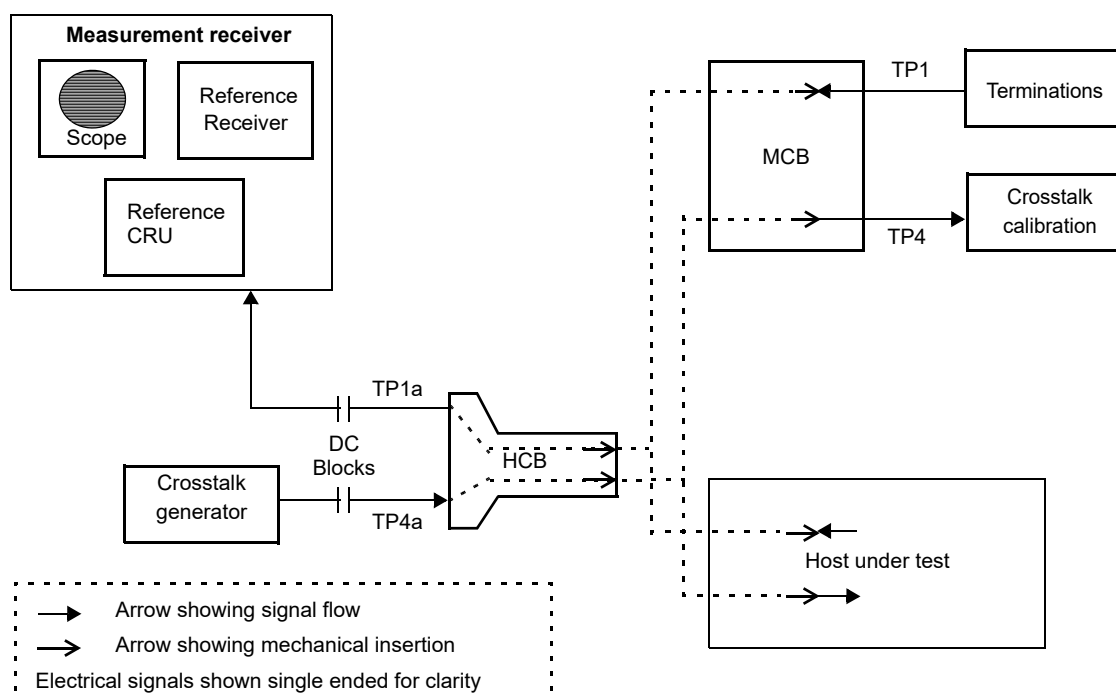


Figure 120G-6—Example host output test configuration

120G.3.2 Module output characteristics

The module output shall meet the specifications given in Table 120G–3. The low-frequency 3 dB cutoff of the output AC-coupling within the module should be less than 100 kHz. A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all output signal measurements, unless otherwise specified.

²¹ PRBS13O is also referred to as Pattern 4 for some PAM4 optical PMDs (e.g., see [Table 121–9](#)).

²² PRBS31Q is also referred to as Pattern 3 for some PAM4 optical PMDs (e.g., see [Table 121–9](#)).

Table 120G–3—Module output characteristics at TP4

Parameter	Reference	Value	Units
Signaling rate, each lane (nominal)		53.125 ^a	GBd
Peak-to-peak AC common-mode voltage (max) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	32 80	mV
Differential peak-to-peak output voltage (max) Short mode Long mode	120G.5.1	600 845	mV mV
Eye height (min)	120G.3.2.2	15	mV
Vertical eye closure, VEC (max)	120G.3.2.2	12	dB
Common-mode to differential-mode return loss, RL_{dc} (min)	120G.3.1.1	Equation (120G–1)	dB
Effective return loss, ERL (min)	120G.3.2.3	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	10	%
Transition time (min)	120G.3.1.4	8.5	ps
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.2.4	2.85 -0.35	V V

^a The signaling rate range is derived from the PMD receiver input.

120G.3.2.1 Module output modes

The module output shall support two modes: short and long. The means of controlling the module output mode is implementation dependent. For each output mode, the module shall meet the requirements for eye height (min) and VEC (max) in Table 120G–3 for both near-end and far-end measurements (see 120G.3.2.2.1).

One method for configuring the module output mode is to configure the module for a particular host electrical interface. The mapping of the combination of IEEE 802.3 interface type and module output mode to a host electrical interface may be determined from Table 120G–4.

Table 120G–4—Module output mode mapping

IEEE 802.3 interface type	Module output mode	Host electrical interface
100GAUI-1 C2M	short	100GAUI-1-S C2M
100GAUI-1 C2M	long	100GAUI-1-L C2M
200GAUI-2 C2M	short	200GAUI-2-S C2M
200GAUI-2 C2M	long	200GAUI-2-L C2M
400GAUI-4 C2M	short	400GAUI-4-S C2M
400GAUI-4 C2M	long	400GAUI-4-L C2M

120G.3.2.2.1 Near-end and far-end eye measurement methodology

The signal measured at TP4 is first convolved with a reference host channel. The reference host channel is the host receiver printed circuit board (PCB) signal path $S^{(\text{HOSPR})}$ defined in 162.11.7.1.1 with the exceptions that the length z_p for each test is provided in Table 120G–5, and C_0 and C_1 are both 0 nF. The eye height and VEC are measured using the method in 120G.5.2.

Table 120G–5—PCB length for module output measurements

Module output mode	Host channel type	PCB length, z_p (mm)
Short	near-end	0
Short	far-end	133
Long	near-end	80
Long	far-end	244.7

120G.3.2.3 Module output ERL

The ERL of the module output at TP4 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8, and with the value of T_{fx} equal to twice the delay between the test fixture test connector and the test fixture module-facing connection minus 0.2 ns.

Module output ERL at TP4 shall be greater than or equal to ERL (min) specified in Table 120G–3.

Table 120G–6—Module output and input ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Tukey window flag	tw	1	—

120G.3.2.4 Module output common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the output specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP4, within the range specified in Table 120G–3.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.3.3 Host input characteristics

The host input shall meet the specifications given in Table 120G–7. Channel equalization is provided by an adaptive equalizer in the host.

Table 120G–7—Host input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.3.1	TP4a	53.125 ± 100 ppm	GBd
Differential peak-to-peak input voltage tolerance (min) for short mode for long mode	120G.5.1	TP4	600 845	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP4	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP4a	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.3.4	TP4a	7.3	dB
Host stressed input tolerance	120G.3.3.5	TP4	See 120G.3.3.5	
Differential termination mismatch (max)	120G.3.1.3	TP4a	10	%
Common-mode voltage ^a Min Max	120G.5.1	TP4a	–0.3 2.8	V

^a Generated by host, referred to host ground.

120G.3.3.1 Host input signaling rate

The host input shall comply with the requirements of 120G.3.3.5 for any signaling rate in the range specified in Table 120G–7.

120G.3.3.2 Host input AC common-mode voltage tolerance

A host input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–7.

120G.3.3.3 Host input differential-mode to common-mode return loss

The host input differential-mode to common-mode return loss shall meet Equation (120G–2) as illustrated in Figure 120G–8.

$$RL_{cd}(f) \geq \begin{cases} 22 - 16(f/53.125) & 0.05 \leq f \leq 26.56 \\ 19 - 10(f/53.125) & 26.56 < f \leq 50 \end{cases} \quad (120G-2)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

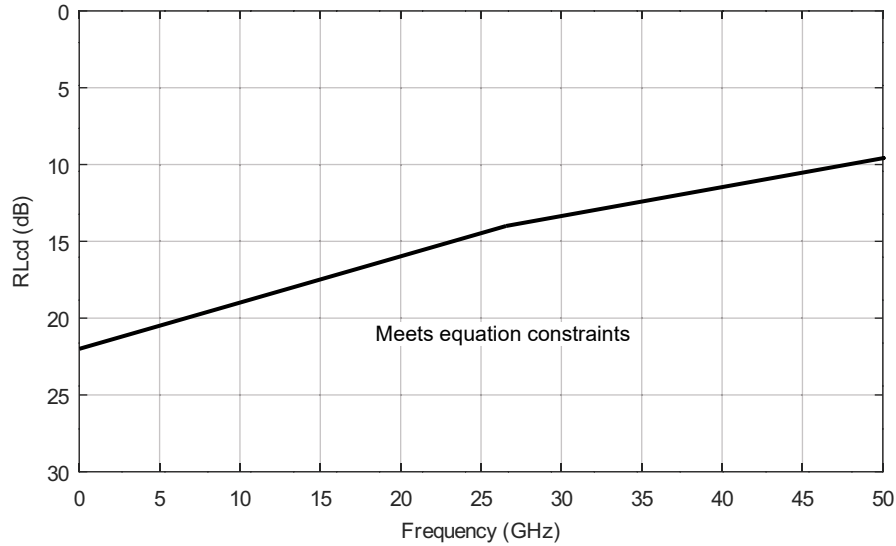


Figure 120G-8—Host input differential-mode to common-mode return loss

120G.3.3.4 Host input ERL

The ERL of the host input at TP4a is computed using the procedure in 93A.5 with the values in Table 120G-2 and Table 120F-8.

Host input ERL at TP4a shall be greater than or equal to ERL (min) specified in Table 120G-7.

120G.3.3.5 Host stressed input tolerance

Host stressed input tolerance is defined by the procedure described in 120G.3.3.5.1 through 120G.3.3.5.3.

The host under test shall meet the BER requirement in 120G.1.1:

- For either the short or long mode.
- With all sinusoidal jitter cases in Table 162-17.
- For any signaling rate in the range given in Table 120G-7.

120G.3.3.5.1 Host stressed input test setup

The host stressed input test setup is illustrated in Figure 120G-9. The stressed signal is applied at TP4a and is calibrated at TP4.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Counter-propagating crosstalk signals.

The pattern generator output equalization functional behavior is equivalent to the model shown in Figure 120F–3. The tap coefficients are not specified with the exception that the “no equalization” state is defined as setting coefficient $c(0)$ to 1 and all other coefficients to 0.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated by driving the pattern generator external jitter modulation input with a low-pass filtered pseudo-random pattern. The pattern should be either PRBS7 or PRBS9 (see 83.5.10) with a signaling rate approximately 1/20 of the stressed pattern signaling rate (e.g., 2.656 GBd). The clock source for the PRBS generator is asynchronous to the pattern generator clock source. The low-pass filter has 20 dB/decade roll-off with a –3 dB corner frequency between 150 MHz and 300 MHz.

A crosstalk generator for counter-propagating signals representing the host output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the host under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference host channel to be configured for short or long mode far-end test calibration or long-mode near end verification as specified in 120G.3.2.2.1.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.2.3 when measured at TP4.

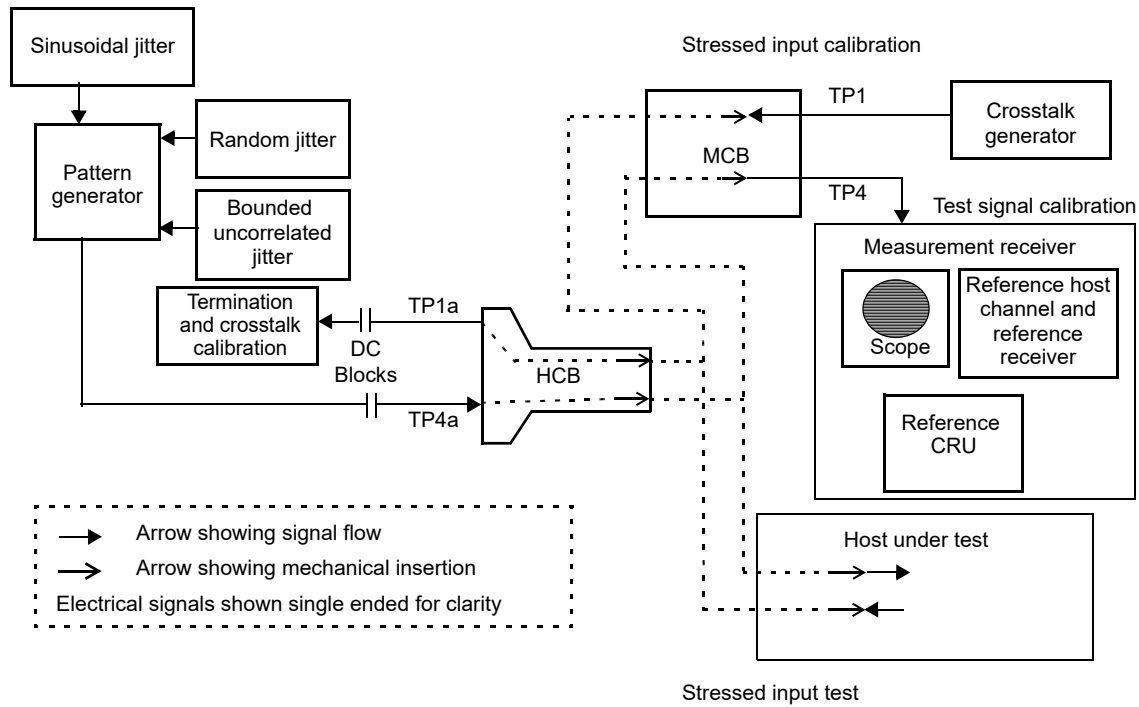


Figure 120G-9—Example host stressed input test

120G.3.3.5.2 Host stressed input test calibration

The host stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at TP4a, with the pattern generator output equalization configured for “no equalization”, is as specified in Table 120G-8. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-7.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at TP4a approximates J_{RMS} (max) and J_{4u} (max) and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP1a using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-8. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11 and 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- The reference host channel is configured in the same way as in 120G.3.2.2.1 using the parameters in Table 120G-5 for host channel type and the requested module mode (short or long).

- g) Eye height and VEC are measured at TP4 as described in 120G.5.2. The pattern generator amplitude and random jitter are adjusted, while the pattern generator output equalization and reference receiver settings are adjusted to minimize far-end VEC, so that the far-end eye height of the smallest eye matches the target value, near-end VEC is within the limits in Table 120G–8, and differential peak-to-peak output voltage, near-end VEC, and eye height are within the limits in Table 120G–3. The differential peak-to-peak voltage measured at TP4 does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–7.

Table 120G–8—Host stressed input parameters

Parameter	Value	Units
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	15	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	750	mV
Crosstalk transition time for short mode for long mode	10 15	ps

120G.3.3.5.3 Host stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and is plugged into the host under test. The host electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162–17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4 Module input characteristics

The module input shall meet the specifications given in Table 120G–9. Channel equalization is provided by an adaptive equalizer in the module.

Table 120G–9—Module input characteristics

Parameter	Reference	Test point	Value	Units
Signaling rate, each lane (range)	120G.3.4.1	TP1	53.125 ± 100 ppm	GBd
Differential pk-pk voltage tolerance (min)	120G.5.1	TP1a	750	mV
Peak-to-peak AC common-mode voltage tolerance (min) Low-frequency, $V_{CM_{LF}}$ Full-band, $V_{CM_{FB}}$	120G.5.1	TP1a	32 80	mV
Differential-mode to common-mode return loss, RL_{cd} (min)	120G.3.3.3	TP1	Equation (120G–2)	dB
Effective return loss, ERL (min)	120G.3.4.4	TP1	8.5	dB
Differential termination mismatch (max)	120G.3.1.3	TP1	10	%
Module stressed input tolerance	120G.3.4.3	TP1a	See 120G.3.4.3	
Single-ended voltage tolerance range (min)	120G.5.1	TP1a	–0.4 to 3.3	V
DC common-mode voltage tolerance (range) Upper limit Lower limit	120G.3.4.5	TP1	2.85 –0.35	V V

120G.3.4.1 Module input signaling rate

The module input shall comply with the requirements of 120G.3.4.3 for any signaling rate in the range specified in Table 120G–9.

120G.3.4.2 Module input AC common-mode voltage tolerance

A module input shall meet all other specifications with both $V_{CM_{LF}}$ and $V_{CM_{FB}}$ (see 120G.5.1) up to the limits specified in Table 120G–9.

120G.3.4.3 Module stressed input tolerance

Module stressed input tolerance is defined by the procedure described in 120G.3.4.3.1 through 120G.3.4.3.3.

The module under test shall meet the BER requirement in 120G.1.1:

- For both the high-loss and low-loss cases (see 120G.3.4.3.2), calibrated separately.
- With all sinusoidal jitter cases in Table 162–17.
- For any signaling rate in the range given in Table 120G–9.

120G.3.4.3.1 Module stressed input test setup

The module stressed input test setup is illustrated in Figure 120G–10. The stressed signal is applied at TP1 and is calibrated at TP1a.

The stressed signal includes the following impairments:

- Sinusoidal jitter, random jitter, and bounded uncorrelated jitter.
- Frequency-dependent attenuation representing the host channel, which may be implemented with PCB traces. The frequency-dependent attenuation is used only for the high-loss case (see 120G.3.4.3.2).
- Counter-propagating crosstalk signals.

The pattern generator output equalization is described in 120G.3.3.5.1.

Bounded uncorrelated jitter may not be available in all stressed pattern generators or bit error ratio testers. It can be generated as described in 120G.3.3.5.2.

A crosstalk generator for counter-propagating signals representing the module output is used for calibration of the stress. The counter-propagating signals are asynchronous to the pattern generator. This crosstalk generator is not present in the test, being replaced by the output of the module under test.

The measurement receiver used for test calibration includes:

- A clock recovery unit (CRU) that acts as a high-pass jitter filter with a 3 dB corner frequency of 4 MHz and a slope of 20 dB/decade.
- A reference receiver as specified in 120G.5.2.

The test system meets the ERL specification given in 120G.3.1.2 when measured at TP1a.

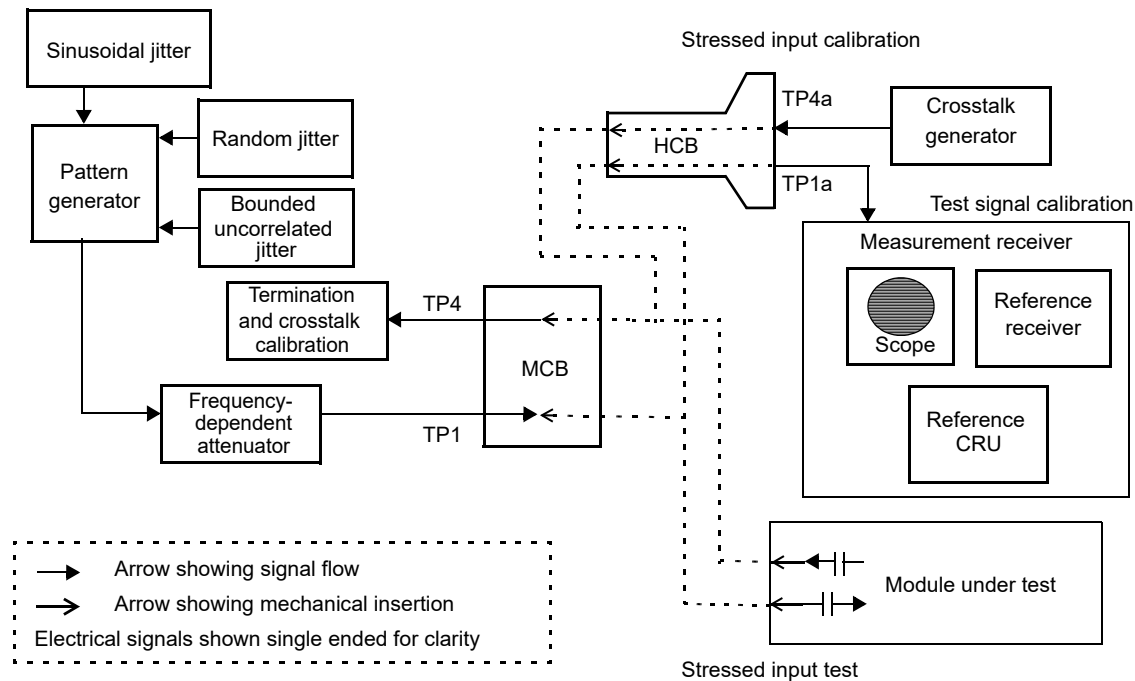


Figure 120G-10—Example module stressed input test

120G.3.4.3.2 Module stressed input test calibration

The stressed input signal is calibrated by the following procedure.

- The pattern generator is set to generate a PRBS13Q pattern (see 120.5.11.2.1). The transition time (see 120G.3.1.4) measured at the output of the pattern generator, with the pattern generator output equalization configured for “no equalization” (see 120G.3.3.5.1), is as specified in Table 120G-10. The initial signal level is set to the differential peak-to-peak input voltage tolerance given in Table 120G-9.
- Sinusoidal jitter is applied with frequency and amplitude per case F in Table 162-17.
- Random jitter and bounded uncorrelated jitter are added to the sinusoidal jitter such that the jitter profile of the signal at the output of the pattern generator approximates $J_{RMS}(\max)$ and $J_{4u}(\max)$ and complies with the even-odd jitter (max) specification in Table 120F-1. These are initial jitter values, that will be modified by the addition of crosstalk in step e) and adjustment of random jitter in step g).
- The HCB is plugged into the MCB.
- The counter-propagating crosstalk signals are calibrated at TP4 using a test system with a response as defined in 120G.3.1, rather than the reference receiver of 120G.5.2, to the differential peak-to-peak voltage and transition time (see 120G.3.1.4) specified in Table 120G-10. The crosstalk signal transition time and amplitude are calibrated with a PRBS13Q pattern. The crosstalk pattern is changed to PRBS31Q (see 120.5.11.2.2), scrambled idle (see 82.2.11, 119.2.4.9), or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R signal for stressed signal calibration (see step g)). If the PRBS31Q pattern is used with a common clock, there is at least 31 UI delay between the PRBS31Q patterns on one lane and any other lane.
- For the low-loss signal calibration, the output of the pattern generator is fed directly to the MCB input (TP1). For the high-loss signal calibration, the frequency-dependent attenuator is configured such that the insertion loss of the test channel from the output of the pattern generator to TP1a

approximates the target insertion loss, $ILDD_{channel}$, given by Equation (120G–3). The target insertion loss is 18.2 dB at 26.56 GHz, representing 16 dB channel loss with an additional 2.2 dB, which provides some allowance for the difference between the effective insertion loss of the pattern generator output and the worst-case host transmitter package loss. To achieve the target test channel insertion loss, the scattering parameters of the frequency-dependent attenuator should approximate those for a PCB transmission line calculated from Equation (93A–13) and Equation (93A–14) using $z_p = 296$ mm in length and the relevant parameter values given in Table 162–21. Additional frequency-independent delay may be included in the test channel. The magnitude of the insertion loss of the target frequency-dependent attenuator scattering parameters, $ILDD_{FDA}$, is represented by Equation (120G–4). $ILDD_{channel}$ and $ILDD_{FDA}$ are illustrated in Figure 120G–11.

- g) Eye height and VEC are measured at TP1a as described in 120G.5.2. For the high-loss case, an exception is made that the reference receiver CTLE is limited to settings where $g_{DC} + g_{DC2}$ is less than or equal to -10.5 dB.
- h) The pattern generator amplitude, output equalization, and random jitter are adjusted together to minimize VEC, so that the eye height of the smallest eye matches the target value, VEC is within the limits in Table 120G–10, and the differential peak-to-peak voltage measured at TP1a does not exceed the differential peak-to-peak input voltage tolerance given in Table 120G–9. The pattern generator output equalization has to be set such that for the resulting signal, the same VEC is achieved with or without the limitations on g_{DC} and g_{DC2} in item g).

Other calibration procedures resulting in a signal that meets these requirements may be used.

Table 120G–10—Module stressed input parameters

Parameter	Value	Value
Pattern generator transition time (target)	9	ps
Applied peak-to-peak sinusoidal jitter	Table 162–17	—
Eye height (target)	10	mV
Vertical eye closure, VEC (min)	12	dB
Vertical eye closure, VEC (max)	12.5	dB
Crosstalk differential peak-to-peak voltage	845	mV
Crosstalk transition time	8.5	ps

$$ILdd_{channel}(f) = 1.7962\sqrt{f} + 0.2463f + 0.003405f^2 \quad (120G-3)$$

where

$ILdd_{channel}(f)$ is the target pattern generator output to TP1a insertion loss in dB
 f is the frequency in GHz

$$ILdd_{FDA}(f) = 0.981 \sqrt{f} + 0.2463f \quad (120G-4)$$

where

$ILdd_{FDA}(f)$ is the target high-loss frequency-dependent attenuator insertion loss in dB
 f is the frequency in GHz

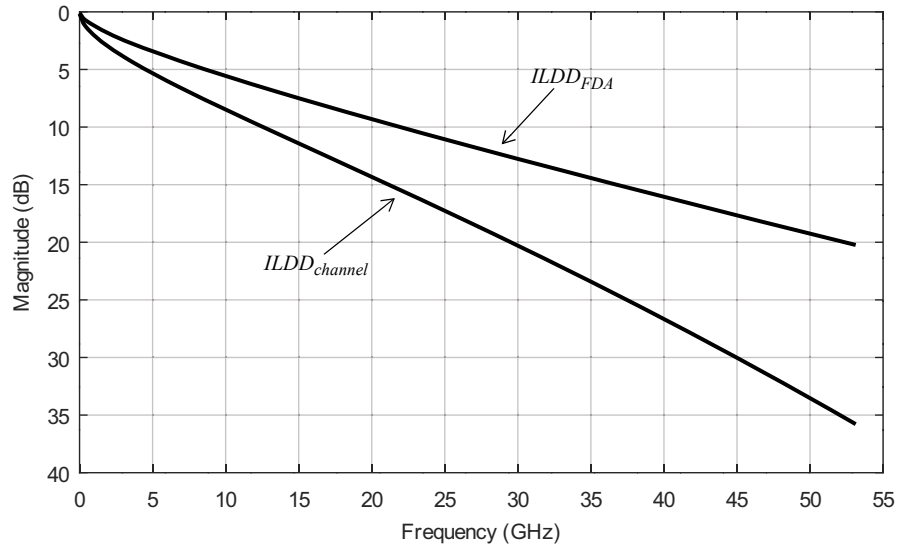


Figure 120G-11—Module stressed input insertion loss targets

120G.3.4.3.3 Module stressed input test procedure

After the stress has been calibrated, the pattern generator is set to generate PRBS31Q, scrambled idle, or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence. The HCB is unplugged from the MCB and the module under test is plugged into the MCB. The module electrical output is enabled on all lanes with any of the patterns above. The test is repeated with sinusoidal jitter set to each of the six cases in Table 162-17.

If the test is performed with PRBS31Q, the BER for the AUI under test may be calculated using the bit error counter in the PMA test pattern checker (see 120.5.11.2.2) as the number of bit errors divided by the number of received bits. The BER for the AUI under test is the average of the BER of each of its lanes.

If the test is performed with scrambled idle or another valid 100GBASE-R, 200GBASE-R, or 400GBASE-R sequence, the BER for the AUI under test may be calculated by placing the module under test into local loopback (see 120.5.9) and feeding the module output into a compliant host or its equivalent. The BER for the AUI under test is calculated using the FEC decoder error counters (see 91.6 and 119.3.1), as the number of FEC symbol errors divided by the number of received bits.

The number of received bits may be estimated based on the test time.

Methods of extracting the received bit pattern and counting errors other than the ones described above may be used if they generate equivalent results.

120G.3.4.4 Module input ERL

The ERL of the module input at TP1 is computed using the procedure in 93A.5 with the values in Table 120G–6 and Table 120F–8.

Module input ERL at TP1 shall be greater than or equal to ERL (min) specified in Table 120G–9.

120G.3.4.5 Module input common-mode voltage tolerance

DC common-mode voltage is generated by the host. A compliant module meets the input specifications with any DC common-mode voltage (see 120G.5.1), as measured at TP1, within the range specified in Table 120G–9.

NOTE—The specified voltages allow for the effects of ground offset voltage.

120G.4 Channel insertion loss (recommended)

For worst-case package loss in the host, the insertion loss of the channel between the host and module C2M components (see Figure 120G–2) is recommended to meet Equation (120G–5), which is illustrated in Figure 120G–12. Note that for this equation the insertion loss at the Nyquist frequency is less than or equal to 16 dB. The insertion loss budget is illustrated in Figure 120G–2.

$$IL_{dd}(f) \leq \begin{cases} 0.05 + 1.8\sqrt{f} + 0.2513f & 0.01 \leq f \leq 26.56 \\ -12.4181 + 1.07f & 26.56 < f \leq 40 \end{cases} \quad (120G-5)$$

where

$IL_{dd}(f)$ is the channel insertion loss in dB
 f is the frequency in GHz

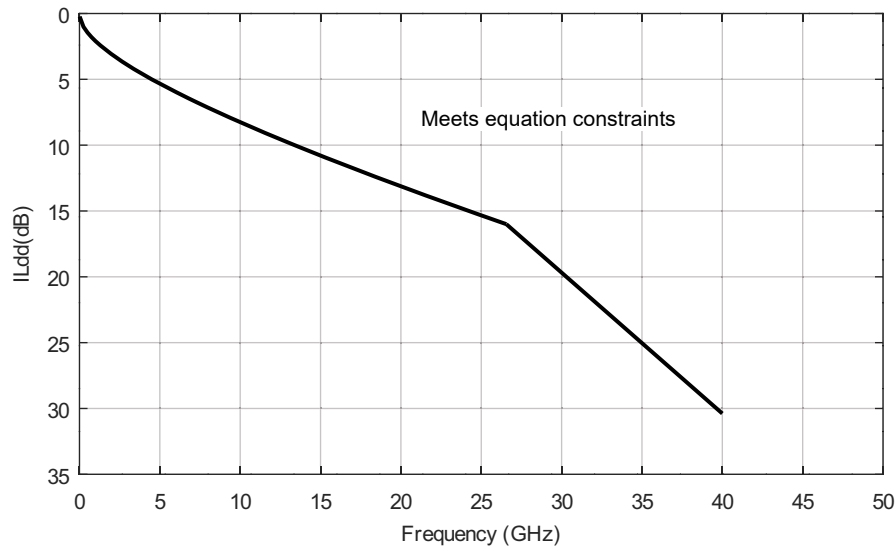


Figure 120G–12—Recommended channel insertion loss

120G.5 Measurement methodology

This subclause describes common measurement tools and methodology.

120G.5.1 Signal levels

The signal levels are as defined in 120E.3.1.2.

Low-frequency and full-band peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ and $V_{CM_{FB}}$, respectively, are defined by the method specified in 162.9.4.4 with the following exceptions:

- The peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP1a or TP4 that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.
- The condition for transmitter equalization to be turned off does not apply.

120G.5.2 Eye opening measurement method

The eye opening parameters eye height and VEC are measured with the effect of a reference receiver that includes receiver input referred noise, a continuous-time filter as defined in 93A.1.4.3, a receiver noise filter as defined in 93A.1.4.1, and a decision-feedback equalizer as defined in 93A.1.6, using the parameters specified in Table 120G–11. All parameters in Table 120G–11 apply to TP1a, TP4 far-end, and TP4 near-end unless indicated otherwise. The pattern used for output eye diagram measurements is PRBS13Q (see 120.5.11.2.1). Eye height and VEC parameters are defined by the following procedure.

Table 120G–11—Eye opening reference receiver parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz
Continuous time filter, DC gain for TP1a Range for $g_{DC2} = 0$ Range for $-1 \leq g_{DC2} < 0$ Range for $-2 \leq g_{DC2} < -1$ Range for $-3 \leq g_{DC2} < -2$ Step size	g_{DC}	–2 to –9 –2 to –11 –4 to –10 –4 to –9 1.0	dB
Continuous time filter, DC gain 2 for TP1a Range Step size	g_{DC2}	–3 to 0 0.5	dB
Continuous time filter, DC gain for TP4 near-end Range Step size	g_{DC}	–5 to –1 1.0	dB
Continuous time filter, DC gain 2 for TP4 near-end Range Step size	g_{DC2}	–2 to 0 0.5	dB
Continuous time filter, DC gain for TP4 far-end Range Step size	g_{DC}	–9 to –2 1.0	dB

Table 120G–11—Eye opening reference receiver parameter values (*continued*)

Parameter	Symbol	Value	Units
Continuous time filter, DC gain 2 for TP4 far-end Range Step size	g_{DC2}	–3 to –1 0.5	dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	12.58	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	20 28	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 40$	GHz
Decision feedback equalizer (DFE) length	N_b	4	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\max}(n)$	0.4 0.15 0.1	— — —
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ or 4	$bb_{\min}(n)$	0.1 –0.15 –0.05	— — —
One-sided noise spectral density	η_0	4.1×10^{-8}	V ² /GHz

Perform the following step once:

- a) Capture the PRBS13Q signal $y_1(k)$ with the effect of low-pass response equivalent to the specified receiver noise filter with associated parameter f_r in Table 120G–11 (instead of the test system response specified in 120G.3.1), and using a clock recovery unit with a corner frequency of 4 MHz and slope of 20 dB/decade. The signal is captured such that samples, after any post-processing as necessary, are evenly distributed throughout the measurement window and are numerous enough for an accurate normalized cumulative distribution function (CDF) to a probability of 10^{-5} without extrapolation.

Perform the following steps for each valid combination of g_{DC} and g_{DC2} as specified in Table 120G–11:

- b) Compute the response $y_2(k)$ by applying the effect of the continuous time filter to $y_1(k)$ using the associated parameters in Table 120G–11.
- c) Compute the linear fit pulse response $p_2(k)$ using the method defined in 162.9.4.1.1 with parameter M the same as for step a), D_p equal to 3, and N_p equal to 200.
- d) Compute the DFE sampling phase t_s and tap weights $b(n)$ for $p_2(k)$ according to the methodology in 93A.1.6 using the associated parameters in Table 120G–11.
- e) Compute the receiver input signal $y_{rx}(k)$ by applying the effect of the DFE to $y_2(k)$ using the sampling phase t_s and tap weights $b(n)$ determined in the previous step.
- f) Compute the variance of the noise at the output of the receive equalizer σ_N^2 based on the one-sided spectral density η_0 , provided in Table 120G–11, referred to the receiver noise filter input per Equation (93A–35).
- g) Compute an eye diagram from $y_{rx}(k)$, including the effect of Gaussian noise with variance calculated in the previous step, but taking into account that some noise from the measurement instrument is already in $y_{rx}(k)$.
- h) Compute Vmid, Vupp, Vlow, VCmid, VCupp, VClow from the eye diagram using 120E.4.2 steps 4) through 6) with the exception that the CDF of the signal voltage is calculated from the set of voltage samples within the time interval $t_s \pm 0.05$ UI and with accumulated probability for each sample

weighted by the function $w(t)$ defined by Equation (120G–6). As an example, $UPPCDF1$ may be calculated using Equation (120G–8) from a set of N samples $\{v_i, t_i\}$.

- i) The eye height is the minimum of V_{mid} , V_{upp} , and V_{low} .
- j) Compute VEC using Equation (120E–4) and the values computed in step h) with the exception that $VM3$, $VM2$, $VM1$, and $VM0$ are the average values over the time interval $t_s \pm 0.05$ UI instead of “within 0.025 UI of time TC_{mid} ”.

The values of eye height and VEC are the values obtained with the combination of g_{DC} and g_{DC2} that produces the minimum value of VEC where eye height also complies with the specification for eye height (min) as specified for the interface.

$$w(t) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma_r} \exp\left(-\frac{1}{2}\left(\frac{x(t)}{\sigma_r}\right)^2\right) & |x(t)| \leq 0.05 \\ 0 & |x(t)| > 0.05 \end{cases} \quad (120G-6)$$

$$x(t) = (t - t_s)f_b \quad (120G-7)$$

$$UPPCDF1(v) = \frac{1}{A} \sum_{i=1}^N w(t_i) u(v_i, VC_{upp}, v) \quad (120G-8)$$

$$u(x, a, b) = \begin{cases} 1 & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad (120G-9)$$

$$A = \sum_{i=1}^N w(t_i) \quad (120G-10)$$

where:

$w(t)$	is the weighting function
$x(t)$	is the deviation from the sampling time t_s in UI
σ_r	is the standard deviation in UI and is equal to 0.02 UI
f_b	is the signaling rate in Hz
t	is time in seconds
A	is a factor to normalize the weighting function over the sample set

120G.5.3 Steady-state voltage

The steady-state voltage v_f is defined as the sum of the linear fit pulse $p(1)$ through $p(M \times N_p)$ divided by M with the specific equalization used by the transmitter. N_v is set equal to N_p . The linear fit procedure for obtaining $p(k)$ and the values of M and N_p are defined in 162.9.4.1.1.

120G.5.4 HCB/MCB characteristics

HCB characteristics are described in 162B.2 where the HCB performs the equivalent function as the TP2 or TP3 test fixture. The MCB characteristics are described in 162B.3 where the MCB performs the equivalent

functionality as the cable assembly test fixture. The mated compliance board characteristics are described in 162B.4.

120G.6 Protocol implementation conformance statement (PICS) proforma for Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)²⁵

120G.6.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120G.6.2 Identification

120G.6.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120G.6.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120G, Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

²⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

120G.6.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120G.1	One, two, or four independent data paths in each direction for 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M, respectively	M	Yes []
BER	Meets BER requirement	120G.1.1	See 120G.1.1	M	Yes []
*PCSPKG	PMA in same package as PCS.			O	Yes [] No []

120G.6.4 PICS proforma tables for Chip-to-module 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2M), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2M), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2M)

120G.6.4.1 Host output

Item	Feature	Subclause	Value/Comment	Status	Support
TH1	Signaling rate, each lane, PMA in same package as PCS	120G.3.1	Per Table 120G–1	PCSPKG:M	Yes [] N/A []
TH2	Signaling rate, each lane, PMA not in same package as PCS	120G.3.1	Derived from PMA input lanes	!PCSPKG:M	Yes [] N/A []
TH3	DC common-mode output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH4	Single-ended output voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH5	Low-frequency peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH6	Full-band peak-to-peak AC common-mode voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH7	Differential peak-to-peak voltage, transmitter disabled	120G.3.1	Per Table 120G–1	M	Yes []
TH8	Differential peak-to-peak voltage, transmitter enabled	120G.3.1	Per Table 120G–1	M	Yes []
TH9	Steady-state voltage	120G.3.1	Per Table 120G–1	M	Yes []
TH10	Eye height	120G.3.1	Per Table 120G–1	M	Yes []
TH11	VEC	120G.3.1	Per Table 120G–1	M	Yes []
TH12	Common-mode to differential-mode return loss	120G.3.1.1	Per Table 120G–1	M	Yes []
TH13	ERL	120G.3.1.2	Per Table 120G–1	M	Yes []
TH14	Differential termination mismatch	120G.3.1.3	Per Table 120G–1	M	Yes []
TH15	Transition time	120G.3.1.4	Per Table 120G–1	M	Yes []

120G.6.4.2 Module output

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	Signal rate, each lane, nominal	120G.3.2	Per Table 120G–3	M	Yes []
TM2	Signal rate, each lane, range	120G.3.2	Derived from PMD receiver input.	M	Yes []
TM3	Low-frequency peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM4	Full-band peak-to-peak AC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM5	Peak-to-peak differential output voltage	120G.3.2	Per Table 120G–3	M	Yes []
TM6	Near-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM7	Near-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM8	Far-end eye height	120G.3.2	Per Table 120G–3	M	Yes []
TM9	Far-end VEC	120G.3.2	Per Table 120G–3	M	Yes []
TM10	Common-mode to differential-mode return loss	120G.3.2	Per Table 120G–3	M	Yes []
TM11	ERL	120G.3.2	Per Table 120G–3	M	Yes []
TM12	Differential termination mismatch	120G.3.2	Per Table 120G–3	M	Yes []
TM13	Transition time	120G.3.2	Per Table 120G–3	M	Yes []
TM14	DC common-mode voltage	120G.3.2	Per Table 120G–3	M	Yes []

120G.6.4.3 Host input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in host	120G.3.3		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.3	Per Table 120G–7	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.3	Per Table 120G–7	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.3	Per Table 120G–7	M	Yes []
RH5	ERL	120G.3.3	Per Table 120G–7	M	Yes []
RH6	Differential termination mismatch	120G.3.3	Per Table 120G–7	M	Yes []
RH7	Common-mode voltage	120G.3.3	Per Table 120G–7	M	Yes []
RH8	Stressed input test	120G.3.3.5		M	Yes []

120G.6.4.4 Module input

Item	Feature	Subclause	Value/Comment	Status	Support
RH1	Adaptive equalizer in module	120G.3.4		M	Yes []
RH2	Signaling rate tolerance, each lane	120G.3.4	Per Table 120G–9	M	Yes []
RH3	Differential peak-to-peak input voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH4	Common-mode to differential-mode return loss	120G.3.4	Per Table 120G–9	M	Yes []
RH5	ERL	120G.3.4	Per Table 120G–9	M	Yes []
RH6	Differential termination mismatch	120G.3.4	Per Table 120G–9	M	Yes []
RH7	Single-ended voltage tolerance	120G.3.4	Per Table 120G–9		
RH8	Common-mode voltage tolerance	120G.3.4	Per Table 120G–9	M	Yes []
RH9	Stressed input test	120G.3.4.3		M	Yes []

Annex 135A

(informative)

50 Gb/s and 100 Gb/s PMA sublayer partitioning examples

135A.2 Partitioning examples of 100GBASE-P PHYs

Replace Figure 135A–6 with the following figure:

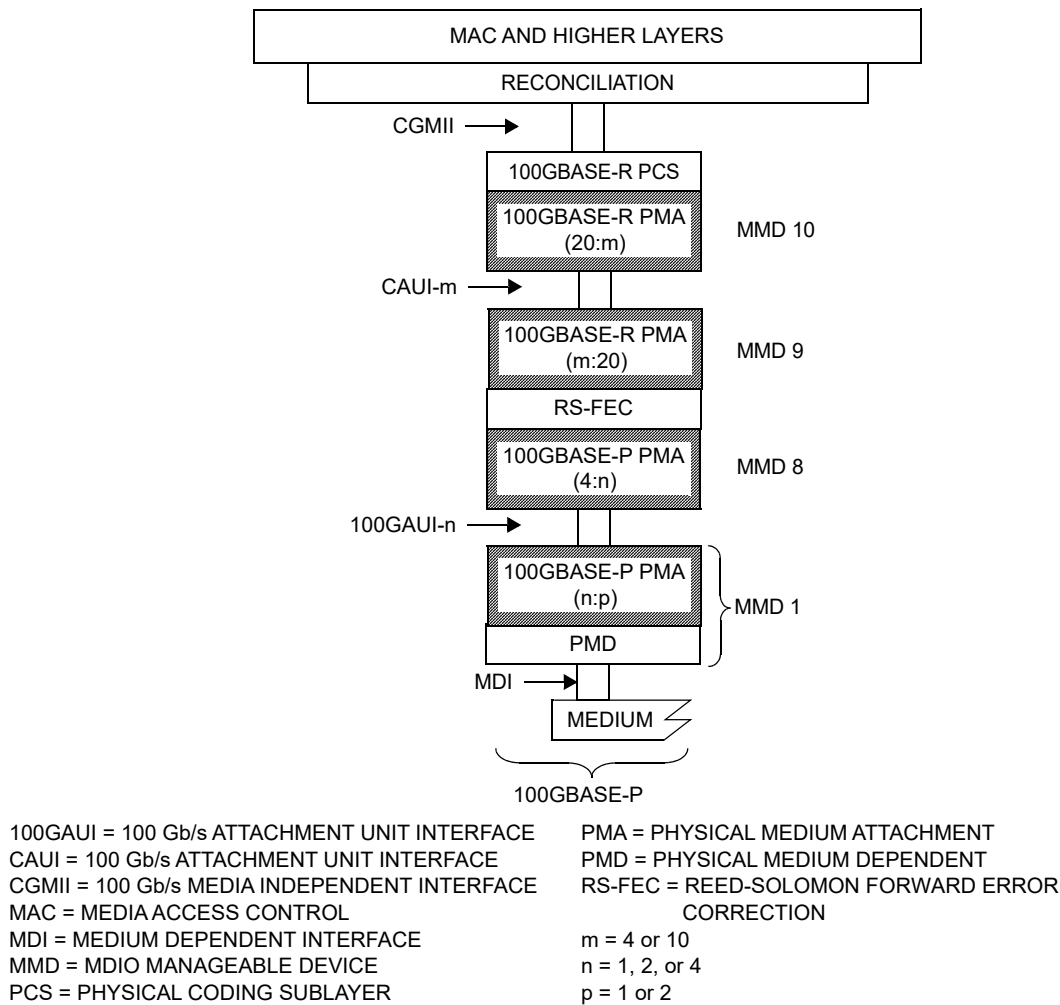
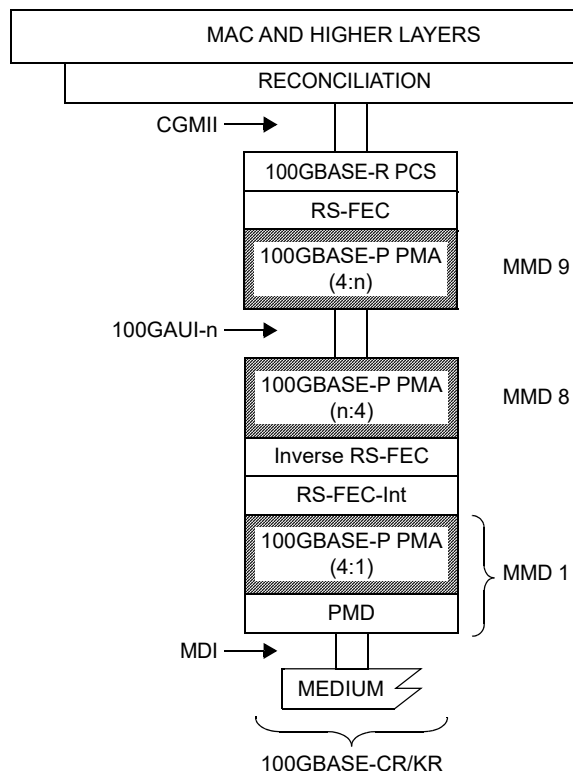


Figure 135A–6—Example 100GBASE-P PHY with CAUI-n and 100GAUI-n

Insert a new paragraph and Figure 135A–7a after Figure 135A–7 in 135A.3.1 as follows:

Figure 135A–7a depicts an example of a 100GBASE-CR/KR PHY with 100GAUI-n interface and Inverse RS-FEC.



100GAUI = 100 Gb/s ATTACHMENT UNIT INTERFACE
CGMII = 100 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
MMD = MDIO MANAGEABLE DEVICE
PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT

PMD = PHYSICAL MEDIUM DEPENDENT
RS-FEC = REED-SOLOMON FORWARD ERROR
CORRECTION
RS-FEC-Int = REED-SOLOMON FORWARD ERROR
CORRECTION CODEWORD-INTERLEAVED
n = 1, 2, or 4

Figure 135A–7a—Example 100GBASE-CR/KR PHY with RS-FEC and RS-FEC-Int

Insert new annexes Annex 162A through Annex 162D corresponding to Clause 162 as follows:

Annex 162A

(informative)

Transmitter, receiver and channel parameters associated with test points TP0 and TP5 for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162A.1 Overview

This annex provides information on transmitter, receiver, and channel parameters associated with test points TP0 and TP5, which might not be testable in an implemented system. TP0 and TP5 test points are illustrated in the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 link block diagram of Figure 162–2. This annex also provides information on channel characteristics.

162A.2 Transmitter characteristics at TP0

The recommended transmitter characteristics at TP0 as measured at TP0v are described in 163.9.2.

162A.3 Receiver characteristics at TP5

The recommended receiver characteristics at TP5 as measured at TP5v are described in 163.9.3.

162A.4 Transmitter and receiver differential printed circuit board (PCB) trace insertion loss

The recommended maximum and minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs are determined using Equation (162A–1) and Equation (162A–2), respectively, and illustrated in Figure 162A–1. Note that the recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs with allowances for ball grid array (BGA) footprint and host connector footprints is 6.875 dB at 26.56 GHz and the recommended minimum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is 2.3 dB at 26.56 GHz.

$$ILdd_{PCB}(f) \leq ILdd_{PCBmax}(f) = 0.9809(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-1)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{PCB}(f)$	is the insertion loss of the transmitter and receiver PCB in dB
$ILdd_{PCBmax}(f)$	is the maximum insertion loss of the transmitter and receiver PCB in dB
f	is the frequency in GHz

$$ILdd_{PCB}(f) \geq ILdd_{PCBmin}(f) = 0.0059 + 0.0361\sqrt{f} + 0.0794f \quad (162A-2)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{PCB}(f)$ is the insertion loss of the transmitter and receiver PCB in dB
 $ILdd_{PCBmin}(f)$ is the minimum insertion loss of the transmitter and receiver PCB in dB
 f is the frequency in GHz

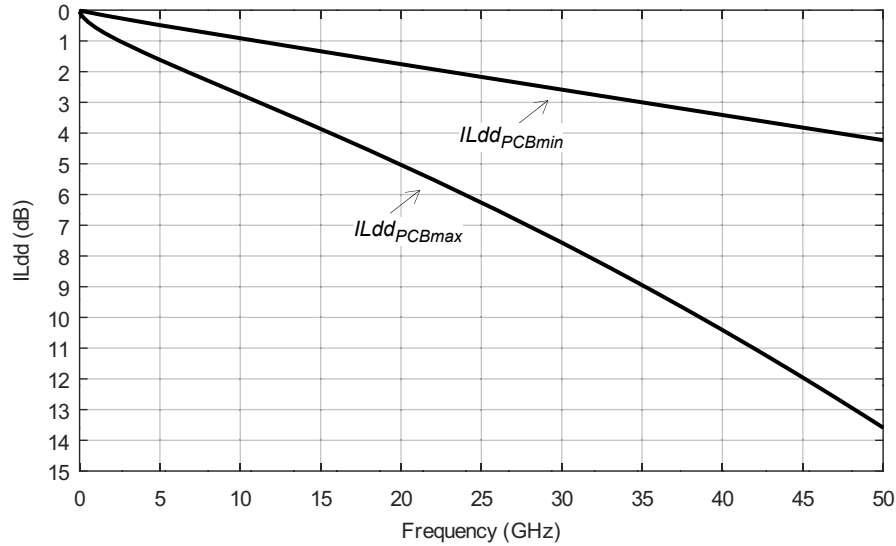


Figure 162A-1—Maximum and minimum insertion loss for transmitter or receiver PCB

The recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture is determined using Equation (162A-3), and illustrated in Figure 162A-2. The recommended maximum insertion loss allocation for the transmitter or receiver differential controlled impedance PCBs is consistent with the insertion loss from TP0 to TP2 or TP3 to TP5 and an assumed mated connector insertion loss of 1.6 dB. Note that the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 is 10.975 dB at 26.56 GHz.

$$ILdd_{Host}(f) \leq ILdd_{HostMax}(f) = 1.5658(0.471\sqrt{f} + 0.1194f + 0.002f^2) \quad (162A-3)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{Host}(f)$ is the insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

$ILdd_{HostMax}(f)$ is the recommended maximum insertion loss from TP0 to TP2 or from TP3 to TP5 including the test fixture in dB

f is the frequency in GHz

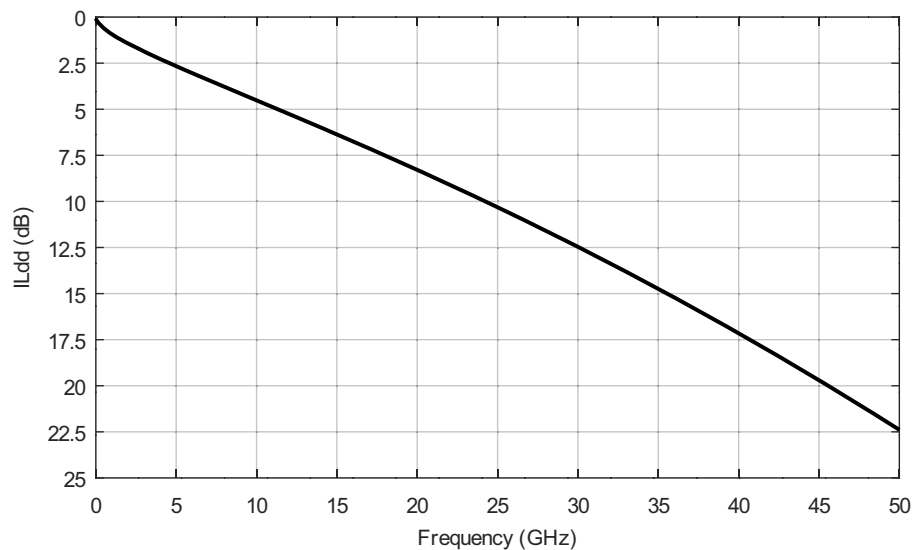


Figure 162A-2—Insertion loss from TP0 to TP2 or from TP3 to TP5

162A.5 Channel insertion loss

This subclause provides information on channel insertion losses for the maximum-loss and minimum-loss cabling topologies.

The maximum channel insertion loss associated with the maximum loss cable assembly is determined using Equation (162A-4). The channel insertion loss budget at 26.56 GHz is illustrated in Figure 162A-3 and Figure 162A-4. Loss budget parameters for all cable topologies are provided in Table 162A-1.

$$ILdd_{Chmax}(f) = ILdd_{Camax}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-4)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmax}(f)$	is the maximum channel insertion loss in dB between TP0 and TP5
$ILdd_{Camax}(f)$	is the maximum cable assembly insertion loss in dB (TP1 to TP4)
$ILdd_{MaxHost}(f)$	is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
$ILdd_{MTFref}(f)$	is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
f	is the frequency in GHz

The channel insertion loss associated with the minimum insertion loss cable assembly and a maximum insertion loss host channel is determined by Equation (162A-5).

$$ILdd_{Chmin}(f) = ILdd_{Camin}(f) + 2ILdd_{MaxHost}(f) - 2ILdd_{MTFref}(f) \quad (162A-5)$$

for $0.05 \leq f \leq 40$

where

$ILdd_{Chmin}(f)$	is the channel insertion loss in dB between TP0 and TP5 representative of a minimum insertion loss cable assembly and a maximum loss host channel
$ILdd_{Camin}(f)$	is the minimum cable assembly insertion loss in dB (TP1 to TP4) given in Equation (162-19) and illustrated in Figure 162-8
$ILdd_{MaxHost}(f)$	is the maximum insertion loss in dB from TP0 to TP2 or TP3 to TP5 using Equation (162A-3)
$ILdd_{MTFref}(f)$	is the reference insertion loss in dB of the mated test fixture using Equation (162B-5)
f	is the frequency in GHz

Table 162A-1—Insertion loss budget values at 26.56 GHz

Parameter	Value	Units
$ILdd_{Chmax}$	28.5	dB
$ILdd_{Camax}$	19.75	dB
$ILdd_{Chmin}$	19.75	dB
$ILdd_{Camin}$	11.0	dB
$ILdd_{MaxHost}$	10.975	dB
$ILdd_{MTFref}$	6.6	dB

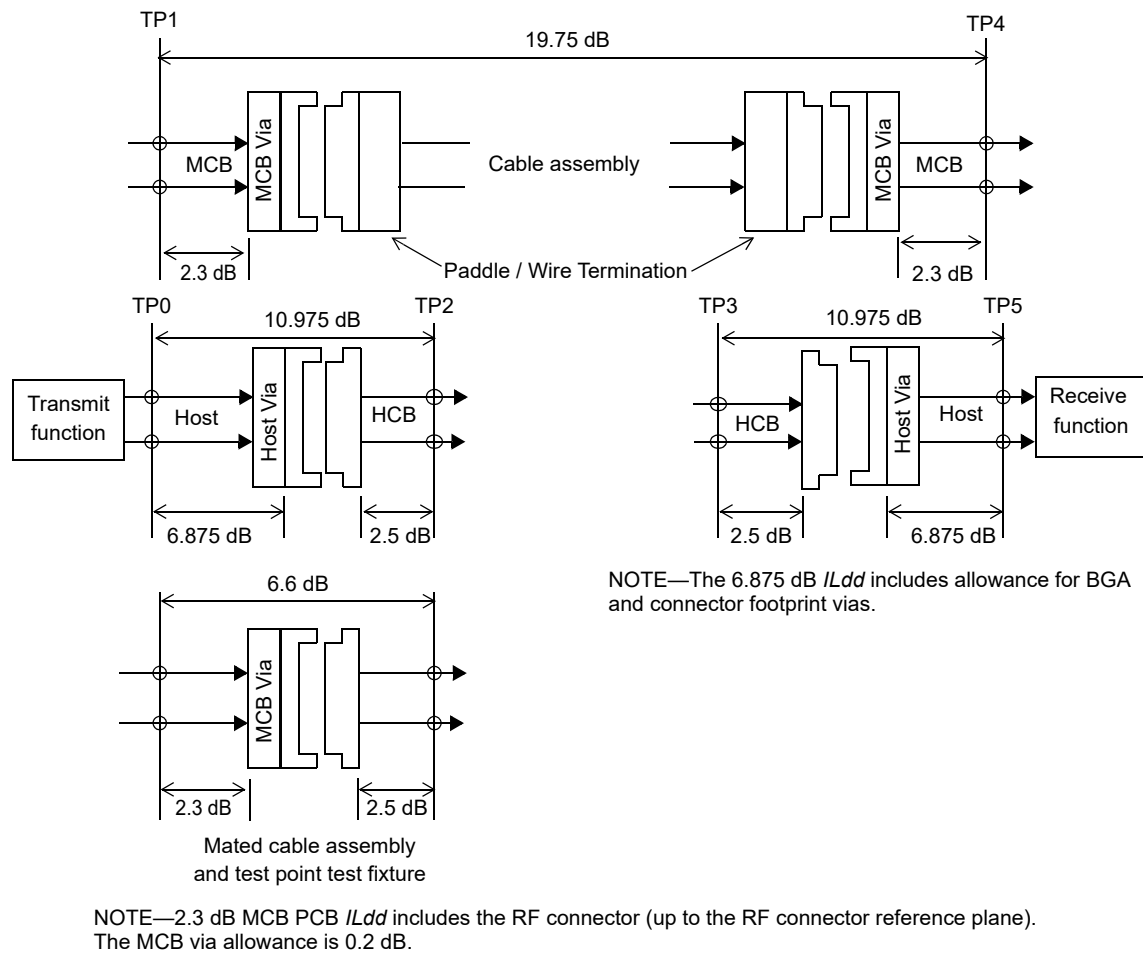
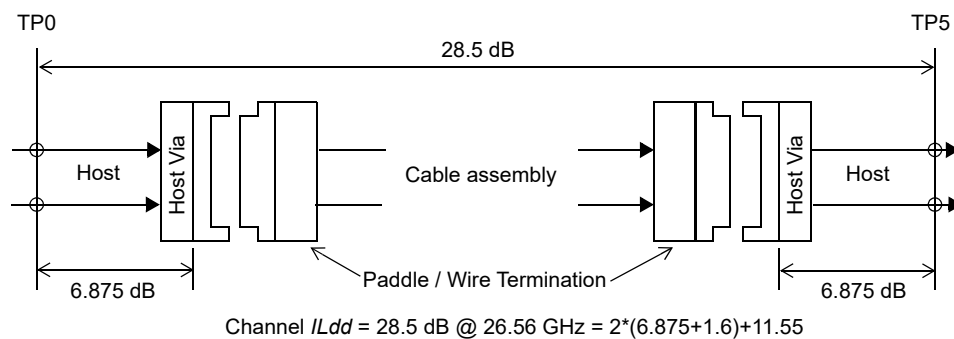


Figure 162A-3—Cable assembly, host, and test fixture insertion loss at 26.56 GHz



NOTE—Channel *ILdd* derived from cable assembly host, and mated test fixture

Figure 162A-4—Channel insertion loss at 26.56 GHz

162A.6 Channel effective return loss (ERL)

The 100GBASE-CR1, 200GBASE-CR2 and 400GBASE-CR4 channels are recommended to meet the ERL specified in 163.10.3.

162A.7 Channel Operating Margin (COM)

The Channel Operating Margin (COM) for the channel between TP0 and TP5, computed using the procedure in 93A.1 and the parameters in Table 163–11, is recommended to be greater than or equal to 3 dB.

Annex 162B

(normative)

Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M

162B.1 Test fixtures

Transmitter and receiver measurements at TP2 or TP3 for the 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 hosts (see Annex 162D) and at TP1a or TP4a for the 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2M hosts (see Annex 120G), are made utilizing the test fixture specified in 162B.2. Cable assembly measurements for the cable assembly types (see Annex 162D) are made between TP1 and TP4 with test fixtures as specified in 162B.3 on both ends. The test fixtures are specified in a mated state to enable connections to measurement equipment. The reference insertion loss of the mated test fixtures is 6.6 dB at 26.56 GHz using Equation (162B–5). The requirements in the referenced subclauses are not the MDI connector specifications for an implemented design.

162B.2 TP2 or TP3 test fixture

The TP2 or TP3 test fixture (also known as Host Compliance Board) is required for measuring the transmitter and receiver specifications at TP2 and TP3. The TP2 and TP3 test points are illustrated in Figure 162–2.

162B.2.1 TP2 or TP3 test fixture insertion loss

The TP2 or TP3 test fixture printed circuit board (PCB) insertion loss values determined using Equation (162B–1) shall be used as the TP2 or TP3 test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{tfref}(f) = 1.02(0.001 + 0.24\sqrt{f} + 0.046f) \quad (162B-1)$$

for 0.01 GHz $\leq f \leq$ 50 GHz

where

$ILdd_{tfref}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The TP2 or TP3 test fixture PCB reference insertion loss is illustrated in Figure 162B–1.

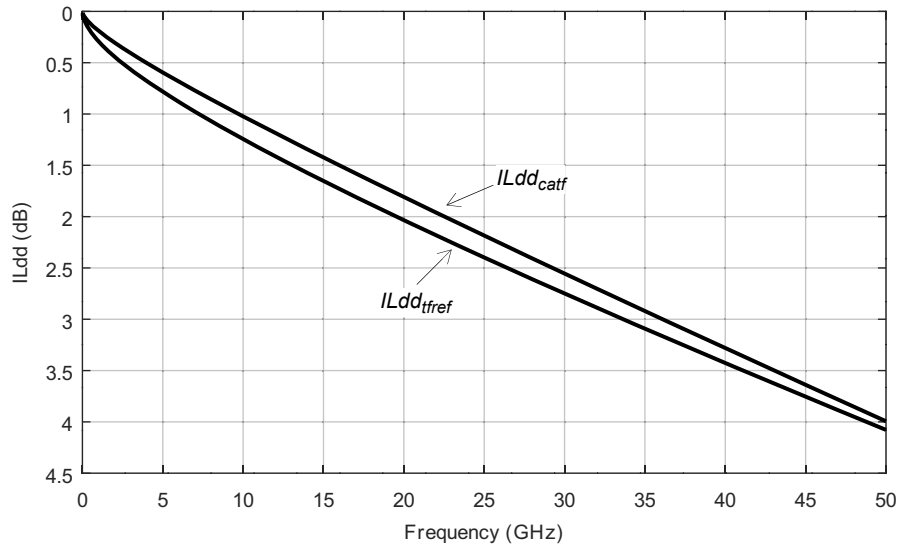


Figure 162B-1—Test fixtures PCB reference insertion losses

162B.3 Cable assembly test fixture

The cable assembly test fixture (also known as Module Compliance Board) is required for measuring the cable assembly specifications in 162.11 at TP1 and TP4. The TP1 and TP4 test points are illustrated in Figure 162-2.

162B.3.1 Cable assembly test fixture insertion loss

The cable assembly test fixture PCB and test point insertion loss values determined using Equation (162B-2) shall be used as the test fixture reference insertion loss. The effects of differences between the insertion loss of an actual test fixture and the reference insertion loss are to be accounted for in the measurements.

$$ILdd_{catf}(f) = 1.073(-0.00125 + 0.12\sqrt{f} + 0.0575f) \quad (162B-2)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{catf}(f)$ is the test fixture PCB reference insertion loss in dB at frequency f
 f is the frequency in GHz

The cable assembly test fixture reference insertion loss is illustrated in Figure 162B-1.

162B.4 Mated test fixtures

The TP2 or TP3 test fixture and the cable assembly test fixture are specified in a mated state illustrated in Figure 92–18. The mated test fixtures specifications are given below.

162B.4.1 Mated test fixtures insertion loss

The insertion loss of the mated test fixtures shall meet the values determined using Equation (162B–3) and Equation (162B–4).

$$ILdd(f) \leq ILdd_{MTFmax}(f) = \begin{cases} 1.185\sqrt{f}-0.072f+0.007f^2 & 0.01 \leq f < 40 \\ 1.915f-60.78 & 40 \leq f \leq 50 \end{cases} \quad (162B-3)$$

$$ILdd(f) \geq ILdd_{MTFmin}(f) = 1.0225(0.0656\sqrt{f}+0.164f) \quad (162B-4)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd(f)$	is the measured insertion loss in dB at frequency f
$ILdd_{MTFmax}(f)$	is the maximum mated test fixture insertion loss in dB at frequency f
$ILdd_{MTFmin}(f)$	is the minimum mated test fixture insertion loss in dB at frequency f
f	is the frequency in GHz

The reference insertion loss of the mated test fixtures is determined using Equation (162B–5).

$$ILdd_{MTFref}(f) = 0.8153\sqrt{f}+0.003405f^2 \quad (162B-5)$$

for $0.01 \text{ GHz} \leq f \leq 50 \text{ GHz}$

where

$ILdd_{MTFref}(f)$	is the reference insertion loss of the mated test fixture in dB at frequency f
f	is the frequency in GHz

The mated test fixtures insertion loss limits are illustrated in Figure 162B–2.

The FOM_{ILD} is calculated according to 93A.4 with $f_b=53.125 \text{ GHz}$, $T_i=8.5 \text{ ps}$, and $f_r=0.75 \times f_b$. The fitted insertion loss and insertion loss deviation are computed over the range $f_{min}=0.05 \text{ GHz}$ to $f_{max}=40 \text{ GHz}$. FOM_{ILD} shall be less than or equal to 0.15 dB.

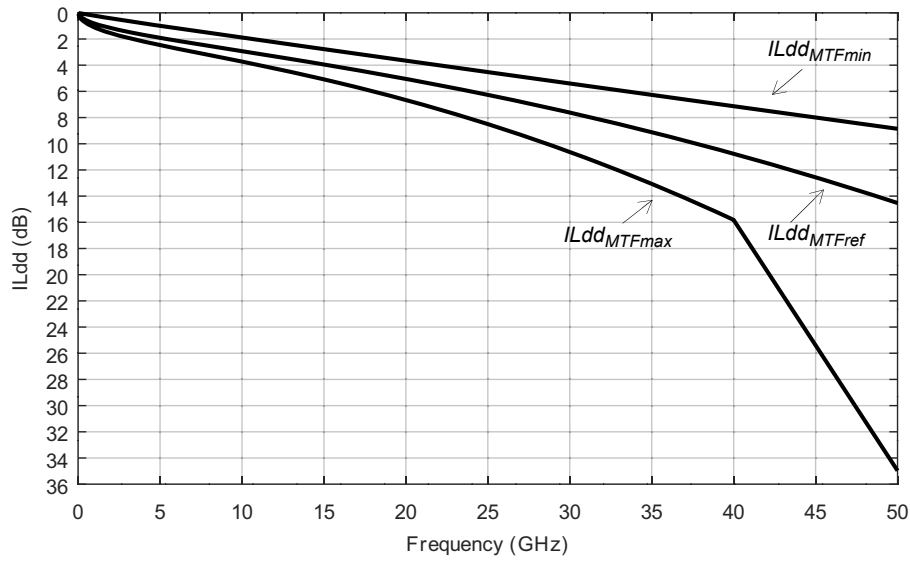


Figure 162B-2—Mated test fixtures insertion loss

162B.4.2 Mated test fixtures effective return loss (ERL)

The values of the mated test fixtures ERL are computed using the procedure in 93A.5 with the parameter values in Table 162B-1. Parameters that do not appear in Table 162B-1 take values from Table 162-20.

The mated test fixture ERL shall be greater than or equal to 10.3 dB.

Table 162B-1—Mated test fixture ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	400	UI
Equalizer length associated with reflection signal	N_{bx}	0	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—
Target detector error ratio	DER_0	10^{-5}	—
NOTE—The mated test fixture test connector and transmission line are not time-gated (by setting T_{fx} to 0) in order to include the entire test fixture.			

162B.4.3 Mated test fixtures common-mode to differential-mode insertion loss

The common-mode to differential-mode insertion loss of the mated test fixtures measured in both directions shall meet the values determined using Equation (162B–6).

$$IL_{dc}(f) \geq \begin{cases} 30 - (21/28)f & 0.01 \leq f < 20 \\ 15 & 20 \leq f \leq 50 \end{cases} \quad (162B-6)$$

where

$IL_{dc}(f)$ is the common-mode to differential-mode insertion loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to differential-mode insertion loss limit is illustrated in Figure 162B–3.

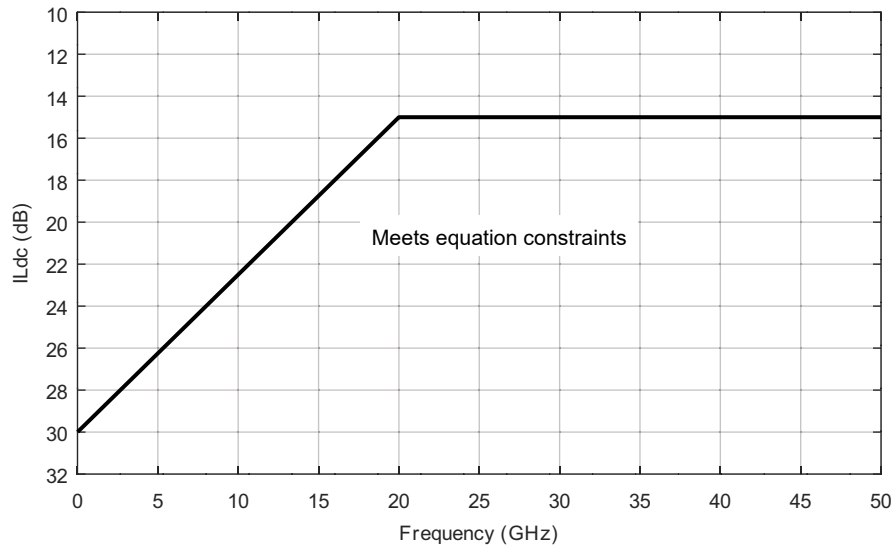


Figure 162B–3—Mated test fixtures common-mode to differential-mode insertion loss

162B.4.4 Mated test fixtures common-mode to common-mode return loss

The common-mode to common-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet the values determined using Equation (162B–7).

$$RL_{cc}(f) \geq \begin{cases} 12 - 9f & 0.01 \leq f < 1 \\ 3 & 1 \leq f < 4 \\ 2.6 + 0.1f & 4 \leq f < 30 \\ 9.5 - 0.13f & 30 \leq f \leq 50 \end{cases} \quad (162B-7)$$

where

$RL_{cc}(f)$ is the common-mode to common-mode return loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to common-mode return loss limit is illustrated in Figure 162B–4.

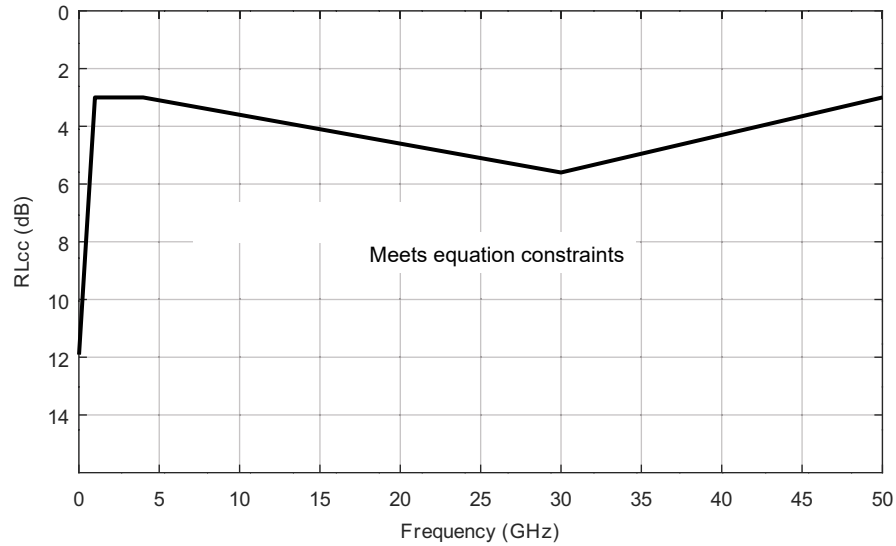


Figure 162B–4—Mated test fixtures common-mode to common-mode return loss

162B.4.5 Mated test fixtures common-mode to differential-mode return loss

The common-mode to differential-mode return loss of the mated test fixtures measured at each test fixture test interface shall meet the values determined using Equation (162B–8).

$$RL_{dc}(f) \geq \begin{cases} 30 - (30/25.78)f & 0.01 \leq f < 12.89 \\ 17.85 - 0.225f & 12.89 \leq f < 35 \\ 10 & 35 \leq f \leq 50 \end{cases} \quad (162B-8)$$

where

$RL_{dc}(f)$ is the common-mode to differential-mode return loss in dB at frequency f
 f is the frequency in GHz

The mated test fixtures common-mode to differential-mode return loss limit is illustrated in Figure 162B–5.

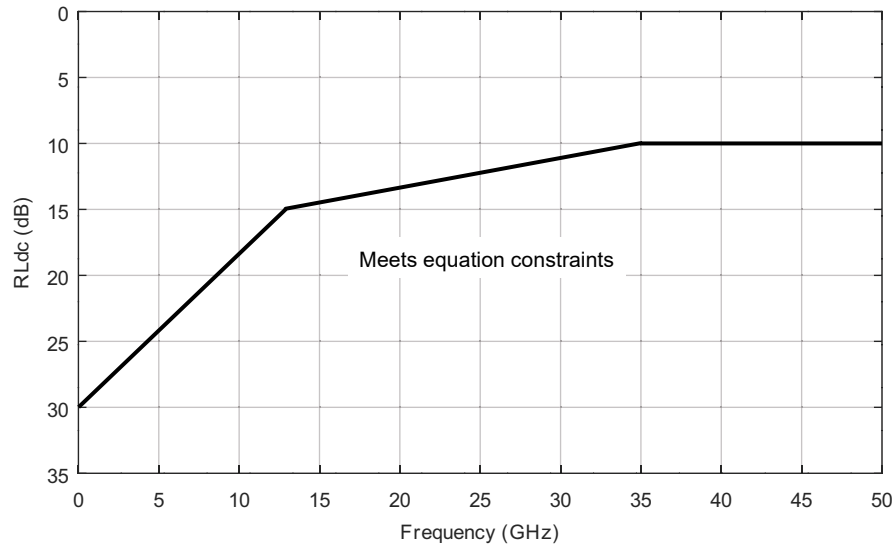


Figure 162B-5—Mated test fixtures common-mode to differential-mode return loss

162B.4.6 Mated test fixtures integrated crosstalk noise

The SFP112 mated test fixture integrated near-end crosstalk noise voltage for the disturber near-end crosstalk loss is determined according to the method in 110B.1.3.6, given the disturber near-end crosstalk loss $NEXT_loss(f)$ is measured over N uniformly spaced frequencies f_n spanning the frequency range 50 MHz to 40 000 MHz with maximum frequency spacing of 10 MHz, and the parameters shown in Table 162B-2. The mated test fixture integrated near-end crosstalk noise voltage shall meet the specification in Table 162B-3.

Table 162B-2—SFP112 mated test fixture integrated near-end crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	53.125	GBd
3 dB reference receiver bandwidth	f_r	39.84	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	8.5	ps

Table 162B-3—SFP112 mated test fixture integrated near-end crosstalk noise voltage

Parameter	Value	Units
Integrated near-end crosstalk noise voltage	Less than 1.6	mV

The multi-lane mated test fixtures integrated crosstalk noise voltages are determined using Equation (92–44) through Equation (92–48), given the multiple disturber near-end crosstalk loss $MDNEXT_loss(f)$ and multiple disturber far-end crosstalk loss $MDFEXT_loss(f)$ measured over N uniformly-spaced frequencies f_n spanning the frequency range 50 MHz to 40 000 MHz with a maximum frequency spacing of 10 MHz, and the parameters shown in Table 162B–4. The multi-lane mated test fixture integrated crosstalk noise shall meet the specifications in Table 162B–5.

Table 162B–4—Multi-lane mated test fixture integrated crosstalk noise parameters

Description	Symbol	Value	Units
Symbol rate	f_b	53.125	GBd
3 dB reference receiver bandwidth	f_r	39.84	GHz
Near-end disturber peak differential output amplitude	A_{nt}	600	mV
Far-end disturber peak differential output amplitude	A_{ft}	600	mV
Near-end disturber 20% to 80% rise and fall times	T_{nt}	8.5	ps
Far-end disturber 20% to 80% rise and fall times	T_{ft}	8.5	ps

Table 162B–5—Multi-lane mated test fixture integrated crosstalk noise

Parameters	Value	Units
MDFEXT integrated crosstalk noise voltage	less than 4.2	mV
MDNEXT integrated crosstalk noise voltage	less than 1.5	mV
Total integrated crosstalk noise voltage	less than 4.4	mV

162B.5 Protocol implementation conformance statement (PICS) proforma for Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M²⁶

162B.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

162B.5.2 Identification

162B.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

162B.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 162B, Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

²⁶Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

162B.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
HTF1	TP2 or TP3 test fixture	162B.2		O	Yes [] No []
CATF1	Cable assembly test fixture	162B.3		O	Yes [] No []

162B.5.4 PICS proforma tables for test fixtures for Test fixtures for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, 100GAUI-1 C2M, 200GAUI-2 C2M, and 400GAUI-4 C2M

Item	Feature	Subclause	Value/Comment	Status	Support
TF1	Mated test fixtures insertion loss	162B.4.1	Per Equation (162B–3) and Equation (162B–4)	M	Yes []
TF2	Mated test fixtures FOM_{ILD}	162B.4.1	Less than or equal to 0.15 dB	M	Yes []
TF3	Mated test fixtures ERL	162B.4.2	Greater than or equal to 10.3 dB	M	Yes []
TF4	Mated test fixtures common-mode to differential-mode insertion loss	162B.4.3	Per Equation (162B–6)	M	Yes []
TF5	Mated test fixtures common-mode to common-mode return loss	162B.4.4	Per Equation (162B–7)	M	Yes []
TF6	Mated test fixtures common-mode to differential-mode return loss	162B.4.5	Per Equation (162B–8)	M	Yes []
TF7	SFP112 Mated test fixtures integrated near-end crosstalk noise	162B.4.6	Meets the requirements of Table 162B–3	M	Yes []
TF8	Multi-lane mated test fixtures integrated crosstalk noise	162B.4.6	Meets the requirements of Table 162B–5	M	Yes []

Annex 162C

(normative)

MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162C.1 Overview

This annex defines the Media Dependent Interface (MDI) for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4. The MDI couples the PMD (specified in 162.8 and 162.9) to the cable assembly (specified in 162.11). The PMDs supportable for each MDI connector type are given in Table 162C–1. The SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP are also referred to as multi-lane connectors.

Table 162C–1—Number of PMDs supportable for each connector type

MDI types	100GBASE-CR1	200GBASE-CR2	400GBASE-CR4	Reference
SFP112	1	—	—	162C.2.1
SFP-DD112	1, 2	1	—	162C.2.2
DSFP	1, 2	1	—	162C.2.3
QSFP112	1, 2, 4	1, 2	1	162C.2.4
QSFP-DD800	1, 2, 4, 8	1, 2, 4	1, 2	162C.2.5
OSFP	1, 2, 4, 8	1, 2, 4	1, 2	162C.2.6

For 100GBASE-CR1, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of 162C.2.1 (SFP112), 162C.2.2 (SFP-DD112), 162C.2.3 (DSFP), 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800), or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD.

For 200GBASE-CR2, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 162C.2.2 (SFP-DD112), 162C.2.3 (DSFP), 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800), or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD.

For 400GBASE-CR4, the mechanical interface between the PMD and the cable assembly is a mated pair of connectors meeting the requirements of 162C.2.4 (QSFP112), 162C.2.5 (QSFP-DD800) or 162C.2.6 (OSFP). The plug connector is used on the cable assembly and the receptacle is used on the PMD. The cable assembly connector shall be the MDI connector plug. The PMD connector shall be the MDI connector receptacle.

The MDI connector electrical performance is consistent with the signal quality and electrical requirements of 162.9 and 162.10.

An MDI connector type may support one or more PMDs. The assignment of PMD signals to connector signals is specified in Table 162C–2, where as an example 0:DL0n refers to the DL0n signal of the first PMD; see 162.8.1 for signal naming definitions. When an MDI connector is not fully utilized the lower PMD numbers in Table 162C–2 should be used.

Table 162C–2—PMD to connector signal assignments

PMD signal <PMD number>:<PMD signal>			Connector signal
100GBASE-CR1	200GBASE-CR2	400GBASE-CR4	
0:DL0n	0:DL0n	0:DL0n	DL0n
0:DL0p	0:DL0p	0:DL0p	DL0p
1:DL0n	0:DL1n	0:DL1n	DL1n
1:DL0p	0:DL1p	0:DL1p	DL1p
2:DL0n	1:DL0n	0:DL2n	DL2n
2:DL0p	1:DL0p	0:DL2p	DL2p
3:DL0n	1:DL1n	0:DL3n	DL3n
3:DL0p	1:DL1p	0:DL3p	DL3p
4:DL0n	2:DL0n	1:DL0n	DL4n
4:DL0p	2:DL0p	1:DL0p	DL4p
5:DL0n	2:DL1n	1:DL1n	DL5n
5:DL0p	2:DL1p	1:DL1p	DL5p
6:DL0n	3:DL0n	1:DL2n	DL6n
6:DL0p	3:DL0p	1:DL2p	DL6p
7:DL0n	3:DL1n	1:DL3n	DL7n
7:DL0p	3:DL1p	1:DL3p	DL7p
0:SL0n	0:SL0n	0:SL0n	SL0n
0:SL0p	0:SL0p	0:SL0p	SL0p
1:SL0n	0:SL1n	0:SL1n	SL1n
1:SL0p	0:SL1p	0:SL1p	SL1p
2:SL0n	1:SL0n	0:SL2n	SL2n
2:SL0p	1:SL0p	0:SL2p	SL2p
3:SL0n	1:SL1n	0:SL3n	SL3n
3:SL0p	1:SL1p	0:SL3p	SL3p
4:SL0n	2:SL0n	1:SL0n	SL4n
4:SL0p	2:SL0p	1:SL0p	SL4p
5:SL0n	2:SL1n	1:SL1n	SL5n

Table 162C–2—PMD to connector signal assignments (*continued*)

PMD signal <PMD number>:<PMD signal>			Connector signal
100GBASE-CR1	200GBASE-CR2	400GBASE-CR4	
5:SL0p	2:SL1p	1:SL1p	SL5p
6:SL0n	3:SL0n	1:SL2n	SL6n
6:SL0p	3:SL0p	1:SL2p	SL6p
7:SL0n	3:SL1n	1:SL3n	SL7n
7:SL0p	3:SL1p	1:SL3p	SL7p

The MDI connectors shall have data signal and signal ground contact assignments as specified in Table 162C–3, Table 162C–4, and Table 162C–5 for the paired transmit and receive lanes (e.g., (SL0, DL0), (SL1, DL1)).

Table 162C–3—MDI connector contact mapping for SFP112, SFP-DD112, and DSFP

SFP112	SFP-DD112	DSFP	Connector signal name	Description
11	11	11	GND	Ground
12	12	12	DL0n	Receiver inverted data output
13	13	13	DL0p	Receiver non-inverted data output
14	14	14	GND	Ground
17	17	17	GND	Ground
18	18	18	SL0p	Transmitter non-inverted data input
19	19	19	SL0n	Transmitter inverted data input
20	20	20	GND	Ground
—	31	10	GND	Ground
—	32	9	DL1n	Receiver inverted data output
—	33	8	DL1p	Receiver non-inverted data output
—	34	7	GND	Ground
—	37	3	GND	Ground
—	38	2	SL1p	Transmitter non-inverted data input
—	39	1	SL1n	Transmitter inverted data input
—	40	22	GND	Ground

Table 162C–4—MDI connector contact mapping for QSFP112 and QSFP-DD800

QSFP112	QSFP-DD800	Connector signal name	Description
1	1	GND	Ground
2	2	SL1n	Transmitter inverted data input
3	3	SL1p	Transmitter non-inverted data input
4	4	GND	Ground
5	5	SL3n	Transmitter inverted data input
6	6	SL3p	Transmitter non-inverted data input
7	7	GND	Ground
13	13	GND	Ground
14	14	DL2p	Receiver non-inverted data output
15	15	DL2n	Receiver inverted data output
16	16	GND	Ground
17	17	DL0p	Receiver non-inverted data output
18	18	DL0n	Receiver inverted data output
19	19	GND	Ground
20	20	GND	Ground
21	21	DL1n	Receiver inverted data output
22	22	DL1p	Receiver non-inverted data output
23	23	GND	Ground
24	24	DL3n	Receiver inverted data output
25	25	DL3p	Receiver non-inverted data output
26	26	GND	Ground
32	32	GND	Ground
33	33	SL2p	Transmitter non-inverted data input
34	34	SL2n	Transmitter inverted data input
35	35	GND	Ground
36	36	SL0p	Transmitter non-inverted data input
37	37	SL0n	Transmitter inverted data input
38	38	GND	Ground
—	39	GND	Ground
—	40	SL5n	Transmitter inverted data input
—	41	SL5p	Transmitter non-inverted data input

Table 162C–4—MDI connector contact mapping for QSFP112 and QSFP-DD800 (*continued*)

QSFP112	QSFP-DD800	Connector signal name	Description
—	42	GND	Ground
—	43	SL7n	Transmitter inverted data input
—	44	SL7p	Transmitter non-inverted data input
—	45	GND	Ground
—	51	GND	Ground
—	52	DL6p	Receiver non-inverted data output
—	53	DL6n	Receiver inverted data output
—	54	GND	Ground
—	55	DL4p	Receiver non-inverted data output
—	56	DL4n	Receiver inverted data output
—	58	GND	Ground
—	59	DL5n	Receiver inverted data output
—	60	DL5p	Receiver non-inverted data output
—	61	GND	Ground
—	62	DL7n	Receiver inverted data output
—	63	DL7p	Receiver non-inverted data output
—	64	GND	Ground
—	70	GND	Ground
—	71	SL6p	Transmitter non-inverted data input
—	72	SL6n	Transmitter inverted data input
—	73	GND	Ground
—	74	SL4p	Transmitter non-inverted data input
—	75	SL4n	Transmitter inverted data input

Table 162C–5—MDI connector contact mapping for OSFP

OSFP	Connector signal name	Description
1	GND	Ground
2	SL1p	Transmitter non-inverted data input
3	SL1n	Transmitter inverted data input
4	GND	Ground
5	SL3p	Transmitter non-inverted data input
6	SL3n	Transmitter inverted data input
7	GND	Ground
8	SL5p	Transmitter non-inverted data input
9	SL5n	Transmitter inverted data input
10	GND	Ground
11	SL7p	Transmitter non-inverted data input
12	SL7n	Transmitter inverted data input
13	GND	Ground
18	GND	Ground
19	DL6n	Receiver inverted data output
20	DL6p	Receiver non-inverted data output
21	GND	Ground
22	DL4n	Receiver inverted data output
23	DL4p	Receiver non-inverted data output
24	GND	Ground
25	DL2n	Receiver inverted data output
26	DL2p	Receiver non-inverted data output
27	GND	Ground
28	DL0n	Receiver inverted data output
29	DL0p	Receiver non-inverted data output
30	GND	Ground
31	GND	Ground
32	DL1p	Receiver non-inverted data output
33	DL1n	Receiver inverted data output
34	GND	Ground
35	DL3p	Receiver non-inverted data output
36	DL3n	Receiver inverted data output

Table 162C–5—MDI connector contact mapping for OSFP (*continued*)

OSFP	Connector signal name	Description
37	GND	Ground
38	DL5p	Receiver non-inverted data output
39	DL5n	Receiver inverted data output
40	GND	Ground
41	DL7p	Receiver non-inverted data output
42	DL7n	Receiver inverted data output
43	GND	Ground
48	GND	Ground
49	SL6n	Transmitter inverted data input
50	SL6p	Transmitter non-inverted data input
51	GND	Ground
52	SL4n	Transmitter inverted data input
53	SL4p	Transmitter non-inverted data input
54	GND	Ground
55	SL2n	Transmitter inverted data input
56	SL2p	Transmitter non-inverted data input
57	GND	Ground
58	SL0n	Transmitter inverted data input
59	SL0p	Transmitter non-inverted data input
60	GND	Ground

162C.2 MDI connector types

162C.2.1 SFP112

For 100GBASE-CR1, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of the SFP-DD MSA. SFP112 supports one lane.

The SFP112 cable assembly connector is the plug illustrated in Figure 162C–1. The SFP112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C–2. The connectors have data signal and signal ground contact assignments as specified in Table 162C–3.

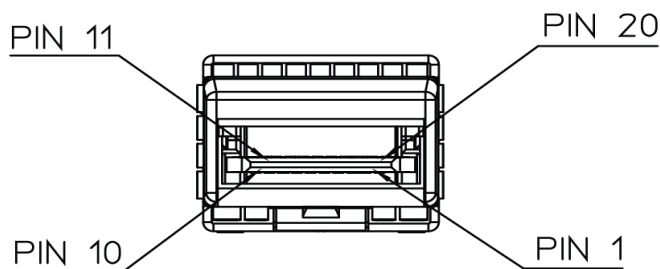


Figure 162C-1—SFP112 cable assembly plug

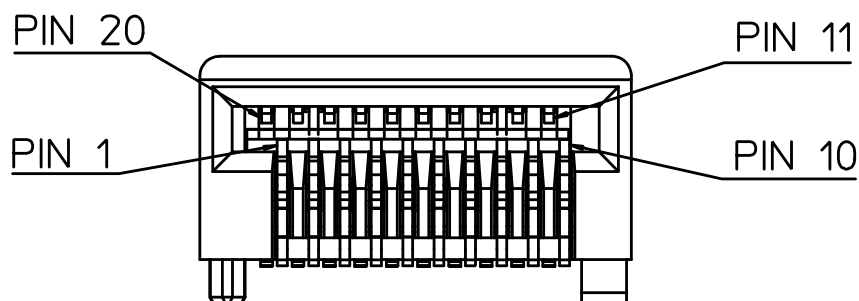


Figure 162C-2—SFP112 PMD receptacle

162C.2.2 SFP-DD112

For 100GBASE-CR1 or 200GBASE-CR2 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the SFP-DD MSA. SFP-DD112 supports up to two lanes.

The SFP-DD112 cable assembly connector is the plug illustrated in Figure 162C-3. The SFP-DD112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-4. The connectors have data signal and signal ground contact assignments as specified in Table 162C-3.

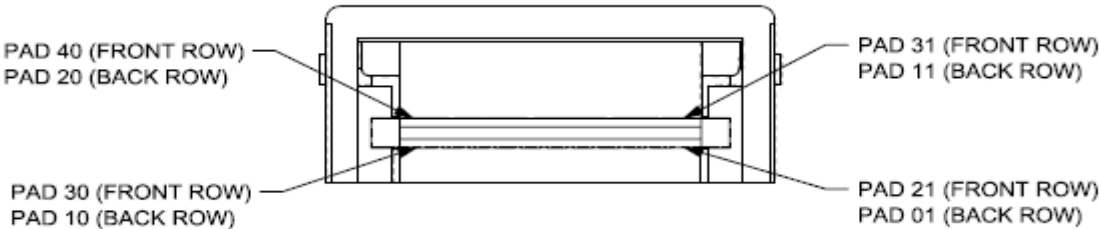


Figure 162C-3—SFP-DD112 cable assembly plug

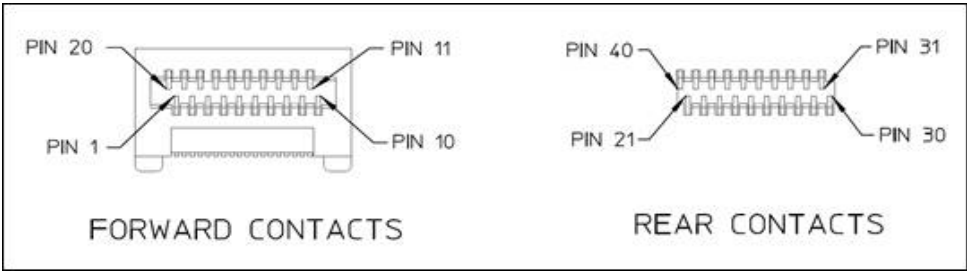


Figure 162C-4—SFP-DD112 PMD receptacle

162C.2.3 DSFP

For 100GBASE-CR1 or 200GBASE-CR2 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the DSFP MSA. DSFP supports up to two lanes.

The DSFP cable assembly connector is the plug illustrated in Figure 162C-5. The DSFP PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-6. The connectors have data signal and signal ground contact assignments as specified in Table 162C-3.

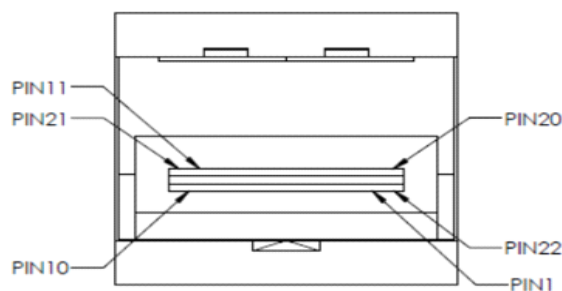


Figure 162C-5—DSFP cable assembly plug

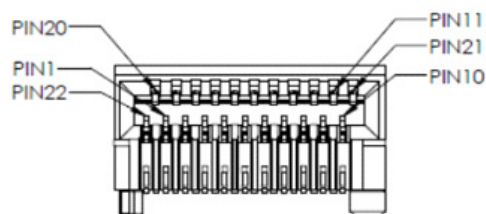


Figure 162C-6—DSFP PMD receptacle

162C.2.4 QSFP112

For 100GBASE-CR1, 200GBASE-CR2 or 400GBASE-CR4 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors meeting the requirements of the QSFP-DD MSA. QSFP112 supports up to four lanes.

The QSFP112 cable assembly connector is the plug illustrated in Figure 162C-7. The QSFP112 PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-8. The connectors have data signal and signal ground contact assignments as specified in Table 162C-4.

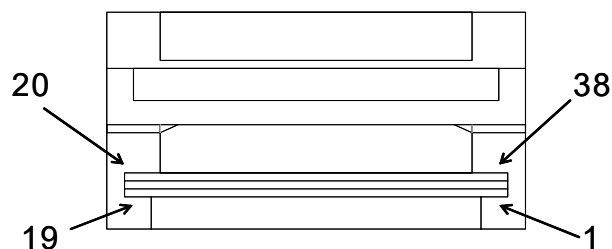


Figure 162C-7—QSFP112 cable assembly plug

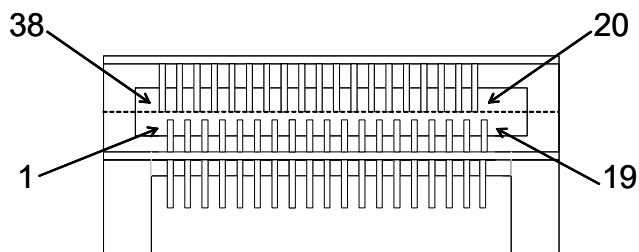


Figure 162C-8—QSFP112 PMD receptacle

162C.2.5 QSFP-DD800

For 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined as defined in the QSFP-DD800 MSA. QSFP-DD800 supports up to eight lanes.

The QSFP-DD800 cable assembly connector is the plug illustrated in Figure 162C-9. The QSFP-DD PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-10. The connectors have data signal and signal ground contact assignments as specified in Table 162C-4.

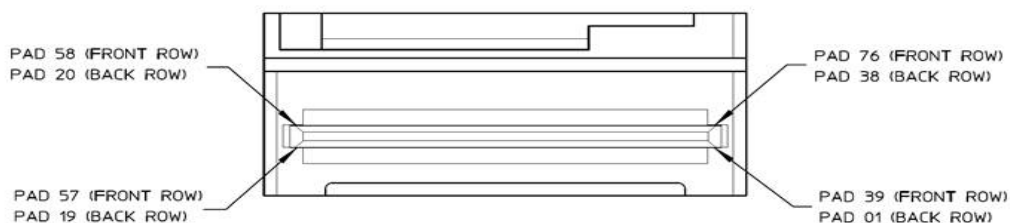


Figure 162C-9—QSFP-DD800 cable assembly plug

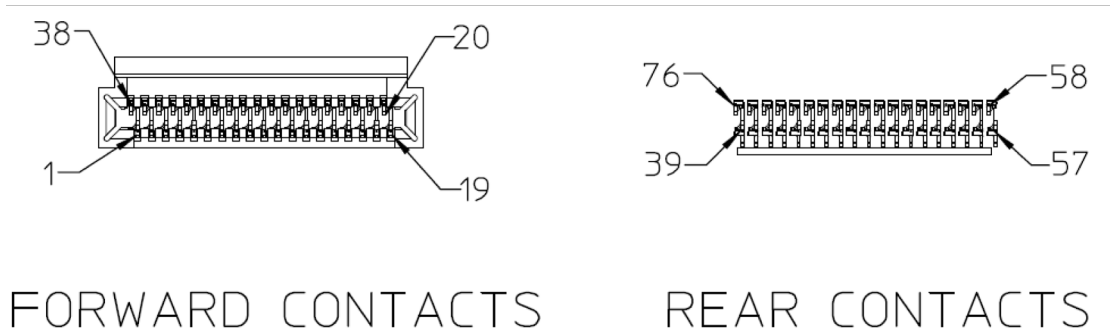


Figure 162C-10—QSFP-DD800 PMD receptacle

162C.2.6 OSFP

For 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4, the mechanical interface between the PMD and the cable assembly may be a mated pair of connectors as defined in the OSFP MSA. OSFP supports up to eight lanes.

The OSFP cable assembly connector is the plug illustrated in Figure 162C-11. The OSFP PMD connector is the receptacle with the mechanical mating interface as illustrated in Figure 162C-12. The connectors have data signal and signal ground contact assignments as specified in Table 162C-5.

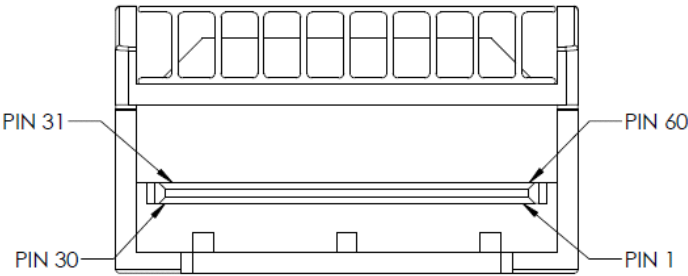


Figure 162C-11—OSFP cable assembly plug

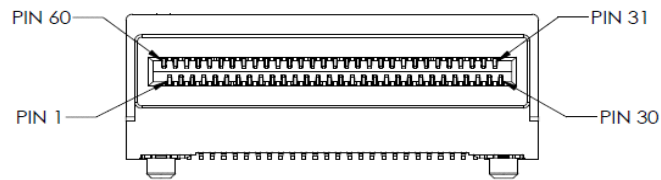


Figure 162C–12—OSFP PMD receptacle

162C.3 Protocol implementation conformance statement (PICS) proforma for Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4²⁷

162C.3.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

162C.3.2 Identification

162C.3.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1, 3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

162C.3.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 162C, MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
-------------------	--

²⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this annex so that it can be used for its intended purpose and may further publish the completed PICS.

162C.3.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MDI1	SFP112 MDI connector	162C.2.1	Device uses SFP112 receptacle as MDI connector	O	Yes [] No []
*MDI2	SFP-DD112 MDI connector	162C.2.2	Device uses SFP-DD112 receptacle as MDI connector	O	Yes [] No []
*MDI3	DSFP MDI connector	162C.2.3	Device uses DSFP receptacle as MDI connector	O	Yes [] No []
*MDI4	QSFP112 MDI connector	162C.2.4	Device uses QSFP112 receptacle as MDI connector	O	Yes [] No []
*MDI5	QSFP-DD800 MDI connector	162C.2.5	Device uses QSFP-DD800 receptacle as MDI connector	O	Yes [] No []
*MDI6	OSFP MDI connector	162C.2.6	Device uses OSFP receptacle as MDI connector	O	Yes [] No []

162C.3.4 PICS proforma tables for MDIs for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162C.3.4.1 Contact Mapping

Item	Feature	Subclause	Value/Comment	Status	Support
CM1	SFP112 MDI connector contact mapping according to Table 162C–3	162C.2.1		MDI1:M	Yes [] N/A []
CM2	SFP-DD112 MDI connector contact mapping according to Table 162C–3	162C.2.2		MDI2:M	Yes [] N/A []
CM3	DSFP MDI connector contact mapping according to Table 162C–3	162C.2.3		MDI3:M	Yes [] N/A []
CM4	QSFP112 MDI connector contact mapping according to Table 162C–4	162C.2.4		MDI4:M	Yes [] N/A []
CM5	QSFP-DD800 MDI connector contact mapping according to Table 162C–4	162C.2.5		MDI5:M	Yes [] N/A []
CM6	OSFP MDI connector contact mapping according to Table 162C–5	162C.2.6		MDI6:M	Yes [] N/A []

Annex 162D

(informative)

Cable assemblies and hosts for 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4

162D.1 Overview

This annex describes cable assembly types specified in 162.11 for hosts with 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 Physical Layers. The six MDI connector receptacles specified are given in Table 162D–1. This enables multiple cable assembly types with different combinations of the plug connectors at each end.

Table 162D–1—Host receptacles and cable assembly plugs

Receptacle/Plugs	Reference
SFP112	162C.2.1
SFP-DD112	162C.2.2
DSFP	162C.2.3
QSFP112	162C.2.4
QSFP-DD800	162C.2.5
OSFP	162C.2.6

162D.1.1 Cable assembly types

Cable assembly types, denoted as 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, have a common set of electrical parameters specified in 162.11, enabling a 2 m length. Cable assembly types with different combinations of the plug connectors for compatible host receptacles and supportable number of PMDs are given in Table 162D–2 for 100GBASE-CR1, Table 162D–3 for 200GBASE-CR2, and Table 162D–4 for 400GBASE-CR4; other combinations are possible.

Table 162D–2—100GBASE-CR1 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
SFP112	1	SFP112	1	1
SFP-DD112	1	SFP112	2	2
DSFP	1	SFP112	2	2
QSFP112	1	SFP112	4	4

Table 162D–2—100GBASE-CR1 cable assembly types and supportable number of PMDs (*continued*)

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
QSFP-DD800	1	SFP112	8	8
OSFP	1	SFP112	8	8
SFP-DD112	1	SFP-DD112	1	2
DSFP	1	DSFP	1	2
QSFP112	1	QSFP112	1	4
QSFP-DD800	1	QSFP-DD800	1	8
OSFP	1	OSFP	1	8

Table 162D–3—200GBASE-CR2 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
SFP-DD112	1	SFP-DD112	1	1
SFP-DD112	1	DSFP	1	1
QSFP112	1	SFP-DD112	2	2
QSFP-DD800	1	112	4	4
OSFP	1	SFP-DD112	4	4
DSFP	1	DSFP	1	1
QSFP112	1	DSFP	2	2
QSFP-DD800	1	DSFP	4	4
OSFP	1	DSFP	4	4
QSFP112	1	QSFP112	1	2
QSFP-DD800	1	QSFP-DD800	1	4
QSFP-DD800	1	OSFP	1	4
OSFP	1	OSFP	1	4

Table 162D-4—400GBASE-CR4 cable assembly types and supportable number of PMDs

One end		Other end(s)		Supportable PMDs
Receptacle/Plug	Number	Receptacle/Plug	Number	Number
QSFP112	1	QSFP112	1	1
QSFP-DD800	1	QSFP112	2	2
OSFP	1	QSFP112	2	2
QSFP-DD800	1	QSFP-DD800	1	2
QSFP-DD800	1	OSFP	1	2
OSFP	1	OSFP	1	2

Insert new annexes Annex 163A and Annex 163B corresponding to Clause 163 as follows:

Annex 163A

(normative)

Specification methods for electrical interfaces

163A.1 Overview

This annex defines measurement methods and test points for backplane and chip-to-chip interfaces. The specific methods defined pertain to the requirements for the transmitter steady-state voltage and pulse peak, and for transmitter and receiver effective return loss (ERL).

The methods specified in this annex define normative measurements at test points that are implementation dependent. Determination of whether the device under test meets requirements is accomplished using the following steps:

- Scattering parameters for the test fixture or replica are measured and reference values for steady-state voltage (v_f), linear fit pulse peak voltage (v_{peak}), and ERL are computed using the methods defined in 163A.3.1 and 163A.4.1.
- Measured values for transmitter v_f and v_{peak} are obtained using the method defined in 162.9.4.1.2. Measured values for transmitter and receiver ERL are obtained using the method described in 93A.5.
- The difference between measured and reference values are computed using the methods defined in 163A.3.2. and 163A.4.1.2.

The methods defined above are summarized in Figure 163A–1.

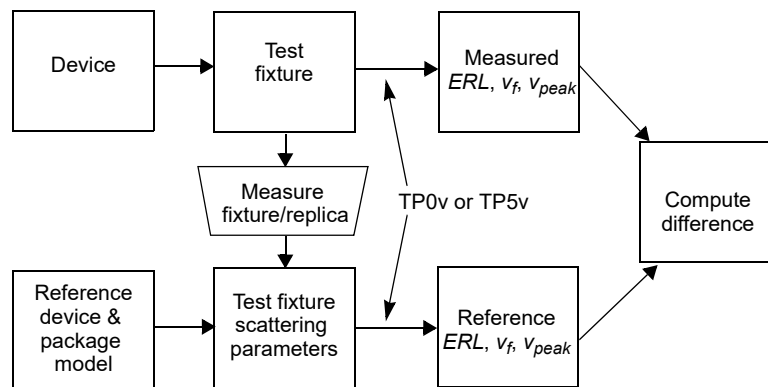


Figure 163A–1—Measurement method for transmitter reference steady-state voltage, pulse peak, and ERL

163A.2 Test fixture

The transmitter test fixture is between test points TP0 and TP0v as shown in Figure 163A–2. The receiver test fixture is between test points TP5v and TP5 as shown in Figure 163A–4. Test fixture requirements are specified by the clause that invokes this method.

163A.3 Test methods for transmitters

Methods for determining the reference parameter values are specified in 163A.3.1. Parameter measurements and determination of the differences between measured and reference parameters are described in 163A.3.2. Specific pass/fail requirements for each parameter are specified by the clause that invokes these methods. An example test fixture and its reference values are provided in 163B.3.

163A.3.1 Reference parameter determination

The methods for obtaining the v_f , v_{peak} , and ERL reference parameters using the measured scattering parameters and the reference transmitter and package models are defined below, and are outlined in Figure 163A–2.

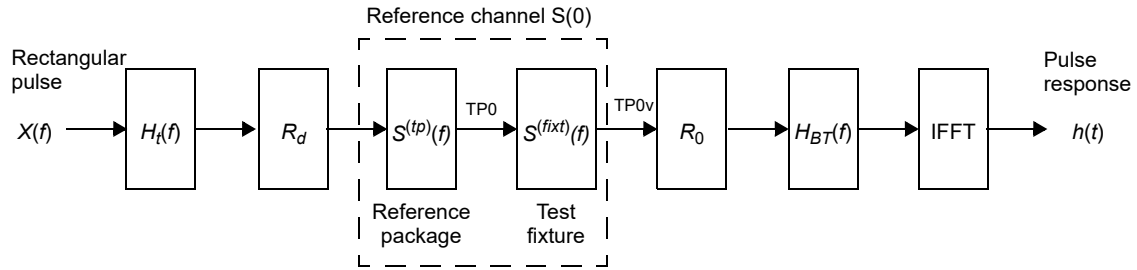


Figure 163A–2—Configuration for transmitter reference steady-state voltage, pulse peak, and ERL

Measure the scattering parameters of the TP0 to TP0v channel for the device and lane under test, $S^{(fixt)}$, using the method specified in 93A.1.1.

Obtain the scattering parameters of the reference channel, $S^{(0)}$, using Equation (163A–1), where the cascade() function is defined in 93A.1.2. The scattering parameters for the reference package, $S^{(tp)}$, are determined using the method in 93A.1.2, with electrical characteristics specified in the clause that invokes this method.

$$S^{(0)} = \text{cascade}(S^{(tp)}, S^{(fixt)}) \quad (163A-1)$$

163A.3.1.1 Steady-state voltage and pulse peak reference values

Calculate the voltage transfer function, $H_{21}(f)$, from the scattering parameters of the reference channel, $S^{(0)}$, using Equation (93A–18) where Γ_1 is given by Equation (93A–17) and Γ_2 is set to 0. In Equation (93A–17), the single-ended reference resistance, R_0 , is set to 50 Ω and the single-ended termination resistance, R_d , is specified by the clause that invokes this method. If the invoking clause lists more than one set of reference package parameters, the calculation is performed with the longest package trace length.

Calculate the voltage transfer function for the full signal path, $H^{(0)}(f)$, using Equation (163A–2).

$$H^{(0)}(f) = H_t(f)H_{21}(f)H_{BT}(f) \quad (163A-2)$$

where

$H_t(f)$	is calculated using Equation (93A–46) with T_r specified by the clause that invokes this method
$H_{BT}(f)$	is calculated using Equation (52–2) with f_r specified by the clause that invokes this method

Obtain the output pulse response, $h(t)$, as defined in 93A.1.5, with $H^{(0)}(f)$ from Equation (163A–2), where A_v and f_b are specified by the clause that invokes this method.

The reference pulse response peak, $v_{peak}^{(ref)}$, is the maximum value of $h(t)$. From the output pulse response calculate the reference value for the transmitter output steady state voltage, $v_f^{(ref)}$, using Equation (163A–3). The values for parameters N_v , M , and D_p are provided by the clause that invokes this method.

$$v_f^{(ref)} = \frac{1}{M} \sum_{i=1}^{MN_v} h\left(t_{max} + \left(\frac{i}{M} - D_p - \frac{1}{2}\right)T_b\right) \quad (163A-3)$$

where

M	is the number of samples per unit interval
t_{max}	is the time where $h(t)$ reaches the maximum value
D_p	is the linear fit pulse delay
T_b	is the unit interval in ps
N_v	represents the number of symbols included in the steady-state voltage calculation

163A.3.1.2 ERL reference value

The reference reflection coefficient at TP0v is given by Equation (93A–7) where $s_{22}^{(x)}$ is Γ_1 as defined by Equation (93A–17) and $s_{ij}^{(y)}$ are the components of the scattering matrix of the reference channel $S^{(0)}$. In Equation (93A–17), the single-ended reference resistance, R_0 , is set to 50 Ω and the single-ended termination resistance, R_d , is specified by the clause that invokes this method. The reference pulse time-domain reflection (PTDR) response is computed from the reference reflection coefficient at TP0v using Equation (93A–58) and Equation (93A–59). The reference ERL value is determined from the reference PTDR response using the method in 93A.5 with parameters specified by the clause that invokes this method. If the invoking clause lists more than one set of reference package parameters, the ERL calculation is performed with each set, and the minimum value is used as the reference value.

163A.3.1.3 Transition time reference value

The method for obtaining the reference transition time using the measured test fixture scattering parameters and the reference transmitter and package models is defined below, and is outlined in Figure 163A–3.

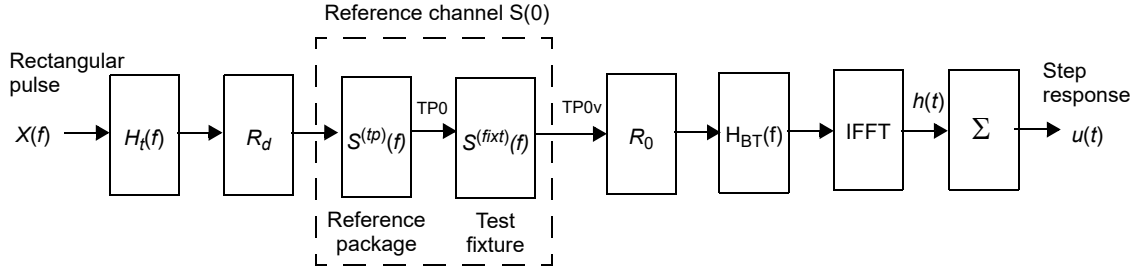


Figure 163A-3—Configuration for transmitter reference transition time

- Obtain the output pulse response, $h(t)$, as defined in 93A.1.5, with $H^{(0)}(f)$ from Equation (163A-2), where A_v and f_b are specified by the clause that invokes this method.
- Obtain the output step response, $u(t)$, using Equation (163A-4).
- From the output step response, find the time to reach 20% and 80% of the reference steady-state voltage, $v_f^{(ref)}$, as T_{20} and T_{80} , respectively.
- From T_{20} and T_{80} , calculate the reference transition time, $T_r^{(ref)}$, using Equation (163A-5).

$$u(t) = \sum_{i=0}^{\infty} h(t - i \times T_b) \quad (163A-4)$$

$$T_r^{(ref)} = T_{80} - T_{20} \quad (163A-5)$$

where

$T_r^{(ref)}$	is the reference transition time
$u(t)$	is the output step response
T_{80}	is the time to reach 80% of the reference steady-state voltage
T_{20}	is the time to reach 20% of the reference steady-state voltage
T_b	is the unit interval in ps
$v_f^{(ref)}$	is the reference steady-state voltage calculated by Equation (163A-3)

If the invoking clause lists more than one set of reference package parameters, the calculation in Equation (163A-5) is performed with the longest package trace length.

163A.3.2 Difference parameter determination

This subclause defines the parameters that quantify the difference between measured values and reference values.

163A.3.2.1 Steady-state voltage and pulse peak ratio

The measured linear fit pulse peak, $v_{peak}^{(meas)}$, and steady-state voltage, $v_f^{(meas)}$, are calculated from a linear fit pulse response, $p(k)$, obtained from measurement at TP0v with the transmit equalizer set to preset 1 (no equalization) using the method in 163A.3.1.1. $v_{peak}^{(meas)}$ is the peak value of $p(k)$. $v_f^{(meas)}$ is defined by

Equation (163A–7) where $p(i)$ and M are defined in 162.9.4.1.1 and N_v is set by the clause that invokes this method.

The difference steady-state voltage, dv_f , is calculated using Equation (163A–6).

$$dv_f = v_f^{(meas)} - v_f^{(ref)} \quad (163A-6)$$

$$v_f^{(meas)} = \sum_{i=1}^{M \times N_v} \frac{p(i)}{M} \quad (163A-7)$$

The difference pulse peak ratio, dR_{peak} , is calculated as the difference between measured pulse peak ratio, $R_{peak}^{(meas)}$, and reference pulse peak ratio, $R_{peak}^{(ref)}$, using Equation (163A–8), Equation (163A–9), and Equation (163A–10).

$$dR_{peak} = R_{peak}^{(meas)} - R_{peak}^{(ref)} \quad (163A-8)$$

$$R_{peak}^{(meas)} = \frac{v_{peak}^{(meas)}}{v_f^{(meas)}} \quad (163A-9)$$

$$R_{peak}^{(ref)} = \frac{v_{peak}^{(ref)}}{v_f^{(ref)}} \quad (163A-10)$$

where

$v_f^{(ref)}$ is defined in 163A.3.1.1
 $v_{peak}^{(ref)}$ is defined in 163A.3.1.1

163A.3.2.2 Difference ERL

The difference ERL, $dERL$, is calculated using Equation (163A–11).

$$dERL = ERL^{(meas)} - ERL^{(ref)} \quad (163A-11)$$

where

$ERL^{(ref)}$ is ERL reference value in dB defined in 163A.3.1.2
 $ERL^{(meas)}$ is the ERL calculated from measurement in dB as defined in 93A.5

163A.4 Test methods for receivers

Methods for determining the reference parameter values are specified in 163A.4.1. Parameter measurements and determination of the differences between measured and reference parameters are described in 163A.4.1.2. Specific pass/fail requirements for each parameter are specified by the clause that invokes these methods.

163A.4.1 Reference parameter determination

The methods for obtaining receiver ERL reference parameters using the measured scattering parameters and the reference receiver and package models are defined below, and are outlined in Figure 163A–4.

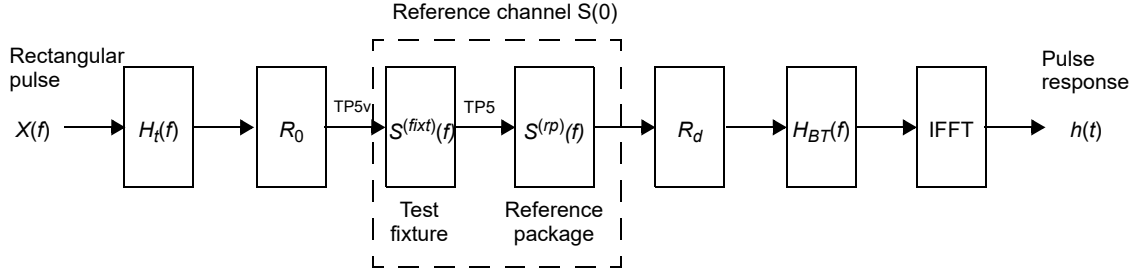


Figure 163A–4—Configuration for receiver ERL

Measure the scattering parameters of the TP5v to TP5 channel for the device and lane under test, $S^{(fixt)}$, using the method specified in 93A.1.1.

Obtain the scattering parameters of the reference channel, $S^{(0)}$, using Equation (163A–12), where the cascade() function is defined in 93A.1.2. The scattering parameters for the reference package, $S^{(rp)}$, are determined using the method in 93A.1.2, with electrical characteristics specified in the clause that invokes this method.

$$S^{(0)} = \text{cascade}(S^{(fixt)}, S^{(rp)}) \quad (163A-12)$$

163A.4.1.1 ERL reference value

The reference reflection coefficient at TP5v is given by Equation (93A–7) where $s_{22}^{(x)}$ is Γ_1 as defined by Equation (93A–17) and $s_{ii}^{(y)}$ are the components of the scattering matrix of the reference channel $S^{(0)}$. In Equation (93A–17), the single-ended reference resistance, R_0 , is set to $50\ \Omega$ and the single-ended termination resistance, R_d , is specified by the clause that invokes this method. The reference pulse time-domain reflection (PTDR) response is computed from the reference reflection coefficient at TP5v using Equation (93A–58) and Equation (93A–59). The reference ERL value is determined from the reference PTDR response using the method in 93A.5.2 with T_{fx} set to 0 and other parameters specified by the clause that invokes this method. If the invoking clause lists more than one set of reference package parameters, the ERL calculation is performed with each set, and the minimum value is used as the reference value.

163A.4.1.2 Difference ERL

Measure the ERL using the method defined in 93A.5.

The difference ERL, $dERL$, is calculated using Equation (163A–11).

Annex 163B

(informative)

Example test fixture

163B.1 Overview

An example test fixture meeting the requirements for TP0v is described in this annex.

163B.2 Characteristics

This example test fixture is defined using the PCB trace model in 162.11.7.1, with $z_p = 71$ mm, and parameter values in Table 162–21, with the exception that C_0 and C_1 are both 0. This results in a TP0 to TP0v insertion loss of 2.8 dB at 26.5625 GHz. The reference values are calculated for the transmitter characteristics of Clause 163. The reference transmitter device and package model uses the parameter values $T_r = 7.5$ ps, $f_r = 0.75 \times f_b = 39.8438$ GHz, $z_p = 31$ mm, and $A_v = 0.413$ V. The values of v_{peak} and v_f are calculated with $f_b = 53.125$ GBd and $N_v = 200$.

Although clauses using the TP0v methodology may require the ERL reference value to be calculated at more than one package length, only one is shown here.

The insertion loss of the example test fixture is approximated by Equation (163B–1) which is illustrated in Figure 163B–1.

$$ILdd(f) = 0.235616\sqrt{f} + 0.059147f \quad 0.05 \leq f \leq 53.125 \quad (163B-1)$$

where

$ILdd(f)$ is the insertion loss at frequency f in dB
 f is the frequency in GHz

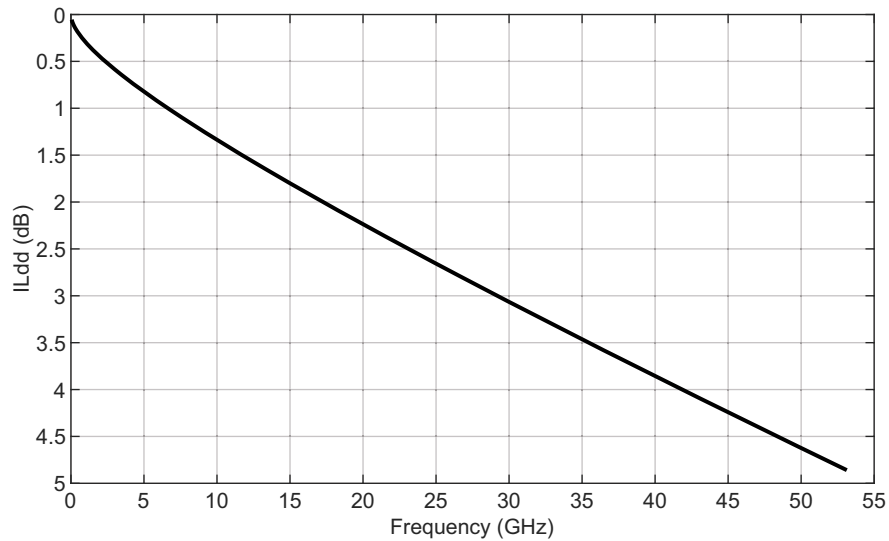


Figure 163B-1—Example test fixture insertion loss

163B.3 Reference Values

For this test fixture, the reference values determined according to the methodology in 163A.3 using the parameters supplied in Clause 163 are listed in Table 163B-1.

Table 163B-1—Summary of transmitter reference values at TP0v

Parameter	Reference	Value	Units
Effective return loss, $ERL^{(ref)}$	163A.3.1.2	12.95	dB
Transmitter steady-state voltage, $v_f^{(ref)}$	163A.3.1.1	0.409	V
Transmitter linear fit pulse peak, $v_{peak}^{(ref)}$	163A.3.1.1	0.237	V
Transmitter pulse peak ratio, $R_{peak}^{(ref)}$	163A.3.2.1	0.580	—

RAISING THE WORLD'S STANDARDS

Connect with us on:



Twitter: twitter.com/ieeesa



Facebook: facebook.com/ieeesa



LinkedIn: linkedin.com/groups/1791118



Beyond Standards blog: beyondstandards.ieee.org



YouTube: youtube.com/ieeesa

standards.ieee.org

Phone: +1 732 981 0060